AERODYNAMIC HEATING AND THERMAL PROTECTION SYSTEMS

Edited by
L. Fletcher

Progress in Astronautics and Aeronautics

Martin Summerfield

Series Editor-in-Chief

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AERODYNAMIC HEATING AND THERMAL PROTECTION SYSTEMS

Edited by
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PREFACE

The interdisciplinary field of thermophysics evolved during the rapid growth of space technology. Initially, this field dealt primarily with the heating and cooling problems of space vehicles and their components. More recently, thermophysics has expanded to include such diverse yet related areas as detection of air and water pollution, energy collection, conversion and storage, resource assessment by satellite, and thermal protection of space vehicles. Of particular importance is the application of thermophysics to energy conversion and the potential for applying space technology to the solution of the expanding energy dilemma.

As space exploration continues, more refined investigations of outer planet environments are required. The fundamentals of thermophysics are essential to a better understanding of the thermal environments of the outer planets and the effects of those environments on planetary probes. The radiation heating problems experienced by planetary probes, and the thermal protection systems designed to deal with these problems, require thorough investigation in order to permit successful space missions.

With the increasing interest in the utility of outer space and the near completion of the Space Shuttle for orbital flight, many new and interesting ideas have been proposed and investigated. Improved methods for predicting and analyzing the aerothermal environment have been proposed, and techniques for the assessment and evaluation of rocket exhaust plume radiation have been established. In the area of thermal protection systems, new materials have been developed for heat shields and nosetips, and the experimental techniques used in evaluation of these materials have been refined. Although recent years have seen significant advances in these areas, many remaining problems invite continued investigation.

The present volume contains a selection of recent studies dealing with aerodynamic heating and thermal protection systems. The papers were drawn from the AIAA 15th Aerospace Sciences Meeting in Los Angeles, Calif. in January 1977 and the AIAA 12th Thermophysics Conference in Albuquerque, N. Mex. in June 1977, and were revised and updated especially for this volume. They have been grouped into three chapters: aerothermal environment, plume radiation, and thermal protection systems.

Chapter I consists of eight papers related to planetary aerothermal environments and aerodynamic heating. The basic concepts and new developments presented in this chapter will be extremely useful in refining current design requirements and analyses of aerodynamic heating. In the first paper, Moss, Zoby, and Sutton define the entry aerothermal environment for four Venus probes through use of both detailed and approximate flow-field analyses, identifying radiation rates and associated absorbing characteristics of attendant boundary layers. The second paper, by Scott, Murray, and Milhoan, reports on the aerodynamic heating of the cove seal area of the Space Shuttle elevon wing junction for simulated Shuttle flight and re-entry. The base heating of a two-dimensional bluff body in supersonic flow is evaluated by Inoue and Page by means of the component analysis method, incorporating the heating between the recirculating flow and base wall and across the mixing layer in the wake. Nagib and Hodson propose the use of vortices induced by wakes as a means for augmenting convective heat transfer from aerodynamic bodies. Hung presents simplified methods for the prediction of peak interference heating for both laminar and turbulent flow which can be applied directly to the Space Shuttle configuration. Nicolet, Waterland, and Kendall describe a numerical method for predicting the heating events of hypervelocity entry into outer planet atmospheres which incorporates radiation-coupled flow fields. Park and Fujiwara present a study of the radiative heat flux from a hot inviscid shock layer composed of ablation-product gases from a silica volume reflecting heat shield, identifying regions of radiation blockage which might occur during entry into the Jupiter atmosphere. The last paper in the chapter, by Fogaroli and Laganelli, presents the development of a correlation for ascertaining the effects of Mach number, wall temperature, and mass injection on heat blockage for turbulent flow over surfaces with negligible pressure gradient.

Rocket plume radiation is the subject of the three papers in Chapter II. The first paper, by Thorpe and Miller, utilizes state-of-the-art afterburning models coupled with infrared (IR) radiation model predictions for comparison with experimental data taken on a subscale hydrocarbon/oxygen-fueled rocket motor. The second paper, by Harwell, Jackson, and Poslajko, extends the measurement of the spatial distribution of infrared radiation in rocket exhausts, reported in Volume 56 of this series, and compares these data with analytical models currently available. The last paper, by Sukanek and Davis, reports on a modification of the NASA band model formulation for missile base heating predictions which allows the calculation of radiance, radiant intensity, and transmission from a gas, gaseous mixtures, or exhaust plumes.

Chapter III consists of ten papers concerned with thermal protection systems for outer planet probes. Included are papers related to ablation testing and material performance and a number of papers on the characteristics of nosetips for atmospheric entry. The first paper, by Kemp, deals with the development of theoretical models for melt-through of metal plates under intense one-dimensional heating and tangential airflow. In the second paper, Kesselring (with co-workers Maurer, Suchsland, Hartman, and Peterson) reports on arc heater tests of candidate outer planet entry probe heat-shield materials to evaluate surface thermochemistry and in-depth thermal response. Arc heater tests also were conducted by Auerbach to study the ablation performance of solid, porous, and copper-infiltrated tungstens, tantalum, and other materials and to establish the recession rates in laminar and turbulent flow. The paper contributed by Medford deals with the development of a method for predicting the oxidation distribution in reinforced carbon-carbon heat shield materials in Earth atmospheric entry for use in the analysis of Space Shuttle leading edges.

The next six papers in Chapter III deal with various investigations of nosetips. The one by Dirling examines the effects of surface roughness, re-entry trajectory, and vehicle configuration on expected shape development of asymmetric nosetips in atmospheric entry. An analysis of film-cooled nosetips in a particle erosion environment is contributed by Grabow. This analysis includes the effects of particle impingement heating, convective heating augmentation, and coolant film boiling. In the next paper, Nestler presents the results of recent high-pressure arc tests of three-dimensional woven carbon-carbon nosetips, identifying two distinct types of nose sharpening. The paper by Donohoe, Blackstock, and Keyes reports on wind-tunnel tests of the GASJET nosetip with an evaluation of the measurement technique used in establishing nosetip recession and erosion characteristics. In the paper by Sherman and Schutzler, the design of a segmented tungsten nosetip for use on high-performance, all-weather flight vehicles is presented, with details on the ablation and thermostructural performance of the segmented design. The last paper deals with the optimization of nosetip shape for minimization of the total trajectory heat transfer for fixed fineness ratio and re-entry parameters. In this paper, Baker and Kramer develop analytic expressions for total integrated heat transfer to a nosetip in order to determine a suitable nosetip shape for minimum transpiration coolant requirements.

Throughout the planning and preparation of this volume, a number of people have provided guidance and assistance. Dr. Allie M. Smith, the chairman of the AIAA Thermophysics Technical Committee in 1977, provided advice and consultation. Dr. Robert K. MacGregor was responsible for organizing the thermophysics sessions at the AIAA 15th Aerospace Sciences Meeting and assisted in reviewing those papers for this volume. As General Chairman of the AIAA 12th Thermophsyics Conference, from which a majority of the papers in this volume were selected, I was supported by Dr. Walter B. Olstad as Technical Program Chairman. He also was a most helpful Session Chairman and assisted me by handling the review of many of the papers. I am also grateful to Dr. Martin Summerfield, Editor-in-Chief of the AIAA *Progress in Astronautics and Aeronautics* series who provided thoughtful advice and consultation, and to Miss Ruth F. Bryans, Administrator of Scientific Publications for AIAA, for invaluable assistance and support in the editing of the volume.

Leroy S. Fletcher
January 9, 1978

EDITORIAL COMMITTEE FOR VOLUME 59

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AEROTHERMAL ENVIRONMENT FOR THE PIONEER VENUS MULTIPROBE MISSION

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Abstract

In December 1978, four Pioneer Venus probe spacecraft are scheduled for almost simultaneous entry into the Venusian atmosphere at widely dispersed points about the planet. In this study, both detailed and approximate flowfield analyses are used to define the entry aerothermal environment for the forebody of each of the four probes. The results show that approximate analyses can be used to predict inviscid radiative and laminar convective heating rates with acceptable accuracy. However, the radiative heating rates obtained with inviscid analyses are significantly greater than those obtained with a nonablating viscous-shock-layer (VSL) analysis, because the VSL analysis includes a strongly absorbing boundary layer. Also, the results show that the radiative heating is sensitive to small variations in atmospheric gas composition, whereas the convective heating is not affected. With carbon-phenolic injection, the convective heating is reduced substantially, whereas the overall radiative heating reduction is very small. Most of the radiative blockage occurs in the atomic line transitions which are significant only in the stagnation region.

Nomenclature

B' = mass addition parameter $B' = \dot{m}/N_{St}$
H = total enthalpy

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H_r = recovery enthalpy
$\dot{m}$ = nondimensional mass injection $(\rho v)_w/(\rho U)_\infty$
n = coordinate measured normal to the body
N_{St} = Stanton number
p = pressure
q_c = convective heat flux to the wall
q_r = radiative heat flux to the wall
r = radial distance from centerline of symmetry to the body surface
R_b = base radius
R_N = nose radius
s = coordinate measured along the body surface
T = temperature
u = velocity component tangent to body surface
U_∞ = freestream velocity
γ = entry angle
δ = boundary-layer thickness
μ = molecular viscosity
ρ = density

Superscripts

j = 0 for plane flow; 1 for axisymmetric flow

Subscripts

A = adiabatic
e = boundary-layer edge
NA = nonadiabatic
0 = stagnation point
s = shock
w = wall
∞ = freestream
ν = radiation frequency

Introduction

To define the nature, composition, and dynamics of the atmosphere of the planet Venus, four probe spacecraft are scheduled to enter the planet's atmosphere at widely dispersed points about the planet in December 1978.[1] The flowfield and heating environment experienced by the forebody of each of the four probes during atmospheric entry are the subject of this paper. The probe forebodies are spherically capped, 45-deg half-angle cones. The aerothermal environment encountered by the probes will be unique for U.S. flights in that the heating will consist of both convective and radiative contributions, each of which will be comparable in magnitude. This aerothermal environment cannot be duplicated[2] completely in any available ground facility. Consequently, theoretical analyses must be relied upon to provide the final predictions of heating.

References 3-6 are examples of the application of both detailed[3,4] and approximate[5,6] analyses to predict the heating environment for the Pioneer Venus probes. While these studies were being made, a more sophisticated flowfield analysis,[7,8] known as the viscous-shock-layer (VSL) analysis, was developed for radiative viscous flows. The VSL analysis provides a direct means of coupling shock-induced vorticity, radiation transfer, and mass injection, all of which are significant for the Pioneer Venus probes. Furthermore, as the Pioneer Venus project evolved, changes in the Pioneer Venus mission (probe configurations, ballistic coefficients, and trajectories) have occured. Therefore, the present study applies the state-of-the-art in both detailed and approximate flowfield analyses to examine the aerothermal environment for the Pioneer Venus probes entering along the trajectories specified by the Pioneer Venus Project Office, NASA Ames Research Center. Also, the present study supports the heat-shield experiments aboard the probes in that particular attention is focused on the heating environment at points on the probe surface where experimental devices are located. Through comparison of measurements and predictions, the Pioneer Venus mission provides a much-needed opportunity to validate radiative flowfield prediction techniques with flight data.

In this study, the nonablating heating environment for each of the four probes is considered. Results are obtained with both approximate and detailed analyses to assess the applicability of approximate analyses for predicting both radiative and convective heating. Also, the sensitivity of the heating, both convective and radiative, to small variations in atmospheric gas composition is examined. Finally, the effect of carbon-phenolic injection on the stagnation and downstream laminar heating is considered.

Analyses

Three different analyses are used in this study for calculating the heating and flowfield environment. Two are detailed radiative flowfield analyses, one viscous and one inviscid, whereas the third is an approximate radiative inviscid analysis with a laminar convective heating prediction capability.

VSL Analysis

The VSL analysis[7,8] includes a detailed description of the radiative transport,[9] equilibrium chemistry, and transport properties. The analysis provides solution capabilities for ablation injection at the surface. Also, the flow may be either laminar or turbulent. The turbulence is modeled with a two-layer eddy-viscosity model that is described in Ref. 8. The governing equations are solved as a parabolic set of equations using an implicit, finite-difference, numerical procedure. Consequently, the VSL analysis provides a direct means of accounting for interactions between the inviscid and viscous flow regions due to radiative transfer, inviscid entropy layer swallowing, and mass injection.

For the solution coupled with ablation injection, the ablation process is assumed to be quasi-steady, and the surface temperature is the sublimation temperature of the ablator surface. The surface temperature was calculated with a chemical equilibrium analysis that accounts for the gas-phase species at the surface and the elemental composition of the ablator. The elemental mass fractions of the carbon-phenolic ablator considered in this study are 0.90 carbon, 0.07 oxygen, and 0.03 hydrogen. Figure 1 depicts the surface temperature of the ablator as a function of surface pressure and ablator composition at the surface. Note that, when the gas compo-

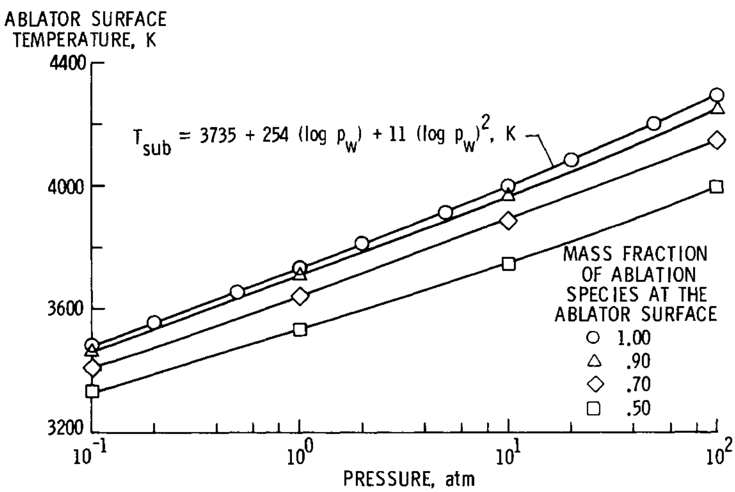

Fig. 1 Surface temperature for the carbon-phenolic ablator (mass fractions are 0.90 carbon, 0.07 oxygen, and 0.03 hydrogen).

sition adjacent to the ablator surface is the same as that of the injected species, the sublimation temperature is

$$T_{sub} = 3735 + 254\ (\log p_w) + 11(\log p_w)^2\ \text{K} \tag{1}$$

where p_w is the wall pressure in atmospheres. The coupled mass injection rate is equal to the surface heating rate (convective plus net radiative) divided by the enthalpy of the species at the surface. The surface emissivity is assumed to be 0.8.

Detailed Inviscid Analysis

The detailed inviscid analysis developed by Sutton[3] is a second-order, time-asymptotic solution method. Detailed radiative transport[9] and equilibrium chemistry are included for the direct solution of the radiative inviscid flow over either an axisymmetric or two-dimensional body.

Approximate Analysis

The approximate radiative inviscid analysis used herein is that developed by Falanga and Olstad[5] for conducting studies of Venusian entries. This is an inverse analysis based upon the inviscid flow technique of Maslen[10] and modified for radiative transport by Olstad.[11] By using a 37-step absorption coefficient model for a high-temperature CO_2-N_2 mixture and thermodynamic correlations, short computational times are possible. In fact, this analysis requires only about 1/400 the computer time required by the more detailed VSL analysis for a Venusian entry calculation. Thus, the analysis can be used easily for defining the radiative inviscid flow over a probe during an entire entry trajectory. At the present time, the thermodynamic correlations and the radiation absorption coefficients incorporated in the computer code for a CO_2-N_2 gas mixture are limited to a 90% CO_2-10% N_2 gas mixture by volume.

During this study, the capability of the approximate analysis was extended to include laminar convective heating predictions. This is accomplished by using approximate heat-transfer expressions in conjunction with the inviscid flow-field solution. The stagnation-point heating was calculated with the empirical equation[12]

$$q_{c,o} = K(H_{r,e,o} - H_{w,o})\ \sqrt{p_o/R_N} \tag{2}$$

where p_o is the stagnation pressure in atmospheres, $H_{r,e,o}$

is the stagnation-point inviscid wall recovery enthalpy, and K is a constant that accounts for the thermodynamic and transport properties of the gas at the wall and external to the boundary layer. The value of K used in this study is computed[12] for the 90% CO_2-10% N_2 gas mixture. Note that this correlation is based on results that do not account for radiation transfer.

The equation used for calculating heating rate distributions is

$$\frac{q_c}{q_{c,o}} = \frac{r^j \left(\frac{p}{p_o}\right)\left(\frac{u_e}{U_\infty}\right)}{\sqrt{2^j + 1 \left[\frac{d(u_e/U_\infty)}{ds}\right]_o \int_o^s \left(\frac{p}{p_o}\right)\left(\frac{u_e}{U_\infty}\right) r^{2j} ds}}\ \phi \tag{3a}$$

With the exception of the parameter ϕ, this is the equation presented by Lees.[13] The parameter ϕ is the ratio of the local enthalpy potential to that of the stagnation-point enthalpy potential:

$$\phi = (H_{r,e} - H_w)/(H_{r,e,o} - H_{w,o}) \tag{3b}$$

The Prandtl number used to calculate the recovery enthalpy is 0.70.

A boundary-layer growth expression for laminar flow also is incorporated in the analysis to account for the effects of entropy layer swallowing. The equation used to define the boundary-layer thickness is similar to that presented by Libby[14]:

$$\delta = 0.606\left(\int_o^s \mu_e \rho_e u_e r^{j+1} ds\right)^{1/2} \Big/ (0.117\ \rho_e u_e r) \tag{4}$$

The constant 0.117 relates the momentum thickness to the boundary-layer thickness.[14] The constant 0.606 is based on the investigation of Marvin et al.[15] The limiting form of Eq. (4) is used at the stagnation point.

Results and Discussion

The data presented show radiative and laminar convective heating rates corresponding to a viscous-shock-layer (VSL) analysis, a detailed inviscid analysis, and an approximate analysis for the Pioneer Venus probe. Both laminar and turbulent solutions are presented for a nonablating surface, and laminar solutions are presented for an ablating surface. The flowfield and heating results are presented for the forebody of each of the four Pioneer Venus probes. The calculated results are for the probe shapes and trajectory data specified by the Pioneer Venus Project Office, NASA Ames Research Center.

The forebody for both the large and the three small probes is a 45-deg spherically blunted cone. The large probe has a nose radius of 0.363 m, a base-to-nose-radius ratio of 1.958, and a ballistic coefficient of 179.5 kg/m^2. Each of the three small probes has a nose radius of 0.191 m, a base-to-nose-radius ratio of 2.0, and a ballistic coefficient of 180.3 kg/m^2. The entry velocity for the four probes is 11.6 km/sec at an altitude of 200 km, whereas the entry angles are -33.8 deg for the large probe and -67.7, -43.0, and -22.7 deg for the small probes. Figure 2 presents the entry trajectories in terms of velocity histories where zero time occurs at an altitude of 200 km. The atmospheric model used for the trajectory calculations is that presented in Ref. 16. Reference 17 presents a detailed summary of the freestream conditions and stagnation streamline results for the four probes.

Comparison of Analyses

In this section, the influence of a strongly absorbing boundary layer upon surface radiative heating rates is

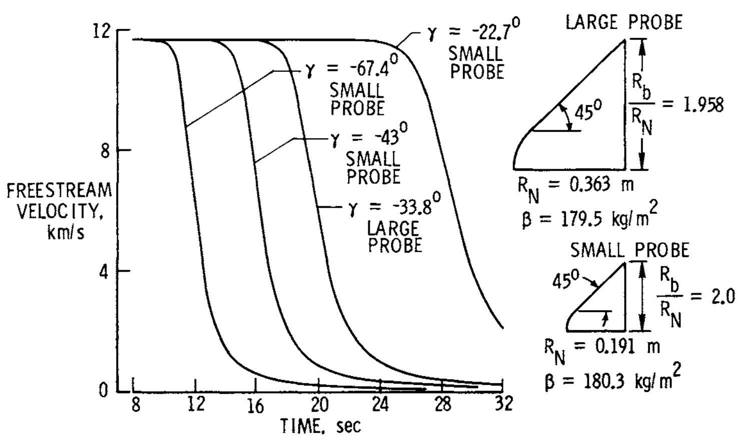

Fig. 2 Entry trajectories for Venus probes.

demonstrated, and some of the difficulties associated with the application of integral boundary-layer analyses to determine the convective heat transfer for radiating blunt-body flowfields are discussed. The large-probe configuration is considered, and the atmospheric gas composition is assumed to be 90% CO_2 and 10% N_2 by volume.

To demonstrate the influence of an absorbing boundary layer upon the radiative heat transfer to the surface, the surface radiative heating rate obtained with the VSL analysis for the large probe is compared with corresponding solutions obtained with the approximate and the detailed radiating inviscid flowfield analyses. Figure 3 shows the stagnation-point radiative heat-transfer rate histories along the entry trajectory as determined by the three methods. The approximate inviscid analysis shows excellent agreement with the detailed inviscid analysis except for the last radiative heating rate value on the trajectory. The radiative heating rates obtained with the VSL analysis are, in general, less than those obtained with the inviscid analyses. This is particularly true near peak radiative heating (18 sec), where the inviscid radiative heating values are 30% greater than the corresponding VSL values. The differences in the VSL and the inviscid results are primarily because the VSL solution includes the strongly absorbing boundary layer, whereas the

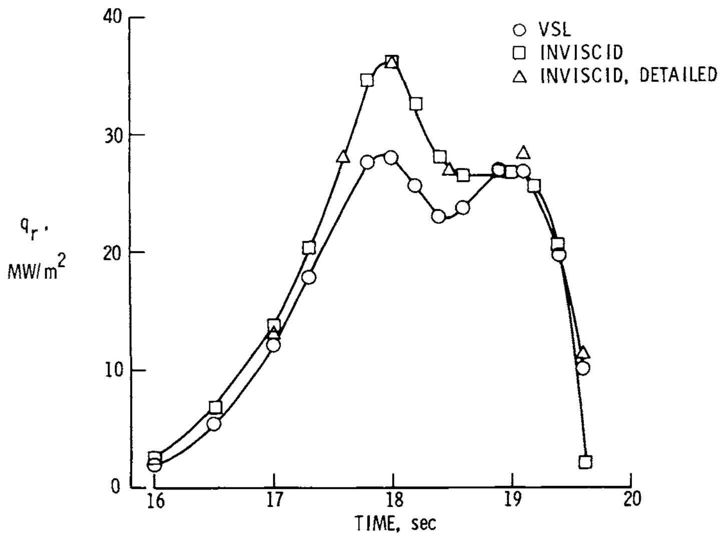

Fig. 3 Comparison of stagnation-point radiative heating results for the large probe (90-10 atm).

inviscid analyses do not. This effect is depicted clearly in Fig. 4, where the net radiative flux profiles calculated with the VSL and the detailed inviscid analyses are shown at the time of peak radiative heating (18 sec). (The inviscid results were adjusted to account for the flux reradiated from the surface.) When comparisons of the flux profiles are made at other entry times, one finds that the extent of the radia-

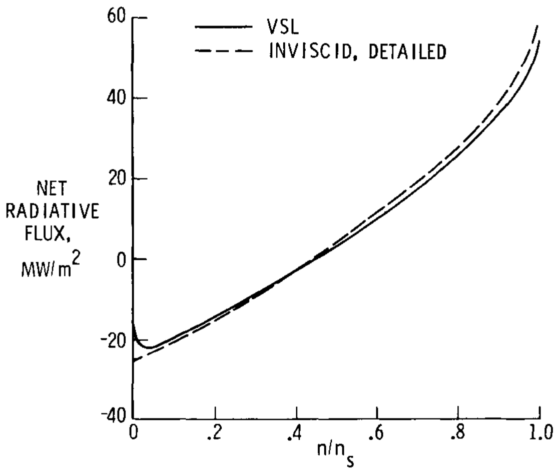

Fig. 4 Comparison of stagnation net radiative flux profiles for the large probe (time = 18 sec, 90-10 atm).

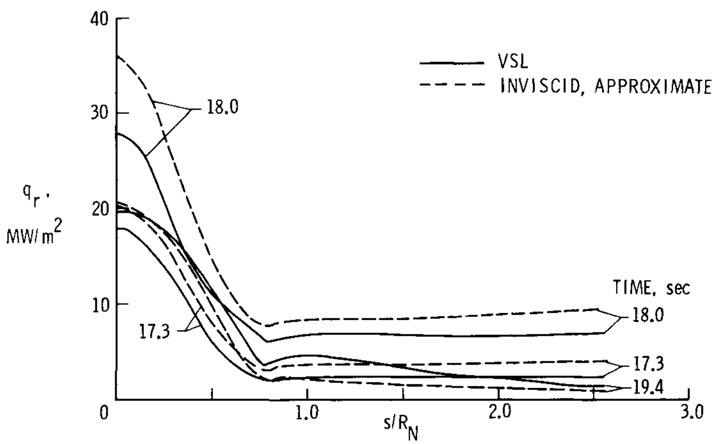

Fig. 5 Comparison of radiative heating distributions for the large probe (90-10 atm).

tion absorption within the boundary layer is very much dependent upon the freestream conditions. These results are consistent with the results of a previous study[4] of Venusian heating where it was shown that, as the velocity for peak radiative heating increases, the unblown boundary layer absorbs more of the radiation.

Figure 5 presents a comparison of the radiative heating distributions as calculated with the VSL analysis and the approximate inviscid analysis. The results are for the large probe at three entry times. The results show that the heating on the conical flank is less than 25% of the corresponding stagnation value. Also, it is clearly evident that substan-

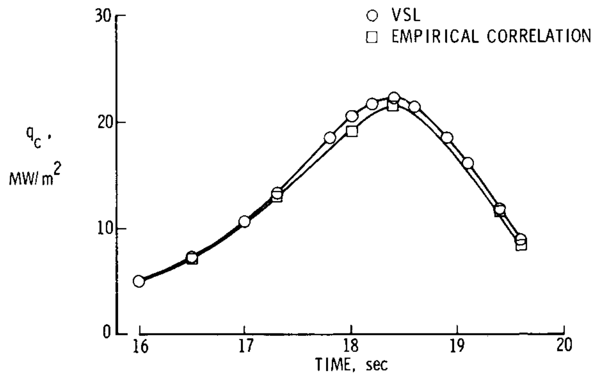

Fig. 6 Comparison of stagnation-point convective heating rates for the large probe (90–10 atm).

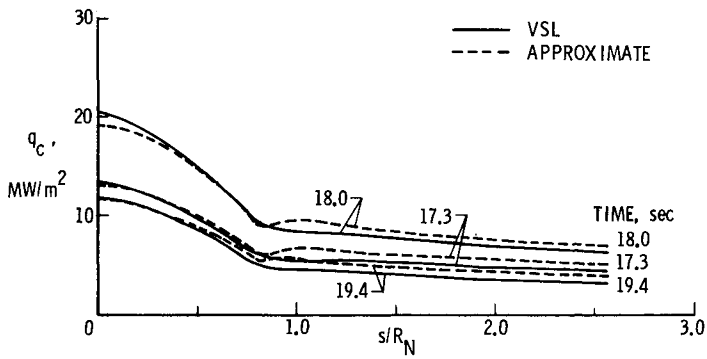

Fig. 7 Comparison of convective heating-rate distributions for the large probe (90–10 atm).

tial differences exist between the VSL solution and the inviscid solution. The differences are due primarily to the absorbing boundary layer, which is accounted for in the VSL solution. For the entry time of 19.4 sec, the absorption within the boundary layer is smaller than that for the earlier entry times. Even so, the agreement between the two solutions is not very good during the latter portion of the radiative heat pulse.

Comparisons of the convective heating-rate values for the VSL analysis and the approximate analysis are presented in Figs. 6 and 7 for the large probe. The approximate convective heating values are obtained by using the stagnation-point empirical correlation given by Eq. (2) and the convective heating distribution given by Eq. (3). The boundary-layer edge values used in these equations are the nonadiabatic inviscid results for the streamline located a distance δ [Eq. (4)] from the wall. For these stagnation conditions, the difference between the approximate solutions and the VSL solutions is less than 7%. The convective heating-rate distributions are in good agreement on the spherical portion of the probe, and the overall agreement is acceptable, as shown in Fig. 7. The same general agreement was obtained for the three small probes (not presented). Results obtained with the approximate turbulent analysis are not presented because a satisfactory comparison with the VSL results has not been obtained.

The attempt to use an approximate analysis to predict convective heating, either laminar or turbulent, presents

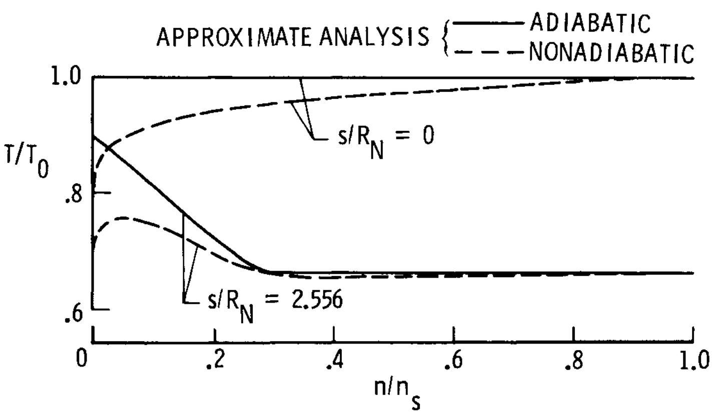

Fig. 8 Radiation coupling effect on inviscid temperature profiles (large probe, 90-10 atm; time = 18 sec).

difficulties for complicated flows such as considered in this study. The significance of radiation cooling, together with the development of the entropy layer, has important implications[18] for the calculation of convective heating rates. The nature of the inviscid flow for the present probe configurations is characterized by large density, temperature, enthalpy, and velocity gradients within the inviscid entropy layer near the body surface. Evidence of the inviscid (approximate solution) temperature and velocity gradients for the large probe (time = 18 sec) is shown in Figs. 8-10. At the aft end of the probe, the adiabatic surface temperature is 35% greater than the temperature outside the entropy layer. When radiation cooling is accounted for (nonadiabatic solution), the temperature, enthalpy, and density values are affected significantly. The influence of radiation cooling on the temperature profiles at the aft end of the probe is shown in Figs. 8 and 9. For the same probe location, the adiabatic and nonadiabatic velocity profiles are approximately the same (Fig. 10). Also, as is shown in Ref. 12, the normal pressure profiles are not influenced by radiation cooling. Consequently, the equations used to calculate the convective heating rates are influenced by radiation cooling primarily through the inviscid recovery enthalpy at the boundary-layer edge.

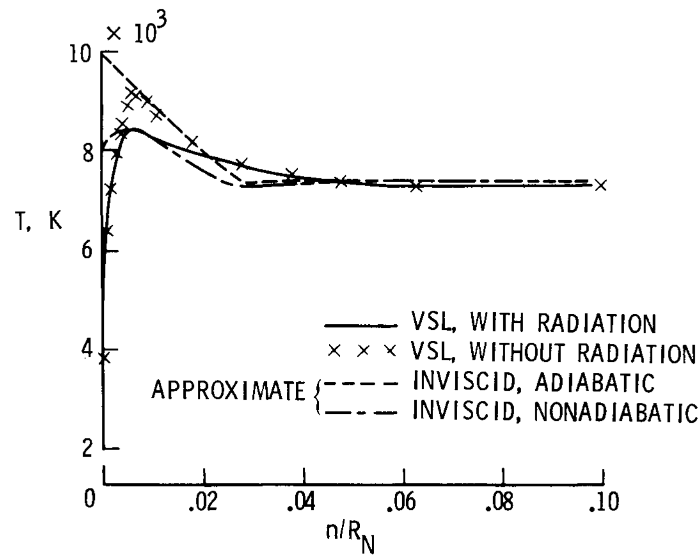

Fig. 9 Comparisons of temperature profiles (large probe, time = 18 sec, s/R_N = 2.556, 90-10 atm).

Figure 9 shows a comparison of the VSL and the approximate inviscid temperature profile at the end of the conical flank. These results are presented both with and without radiation. For the VSL results, the boundary-layer edge location occurs near the location of maximum temperature. Note that the inviscid results are in good agreement with the VSL results at the boundary-layer edge location.

With the approximate convective heating analysis, it also was found that good agreement with the VSL solutions was obtained if the boundary-layer edge conditions were taken to be the inviscid wall conditions computed with the approximate adiabatic inviscid flowfield analysis. With the VSL analysis, the convective heating-rate values (both laminar and turbulent) are essentially the same, whether radiation is or is not included in the calculation. Radiation transfer reduces the enthalpy potential across the boundary layer due to radiation cooling and tends to increase the enthalpy gradient adjacent to the surface due to the radiant energy absorbed in the relatively cool boundary layer near the surface. For the present conditions, the effects of radiation transfer on convective heating tend to counterbalance each other. The extent to which the convective heating rate is influenced by radiation transfer is dependent upon the gas composition and entry conditions. Reference 8 shows that the convective

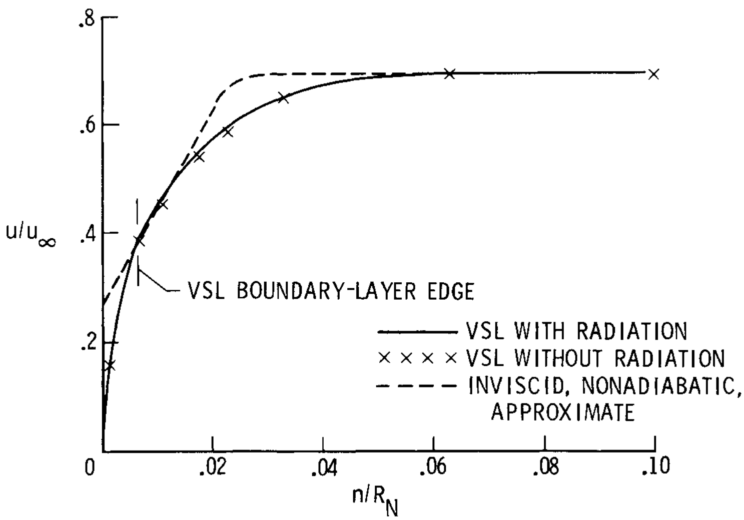

Fig. 10 Comparisons of tangential velocity profiles (large probe, time = 18 sec, s/R_N = 2.556, 90-10 atm).

heating results with and without radiation transfer are significantly different for other planetary entry conditions (different atmospheric gas composition).

Approximate Analysis Results

Figure 11 presents both the radiative and the laminar convective heating rate histories for each of the three small probes at an s/R_N location of 0.3. The body location

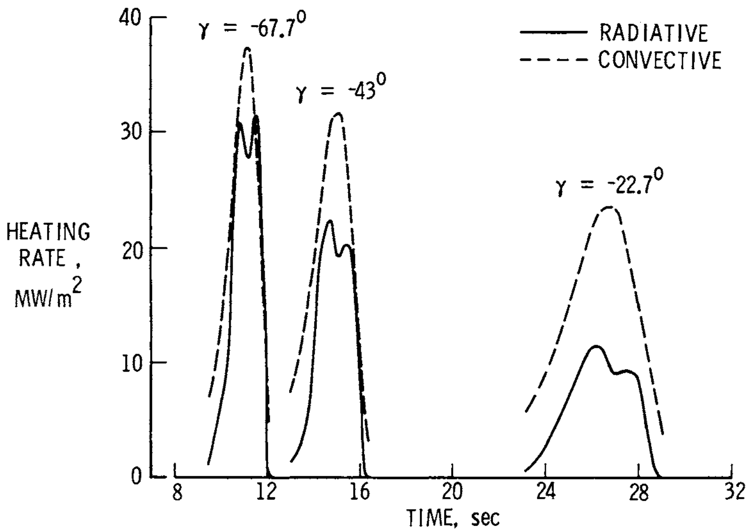

Fig. 11 Heating-rate histories for the small probes at an s/R_N location of 0.3 (90-10 atm, approximate analysis).

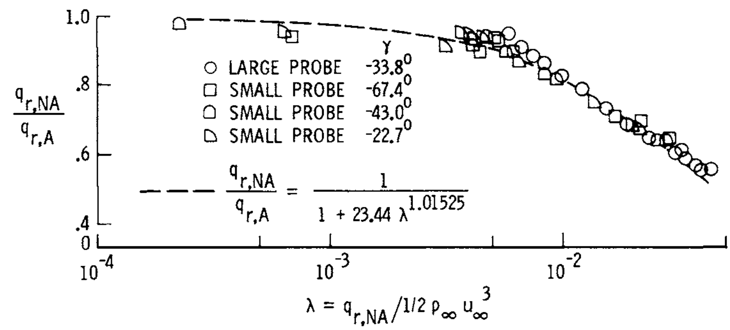

Fig. 12 Stagnation radiative cooling correlation for a 90-10 atm (approximate analysis).

corresponds to the location where a thermocouple will be installed on each of the three small probes as part of the heat-shield experiment. The nonablating results show that the laminar convective heating is generally greater than the radiative heating, with the difference between the convective and the radiative values becoming more pronounced with decreasing entry angle.

Figure 12 presents a stagnation radiative cooling correlation for the 90-10 atmosphere where the data correspond to various entry times for each of the four probes. An attempt to develop a similar correlation of the results on the conical flank was not successful.

No-Injection VSL Results

The previous comparisons are for a freestream gas composition of 90% CO_2 and 10% N_2 by volume. The effect of changing the freestream gas composition from a 90-10 mixture to a 97-3 mixture on the convective heating rates is negligible. However, the same change in gas composition can have an appreciable effect on the predicted radiative heating rates. This is clearly evident by the VSL results shown in Fig. 13 for the large-probe stagnation-point heating pulse. The radiative heating rates for the 90-10 mixture are significantly greater than the corresponding values for the 97-3

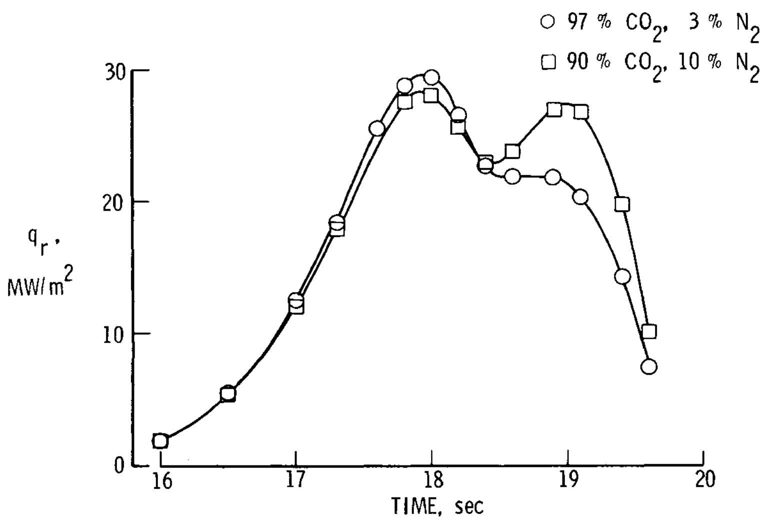

Fig. 13 Effect of freestream gas composition on stagnation-point radiative heating for large probe (VSL results).

mixture during the latter portion of the radiative heating pulse. This difference is due primarily to the emission of the CN molecule, which is more prevalent in the 90-10 mixture. Note that all of the subsequent results are for a 97-3 freestream gas mixture.

The radiative flux incident on the forebody of the probes is due primarily to continuum radiation. In the stagnation region, the radiative heating due to line transitions is significant; however, on the conical flank, where most of the forebody surface area is located and the shock-layer temperature is low relative to the stagnation value, the heating is almost entirely due to continuum and molecular band transitions.

Figures 14-17 present VSL results depicting heating-rate histories and distributions for each of the four probes without ablation injection. The wall-temperature distributions are constant values equal to the stagnation-point sublimation values as obtained with Eq. (1). Figure 14 presents both convective and radiative stagnation-point heating pulses for each of the four probes. The results show that the radiative and convective stagnation-point heating-rate values are comparable for the three probes entering at the steeper entry angles. For the probe entering at the shallowest entry angle, the stagna-

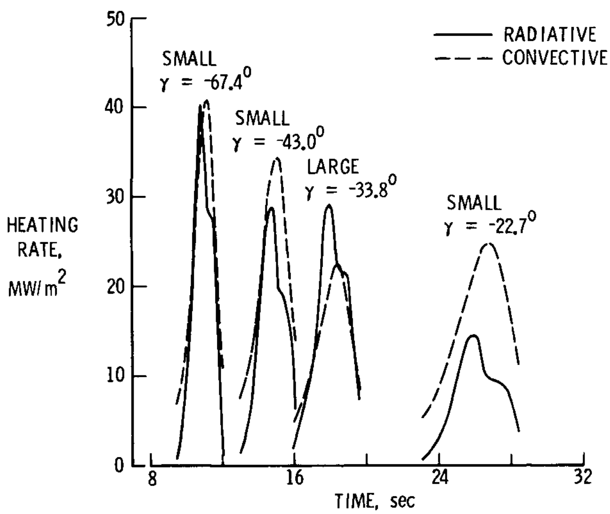

Fig. 14 Stagnation-point heating rate histories for each of the four probes (97-3 atm, VSL results).

tion-point convective heating-rate values are substantially greater than the radiative heating-rate values.

Figure 15 displays the radiative heating-rate distributions at various times during entry for the large probe (results for the small probes are given in Ref. 17). The radiative distributions shown are those obtained assuming the flow to be laminar. When the flow is assumed to be turbulent, the radiative heating-rate distributions are approximately the same as those for laminar flow. The results show that the radiative heating-rate distributions are nonsimilar. That is, the heating on the flank, when expressed as a percentage of

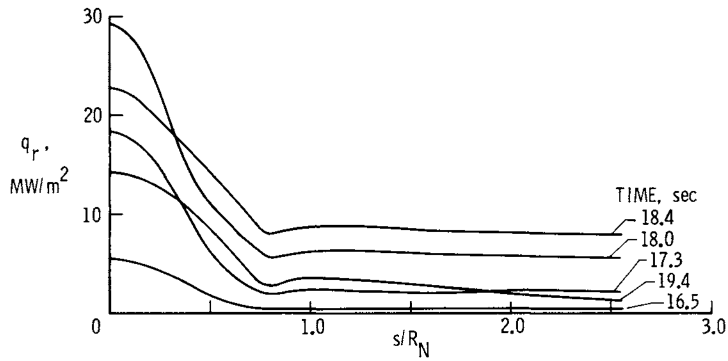

Fig. 15 Radiative heating-rate distributions during entry (VSL results, large probe).

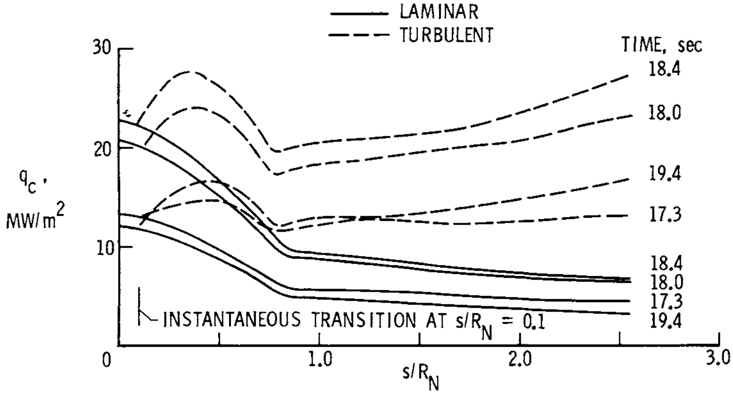

Fig. 16 Convective heating-rate distributions during entry (VSL results, large probe).

the stagnation-point value, varies substantially during the heating pulse.

Figure 16 presents both laminar and turbulent convective heating-rate distributions at various times during entry for the large probe (results for the small probes are given in Ref. 17). For the turbulent calculations, instantaneous transition is initiated arbitrarily at an s/R_N value of 0.1. The turbulent heating-rate values at the aft end of the probes range from about two to five times the corresponding laminar values. Consequently, if the flow is turbulent for most of the heat pulse, then the nonablating energy input to the surface would be dominated by the turbulent convective input. This is evidenced clearly by the results shown in Fig. 17, where the radiative, convective laminar, and convective turbulent heating-rate pulses are presented for the large probe at $s/R_N = 1.6$. The corresponding results for the small probes are presented in Ref. 17 for $s/R_N = 0.3$. Each of the small probes will have a thermocouple installed in the heat shield at $s/R_N = 0.3$.

Results with Ablation Injection

The previous results show the calculated heating environment for the Pioneer Venus probes with a nonablating

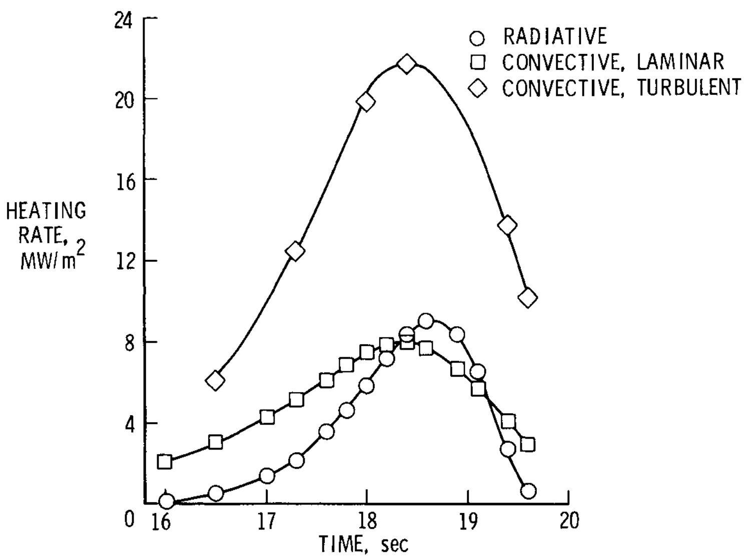

Fig. 17 Heating-rate histories for the large probe at $s/R_N = 1.6$ (VSL results).

surface. However, adequate heat-shield design and successful interpretation of heat-shield engineering measurements require reliable estimates of the effectiveness of the gaseous products of ablation in reducing the surface heating rates and the ablative mass loss. This problem has been addressed by computing solutions with arbitrary ablation injection rates for the large probe using the viscous-shock-layer analysis. These results are presented in Figs. 18-20.

Figure 18 shows the effect of carbon-phenolic injection on the stagnation-point convective heating for the large probe. The nondimensional injection rate $\dot{m}$ is varied arbitrarily from 0.001 to a value exceeding that required for a coupled solution for each of the five entry times considered. Note that the results correlate well when expressed as a Stanton number ratio vs the mass injection parameter B'. The surface temperature for all of the calculations, with the exception of the coupled solution, is that given by Eq. (1). However, the surface temperature is actually a value less (not more than 400 K less, with a maximum effect on convective heating of about 12%) than that given by Eq. (1), as is the coupled solution, because the mass fraction of ablation species at the ablator surface is less than one (see Fig. 1).

The corresponding results, which show the effect of carbon-phenolic injection on the stagnation-point radiative heating, are presented in Fig. 19. The results are presented in terms of the radiative heating ratio $q_r/(q_r)_{\dot{m}=0}$ vs the dimensional mass injection rate. The results show that the radiative heating correction due to carbon-phenolic injection

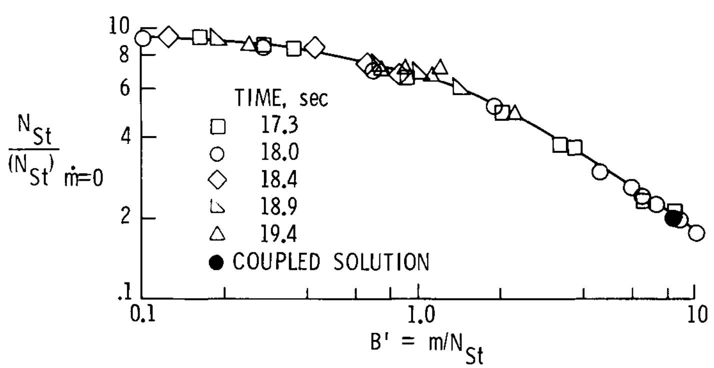

Fig. 18 Effect of carbon-phenolic injection on stagnation-point convective heating for the large probe (VSL results).

is very sensitive to entry conditions. For example, the radiative heating correction is about 0.72 during the earlier portion of the heat pulse. This heating correction remains at approximately this value up to the time of maximum radiative

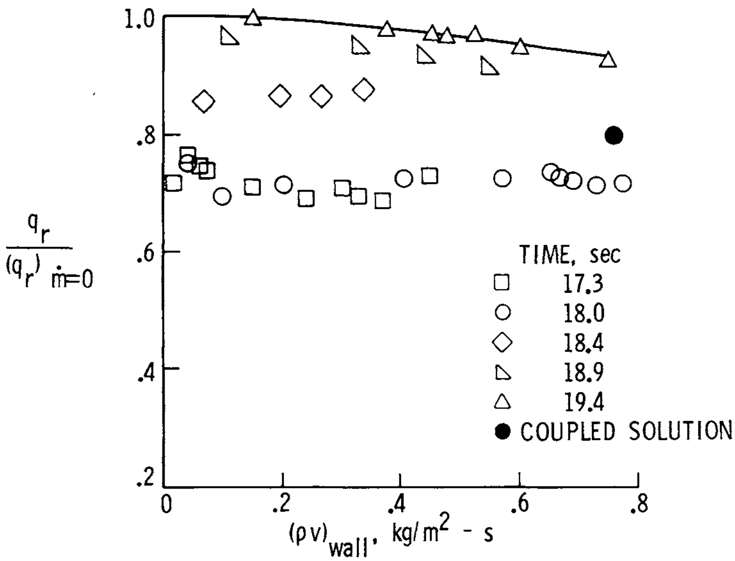

Fig. 19 Effect of carbon-phenolic injection on stagnation-point radiative heating for the large probe (VSL results).

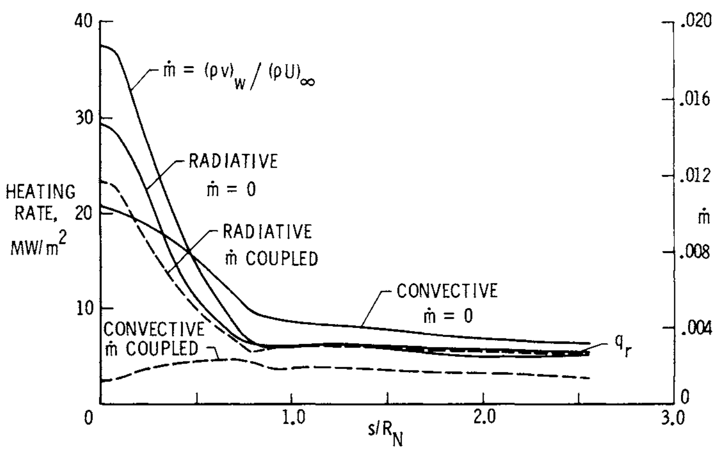

Fig. 20 Effect of coupled carbon-phenolic injection on heating rate (large probe, time = 18 sec, VSL results).

heating; however, as the entry time increases beyond peak radiative heating, the radiative heating correction becomes very small, as indicated by the results at 19.4 sec.

The results in Fig. 19 also show that, for most of the radiative heating pulse, the radiative blockage due to carbon-phenolic injection reaches a maximum value for a very small mass injection rate and then remains approximately constant as the injection rate is increased. The injection rate at which the radiative blockage reaches a maximum can be substantially less than the coupled mass injection rate, as is shown for the coupled solution at 18 sec. Also, the stagnation-point coupled solution, which includes the influence of downstream effects (3% greater shock-layer thickness), shows only a 20% blockage of radiative flux to the wall as compared with the 28% blockage for a comparable blowing rate where the downstream influence was not accounted for.

Figure 20 presents downstream coupled results for the large probe at 18 sec. The radiative and convective heating-rate distributions are compared with the corresponding no-injection values. Also, the coupled carbon-phenolic injection-rate distribution is presented. The coupled carbon-phenolic injection rate causes a 20% reduction in the radiative heating at the stagnation point; however, the reduction decreases with increasing distance from the stagnation point until the sphere-cone tangency point is reached. Along the conical flank, the radiative heating-rate reduction is almost negligible. Therefore, these results, along with the stagnation-point results (Fig. 19), suggest that the nonablating radiative heat load predicted with the VSL analysis would experience a very small reduction due to coupled carbon-phenolic injection. Remember, however, as was discussed previously, that the VSL radiative heating rates were significantly less than the corresponding inviscid values because of the strongly absorbing boundary layer. The reason that the boundary-layer absorption and the ablation-layer absorption are so similar is that the concentration of CO is substantial in both cases.

The effects of coupled carbon-phenolic injection on the laminar convective heating rates are significant. At the stagnation point, the convective heating rate is reduced by 88%, whereas, at the aft end of the conical flank, the reduction is 58%.

The coupled mass injection rate is not sufficiently large to cause the elemental gas composition at the surface to be the same as that of the ablator. For example, the mass fractions of the gas species at the surface due to ablation injection are 0.95 and 0.62 at the stagnation point and the aft end, respectively. Consequently, the surface temperature variation for the coupled solution was 3875 to 3650 K (see Fig. 1).

Concluding Remarks

Heating results are presented for the forebody of each of the four Pioneer Venus probes entering the atmosphere of Venus at four different entry angles. Solutions, both laminar and turbulent, are obtained with a viscous-shock-layer (VSL) analysis where the flow is assumed to be in chemical equilibrium. Also, radiative and laminar convective heating results obtained with an approximate analysis are presented. The approximate convective heating results are obtained by using nonadiabatic inviscid values in the convective heating correlations. Comparisons show that the approximate analysis provides acceptable agreement with the more detailed VSL results. However, the agreement between inviscid and VSL radiative heating results is acceptable only when the effect of the strongly absorbing boundary layer (which is included in the VSL analysis) is accounted for. Stagnation-point results for the large probe show that as much as 30% of the radiative flux incident on the wall is absorbed in the unblown boundary layer.

The nature of the radiative heating environment is such that the stagnation region heating is due to both continuum (molecular band transitions are included) and line transitions. However, during the latter portion of the heating pulses, the stagnation radiative heating is dominated by the continuum transitions. Along the conical flank, where most of the forebody surface area is located, the radiation is predominantly continuum throughout the trajectory. The results show that the continuum radiation flux incident on the wall increases significantly for a small increase in the N_2 composition of the freestream gas, whereas the convective heating is not affected by the same freestream gas variations.

The results with carbon-phenolic injection show that the laminar convective heating is reduced substantially both in the stagnation and downstream flow regions. However, the overall radiative reduction from that of the nonablating VSL results is very small. In fact, most of the radiative blockage is due to the blockage of the line radiation which is significant only in the stagnation region.

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SHUTTLE ELEVON COVE AERODYNAMIC HEATING BY INGESTED FLOW

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Abstract

The thermal response of the cove seal area of the Shuttle elevon wing junction is investigated in an arc-heated supersonic duct. The temperature response of the seal area is measured for various seal gap widths, ambient pressures, and enthalpies to determine the dependence of internal heating on these parameters. A correlation of the results shows that the internal heating to the seal is proportional to the product of the external pressure, the gap width, and the square root of the total enthalpy. Temperatures at the end of a simulated Shuttle entry compare well with predictions based on this correlation for the narrower gap widths, where there is only a small pressure drop at the cove inlet. Similar predictions are made for Shuttle flight as a function of seal gap widths to determine the maximum gap width allowable, not exceeding design temperatures on the structure. Reasons for the pressure drop at the cove inlet also are discussed.

Nomenclature

A = area of seal gap
c = constant
C_D = orifice discharge coefficient
C_p = specific heat at constant pressure
D = seal gap width
D_c = cove gap width
$\bar{f}$ = mean friction coefficient
h = specific enthalpy
h_T = total specific enthalpy in nozzle freestream
K = constant

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L = length of seal and cove (width of test article)
ℓ = length of seal gap
$\dot{m}$ = mass flow rate through seal
m_T = thermal mass of structure = $\rho C_p V$
M = Mach number
p = pressure
q = heat load on surface $\int \dot{q} dt$
$\dot{q}$ = heat flux
$\dot{Q}$ = heating rate to rub tube
$\dot{Q}_o$ = total energy convected into cove
Re = Reynolds number based on hydraulic diameter
Re_x = Reynolds number of local freestream based on distance from leading edge or nozzle throat
s = distance along cove from $\theta=0$
St = Stanton number
t = time
T = temperature
v = bulk velocity
V = Volume
γ = ratio of specific heats
ρ = density
θ = angle around cove from horizontal tangent

Subscripts

o = conditions at cove inlet
s = surface upstream of cove inlet
w = wall condition
cav = cavity condition

Introduction

During atmospheric entry of the Shuttle Orbiter, a significant part of the heat load to the internal structure of the wings may be caused by the leakage of hot gases through discontinuities such as the interface gaps at the wing elevons. These wing elevons are fitted with gas seals that allow free movement of the elevons, while reducing the flow of high-enthalpy gases through the internal structure. The consequences of a mechanical failure of these seals is difficult to predict analytically, since convective heating within the gap is coupled with cross-radiation effects, and little high-temperature experimental data have been available to support an analysis.

A number of investigators have studied heating to the internal surfaces of an elevon-wing cove. Rowe[1] made detailed and extensive measurements of pressure and cold-wall heat-flux distributions over a blunt delta wing equipped with a trailing-edge flap. Measurements were made inside a constant-width cove gap between the wing and the flap for various angles of attack, flap deflections, freestream Reynolds numbers, and gap widths. Stern and Rowe[2] analyzed the data of Ref. 1 and found that the heat flux to the cove walls increased with gap width, pressure, and Reynolds number. The pressure generally increased with flap deflection and angle of attack. However, the pressure in the cove was less than the upstream pressure except in cases of large flap deflection.

Dearing and Hamilton[3] also made pressure and cold-wall heat-flux measurements on the cove between a control flap and a sharp-leading-edged wedge. Although their measurements were not as detailed as Rowe's were, the results are all essentially consistent. Cooper and Putz[4] compared measurements of Refs. 1-3 with calculations of heat flux and pressure in the cove based on one-dimensional flat-plate heating correlations, pipe flow analysis, and nonsimilar boundary-layer computer solutions. The mean enthalpy of the ingested flow was obtained by an integration of the external boundary-layer flow. These calculations agreed reasonably well with the measurements. The applicability of their technique to the present study is possible but difficult because of the complex geometry of the cavity and the seal gap.

Kirlin and Schmitt[5] investigated possible cove seal designs and measured cove heat-flux distributions on a cold prototype. In addition, they measured thermal response of a particular elevon cove seal design for the Space Shuttle Orbiter. They found that the heat flux to the cove increases with cove width, gas flow leakage rate, and elevon deflection. The thermal response of the cove seal design was measured only with a sealed gap; however, they determined that thermal radiation was the major contributor to the heat flux in the cove.

Duffy[6] inferred heat-flux distributions for the present experiments using a trial-and-error procedure in which a constant relative distribution in the cove was assumed and the time-dependent thermal response of the elevon cove seal test article was calculated. The heat-flux distribution was adjusted until the calculated and measured temperatures agreed reasonably well.

References 1-3 and 5 have limited application to the current problems in that the heat-flux distributions were measured on cold surfaces and on geometries that differed from the proposed Orbiter elevon cove seal design. Since the Orbiter structure is covered with an insulating material that is expected to reach very high temperatures, cold-wall measurements of convective heat fluxes in the cove may not represent the actual heat-flux distributions. Nonadiabatic effects for hot and cold walls can be quite different, and the internal flow in narrow cove gaps can be affected strongly by viscous effects. Finally, thermal radiation is the dominant heat-transfer mechanism in some regions of the cove. Experiments with a realistic hot test article were required to combine these effects in an adequate simulation.

An experimental program was designed to acquire information that would relate the thermal performance of the elevon cove seal to predictable parameters of the entry environment. The specific objectives of the experiments were to determine the maximum seal gap width for which the seal and other structure would not be degraded thermally and to correlate the thermal response of the seal with parameters that are measurable in the test environment, as well as predictable in the flight environment. Results of these experiments are presented which relate the temperature rise of the seal rub tube (Fig. 1) to the pressure and enthalpy of the test environment, and an extrapolation is made to the flight case.

Experiments

The test article, shown in Fig. 1, represented a 61- X 61-cm full-scale section of the elevon cove seal at a fixed deflection angle of 0 deg and was instrumented with thermo-

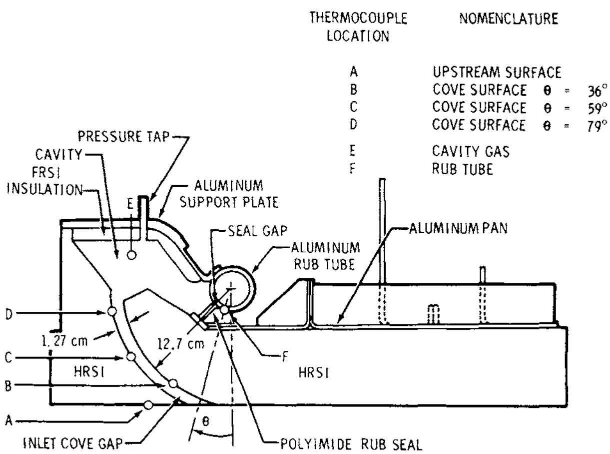

Fig. 1 Test article cross section showing thermocouple locations and nomenclature.

couples and a cavity pressure tap. The test article consisted of high-temperature reusable surface insulation (HRSI) thermal protection tiles bonded to an aluminum support structure or pan. These 7.1-cm-thick tiles had a density of 144 kg/cm^3 and were coated with a pigmented wash to match the surface emittance of the HRSI tiles on the Shuttle Orbiter. A curved slot (cove) in the HRSI simulated the wing-elevon junction near the lower surface. The seal was simulated by a 5.08-cm-diam aluminum rub tube 50 cm long in contact with a polyimide rub seal or wiper shown in Fig. 1. Although the rub tube was coaxial with the hinge axis, the test article did not have a deflection capability. To simulate seal failures, the aluminum rub tube and the polyimide rub seal were held apart to form seal gaps of from 0.076 to 9.53 mm. During the experiments, the test article was installed in the supersonic duct at the Johnson Space Center (JSC) Atmospheric Reentry Materials and Structures Evaluation Facility (ARMSEF). This duct has a half-angle of 10 deg and a rectangular cross section that expands from a 5- X 5-cm throat and extends to a 5- X 91-cm exit (see Fig. 2). The duct is mounted inside a test chamber that can be evacuated to a pressure level below 40 N/m^2 with a flow rate up to about 0.5 kg/sec. The test article was installed in one wall of the duct with the pan side of the test article open to the interior of the test chamber, which allowed the critical area around the rub tube to be viewed during testing and provided a pressure difference across the seal (since the flow in the duct was at a higher pressure than in the chamber). A calibration panel, mounted in the wall of the duct opposite the test article, contained an array of heating rate sensors and pressure transducers to measure experimental conditions. The geometry of the supersonic duct and location of the calibration panel transducers are shown in Fig. 2.

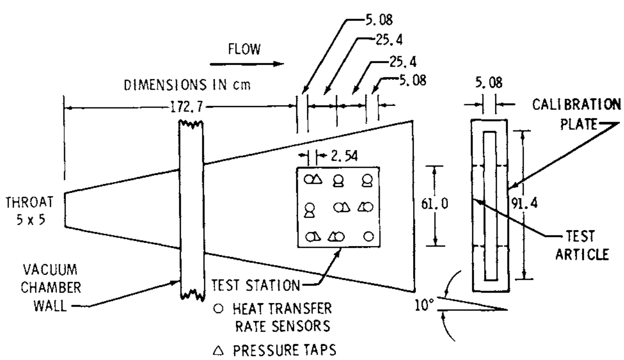

Fig. 2 Schematic of duct giving pertinent dimensions and locations of heat-transfer rate sensors and pressure taps.

Analysis

There are two aspects of the analysis of the cove seal measurements. One is the determination of a correlation parameter for the internal heating rate to the rub tube, and the other is the prediction of the overall energy flow into the elevon cove junction and the mean temperature rise of the entire wing/elevon junction structure.

Correlation Parameter

The heat transfer to the internal components of the cove seal is related directly to the rate at which energy flows through the cove due to a pressure difference. Except possibly for the zero gap case, free convection may be neglected. According to the steady-state inviscid energy equation, the flow of energy equals the product of the mass flow rate and the energy content per unit mass, or

$$\dot{Q}/L = (\dot{m}/L)h_o \tag{1}$$

where $\dot{Q}/L$ is the energy flow rate per unit length of elevon, $\dot{m}/L$ is the mass flow rate per unit length, and h_o is the enthalpy of the gas flowing into the cove. Since the seal gap is small compared with the inlet cove gap and the pressure ratio across the seal is large, the mass flow can be obtained approximately from the sonic (choked) flow relation

$$\dot{m} = K(Ap_{cav}/\sqrt{T_{cav}})C_D \tag{2}$$

where A=LD is the seal gap area; p_{cav} and T_{cav} are the total pressure and temperature, respectively, upstream of the gap; C_D is the discharge coefficient that accounts for local viscous effects; and K is a numerical constant that depends on the ratio of specific heats and the gas constant.

Substituting Eq. (2) into Eq. (1) and assuming that T_{cav} is proportional to h_{cav},

$$\dot{Q}_o/L \propto p_{cav}C_D D(h_o/\sqrt{h_{cav}}) \tag{3}$$

To quantify further the heat absorbed from the gas, the rub tube was assumed to be a calorimeter of a known thermal mass. The net heat transferred to it is then

$$\dot{Q} = m_T(dT/dt) \tag{4}$$

where m_T is the thermal mass per unit length of elevon, and dT/dt is the rate of temperature rise.

The heat transfer to the rub tube is due to convection and radiation from the cavity surfaces, and it is assumed to be proportional to the influx of heat into the cove, i.e.,

$$\dot{Q} \propto \dot{Q}_0 \tag{5a}$$

or

$$m_T(dT/dt) \propto (\dot{m}/L)h_0 \tag{5b}$$

Combining this with Eq. (3),

$$m_T(dT/dt) \propto (p_{cav}C_D D h_0/\sqrt{h_{cav}}) \tag{6}$$

Total Convected Heat Flux Estimate

As mentioned before, the convected heat flow into the cove is $\dot{m}h_0$. However, no direct measurements of these quantities were made; therefore, it is necessary to compute them from other measurements. The mass flow is obtained by solving the one-dimensional viscous flow relation for choked flow from Ref. 7, along with a heat balance relation and one for the heat transfer to inlet of the cove. The following three equations express these relations and are solved simultaneously to obtain h_0, T_{cav}, and $\dot{m}$:

$$\frac{4\bar{f}\ell}{D} = \frac{1-M^2}{\gamma M^2} + \frac{\gamma+1}{2\gamma}\ln\frac{(\gamma+1)M^2}{2\{1+[(\gamma-1)/2]M\}} \tag{7}$$

$$h_0 = C_p T_{cav} + [\int \dot{q}(s)ds/\dot{m}/L] \tag{8}$$

$$\dot{q}_0 = St_0\rho v(h_0 - h_{w_0}) \tag{9}$$

where $\bar{f}=24/Re$ is the mean friction coefficient for the seal gap, ℓ is length of the seal gap, M is the Mach number at the inlet to the seal gap, h_0 is the total enthalpy at the inlet to the cove, and T_{cav} is the bulk temperature of the gas in the cavity, which is assumed to be equal to the total temperature at the inlet to the seal gap. The heat-flux distribution along the cove is $\dot{q}(s)$, the heat-flux to the wall of the cove at the inlet is $\dot{q}_0$, and the mass flux $\rho v=\dot{m}/LD_C$. This set of simultaneous equations was solved by a combination of numerical iterations and a graphical technique. The measured cavity pressure p_{cav} was used in the solution to obtain ρ, Re, etc. From the solution, one could calculate the total energy convected into the elevon/cove region per unit time, $\dot{Q}_0=\dot{m}h_0$. Knowing $\dot{m}$ from the solution of Eqs. (7), (8), and (9), the discharge coefficient is obtained through Eq. (2).

Results and Discussion

Two categories of experiments were performed on the test article in the supersonic duct. In the first category, a series of short experiments was conducted in which the external flow parameters and the seal gap width were varied to determine their effects on the internal heating in the test article. From these measurements, the internal heating rate could be correlated with environmental parameters. In the second category, the test article was subjected to an environment that simulated the local conditions at the elevon/wing junction during a flight trajectory. The overall temperature response of the rub tube and other structure for various gap widths was obtained. These experiments determined the maximum gap width for which temperature limits of the structure would not be exceeded.

Parameter Variation Experiments

The duration of each parameter variation experiment was about 200 sec. The set of conditions to which the test article was subjected in this phase is given in Table 1. The range of the total enthalpy h_T extended from about 4 to 30 MJ/kg, and the local external freestream static pressure p extended from about 600 to 2700 N/m^2. The range of seal gap widths covered two orders of magnitude from 0.076 to 9.53 mm. The mean rub tube temperature rise rates dT/dt were obtained from the temperature measurements for each case and are given in Table 1, along with other run conditions.

As discussed previously, a measure of the internal heating rate is the rate of temperature rise of the rub tube. To determine how the internal heating depends on external flow parameters, various correlations of dT/dt were attempted. As indicated by Eq. (6), dT/dt should correlate with $p_{cav}\, C_D\, Dh_o/\sqrt{h_{cav}}$. However, p_{cav}, which was measurable directly in the experiments, can only be approximated for flight. It was considered desirable to correlate only in terms of known external flow properties and the gap widths.

At first, correlation should be made in terms of known quantities to justify Eq. (6) experimentally; however, h_o and h_{cav} are not measurable directly. These quantities were obtained for two gap widths by solving Eqs. (7-9), using the cove wall heat-flux distributions $q(s)/q_s$ inferred by Duffy[6] and shown in Fig. 3. Using the results given in Table 2 for which dT/dt is available, it was found that $dT/dt \propto (\dot{m}h_o)^{1.11}$.

Table 1 Measured test conditions for parameter variation

Run no.	Gap width D, mm (mil)	Arc heater gas flow rate $\dot{m}$, kg/sec	Arc total enthalpy h_T, MJ/kg	Freestream static pressure p, N/m^2	Static pressure in cavity p_{cav}, N/m^2	Surface temp. T_s, K	Rub tube temp. rise rate dT/dt, K/sec
45	0.076 (3)	0.118	16.7	1048	910	1298	0.0119
49	0.254 (10)	0.109	14.4	962	858	1296	0.0150
50	0.635 (25)	0.115	13.0	870	790	1222	0.0336
51	0.889 (35)	0.114	13.2	877	732	1206	0.0520
47	1.143 (45)	0.114	16.6	995	742	1308	0.171
59	4.57 (180)	0.112	12.2	910	425	1176	0.280
53	9.53 (375)	0.109	14.4	896	215	1208	0.233
48A	1.143 (45)	0.114	15.2	775	495	1097	0.104
48B	1.143 (45)	0.117	18.4	903	667	1191	0.129
48C	1.143 (45)	0.115	23.7	1007	853	1332	0.187
48D	1.143 (45)	0.117	25.9	1040	932	1366	0.196
60	4.57 (180)	0.0545	20.3	611	290	1269	0.35
61	4.57 (180)	0.110	9.67	769	353	1045	0.258
62	4.57 (180)	0.110	11.4	862	414	1165	0.322
63	4.57 (180)	0.109	16.3	959	515	1267	0.444
66	4.57 (180)	0.112	22.3	1025	496	1308	0.533
67	4.57 (180)	0.0558	30.0	675	317	1340	0.436
68	4.57 (180)	0.112	26.2	1050	510	1384	0.672
69	4.57 (180)	0.182	18.5	1766	703	1322	0.700
70	4.57 (180)	0.259	12.1	2055	714	1340	0.639
43	No gap	0.114	16.3	949	977	1269	0.0101
77	4.57 (180)	0.504	3.82	2188	680	962	0.339
78	4.57 (180)	0.481	7.00	2676	927	1292	0.75

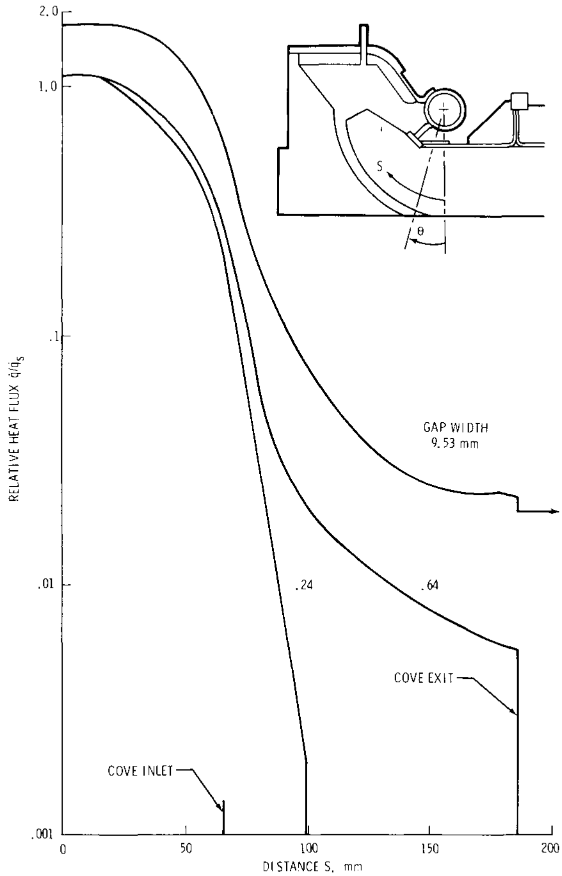

Fig. 3 Inferred heat flux distributions based on temperature response of test article.[6]

In addition, it was found that $\dot{m}h_o \propto (p_{cav}\ C_D\sqrt{h_T}\ D)^{0.815}$.

Therefore,

$$(dT/dt) \propto (p_{cav}\ C_D\sqrt{h_T}\ D)^{0.9} \tag{10}$$

which for practical purposes is linear, considering the uncertainty in $\dot{m}$ and h_o. The adequacy of this correlation is confirmed by noting the goodness of fit in Fig. 4, where dT/dt is plotted against $p_{cav}\ C_D\sqrt{h_T}\ D$. The slope in Fig. 4 is not exactly 0.9 from Eq. (10) because they are derived from different sets of data.

Table 2 Results of solving energy and mass balance equations

	0.64-mm gap		9.53-mm gap	
	Step 1*	Step 2*	Step 1	Step 2
Gap mass flow rate $\dot{m}$, kg/sec-m	5.69×10^{-4}	7.31×10^{-4}	2.26×10^{-3}	3.12×10^{-3}
Surface temperature T_s, K	1222.	1209.	1208.	1337.
Inlet temperature T_o, K	1545.	1918.	2090.	2355.
Surface enthalpy h_s, MJ/kg	1.43	1.42	1.41	1.57
Inlet enthalpy h_o, MJ/kg	1.86	2.42	2.69	3.12
Total enthalpy h_T, MJ/kg	13.0	12.4	14.4	11.1
Cavity temperature T_{cav}, K (calculated)	264.	440.	1055.	1300.
Cavity temperature T_{cav}, K (measured at 350 and 1100 sec, respectively)	456.	805.	805.	980.
Cavity pressure p_{cav}, N/m^2	790.	1412.	215.	323.
Inlet Mach number M_o	0.032	0.045	0.54	0.53

*Step 1: t = 0 to 700 sec.
Step 2: t = 800 to 1800 sec.

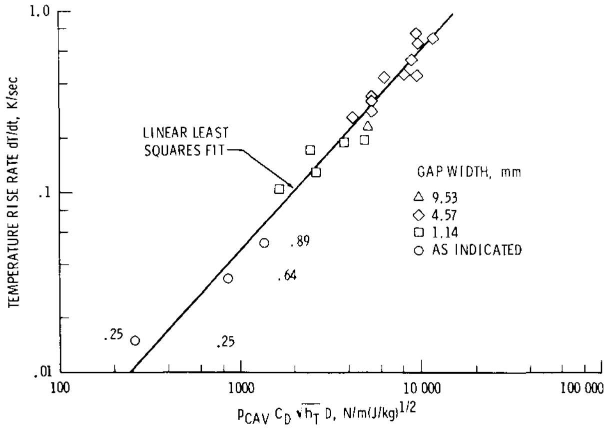

Fig. 4 Rub seal heating rate correlation based on cavity pressure and total enthalpy.

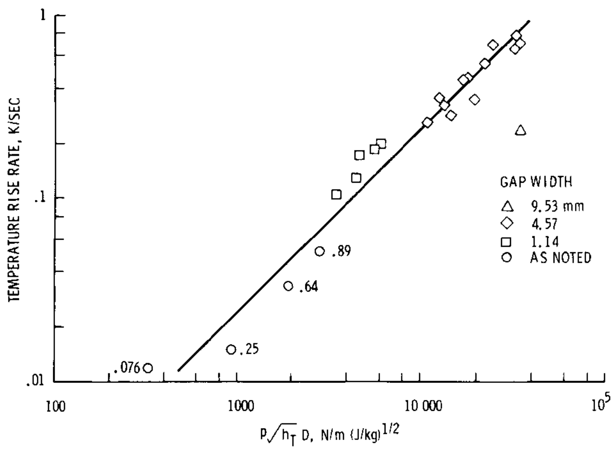

Fig. 5 Rub seal heating rate correlation based on freestream static pressure and total enthalpy.

If there were no pressure drop at the cove inlet and in the cove, and if C_D were constant, we could correlate dT/dt based on external flow properties, e.g., $p\sqrt{h_T}\ D$. This correlation is given in Fig. 5. It is seen that the fit is good for the intermediate gap widths, where C_D is approximately constant and where the pressure drop in the cove (discussed in the Appendix) is small. The discharge coefficient of the seal gap is small for narrow seal gaps, whereas the pressure drop at the cove is large for large gaps because of the large flow or suction. When these factors are considered, a reasonable correlation in terms of external flow parameters is justified. It then is possible to use the correlation to predict the final temperature of the rub tube.

Flight Simulation Experiments

To determine the overall temperature response of the wing elevon junction during an Orbit entry, the test article was subjected to a simulated entry environment. The flight simulations consisted of a constant two-step profile whose nominal conditions are given in Table 3. This profile is compared with the profile of Shuttle Orbiter design trajectory in Fig. 6, where the local static pressures and surface heat fluxes are plotted against time. These two constant heating rates were chosen because they matched the maxima in the flight heating rate profile. The duration of the simulation was chosen so that the total internal heat load was simulated approximately, the experiment being about 93% of flight in terms of the correlation parameter $\int p\sqrt{h_T}\,dt$. The effect of variations in the flight elevon deflection is included in the pressure, since the test article elevon was not deflectable. The total external surface heat load during the experiment reached the flight value after 1100 sec. Since the total simulation of 1800 sec exceeded 1100 sec, the total surface heat load was about twice the flight external heat load. However, this had little effect on the results of internal heating for a nonzero seal gap.

Table 3 Nominal flight simulation test conditions

	1	2
Total enthalpy h_T, MH/kg	14.4	12.1
Arc heater gas flow rate $\dot{m}$, kg/sec	0.113	0.206
Freestream static pressure p, N/m^2	896.0	1819.0
Surface temperature T_S, K (ahead of cove)	1245.0	1400.0
Radiation equilibrium heat flux $\dot{q}_S$, W/cm^2	13.3	23.8
Flow state	Laminar	Probably transitional

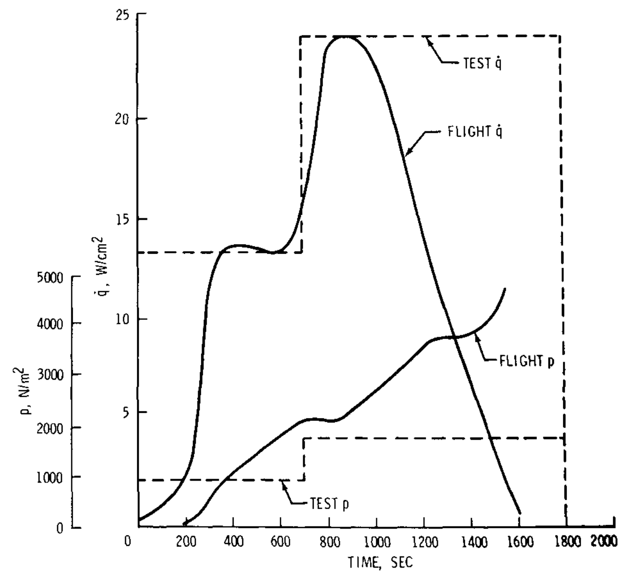

Fig. 6 Local environment for elevon seal during orbiter flight and simulation.

A typical response of the test article temperatures is given in Fig. 7, where temperatures are plotted against time. The test conditions were changed from step 1 to step 2 (see Table 3) at 700 sec. Note the rapid change and then leveling off of temperatures near the inlet of the cove as compared with the slower response within the cavity and on the rub tube. The final temperatures obtained during the simulations are discussed in the next section.

Final Structural Temperatures

Determinations of structural temperatures are obtained by three techniques. The first is by extrapolating the correlation parameter results, the second is from direct measurements in the trajectory simulations, and the third is by solving the three simultaneous equations (7-9) and calculating the temperature rise from the relation

$$\Delta T = (1/m_T) \int (\dot{m}/L) h_o \, dt \qquad (11)$$

The first two techniques will give local temperatures, whereas the latter technique will give only an approximate average over the entire structure of the wing/elevon junction.

The mean temperature rise of the rub tube may be obtained

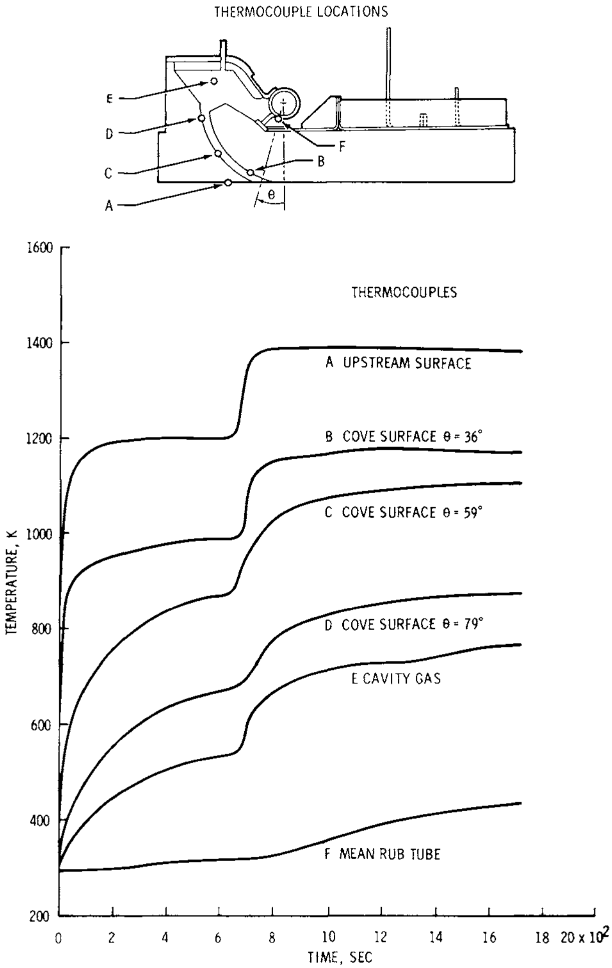

Fig. 7 Temperature history for 0.89-mm gap configuration during re-entry simulation.

by integrating the correlation parameter in time:

$$\Delta T = \int (dT/dt)dt = C\ D \int p\sqrt{h_T}\ dt \tag{12}$$

where C is a proportionality constant determined from the correlation. Plots of $p\sqrt{h_T}$ for both the Orbiter design entry trajectory and the simulation are given in Fig. 8. The dashed curve is an extrapolation to account approximately for the reduction in the enthalpy potential late in the trajectory. Eq. (12) is plotted in Fig. 9 against the seal gap width D, for both cases, together with the measured final mean temperature

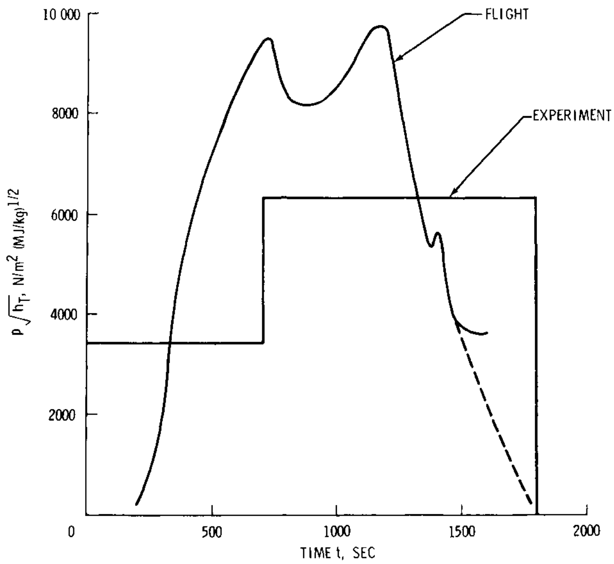

Fig. 8 Local flight and experimental environment in terms of correlating parameter $p\sqrt{h_T}$.

of the rub tube. For some gap widths, the points are obtained by extrapolating the temperatures to 1800 sec, where the full simulation was not achieved because of premature termination of the run. Good agreement between the measured and predicted temperature is seen in this figure except at large gap widths.

The measured ΔT for the 4.57- and 9.53-mm gap cases falls far below the values predicted by the correlation. This is due in part to a pressure drop in the cove or at the entrance to the cove. The driving pressure is, therefore, less than the external static pressure used in the correlation; hence, the actual flow through the gap is less than had there been no pressure drop. The reasons for this pressure drop, which is consistent with previous measurements,[1, 3, 6] are discussed in the Appendix.

From Fig. 9, one may determine the overall temperature rise of the rub tube due to a seal opening of magnitude D. The designer then may associate a maximum temperature expected with various gap widths. In the example shown in Fig. 9, if there is a requirement not to exceed a temperature of 450 K, then the maximum ΔT may be 133 K if the structure begins entry at 317 K. Therefore, the maximum permissible gap width is about 0.53 mm based on the correlation or about 0.70 mm based directly on the simulation measurements.

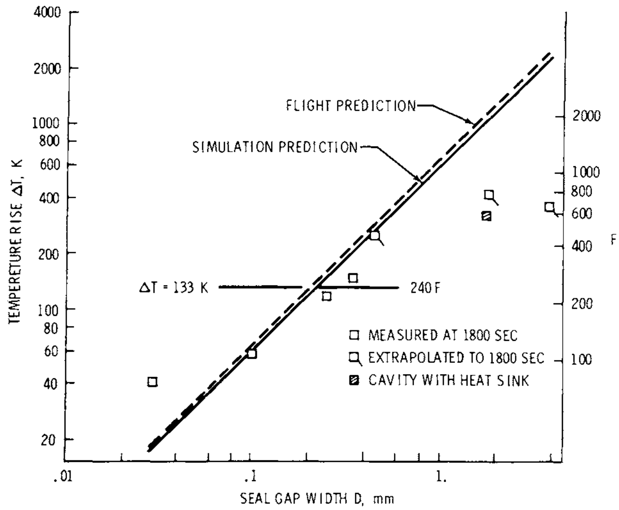

Fig. 9 Temperature rise of rub tube predicted for flight and arcjet simulation using correlation parameter $p\sqrt{h_T}\,D$.

It appeared from the measurements that the most critical temperature was that of the rub tube. However, other areas were considered. Final temperatures measured on the elevon structure pan and spar are plotted against seal gap width D in Fig. 10. These temperatures have a linear dependence up to a gap width of about 4.57 mm. Beyond this value, ΔT is constant or decreases slightly in the same trend as on the rub tube. If there were no pressure drop in the cove, temperatures would be very high compared to actual measurements. The dashed curves are extrapolations of the narrow-gap results and indicate the trends for which there was no pressure drop at the cove. For a seal gap width of 4.57 mm, a maximum temperature rise of about 380 K is seen on the pan near the rub seal; this is near the melt limit of the adhesive holding the RSI to the aluminum.

Total Heat Flow into Cove

The total energy flow into the cove and the final mean temperature of the wing/elevon junction are estimated by solving the heat and mass flow accounting Eqs. (7-9). Results are given in Table 2 for gap widths of 0.653 and 9.53 mm. The heat-transfer distributions in the cove were taken from Duffy[6] and are plotted in Fig. 3. The Stanton number St was calculated using fully developed pipe-flow correlations.[8] Note that the inlet enthalpy h_0 lies between the wall enthalpy h_S and the total enthalpy h_T, as would be expected. The mea-

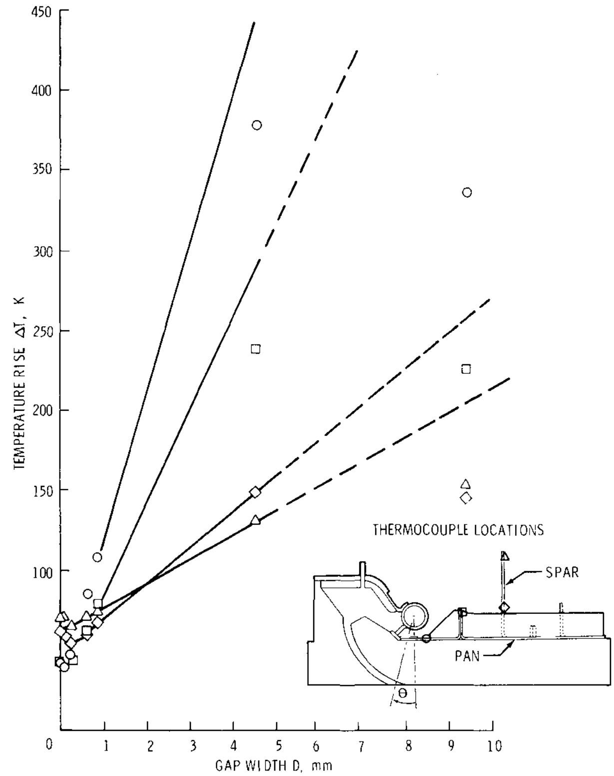

Fig. 10 Temperature rise of aluminum structure during simulation at 1800 sec.

sured cavity temperatures are not accurate because of radiation heat transfer to the thermocouple, differing from the calculated values by as much as 350 K. The mean temperature rise of the wing/elevon junction was found to be 115 K for D = 0.64 mm and 640 K for D = 9.53 mm, assuming that a thermal mass of the structure of 23 kJ/m-K. Although this approach is suitable for rough estimates, it would be desirable to improve the accuracy by obtaining more detailed measurements and a better understanding of the inlet flow conditions, such as the cove inlet enthalpy h_0, the inlet pressure p_0, and the cove heat flux distributions, as functions of local external flow parameters, gap widths, and elevon deflection.

Conclusions

Experiments were conducted in an arc-heated supersonic duct at the Johnson Space Center using a test article that approximated the design features of the Shuttle Orbiter elevon cove and seal with zero deflection. These experiments measured the influence of the two-dimensional flow of hot gases through the Orbiter wing/elevon junction on the temperatures of the elevon cove seal and internal structures. Analysis of the experimental results yielded the following major conclusions:

1) A correlation of the temperature response data, which is supported by an approximate analysis, indicates that a significant parameter for the internal heating on the rub tube is $p\sqrt{h_T}\,D$ over a wide range of environments and gap widths, provided that the pressure drop in the cove is accounted for.

2) The simulation of the trajectory closely approached that of the flight design trajectory in terms of the correlation parameter, the internal heat load in the experiments being about 93% of flight. The simulation was especially good for narrow seal gaps where there was a small pressure drop at the cove.

3) The temperature rise of the rub tube will not exceed the design limit of 133 K for seal gaps up to about 0.6 mm.

4) There was a significant pressure drop at the cove for the large seal gap widths in the experiments. This pressure drop seems to be related to the mass flow through the cove and to external boundary-layer properties. There is evidence in the literature which suggests that there probably will not be as much pressure drop in flight because of a thicker boundary layer.

5) The measurements indicated that regions on the structure such as the support structure pan would reach temperatures as large as 778 K if a seal failure resulted in a 2.67-mm gap.

6) It was estimated that the mean temperature increase of the elevon-wing junction is 115 K for a 0.69-mm seal gap and 650 K for a 9.57-mm seal gap.

Appendix

In these experiments, a pressure drop existed between the freestream static pressure in the supersonic duct and the static pressure in the cavity of the test article. This pressure drop did not seem to be consistent with a pressure drop due to wall shear and choking in the inlet cove for mass flows through the seal gap calculated using sonic flow relations. Even though pressures were not measured at the cove inlet, there are strong indications that this pressure drop occurs at the inlet of the cove. This pressure drop may be caused by the formation of a viscous shear layer in an expansion region in the vicinity of the cove inlet. The detailed pressure measurements of Rowe,[1] Dearing and Hamilton,[3] and the pressure measurements of Deveikis[9] on a similar elevon cove test article in the Langley Research Center 8-ft high-temperature structures tunnel are consistent with the hypothesis that a significant pressure drop occurs at the cove inlet. Pressures measured at the cove inlet of Refs. 1 and 9 are compared with the cavity pressures measured in the present experiments in Fig. 11, where the independent variable is the ratio of boundary-layer thickness to gap width. The gap width in Rowe's case is the cove gap width, which he varied over a range of 0.05 to 0.35 in. The gap width in Deveikis' data and the present results (JSC) refers to the widths of the seal gap rather than the cove gap, since the seal gap governs the mass flow through the cove. It can be seen that the pressure drops for all three sets of data are reasonably consistent. Likewise, when plotted against the gap width D (Fig. 12), there is consistency in the results. It is probable that whether the flow is laminar or turbulent also affects the cove inlet pressure drop. The aforementioned results are similar to those obtained by Brown and Jakubowski[10] and Jakubowski and Lewis[11] in the region downstream of a rearward-facing step with suction. The theory of Inger[12] supports the results of Refs. 10 and 11. The geometry near the inlet to the elevon cove is similar to a rearward-facing step except that there is a gentle ramp extending from the back of the step up to the surface of the elevon. This circumstantial evidence strongly

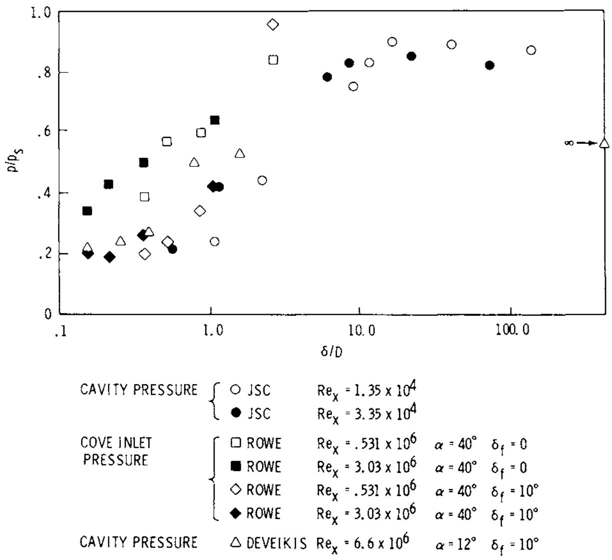

Fig. 11 Pressure ratio across cove vs ratio of boundary-layer thickness to seal gap width (cove width in Rowe's results).

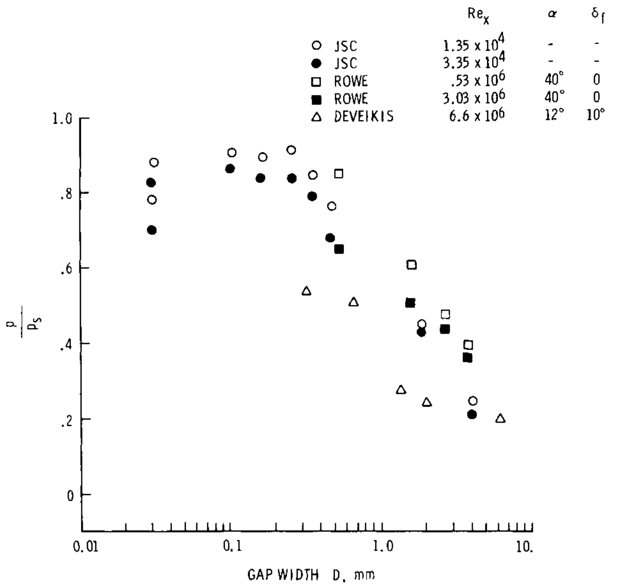

Fig. 12 Pressure ratio across cove vs seal gap width (width of cove in Rowe's data).

suggests the possibility of a pressure drop at the cove inlet in the present experiments. Figure 12 and the results of Refs. 10-12 indicate that the pressure drop increases with smaller δ/D.

This reduced pressure may not be as significant in the flight case where the boundary layer is thicker than in the experiments. For example, in flight the boundary-layer thickness δ near the elevon hinge line is estimated to be from 5- to 15 cm, depending on the time during the trajectory and the particular location. For a partially open seal gap where D = 0.457 cm, the ratio δ/D in flight is from 10 to 30, and δ/D in the experiments is about 1 to 2. From Fig. 12, the corresponding pressure ratio in the experiments is about 0.45, whereas, in the flight cases, the pressure ratio is about 0.85. Therefore, the inlet pressure in flight may be about twice that in the experiments for large gap widths. This would result in correspondingly larger energy fluxes and temperatures in flight.

Acknowledgments

The authors are grateful to W. D. Goodrich for suggesting possible reasons for the pressure drop at the gap inlet.

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AEROTHERMODYNAMIC BASE HEATING

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Abstract

Base heating of a two-dimensional bluff body in a supersonic or hypersonic flow is analyzed by the component analysis method, paying special attention to the effect of the approaching turbulent boundary-layer thickness on the heating rate. The heat-transfer analysis takes account of two resistive elements to heat transfer in the separated region: 1) the heating between the recirculating flow and the base wall, and 2) heating across the mixing layer in the wake. The turbulent Prandtl number is assumed to be unity, and radiation is neglected. Calculated results of base heating rate vs Mach number, wall temperature, and boundary-layer thickness are shown and discussed.

Nomenclature

C	=	Crocco number $(u^2/2c_pT_o)^{1/2}$
c_p	=	specific ratio at constant pressure
E	=	heat-transfer function used in Eq. (31)
h	=	half-height of the base
k	=	specific heats ratio
M	=	Mach number
p	=	pressure
q	=	average heat-transfer rate to the base per unit area per unit time, $= -\Omega_c/h$
R	=	gas constant

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T = temperature
T_b = bulk temperature in the wake
u = flow velocity
(X,Y) = reference coordinate system
(x,y) = intrinsic coordinates system
(x_o,y_o) = displaced origin of the equivalent shear layer
y_m = separation between reference X axis and intrinsic x axis
δ = physical thickness of the approaching boundary layer
δ^* = displacement thickness of the approaching boundary layer
δ^{**} = momentum thickness of the approaching boundary layer
η = similarity variable of the mixing layer
η_m = value of η corresponding to y_m
θ = flow angle
ρ = density
σ = similarity parameter; see Eqs. (12) and (9)
ϕ = nondimensionalized velocity
ψ = length of the recirculation region measured along the centerline
Ω_c = heat flow across the mixing layer from the wake

Subscripts

1,2,3,4 = regions indicated in Fig. 1
o = stagnation condition
a = adjacent inviscid flow region
d = discriminating streamline condition
J = jet boundary streamline condition when $\delta \neq 0$
j = jet boundary streamline condition when $\delta \approx 0$
m.r. = mixing region
R = location in uniform flow mixing region
refl. = reflected or mirror image condition
w = base-wall condition
y = condition after the normal shock wave

Introduction

Knowledge of the aerothermodynamic base heat transfer as well as the base pressure of a bluff body is necessary for the design of high-speed vehicles such as hypersonic transporters and space shuttles. Since the 1950's, two major approaches have been developed and successfully applied to many problems involving separated and reattached flow. One is the so-called Crocco-Lees method,[1,2] which is based on the moment of momentum idea. The other one is the component analysis method, based on the flow model proposed by Chapman[3] and Korst.[4] In the latter method, the flowfield is divided into distinguishable regions, and each individual component is analyzed quantitatively taking into account the possible interactions with other components. Finally, all of the components are unified to give a unique flow solution.

Base pressure of two-dimensional bluff bodies has been analyzed by use of the component analysis method both in the case of zero boundary-layer thickness and in the case of finite thickness.[4-6] Page and Dixon[7] analyzed turbulent base heat transfer of a two-dimensional body with negligible incoming boundary-layer thickness and obtained good agreement with the available experimental data.

Although there is a large body of literature on base heating,[8] little attention has been focused on the effect of the approaching boundary layer. In this paper, heat transfer to the base of a two-dimensional body with steady approaching turbulent boundary layer with finite thickness and uniform supersonic main flow is considered. The component analysis method is used while taking into account the dissipative free shear layer involved with the recirculation region, and Page's "reflected image" approximation.[9]

Analysis

The body considered (Fig. 1) is symmetrical about the plane through its centerline. The flow configuration, therefore, is also symmetrical about the plane, so that we restrict our discussion to the upper half-plane. The flowfield is considered to be composed of the following components: 1) the expansion of approaching flow around the corner 1-2, 2) the constant-pressure turbulent mixing region 2-3, 3) the recompression of the flow 3-4, and 4) the recirculation (or near-wake) region.

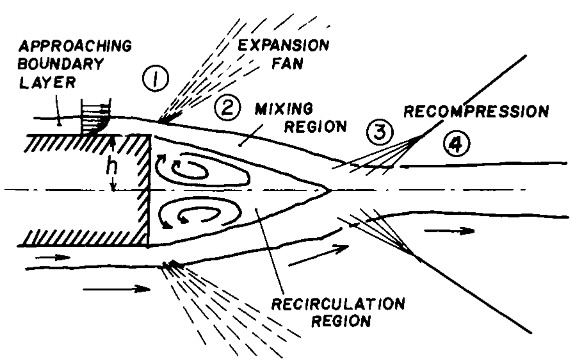

Fig. 1 Sketch of the flow model.

These components are not independent of each other in that they interact, and the boundary conditions of one component are given by the other components. We assume the following conditions in this analysis:

1) The flows in the mixing layer, recirculation flow, and approaching boundary-layer flow are turbulent.
2) The approaching outer flow is uniform.
3) The gas is considered a perfect gas.
4) There is no mass bleed or suction to the wake.
5) The Prandtl number of the turbulent flow is unity.
6) The base-wall temperature T_w is constant.
7) The radiation is neglected.

A. Freestream Component

The expansion process of the freestream from region 1 to 2 is governed by the well-known Prandtl-Meyer relations. Therefore, in functional notation,

$$M_{2a} = M_{2a}(M_{1a}, \theta_2 - \theta_1, k) \tag{1}$$

$$p_2/p_1 = p_2/p_1(M_{1a}, \theta_2 - \theta_1, k) \tag{2}$$

Since there are no incoming waves to the region from 2 to 3, the states 2 and 3 are identical. The flow angle at 4 should be the same as the original flow direction ($\theta_4 = \theta_1$):

$$\theta_2 - \theta_1 = \theta_3 - \theta_4 \tag{3}$$

The recompression is assumed to take place through isentropic compression waves. Then, in functional notation,

$$p_4/p_3 = p_4/p_3(M_{3a}, \theta_4 - \theta_3, k) \tag{4}$$

From Eqs. (2-4),

$$p_2/p_1 = p_3/p_4 \tag{5}$$

B. Mixing Component

In the case that the thickness of the approaching boundary layer is negligible, the velocity profile across the turbulent mixing region under constant pressure is represented by an error function[10]:

$$u/u_{2a} = \phi_2 = (1/2)(1 + \mathrm{erf}\ \eta) \tag{6}$$

$$\mathrm{erf}\ \eta = (2/\sqrt{\pi}) \int_0^{\eta} e^{-\beta^2} d\beta \tag{7}$$

$$\eta = \sigma(y/x) \tag{8}$$

σ is a proportional constant, and we hereafter use the following empirical relation[11]:

$$\sigma = 12.0 + 2.758\ M_{2a} \tag{9}$$

x,y are intrinsic coordinates and related to the reference coordinates X,Y by the following relations:

$$x = X,\ y = Y + y_m(x)$$

and $y_m(x)$ is determined from the conservation of momentum as

$$\sigma y_m/x = \eta_m = \eta_R - (1 - C_{2a}^2) \int_{-\infty}^{\eta_R} \frac{\phi_2^2}{(T_o/T_{o_{2a}}) - C_{2a}^2 \phi_2^2} d\eta \tag{10}$$

In the present case, with finite boundary-layer thickness, the error function profile still is considered a good approximation if its origin is displaced so that the conservation of mass and momentum flux between error function profile and real boundary-layer flow is satisfied at the corner.[12] Then we get

$$u/u_{2a} = \phi_2 = (1/2)(1 + \mathrm{erf}\ \eta) \tag{11}$$

where

$$\eta = \sigma(y/x) = \sigma(Y + y_m - y_o)/(X + x_o) \tag{12}$$

$$x_o = \frac{\sigma\delta^{**}}{\int_{-\infty}^{\eta_R} (\rho u/\rho_{2a} u_{2a})(1 - \phi_2) d\eta} \tag{13}$$

$$y_o = \delta^* + \delta^{**} \tag{14}$$

It should be noted that x_o and y_o depend upon the shape of the boundary-layer velocity profile and go to zero if the boundary-layer thickness is negligible.

The total temperature and density profiles across the mixing layer are given by the Crocco integral, since Pr = 1.0 and p = constant. Thus, between regions 2 and 3,

$$T_o/T_{o_{2a}} = (T_b/T_{o_{2a}}) + [1 - (T_b/T_{o_{2a}})]\phi_2 \tag{15}$$

$$\rho/\rho_{2a} = (1 - C_{2a}^2)/[(T_o/T_{o_{2a}}) - C_{2a}^2\phi_2^2] \tag{16}$$

where T_b is the bulk temperature of the stagnant gas in the middle of the wake.

C. Corner Flow Component

The velocity profile of the approaching turbulent boundary layer is approximated by a power-law relationship:

$$\phi_1 = u/u_{1a} = (y/\delta)^{1/n} \tag{17}$$

It is assumed that the gas within the boundary layer follows an isentropic one-dimensional stream-tube expansion from p_1 to p_2.[13] Here, we assume that the flow direction is the same ($\theta = \theta_2$) for all stream tubes after expansion.

The velocity profile after expansion is given as

$$\phi_2 = \frac{u}{u_{2a}} = \left[\frac{C_{1a}^2}{C_{2a}^2}\left\{\phi_1^2 + \left(\frac{C_{2a}^2}{C_{1a}^2} - 1\right)\frac{1 - C_{1a}^2\phi^2}{1 - C_{1a}^2}\right\}\right]^{1/2} \tag{18}$$

and the corresponding location within the viscous layer after the expansion can be determined by mass conservation as

$$\frac{y_2}{\delta_1} = \left(\frac{1 - C_{1a}^2}{1 - C_{2a}^2}\right)^{1/(k-1)} \frac{C_{1a}}{C_{2a}} \int_0^{y_1} \frac{\phi_1}{\phi_2}\frac{dy_1}{\delta_1} \tag{19}$$

By using this profile ϕ_2, we can calculate the location of the displaced origin of the equivalent mixing layer profile (error function profile) x_o and y_o. Also, $\delta_1^{**}/\delta_2^{**}$ can be evaluated for these velocity profiles.

D. Mass Conservation

We define a jet boundary streamline as a streamline that starts at the separation point and lies between the mass flux entrained in the mixing layer and the mass flux of freestream which passes through the reattachment (realignment) location. Therefore, across the jet boundary streamline, there is no net mass transfer into the wake. The location of this streamline η_J at X = X is given[12] by the control volume technique as

$$\int_{\eta_J}^{\eta_j} \frac{\rho}{\rho_{2a}} \phi_2 d\eta = \frac{\sigma \delta_2^{**}}{X + x_o} \tag{20}$$

where η_j is the location of a jet boundary streamline when δ is negligible and is determined by the relation

$$\int_{\eta_j}^{\eta_R} \frac{\rho}{\rho_{2a}} \phi_2 d\eta = \int_{-\infty}^{\eta_R} \frac{\rho}{\rho_{2a}} \phi_2^2 d\eta \tag{21}$$

Energy may be transferred across the jet boundary streamline. In this case of a steady, closed wake, the jet boundary streamline coincides with the dividing streamline.

E. Realignment Process

The separated flow is forced to realign to the original direction ($\theta_4 = 0$). The discriminating streamline discriminates between the flow that penetrates the pressure increase caused by recompression and the flow that recirculates. It coincides with the jet boundary streamline when there is no mass bleed or suction.

The escape criterion is stated as follows. If the Mach number of the flow on the discriminating streamline is less than unity, the flow stagnates and regains its isentropic pressure, which is just sufficient to match the static pressure in region 4. Thus,

$$(p_{o3})_d = p_4 \quad \text{if } M_{3d} \leq 1.0 \tag{22}$$

If M_{3d} is greater than unity, the flow passes through a normal shock and then regains its total pressure at stagnation point.[14] Therefore,

$$\left(p_{o_{3y}} \right) = p_4 \quad \text{if } M_{3d} > 1.0 \tag{23}$$

Fluid within the mixing region having a pitot pressure greater than the static pressure in region 4 will pass on into

4, whereas fluid with a pitot pressure less than the static pressure of region 4 will turn upstream to recirculate. Therefore,

$$\frac{p_1}{p_2} = \frac{p_4}{p_3} = \frac{\left(p_{o_{3y}}\right)_d}{p_3} = \frac{\{[(k+1)/2]M_d^2\}^{k/(k-1)}}{\{[2kM_d^2-(k-1)]/(k+1)\}^{1/(k-1)}} \text{ for } (M_d > 1) \tag{24}$$

or

$$\frac{p_1}{p_2} = \frac{p_4}{p_3} = \frac{p_{o_{3d}}}{p_3} = \left(1 + \frac{k-1}{2} M_d^2\right)^{k/(k-1)} \quad \text{for } (M_d^2 \leq 1) \tag{25}$$

F. Heat-Transfer Component

There are two resistances to heat transfer between the freestream and the base wall: 1) the freejet mixing region, and 2) the recirculating flow (see Fig. 2). The relation between T_{oa} and T_b is given in Eq. (15) as a function of ϕ_2. In order to get the relation between T_b and T_w, we need to know the velocity profile of the upstream recirculating flow. However, it depends upon many undeterminable parameters. Therefore, as an approximation, the recirculating flow is assumed to be a mirror image of the main mixing from the zero velocity line to the jet boundary streamline.[9] It therefore satisfied the mass conservation between the mass entrained and mass that is recirculated. In the mirror image,

$$T_w/T_b = (T_{oj}/T_b)^{-1}_{m.r.} \tag{26}$$

Thus, finally,

$$\frac{T_w}{T_{o_{2a}}} = \frac{\left(T_b/T_{o_{2a}}\right)^2_{refl.}}{\left(T_b/T_{o_{2a}}\right)_{refl.} + [1 - \left(T_b/T_{o_{2a}}\right)_{refl.}](u_j/u_{2a})} \tag{27}$$

It is assumed that the wake temperature varies. Near region 3 between the centerline and the freestream, it is the same as the freestream. Yet the wake temperature between the freestream and the base wall can be calculated by the reflected image approximation, where the value $(T_b/T_{oa})_{m.r.}$ is calculated from the mixing layer profile just before the realignment.

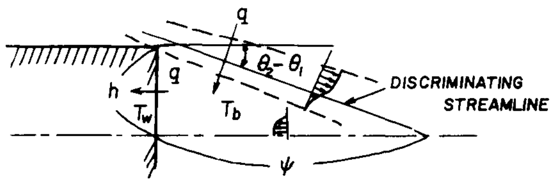

Fig. 2 Sketch illustrating two resistances to heat transfer.

Then, the overall bulk temperature is an average of these two wake temperatures, which are weighted according to the path length

$$\left(\frac{T_b}{T_{o_{2a}}}\right)_{m.r.} = \frac{h\left(T_b/T_{o_{2a}}\right)_{refl.} + \psi}{h + \psi} \tag{28}$$

Finally,

$$\left(\frac{T_b}{T_{o_{2a}}}\right)_{refl.} = \frac{\left(T_b/T_{o_{2a}}\right)_{m.r.} - 1 + \sin(\theta_2 - \theta_1)}{\sin(\theta_2 - \theta_1)} \tag{29}$$

Thus, T_w/T_{oa} is obtained from Eqs. (27) and (29).

The heat-transfer rate to the base must equal the heat-transfer rate through the jet boundary streamline. It is evaluated easily by applying the control volume analysis:

$$\frac{\sigma\Omega_c}{c_p\rho_{2a}u_{2a}T_{o_{2a}}X} = \left(1 + \frac{x_o}{X}\right)\int_{\eta_J}^{\eta_R}\Bigg|_{x=x} \frac{\rho}{\rho_{2a}}\,\phi_2\left(1 - \frac{T_o}{T_{o_{2a}}}\right)d\eta$$

$$- \frac{x_o}{X}\int_{\eta_J}^{\eta_R}\Bigg|_{x=0} \frac{\rho}{\rho_{2a}}\,\phi_2\left(1 - \frac{T_o}{T_{o_{2a}}}\right)d\eta \tag{30}$$

where Ω_c is the heat flow per unit width through the jet boundary streamline from wake to surroundings.

After rearrangement, we get the solution for the average heating rate q:

$$\frac{q}{p_{o1}\sqrt{T_{o1}}} = \frac{-\Omega_c}{hp_{o1}\sqrt{T_{o1}}} =$$

$$\frac{c_p M_{2a}(C_{2a}^2 - 1)E}{\sigma\sqrt{R/k}\{1 + [(k - 1)/2]M_{2a}^2\}^{(k+1)/2(k-1)} \sin(\theta_2 - \theta_1)} \tag{31}$$

where $(1 - C_{2a}^2)E$ is the right-hand side of Eq. (30), and $q = -\Omega_c/h$ stands for the average heat-transfer rate to the base per unit area and unit time.

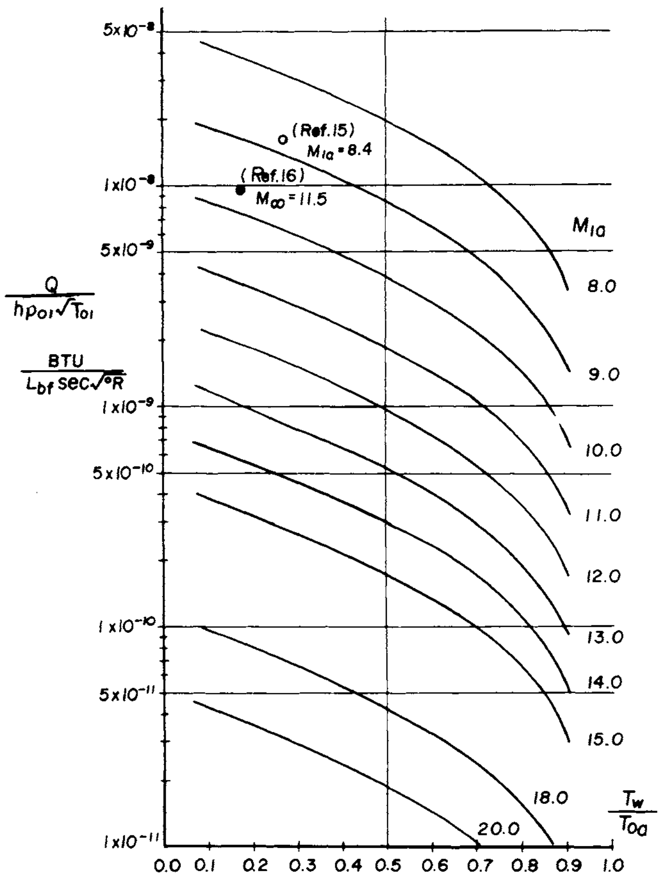

Fig. 3 Base heat-transfer variation with wall temperature.

G. Calculation Procedure

It should be noted that, if we choose the conditions of region 2, such as M_{2a}, δ_2^{**}, and $T_b/T_{o_{2a}}$, as independent variables, we can avoid the trial-and-error approach. The procedure of the calculation is described as follows:

1) M_{2a}, $T_b/T_{o_{2a}}$, and δ_2^{**}/X are given.

2) η_m, η_j, and σ are calculated from Eqs. (9-11, 16, and 21).

3) x_o is obtained from Eq. (13).

4) η_J is calculated by Eq. (20).

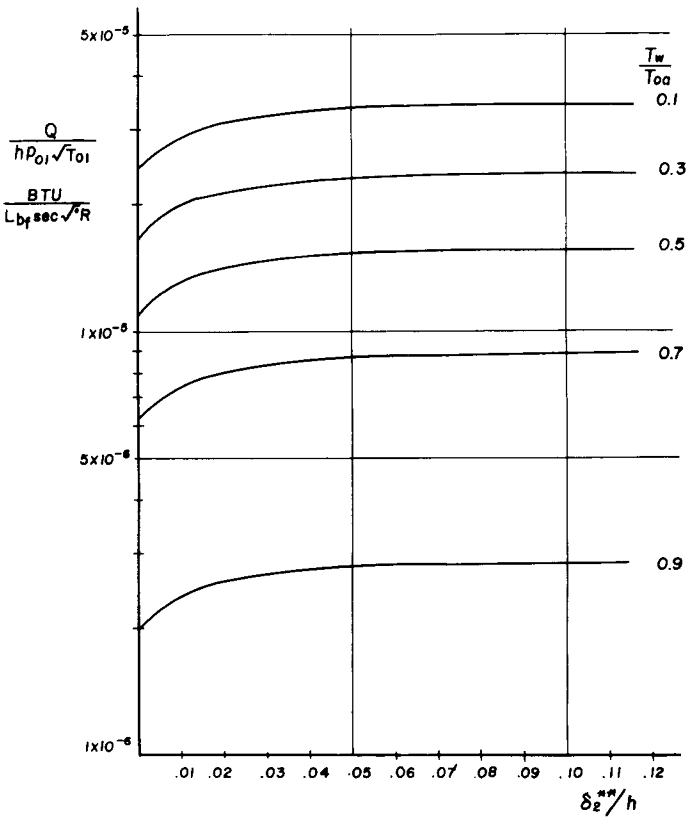

Fig. 4 Base heating (M_1 = 3).

5) p_1/p_2 is evaluated by Eq. (24) or (25).

6) $\theta_2 - \theta_1$ and M_{1a} are calculated from Eqs. (1) and (2).

7) $\left\{T_b/T_{o_{2a}}\right\}_{refl.}$ is calculated from Eq. (29), and $\left\{T_w/T_{o_{2a}}\right\}$ is obtained from Eq. (27).

8) Heat transfer $-\Omega_c$ is calculated from Eqs. (30) and (31).

9) $\delta_1^{**}/\delta_2^{**}$ is calculated from Eqs. (17-19).

Results and Discussion

Some calculated results are shown in Figs. 3-10. In Fig. 3, heat-transfer rate to the base is shown vs T_w/T_{oa}, when the

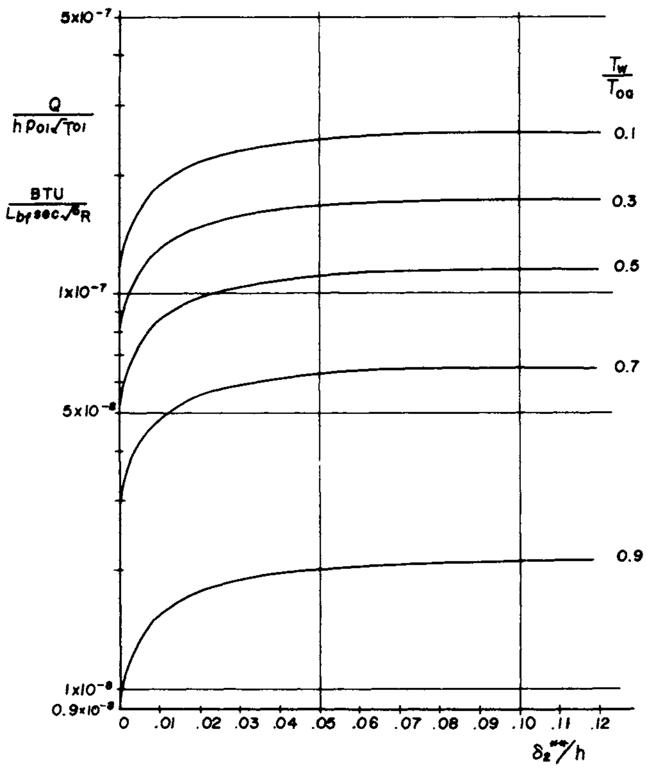

Fig. 5 Base heating (M_1 = 7).

boundary-layer thickness is negligible. Results for the range of M_{1a} from 8 to 20 are shown here. (For M_{1a} from 1 to 7, see Ref. 7.) Experimental data also are shown in this figure. The data of Ref. 16 were calculated using freestream conditions, since the local properties were not provided in the report. Agreement between the theory and the experiment seems reasonable. The authors searched for additional experimental data in the literature, but they turned out to be very scarce for Mach numbers greater than 8.

The variation of the base heating rate vs δ_2^{**}/h is shown in Figs. 4-6 for M_{1a} of 3, 7, and 15, respectively. The initial rapid increase of heating rate and tendency to level off with the boundary-layer thickness are noticed in all cases. Moreover, the higher the Mach number, the greater will be the relative increment $\Delta q/q$ and the smaller the δ_2^{**}/h at which the variation becomes negligible. The tendency of increasing heating rate with the boundary-layer momentum thickness agrees

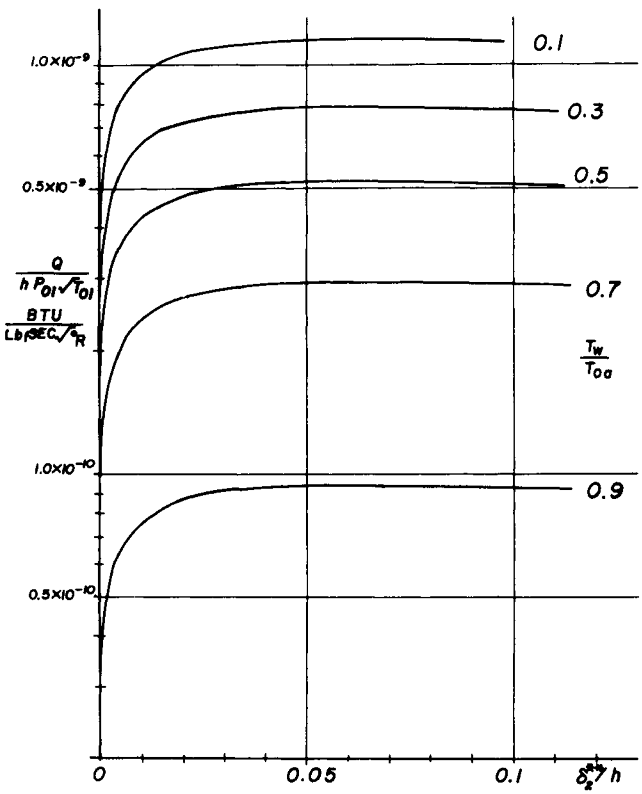

Fig. 6 Base heating (M_1 = 15).

qualitatively with the free-flight testing results[17] of a re-entry vehicle at M = 18.

In Figs. 7-9, the base pressure ratio p_2/p_1 vs δ_2^{**}/h is shown for M_{1a} of 3, 7, and 15, respectively. It is seen that the base pressure increases with the boundary-layer momentum thickness and also that it is insensitive to a very large variation in the ratio of T_w/T_{oa}.

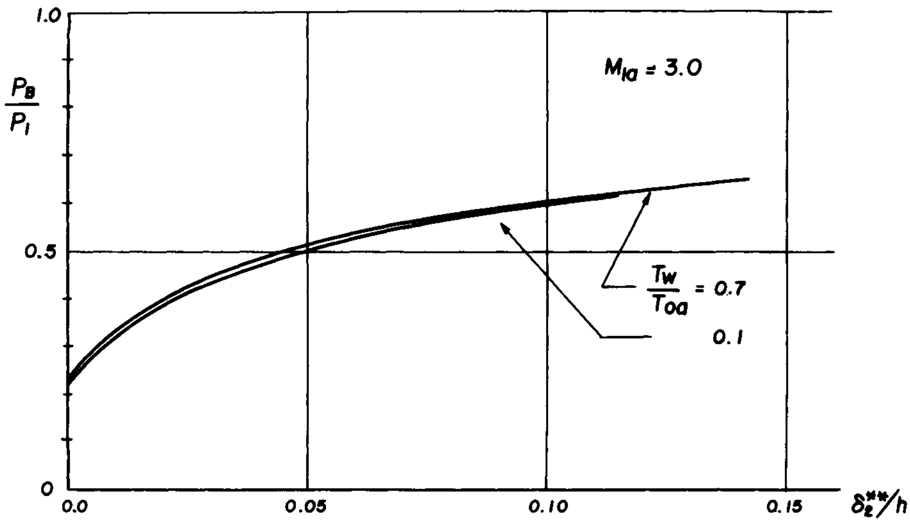

Fig. 7 Base pressure variation.

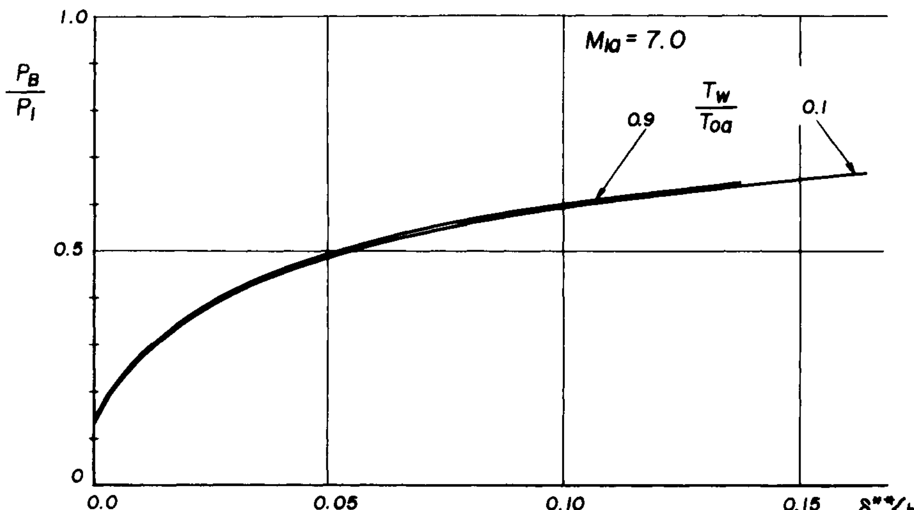

Fig. 8 Base pressure variation.

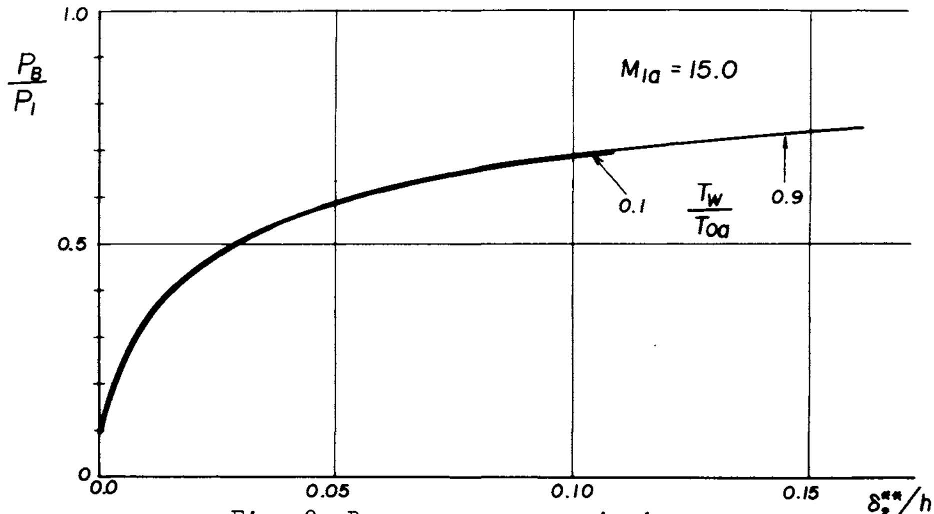

Fig. 9 Base pressure variation.

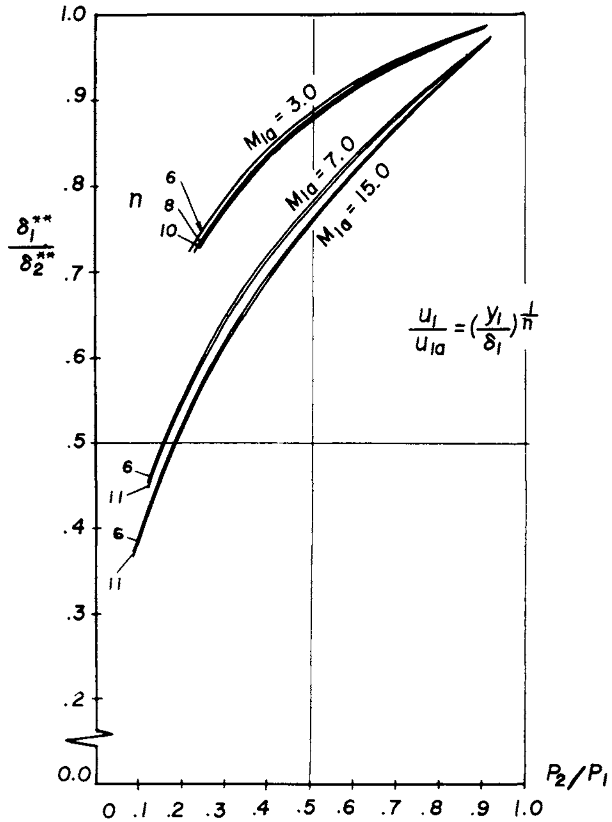

Fig. 10 Boundary-layer thickness variation.

In Fig. 10, the momentum thickness ratio across the expansion, $\delta_1^{**}/\delta_2^{**}$, is shown for the cases of M_{1a} = 3, 7, and 15 vs the pressure ratio p_2/p_1. These values are evaluated assuming $\phi_1 = (y/\delta_1)^{1/n}$ and using the isentropic stream-tube approach. Since the pressure ratio p_2/p_1 may be calculated for any M_{1a}, T_{oa}, and T_w, the abscissa of Figs. 4-9 easily can be transformed to δ_1^{**}/h from the results of Fig. 10.

The flow deflection angle, $\theta_2 - \theta_1$, also may be obtained from the pressure ratio and the approach Mach number by means of the Prandtl-Meyer relations. The most significant conclusion of these studies is that the aerothermodynamic base heating rate is significantly dependent upon the approach boundary-layer thickness, in addition to the wall temperature ratio.

Acknowledgment

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VORTICES INDUCED IN A STAGNATION REGION BY WAKES

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Abstract

Horseshoe-like vortices, induced by wakes in the stagnation region of bluff bodies, are proposed as an efficient mechanism for augmentation of convective heat transfer. The vortex "flow module" induced by single or multiple wakes, which had not been observed previously, was documented first, and the resulting flowfield was studied using various visualization techniques and hot-wire anemometry. In an attempt to understand the driving force behind this flow module, the conditions at which incipient formation of the vortices occurs were investigated. Existence of such a threshold is essential and was hitherto an open question in analytical studies of stability of flow in stagnation regions. Finally, effects of the flow module on heat transfer from a cylinder were measured.

Nomenclature

A_e = effective area of heated cylinder under surface strip
A_s = area of surface strip
D = characteristic dimension of bluff body on which wakes are impinging; diameter of circular cylinder or width of rectangular cylinder
d = diameter of wake-generating cylinder
h_a = local coefficient of convection heat transfer for air
M = mesh size, i.e., distance between neighboring centerlines of multiple wakes of cylinders
Nu = Nusselt number of cylinder in absence of vortex flow module

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Nu' = Nusselt number of cylinder in presence of vortex flow module
$q_{s \to a}$ = heat transferred from surface strip to air
$q_{w \to s}$ = heat transferred from water to strip
Re_D = Reynolds number of bluff body or downstream cylinder $\equiv U_\infty D/\nu$
Re_d = Reynolds number of wake-generating cylinder $\equiv U_\infty d/\nu$
Re_M = Reynolds number of mesh size $\equiv U_\infty M/\nu$
T_a = freestream air temperature
T_s = surface temperature of cylinder at location of strip, i.e., temperature of strip operating in constant current mode
T_w = temperature of water inside cylinder
U_{w+c} = effective coefficient of heat transfer by convection from water to cylinder and by conduction through the cylinder wall to the strip
U_∞ = average freestream velocity
ΔU = maximum velocity defect in wake
Δx = separation distance between wake-generating cylinders and stagnation line of bluff body or downstream cylinder
y = direction perpendicular to flow direction and axis of cylinder which generates wake
y_o = half-width of wake
δ = stagnation region boundary-layer thickness
ξ = distance measured upstream from stagnation line of bluff body or downstream cylinder
θ = angle of mean position of surface strip with freestream direction
σ = solidity of array of cylinders generating multiple wakes = d/M

I. Introduction

This study was initiated to confirm the conjecture that velocity defects approaching the stagnation region of a bluff body can lead to the formation of horseshoe-like vortices. This conjecture was made first by Morkovin, based on his experience with flow over cylinders[1] and near obstacles placed on the floor of a boundary layer.[2] The presence of such horseshoe vortices near the base of a building exposed to the atmospheric boundary layer is well known in the field of wind engineering and has been observed and documented in various experimental studies (e.g., see Corke and Nagib[3] and references therein). However, the presence of horseshoe-like vortices, induced in a stagnation region by wakes or velocity defects, had not been documented previous to the present study.

These vortices may be induced by the impingement of a wake on a bluff body perpendicular to its axis, as illus-

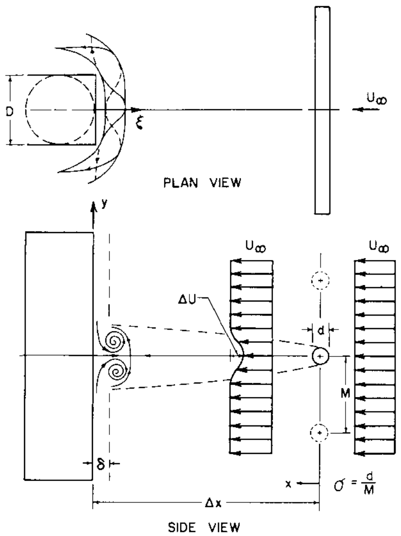


Fig. 1 Schematic of vortex flow module.

trated schematically in Fig. 1. When the pressure gradient caused by the velocity defect of the wake is sufficiently strong, it may result in the establishment of a pair of counter-rotating vortices (or cells). Such vortices turn from an initial direction perpendicular to the mean flow to a direction parallel to U_∞.

Objectives

The present investigation documents the existence of such a phenomenon for single and multiple wakes with the aid of flow visualization and hot-wire anemometry. In an attempt to establish and to understand the driving forces behind this flowfield, the conditions of incipient formation of the vortices also have been investigated. The presence of such a threshold is an essential ingredient and hitherto open question in analytical studies of stability of the flow in the stagnation region of bluff bodies, e.g., circular cylinders. Tani's recent suggestion[4] about the existence of a subcritical instability implies that the flowfield has a critical threshold of instability, a hypothesis that has not been confirmed prior to the present study (which was carried out without knowledge of Tani's theory).

The horseshoe-like vortices also are proposed as an important "flow module" arising in connection with interacting groups of bluff bodies and as an efficient mechanism for the augmentation of convective heat transfer from the body. In studying complex flowfields, Morkovin proposes that the focus should be on salient features that are grouped into coherent modules, i.e., "identifiable, morphologically invariant, mildly interacting flow structures."[5] The augmentation of convective heat transfer from bodies via the introduction of three-dimensional flow modules in their flowfield is a most promising idea. The horseshoe-vortex flow module is used here in the first application of this idea. Other applications presently are contemplated at the Illinois Institute of Technology (IIT).

In the last part of the present study, the effect of the vortex flow module on the heat transfer from a circular cylinder is measured. Multiple wakes generated by an array of parallel cylinders are used to induce the flow module along the stagnation line of the cylinder. A circular cylinder was used in order to be able to evaluate the effects of the flow module and of its carrying velocity field at various streamwise locations. The circular cross section was selected also because of its simple geometry and the availability of heat-transfer data regarding it.

Comments on Grid-Generated Turbulence

Various analytical and experimental studies have been conducted in order to understand the mechanisms leading to augmentation of heat transfer from bluff bodies. In particular, many of these studies have been devoted to the effects of freestream turbulence and vorticity on the heat transfer. To introduce turbulence of varying intensities into the freestream, several experimenters have relied on grids of various mesh sizes and solidities. Although grid-generated turbulence in wind tunnels is the most accepted means of simulating isotropic turbulence, care must be exercised in using it. First, the grid must be manufactured homogeneously, and it should cover the entire test section of the tunnel. Second, the solidity of the grid must be low and should not exceed 40%. And, finally, even with the preceding two conditions satisfied, one should not position the test model closer than 20 to 30 mesh sizes downstream of the grid. Nearer to the grid, the turbulence has not achieved adequate homogeneity, and the spatial variations of the mean velocity are larger than 0.1%. The significance of such small velocity defects will be evident to the reader from the results of the present study. On the other hand (again in the presence of the preceding two conditions), one must locate the bluff body under investigation at a downstream distance less than 15 mesh sizes from the grid in order to expose it to turbulence intensities equal to or in excess of approximately 5%.

The preceding conditions, which impose several restrictions on the experiment, are based on detailed measurements of grid turbulence and have been presented in the literature, by various investigators, e.g., Nagib et al.[6,7] They suggest that in some of the heat-transfer experiments reported in the literature velocity defects in excess of 0.1% may well have been present. In such cases, the measured augmentation of heat transfer from the bluff body may be attributed to the turbulence, as reported by the original investigators, or may be due to the flow module proposed here. The latter mechanism is based on motions of much larger scales compared to the turbulence eddy size and is particularly plausible in view of the results presented in this paper. One would expect the larger scales to be more efficient in carrying heat away from the surface of the cylinder. It is hoped that this study will shed some light on this question.

Some Relevant Literature

As early as 1955, Kestin et al.[8,9] studied the influence of turbulence on heat transfer from cylinders. Various other studies are available in the literature, including the work of Smith[10] and of Kayalar.[11] In particular, Kayalar's results indicate large increases in heat transfer as the turbulence intensity is increased above 1%. The existence of streamwise (longitudinal) vortices in the stagnation region of cylinders has been documented by Hassler,[12] Colak-Antić[13] and others. Similar vortices were observed by Sadeh and Cermak[14] on the upstream face of other bluff bodies. Many of these investigations report a spanwise characteristic wavelength of these vortices. However, the role of such vortices in the heat transfer from the bluff body has not been assessed by any of the foregoing investigators.

Görtler[15] and Hämmerlin[16] were the first to investigate analytically the stability of the two-dimensional, inviscid, irrotational stagnation flow on a flat plate. Substituting the basic Hiemenz similarity solution plus an infinitesimal perturbation proportional to $e^{\alpha t} \cos(kz\sqrt{a/\nu})$, where a is a constant of the stagnation flow, into the Navier-Stokes equation, they obtained a boundary-value problem with k as the eigenvalue. When they applied the principle of exchange of stabilities, that is, nonoscillatory growth or decay (i.e., α is real), they could not find solutions to the boundary-value problem. Therefore, following Taylor's analysis[17] of a completely different problem, they, as well as all investigators following them, have assumed $\alpha = 0$, i.e., steady solutions. The question of how suitable this assumption is in representing random, unsteady turbulent flowfields has been pointed out for the first time and discussed in detail recently by Morkovin.[18]

For the neutral stability case ($\alpha = 0$), Hämmerlin[16] obtained the following vorticity equation:

$$\hat{\omega}'' + (\eta-\beta)\hat{\omega}' + (1-k^2)\hat{\omega} = 0 \tag{1}$$

where β is a constant. The solution of this equation is in the form of hypergeometric functions and yields a continuous spectrum of eigenvalues between 0 and $\sqrt{a/\nu}$, i.e., $0<k<1$. He found no preferred or discrete wave numbers within this spectrum. Sutera et al.[19] postulated the vorticity amplification theory based on the asymptotic solutions of Eq. (1). One of the major problems with their approach is the unbounded vorticity generated in the field for $k > 1$. Kestin and Wood[20] incorporated a finite radius of the stagnation region, instead of the flat plate with infinite radius, and reduced Eq. (1) to a Sturm-Liouville problem whose solution possesses a single eigenvalue of $k = 0.58$. Tani[4] and Iida point out that the vorticity profile assumed to have no zeros by Kestin and Wood is not appropriate, and repeat the finite radius solution without this assumption, obtaining the eigenvalue $k = 0.35$. Recently, Inger[21] considered the stability of the flat-plate stagnation flow to infinitesimal disturbances for a nonadiabatic compressible flow. He found that the maximum wavelength of the eigenvalue spectrum is dependent on the wall temperature ratio.

Although fair agreement is found between some of the experiments and the theories,[4] we do not believe that such an agreement validates the approaches. We agree with Morkovin[18] that "interpreting unsteady heat transfer due to turbulence through steady equations is questionable until proven otherwise." The role of the analysis in predicting the incipient formation of horseshoe vortices, caused by vorticity or velocity defects in the flow, is of course very important by itself, since the vortices will have some effects on the heat transfer, as demonstrated in the following sections.

II. Experimental Approach

Water Channel

The IIT water channel was used for visualization purposes in this investigation. The channel, which is 3x6x192 in. in dimension, has walls made of plexiglass. Two independent traversing mechanisms enable precise positioning of the dye probes. Preceding the test section is a series of five turbulence manipulators made of different mesh honeycombs. Following the test section are two more honeycomb manipulators.

The turbulence manipulators are arranged to insure a steady uniform laminar flow in the test section.

Wind Tunnels

Two wind tunnels were used in this study. The IIT Flow Visualization Wind Tunnel was used first to establish the existence of the vortex flow module and to determine velocity profiles in the region of interest. To insure good visibility, the test section, 6x29x96 in., is completely made of plexiglass. This tunnel operates in the open return mode, and it has a turbulence intensity of less than 0.1% for the range of flows used here. Visualization is achieved by using smoke from a Nichrome ribbon, 0.0126 in. thick and 0.063 in. in chord, which is coated with paraffin wax and then heated electrically during the experiment. Although this approach can be applied in many visualization experiments, in this investigation the ribbon was either used in place of the upstream wake-casting cylinder or mounted along the stagnation line of the downstream cylinder.

The second wind tunnel used in this study was the IIT Environmental Wind Tunnel, which operates in the closed return mode, permitting the utilization of two test sections. All of the heat-transfer measurements were carried out in the high-speed test section, which has the dimensions 2x3x10 ft. Uniform and steady velocities as low as 3 fps can be obtained in this section. The maximum spatial variation and drift in the freestream are approximately 0.1% and 1% of the set velocity, respectively.

Multiple Wakes Generating "Zither"

Before the heat-transfer measurements could be carried out, a frame upon which very fine wires can be strung had to be constructed. This frame is referred to as the "zither," a name coined by Kellogg[22] for a similar device. A number of requirements for the zither had to be met by the design. The zither should be movable inside the test section, so that changes in its position with respect to the heat-transfer model can be made during each test run. This should be possible without changing any other parameters, including the exact freestream conditions, which would have to be changed if the tunnel had to be opened to reposition the zither. It is also important, for calibration of instrumentation and verification against benchmark conditions, that the zither can be removed from the test section. The first of these requirements leads to a zither that is totally contained inside the test section. The side members of the zither are

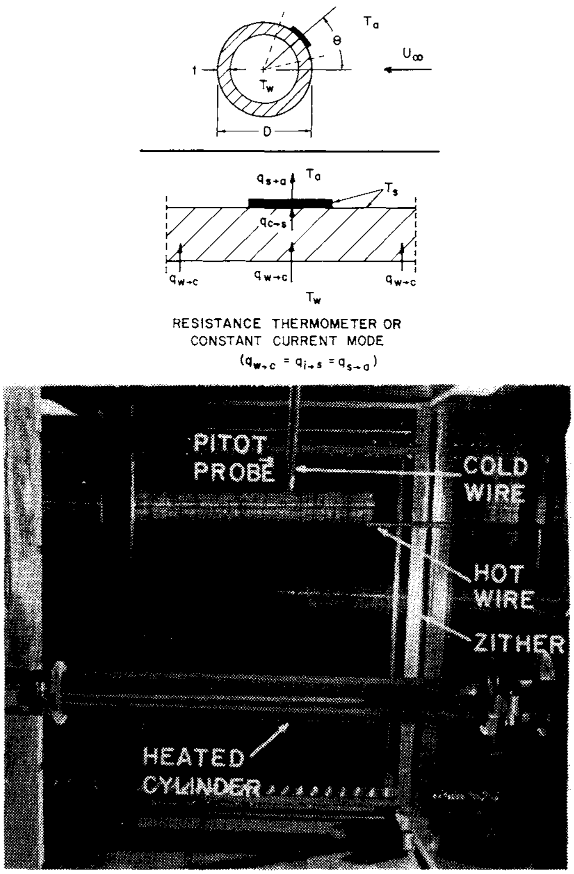


Fig. 2 Schematic diagram and photograph of heat transfer experimental setup.

only 1 in. thick, and the effective thickness of the top and bottom is 1.5 in. All of the upstream corners are rounded off. This design resulted in minimum aerodynamic disturbances which were checked and evaluated carefully[23]; see Fig. 2 for a general indication of the blockage.

In order for the wires to be tightly strung, "tuning pins" are used to adjust the tension in the wires properly. These pins are made from ordinary 1/4-20 nuts and bolts, and each pin holds a single wire strung up and back three times. A wire with a 0.005-in. diam. was chosen so that laminar wakes could be generated within the operating range of the wind tunnel. Heat-transfer experiments in this study were carried out using a solidity of the zither of 5%, i.e., a separation distance between wires of 0.100 in. For details of the construction of the zither, as well as all other aspects of the experiments, the reader is referred to the report[23] by the same authors.

Heat-Transfer Model and Measurement Technique

Various surface boundary conditions are imposed in the analysis of heat transfer by convection. Some of these are constant heat flux conditions, constant surface temperature conditions, and mixed or other boundary conditions. Experimentally it is not easy to duplicate any of the exact boundary conditions, and quite often one must be satisfied with some intermediate ones. However, the resulting ambiguity can be minimized if the measurement techniques are selected carefully and proper "management" of the data acquisition is employed.[23] In particular, the uncertainty in the results can be reduced considerably for a study where comparative measurements are sought, i.e., when the results can be presented as changes in a variable due to a change in a condition or a parameter. All other parameters must be held exactly the same during each comparative measurement.

The objective here is to measure the change (e.g., augmentation) in the heat transfer from the bluff body when the vortex flow module is introduced into the flowfield of the cylinder. As the threshold results demonstrate, the presence of the vortex flow module depends on the separation distance between the "zither" and the cylinder; hence, the requirement for the movable zither was incorporated in the design. A schematic of the heated cylinder and the arrangement with respect to the zither are shown in Fig. 2, in a view looking upstream from inside the test section. The cylinder (2.0 in. o.d., 1.75 in. i.d., and 24 in. long) is made of pyrex glass and has a thin ($\approx 10^{-5}$ in. thick) gold strip baked onto its surface over an arc of 22 deg. The strip is used to measure the surface temperature on the cylinder. The cylinder and strip can be rotated easily to any angle θ during a test. Heating is supplied to the cylinder by a recirculating water system. The temperature of the water can be held to within $0.5^{\circ}F$ indefinitely.

If conduction from the strip to the cylinder (or vice versa) can be neglected with respect to the convection from the strip to the air, direct measurement of the convection heat transfer from the surface of the cylinder, at the position of the strip, can be made by operating the strip in a constant-temperature mode. The electrical heating required to maintain the strip at the surface temperature of the heated cylinder, or slightly above it, would be directly proportional to the convective heat transfer. Several modifications[23] of the present strip design are required before this approach can be utilized.

On the other hand, operating the strip in a constant low-current mode provides direct and accurate measurement of the surface temperature at the strip position, after careful calibration. Changes in the surface temperature then can be measured and related to changes in the heat transfer from the cylinder caused by the vortex flow module. The only assumption required is that near the strip position $\partial q/\partial\theta=0$, i.e., local constant heat flux boundary conditions. This assumption is quite valid for $\theta=0$ and 180 deg. Hence, the only measurements presented here which may suffer from this assumption are those made at $\theta=60$ deg.

Based on the preceding assumption, one can write

$$q_{s\to a} = q_{w\to s} \tag{2}$$

Therefore,

$$h_a A_s (T_s - T_a) = U_{w+c} A_e (T_w - T_s) \tag{3}$$

and

$$h_a = (U_{w+c} A_e / A_s)[(T_w - T_s)/(T_s - T_a)] \tag{4}$$

In Eq. (4), all of the variables, except T_s, remain unchanged for a particular θ and Reynolds number when the vortex flow module is introduced by bringing the zither closer to the cylinder. Therefore, accurate measurement of the change in T_s and good measurements of T_w and T_a can result in calculating the change in the Nusselt number with the aid of Eq. (4), i.e.,

$$Nu'/Nu = h_a'/h_a \tag{5}$$

Returning the zither to its original upstream position, the measurements can be checked.

Instrumentation

Also shown in the bottom photograph of Fig. 2 are the relative positions of the hot and cold wires used in monitoring the freestream velocity and temperature. The pitot probe shown is used to determine the undisturbed freestream velocity U_o upstream of the zither. The hot and cold wires were operated using standard DISA anemometer units and associated signal-conditioning and measurement instrumentation. The cylinder strip was operated and its output measured using a broken bridge circuit utilizing a high-gain operational

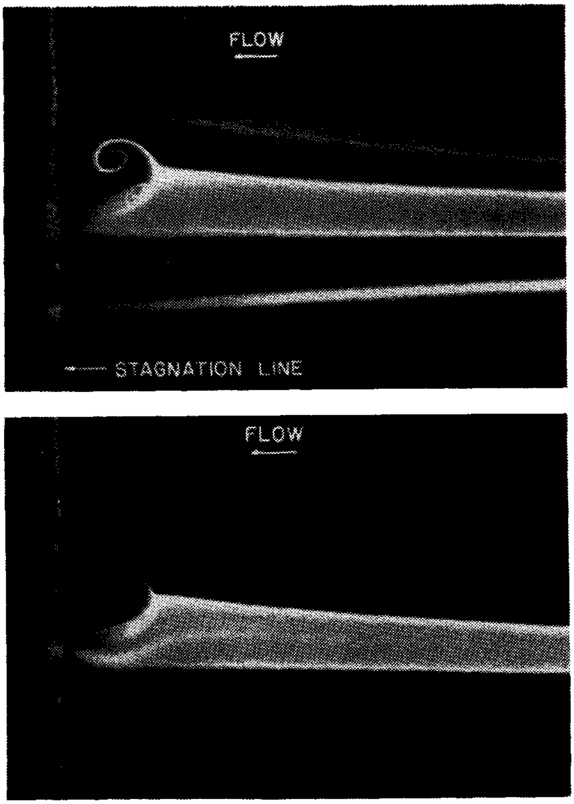


Fig. 3 Smoke visualization of vortex flow module formed by an asymmetric wake.

amplifier. Employing this instrumentation, the described experimental approach leads to a very high confidence level within the symbols used to plot the figures.

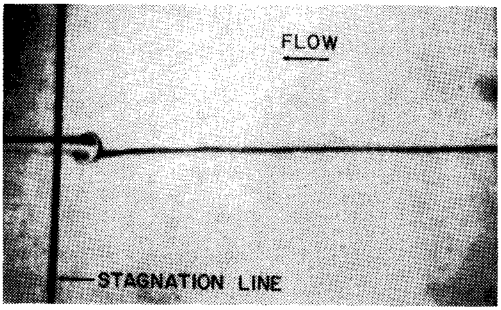
III. Results and Discussion

Documentation of Vortex Flow Module

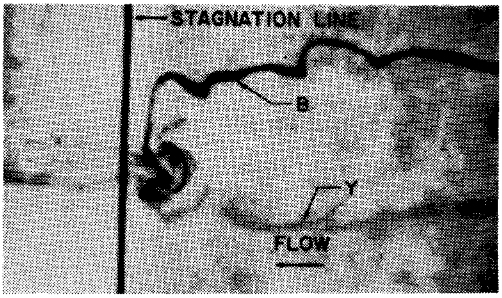
The first documentation of the flow module schematically represented in Fig. 1 was achieved[23] by impinging, on a rectangular cylinder, the wake of the smoke-generating ribbon, which was set at an angle of attack of +45 deg. The asymmetric wake permitted the recording of the photographs of Fig. 3 at a freestream velocity slightly over 2 fps ($Re_d \approx 50$). A sheet of light parallel to the cylinder was used for illumination.

In order to improve the quality of the freestream and of the visualization, and to permit investigation at lower Reynolds numbers, the water channel equipped with multiple dye streaks of different colors was used. With the help of a mirror, positioned above the test section and oriented at 45 deg to the horizontal plane, both the top and side views of the flowfield can be observed simultaneously. Conclusions drawn from any of the observations always were based on the two-view monitoring of the flow in the test section.

Figure 4 is a side view of a single upstream wake impinging on a rectangular cylinder. In the top photograph (Re_d=30) the dye first is made to strike the upstream cylinder before rolling up into the vortical cells at the stagnation line. As can be seen, the wake and the vortices are steady and laminar at this low Re_d. In contrast, the lower photograph shows the vortical cells at a much higher wake Reynolds number (Re_d=365). In this case, a larger upstream cylinder was used, and two dye streaks (B for blue and Y for yellow) were intro-



Re_d=30; Re_D=1040; $\Delta x/d$=111; d=.027 in.; D=.935 in.



Re_d=365; Re_D=2730; $\Delta x/d$=32; d=.125 in.; D=.935 in.

Fig. 4 Side-view dye visualization of a single wake impinging on a rectangular cylinder.

duced upstream of the last turbulence manipulator. Although on the average the vortex pair is in the same place, many disturbances convected in the wake are strong enough to kick it out of position at these higher Re_d values.

Further documentation of this vortex flow module was achieved by using hot-wire anemometry in the wind tunnel. Shown on top of Fig. 5 are velocity profiles at various positions upstream of the rectangular cylinder with no wake-casting cylinder present. The bottom part of Fig. 5 displays similar profiles in the presence of a wake impinging on the stagnation region. Although the flowfield within the vortex module is highly three-dimensional and hence difficult to map with a single hot-wire sensor, the measurements in Fig. 5 give an indication of the size of the vortices and of the magnitude of velocities within them. The profiles indicate a more active flow within the module compared to the undisturb-

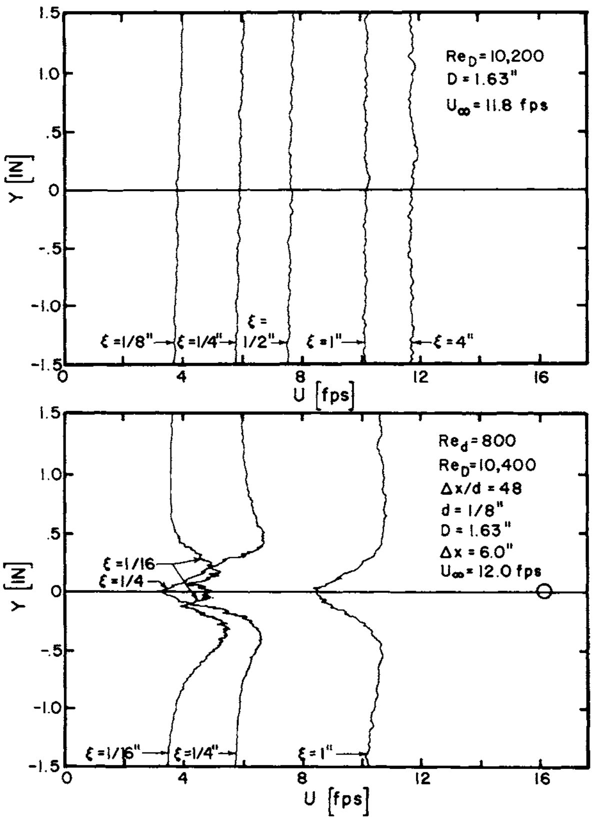
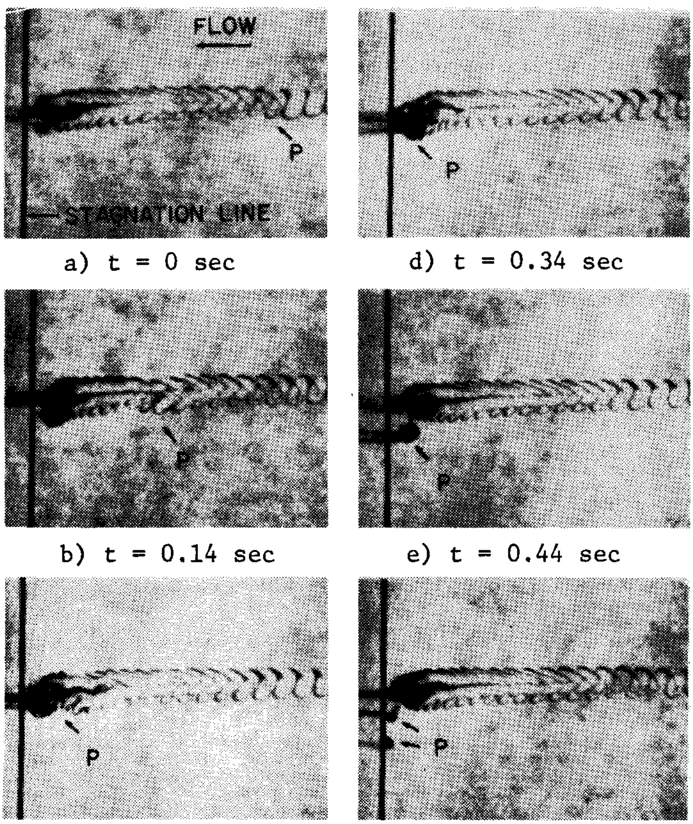


Fig. 5 Velocity profiles in stagnation region of a rectangular cylinder without and with an upstream cylinder.

ed stagnation region. No definite conclusion can be made regarding the size of the vortices. However, we surmise that the vortices scale with the size of the wake rather than the boundary-layer thickness in the stagnation region.

The vortices are steady for $Re_d \lesssim 30$ but are sensitive to random freestream disturbances at higher values of the wake Reynolds number. The wake in Fig. 6 is in the unsteady Kármán vortex sheet range, $Re_d=90$, and the incidental freestream disturbance results in a perturbation P to the periodic wake flow. Again, the vortex flow module is always present on the average but is kicked out of its position into the "freestream" by the perturbation. When this occurs, a



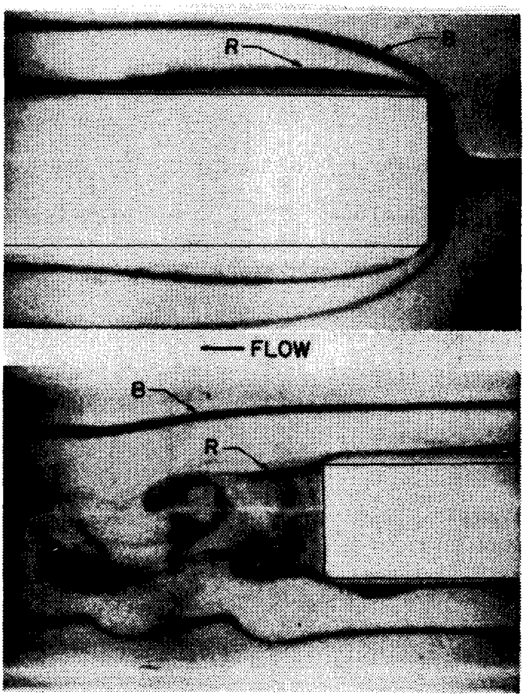
a) t = 0 sec
d) t = 0.34 sec
b) t = 0.14 sec
e) t = 0.44 sec
c) t = 0.24 sec
f) t = 0.55 sec

Re_d=90; Re_D=3120; $\Delta x/d$=111; d=.027 in.; D=.935 in.

Fig. 6 A time sequence of side-view dye visualization showing effects of a freestream perturbation (P) on vortex flow module.

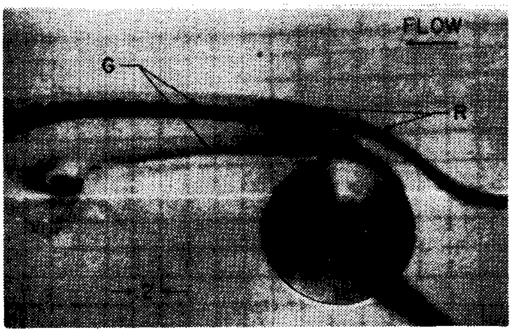
large amount of the fluid in the vortices, which is marked by the dye, is ejected away, providing a very efficient mechanism for the mixing of the flow in the stagnation region with that in the freestream. It is conjectured that this mechanism is responsible for the increased augmentation of heat transfer by the flow module when $25<Re_d<40$. Behavior similar to that in Fig. 6 is observed in high-speed movies of the flow in the case of multiple wakes, where the exchange of vortical fluid occurs not only between the cells and the freestream but also between neighboring pairs of vortices.

The visualization records demonstrate that these longitudinal vortices have an appreciable effect on the outer part of the boundary layer of the bluff body. This results in altering the separation of the flow around it and in formation of the vortices in its near wake, as demonstrated by Figs. 7 and 8. In the photograph of Fig. 7, we are looking at the plan view corresponding to the top photograph of Fig. 4. Here, two dye streaks (B for blue and R for red) are utilized to

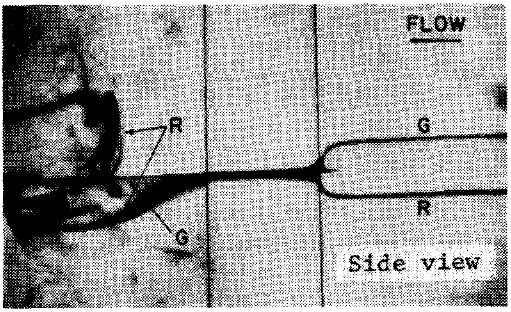


$Re_d=25$; $Re_D=375$; $\Delta x/d=40$; d=.062 in.; D=.935 in.

Fig. 7 Plan-view dye visualization of undisturbed flow (R) and a single wake (B) impinging on a rectangular cylinder.



Re_d=30; Re_D=480; $\Delta x/d$=40; d=.062 in.; D=.988 in.



Re_d=30; Re_D=360; $\Delta x/d$=40; d=.062 in.; D=.747 in.

Fig. 8 Dye visualization showing how vortex flow module affects separation on a circular cylinder.

demonstrate the effects of the vortex flow module on the flowfield around the bluff body. The blue dye is caught in the vortical cell, whereas the red dye is below it away from the flow module. In the plane of the red dye, the normal flow separates and then reattaches, forming a laminar bubble on the sides of the block. Immediately downstream of the rectangular cylinder, the bottom photograph of Fig. 7 shows the effect of the counter-rotating vortices on shedding. The fluid caught in the vortical cells (B) does not get sucked into the low-pressure region behind the cylinder. It appears that there is a much larger wakelike region enclosed by the blue dye than by the red.

In order to understand the photographs of Fig. 7 better, one must remember that the blue dye is marking the cores of the two vortices. Each of the vortices will tend to induce the other away from the surface of the bluff body (even when the image vortices are considered). Once the vortices are away from the body, no other forces will tend to bring them back, as displayed in Figs. 7 and 8, where the green dye is released from the cylinder near the separation point. The resulting modification in the bluff-body separated region and near wake may lead to a reduction in the heat transferred from the downstream part of the cylinder in the presence of the flow module. However, it is conjectured that, in the case of multiple wakes when neighboring pairs of vortices are sufficiently close to each other to permit their mutual interaction, the preceding mechanism may be minimized. It is possible that the multiple vortex flow module will remain more closely attached to the bluff body.

In the case of multiple wakes, a one-to-one correspondence is found to exist between the pairs of vortical cells and the upstream, wake-generating cylinders. This was always the case regardless of the shape or size of the bluff body generating the stagnation region.

Threshold of Incipient Formation

The incipient formation conditions were determined in two ways: either by increasing Δx while holding U_∞ relatively

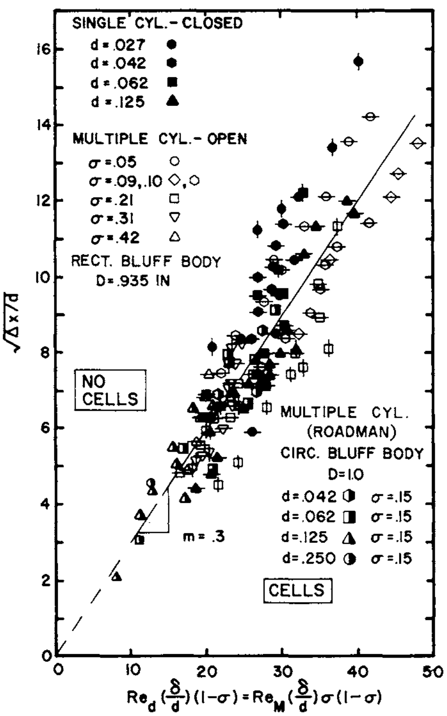


Fig. 9 Empirical correlation of threshold data for single and multiple wakes.

unchanged, or by maintaining Δx constant while increasing U_∞. In the former case, one starts at a condition where a vortex pair is well formed and easy to recognize, as in the top of Fig. 4, and increases Δx until the vortices disappear. In the second approach, Δx is fixed and U_∞ is increased slowly. The beginning of the formation of the vortical cells is reached when some of the dye is drawn upstream, as shown in the bottom photograph of Fig. 8. After several attempts were made to correlate the data empirically, the plot of Fig. 9 was reached. The figure represents both single- and multiple-wake data. The vertical and horizontal lines through the data points indicate the variable parameter during the establishment of the threshold conditions. They correspond to changing Δx or U_∞, respectively. To the left and above the data points, the vortex flow module is not present. The vortices are formed just to the right and below the points. Sufficiently away from the curve, the flow module is fully developed. The boundary-layer thickness at the stagnation line δ was estimated using classical boundary-layer theory, i.e., $\delta/D=6.59(\nu/DU_\infty)^{\frac{1}{2}}$.

The correlation of Fig. 9 is surprisingly good in view of the limited information used to obtain it. The curve establishes the existence of a threshold and indicates that some correlation of the incipient formation conditions may be possible. These results were verified independently by Roadman[24] at very low Reynolds numbers using a tow tank and different visualization techniques. The significance of Fig. 9 rests more on the demonstration of the threshold than on the specific attempt at correlating this complex, nonlinear, three-dimensional phenomenon. However, a preliminary attempt was made to correlate the data based on estimates of some of the important parameters of the problem, i.e., the velocity defect of the wakes, the vorticity in the wakes, and the stagnation-region boundary-layer thickness. Figure 10 depicts the best of these correlations for a single wake; others are given in Ref. 23. In all cases, we find that the magnitudes of the nondimensional velocity defect and vorticity at the threshold conditions are much smaller for multiple cylinders, e.g., an order of magnitude smaller. This should not be surprising if one considers the mutual interaction between neighboring vortex pairs as they are being formed.

Heat-Transfer Results

The effects of the vortex flow module on heat transfer from a circular cylinder at different azimuthal positions were determined using the 5% solidity "zither" described in Sec. II. The results are shown in Fig. 11, which displays the ratio

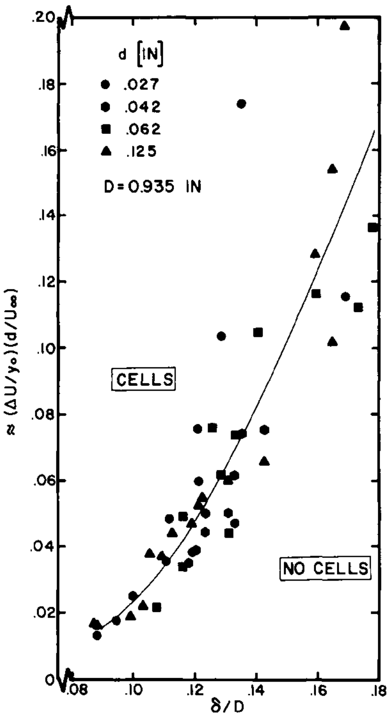


Fig. 10 Correlation of threshold data for single wakes based on estimated nondimensional wake vorticity at stagnation line.

Nu'/Nu as a function of the position around the cylinder surface θ and the Reynolds number of the cylinder Re_D. When Nu'/Nu is larger than one, the vortex flow module is responsible for the augmentation of the heat transfer. When it is smaller than unity, a suppression of the heat lost from the cylinder to the air is caused by the module.

It is most gratifying that, based on the flow visualization results discussed earlier, the effects quantitatively documented in Fig. 11 are predicted qualitatively. In particular, the decreased heat transfer at θ=180 deg and the increased augmentation due to the module when the wakes become unstable are predicted in the discussion of Figs. 6-8. The cross-hatched region of Fig. 11 marks the transition boundary between stable and unstable zither wakes. The boundary is determined from observations of the hot-wire output on the oscilloscope screen during the heat-transfer measurements.

This instability of the multiple wakes is viewed as a shear-layer instability in the manner discussed in detail by Loehrke and Nagib.[6] The hot wire is located in the same

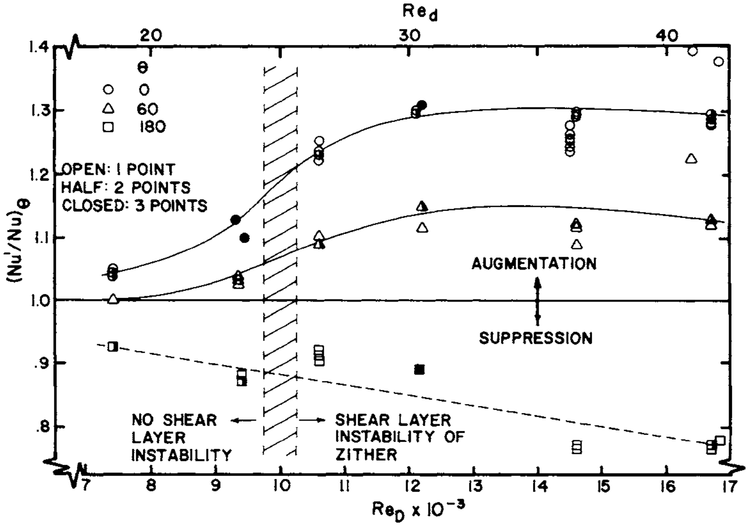


Fig. 11 Effects of vortex flow module on heat transfer from a cylinder.

plane as the stagnation line of the cylinder. Its position, however, is out of the flowfield of the cylinder, which includes the deceleration of the flow as it enters the stagnation region. It is conjectured that the measured transition boundary corresponds to the initiation of instability of the wakes as they are impinging on the stagnation region. This conjecture is based on two independent observations. First, when the freestream velocity is increased slowly through the transition boundary, the initiation of the shear-layer instability, as observed on the oscilloscope screen, always results in a jump in the heat transfer from the cylinder at $\theta=0$ deg (i.e., a sharp decrease in the surface temperature of the cylinder as measured on the strip chart recorder). This effect is not indicated on Fig. 11, since the curves are smoothed through the transition region. The second observation is reported from the same experimental setup by Roadman.[24] He points out that the shear-layer instability is triggered at a different velocity when the zither wires are within the range of influence of the stagnation region (Δx from 0.5 to 1 in.) than when they are sufficiently upstream of that region ($\Delta x \geq 5$ in.). A difference of approximately 10% in the critical velocity is reported by him between the two cases, indicating that the instability may be triggered by the conditions in the stagnation region and propagated along the axis of the wakes.

Figure 11 demonstrates that augmentation as large as 30% above the normal heat transfer is achieved for $30<Re_d<40$.

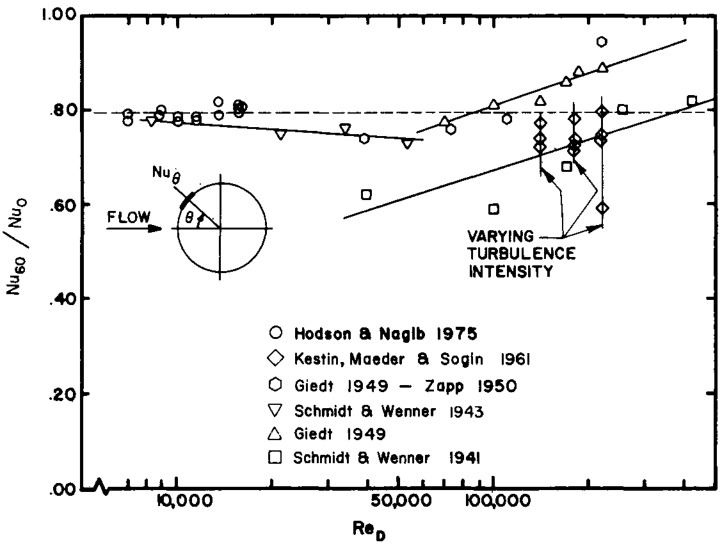


Fig. 12 A benchmark comparison of present heat transfer data.

On the other hand, some reduction of the heat transfer is found on the back side of the cylinder. Although no detailed surveys have been made yet, we expect a net augmentation for $Re_d>25$. In this range, the effectiveness of the module is increased by the unsteadiness of the wakes. The augmentation for $Re_d<25$ is very small at $\theta=0$ deg and negligible at $\theta=60$ deg. The flow module actually may result in a net suppression of the total heat transfer from the cylinder at $Re_d<20$.

As stated in Sec. II, the least accurate data are those for $\theta=60$ deg. The constant heat flux assumption made in the measurements holds best at $\theta=0$ deg. Figure 12 displays the ratio of Nu at these two positions as a function of Reynolds number. Data available in the literature also are reproduced on the figure (from Knudsen and Katz[25]). The agreement is certainly better than we expected, supporting the results of Fig. 11.

IV. Conclusion

The present experiments demonstrate that horseshoe-like vortices can be induced in the stagnation region of bluff bodies by single or multiple wakes and that these vortices can be utilized efficiently in augmentation of convective heat transfer from the body. In order to facilitate this augmentation through simple means, the characteristics of the vortices were investigated.

The number of the vortex pairs induced corresponds to the number of wakes, i.e., velocity defects, and their size appears to scale with the width of the wake rather than the boundary-layer thickness in the stagnation region. The visualization records demonstrate that these streamwise vortices have an appreciable effect on the outer part of the boundary layer of the bluff body which results in altering the separation of the flow around it and the formation of vortices in its near wake (eddies). The vortices are steady for $Re_d<30$ but are sensitive to random freestream disturbances at higher values of the wake Reynolds number. Even small disturbances may lead to the vortices being "kicked" out of position, providing a very efficient mechanism for the mixing of the flow in the stagnation region with that in the freestream.

Quantitative visualization techniques were used to establish the incipient formation conditions of the vortex flowfield for single and multiple wakes. At a given separation distance between the wake-generating cylinders and the bluff body, Δx, an increase in Reynolds number leads to the formation of cells. Decreasing Δx while holding the Reynolds number constant also results in establishing the vortices. The existence of this threshold of vortex formation substantiates Tani's hypothesis[4] and strongly suggests the need for the inclusion of nonlinear effects in future stability analyses of stagnation regions. The observed unsteadiness of the vortices in the presence of upstream disturbances (or turbulence) supports Morkovin's recommendation[18] for using, in the analysis, an imaginary (or a complex) nonzero coefficient of the time term in the exponential form of the disturbance.

By using controlled low Reynolds number wakes in a low-turbulence environment, the preceding experiments bring out the importance of distinguishing between transport processes in the presence of steady vortex-cell flows and those in the presence of unsteady, rather random formations. They also provide clues regarding possible mechanisms that may be responsible for augmentation of heat transfer from bluff bodies exposed to laboratory and practical turbulent flows. Both laminar and unsteady multiple wakes were generated by a 5% solidity "zither" in a wind tunnel and were used to study the effects of the vortex flow module on the heat transfer from a circular cylinder at different azimuthal positions. A small amount of heat-transfer augmentation was achieved on the front part of the cylinder in the laminar conditions of the multiple wakes. At higher Reynolds numbers of the wakes, $25<Re_d<40$, the augmentation is as much as 30% at $\theta=0$ deg , and 15% at $\theta=60$ deg. The vortex ejection mechanism may be responsible for this increase in the effectiveness of the flow module in the transfer of heat from the cylinder. A reduction of heat transfer occurs on the rear part of the cylinder, as much as 10 to 20% at θ=180 deg. It is conjectured that this is due to the change in separation caused by the longitudinal vortices. The overall heat transfer from the cylinder has a net augmentation for $Re_d \gtrsim 25$, which is caused primarily by the unsteadiness of the horseshoe vortices.

Acknowledgments

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THREE-DIMENSIONAL SHOCK-WAVE INTERFERENCE HEATING PREDICTION

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Abstract

Experimental heating and pressure data generated with simple-geometry two- and three-dimensional shock-wave interference models have been collected and analyzed. Simplified methods have been developed to predict peak interference heating for both laminar and turbulent flow cases. For 2-D flow, interference heating factors can be computed once the upstream Mach number and shock-wave generator deflection angle are specified. Simple design curves are provided for practical thermal design purposes. For 3-D flow, however, measured pressure data are needed to compute heating values. In this study, interference heating, pressure, and oil flow patterns of a mated Space Shuttle configuration also were investigated experimentally at a freestream Mach number of 5.3 and a number of Reynolds numbers. Data analysis indicates that heating prediction methods derived from simple-geometry 2-D and 3-D cases can be applied directly to the complicated Space Shuttle configuration, which includes both shock-wave impingement and protuberance interference effects.

Nomenclature

A, a, b, c, F	=	constants
H	=	convective heat-transfer coefficient
L	=	Space Shuttle orbiter fuselage, ET or SRB reference length
ℓ	=	axial distance measured from Space Shuttle orbiter, ET or SRB nose
M	=	Mach number

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n	=	constant
P	=	pressure
Pr	=	Prandtl number
Re	=	Reynolds number
Re/ft	=	unit Reynolds number
St	=	Stanton number
T	=	temperature
V	=	velocity
x	=	streamwise distance measured from flat-plate leading edge
α	=	Space Shuttle angle of attack
β	=	Space Shuttle yaw angle
δ	=	shock generator deflection angle (wedge angle or cone half-angle, Fig. 1)
μ	=	dynamic viscosity
ρ	=	density
ϕ	=	circumferential angle measured from cone or cylinder top centerline

Superscript

$*$	=	Eckert reference conditions

Subscripts

∞	=	freestream
1	=	flow region upstream of impinging shock wave (Fig. 1)
3	=	flow region downstream of reflecting shock wave (Fig. 1)
ET	=	Space Shuttle external tank
FS	=	Space Shuttle orbiter fuselage
i	=	interference
L	=	laminar
PK	=	peak
ref	=	stagnation point of a sphere with scaled 1 ft radius
SRB	=	Space Shuttle solid rocket booster
trans	=	transitional
T	=	turbulent
u	=	undisturbed flow

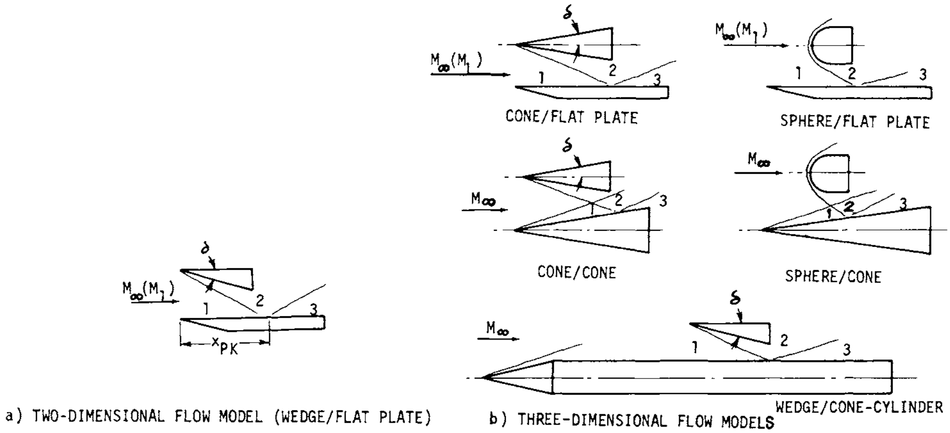

Fig. 1 2-D and 3-D interference flow models: shock generators and receivers.

I. Introduction

The severe interference heating caused by impinging shock waves has been a major concern in the analysis of high-speed aircraft or spacecraft aerodynamic heating and the subsequent design of the thermal protection system (TPS). The thermal environment determines both the TPS materials and the TPS thicknesses, which, in turn, affect the TPS development, fabrication, reuse, refurbishment, and, most importantly, the weight of the vehicle.

Shock-wave/boundary-layer interactions can be two-dimensional (2-D) or three-dimensional (3-D), as indicated in Fig. 1. The oncoming flow (upstream of the interference region) can be laminar or turbulent. The impinging shock wave may trip the laminar oncoming flow and cause flow transition. Also, the flow can either remain attached or be separated, depending on local flow conditions, shock strength, and boundary-layer state. The analytical approach for predicting shock impingement heating is very limited even for the simplest case (2-D laminar flow without boundary-layer transition). Unfortunately, for practical design, such as in the case of the Space Shuttle, most of the interference heating is associated with the more complicated 3-D flow, for which even the inviscid flowfield is not generally well known.[1] The heating prediction can be complicated further by boundary-layer transition. This indicates that simplified methods have to be developed to solve practical engineering problems.

In this study, a series of wind-tunnel tests was conducted with the mated Space Shuttle configuration (Fig. 2). Heating measurements were made using the thin-skin/thermocouple technique. Pressure measurements also were made with identical model configurations and flow conditions and were correlated with heating data. Thermal paint[2] and oil flow tests also were conducted to study the flowfield and obtain qualitative heating information.

In order to derive generalized heating prediction methods the existing interference heating and pressure data generated with simple-geometry 2-D and 3-D interference models (Fig. 1) first were collected and analyzed. A "building block" approach was chosen initially to analyze the simplest case, 2-D turbulent flow. This then was followed by the more complicated 2-D laminar case, in which flow transition may have occurred due to shock impingement. Finally, the simple-geometry 3-D heating/pressure correlations were derived, and results were applied to the more complicated Space Shuttle case.

II. Analysis

For 2-D interference, Eckert's reference method[3] first was applied to predict peak interference heating by the author[4,5]. Interference Stanton number has been correlated with Reynolds number and Prandtl number based on flow proper-

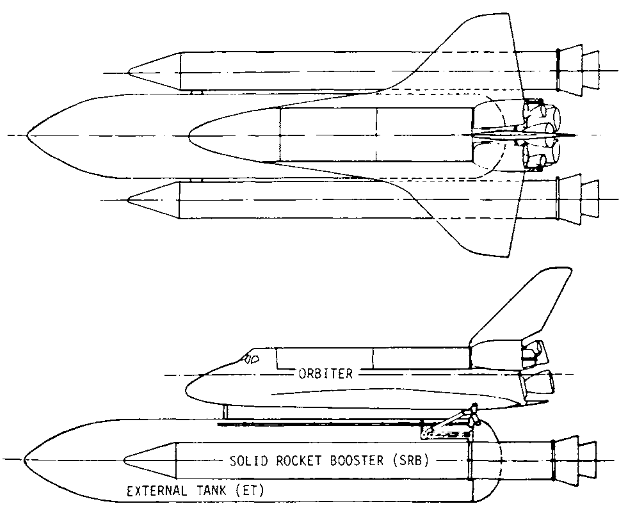

Fig. 2 Mated Space Shuttle configuration.

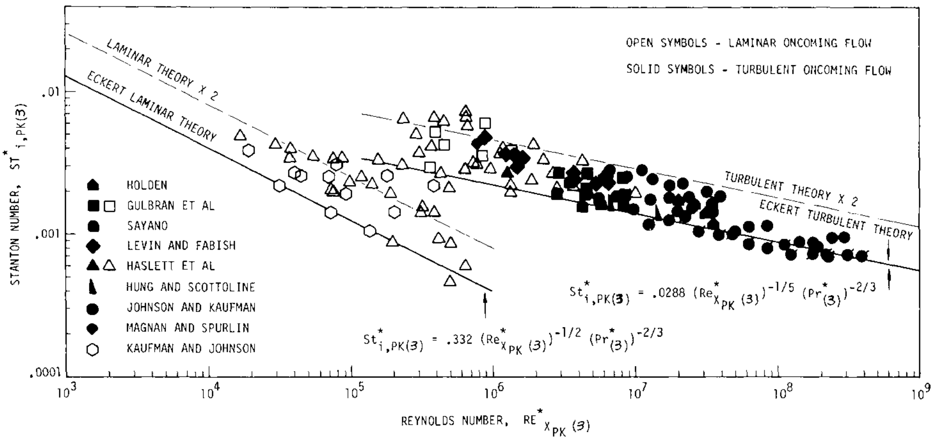

Fig. 3 2-D laminar and turbulent interference heating correlation.[5]

ties behind the impinging/reflecting shock waves, as shown in Fig. 3. Results can be summarized as

$$St^*_{i,PK(3)} = A\left[(Re^*_{x_{PK}}(3))^a\ (Pr^*_{(3)})^b\right]F \qquad (1)$$

where A = 0.332, a = - 1/2, and b = - 2/3 for laminar flow, and A = 0.0288, a = - 1/5, and b = - 2/3 for turbulent flow, and $1 \leq F \leq 2$ with F(mean) = 1.4

In Eq. (1), superscript asterisk denotes reference value based on the reference temperature T*, as defined in Ref. 3. x_{PK} is the streamwise distance from the flat-plate leading edge to the peak interference heating location, as indicated in Fig. 1. It should be noted that values of A, a, and b in Eq. (1) are identical to the values used in Eckert's undisturbed flat-plate theory:

$$St^*_{u(1)} = A(Re^*_{x(1)})^a\ (Pr^*_{(1)})^b \qquad (2)$$

where subscripts u and (1) denote the undisturbed and upstream flow region of the impinging shock wave, respectively, as shown in Fig. 1. Constant F in Eq. (1) can have a value ranging from 1.0 to 2.0, as indicated in Fig. 3. A mean value of 1.4 is chosen for design purposes. The value of F is larger than 1.0 because of the pressure gradient and the complicated viscous-inviscid interaction effects involved in the interference flow. These effects are not included in Eckert's flat-plate theory.

Table 1 Wedge flat-plate interference heating test summary (2-D)

Ref.	M_∞	Re/ft	X_{PK}, ft	P_0, psia	T_0, °R	δ, deg
a) Turbulent oncoming flow						
15	8.6-13.0	$4.7x11x10^6$	2.33-3.25	4570-19900	1820-3400	12.5-20.0
10	6-10	$2.16\text{-}3.4x10^6$	2.0-2.83	190-1800	850-1890	10-15
8	2.4-5.0	$3.3\text{-}4.7x10^6$	1.7	25-64	565-605	8-15
9	2.95-5.02	$3.6\text{-}7.4x10^6$	1.06-1.12	49-64.2	560-575	6-12
6	8	$3.7x10^6$	1.87-1.92	850	1341	5-15
11	5.24	$3.5x10^6$	3.58	350.9-352	1473-1529	5-10
12	6	$2.47\text{-}36.6x10^6$	1.03-10.7	140-2615	855-1022	10-15
13	10.2	$2.16x10^6$	2.33	1800	1890	5
b) Laminar oncoming flow						
10	6-10	$1x10^6$	1.55-1.83	50-750	850-1800	5-10
14	8	$0.4\text{-}2.0x10^6$	0.89-1.22	70-429	1265-1395	1-5
6	8	$0.41\text{-}3.7x10^6$	0.375-1.93	75-850	1200-1341	1.5-15
7	8-10	$0.15\text{-}0.6x10^6$	0.8-1.3	...	...	1-8

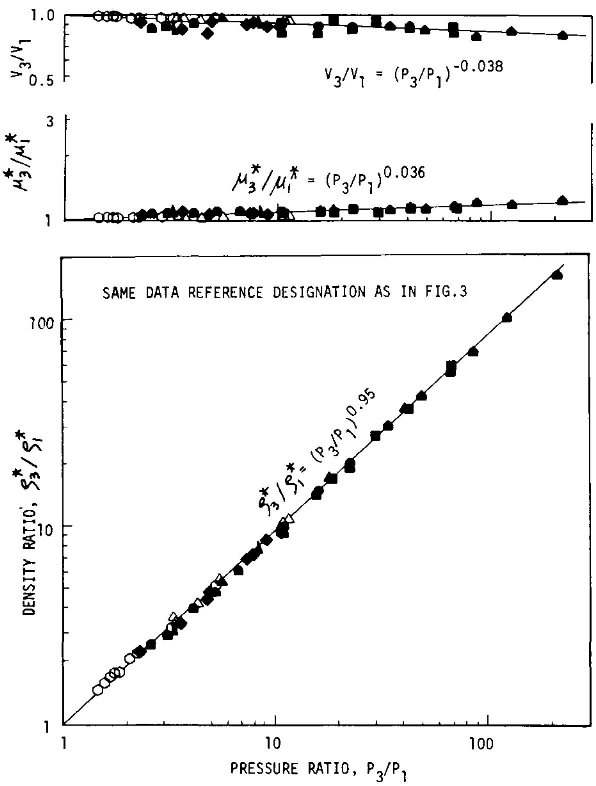

Fig. 4 Velocity, viscosity, and density variation with pressure.

For design purposes, however, it is sometimes more convenient to correlate interference heating factor (heat-transfer coefficient ratio) with shock strength (computed inviscid pressure ratio) as

$$H_{i,PK}/H_u = c(P_3/P_1)^n \tag{3}$$

where n and c are constants. $H_{i,PK}$ is peak interference heat-transfer coefficient. H_u is undisturbed value (flat plate alone) at the location of peak interference heating. Equation (3) can be derived by combining Eqs. (1) and (2) as

$$\frac{St^*_{i,PK(3)}}{St^*_{u(1)}} = F\left[\frac{Re^*_{x_{PK}(3)}}{Re^*_{x(1)}}\right]^a \left[\frac{Pr^*_{(3)}}{Pr^*_{(1)}}\right]^b \tag{4}$$

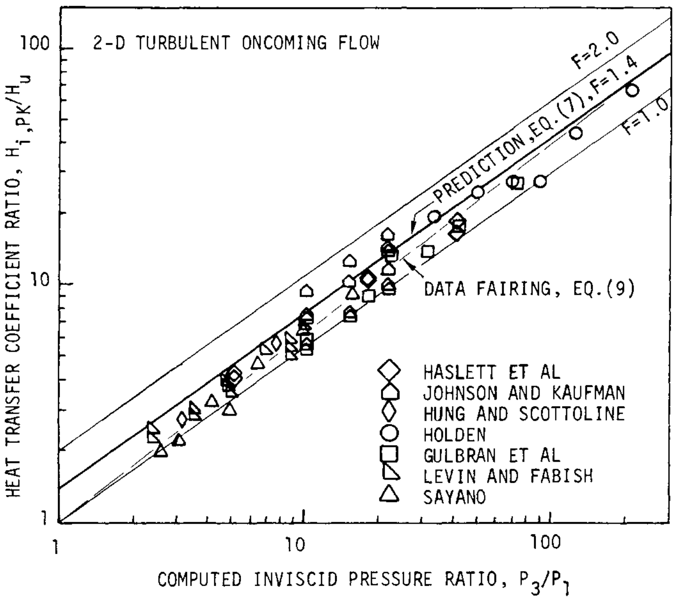

Fig. 5 2-D turbulent peak interference heating correlation with computed inviscid pressure values.

where $x = x_{PK}$. Equation (4) can be simplified further as

$$\frac{H_{i,PK}}{H_u} = F\left[\frac{\rho_3^* \ V_3}{\rho_1^* \ V_1}\right]^{1+a}\left[\frac{\mu_3^*}{\mu_1^*}\right]^{-a} \tag{5}$$

In Eq. (5), it has been assumed that $Pr^*_{(3)} = Pr^*_{(1)}$. The density, velocity, and viscosity ratios in Eq. (5) can be expressed in terms of pressure ratios in order to finalize the heating/pressure correlation, Eq. (3). Test conditions of the existing 2-D data[6-15] collected in this study are summarized in Table 1. Eckert reference flow properties, V_3/V_1, μ_3^*/μ_1^* and ρ_3^*/ρ_1^*, are computed and correlated with P_3/P_1 as shown in Fig. 4, which then is combined with Eq. (5) as

$$H_{i,PK}/H_u = F(P_3/P_1)^{0.912+0.876a} \tag{6}$$

Values of the constant a in Eqs. (1) and (2) then are substituted in Eq. (6) as

$$H_{i,PK}/H_u = F(P_3/P_1)^{0.737} \tag{7}$$

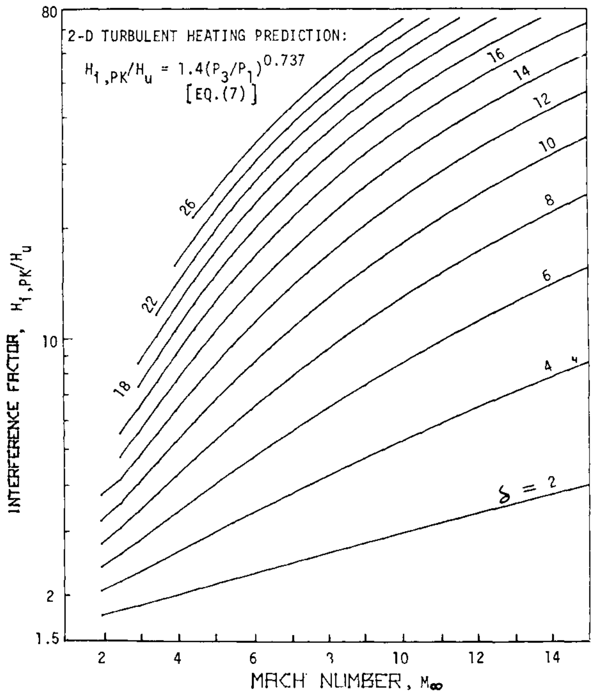

Fig. 6 Turbulent interference heating design curves.

for turbulent interference and

$$H_{i,PK}/H_u = F(P_3/P_1)^{0.474} \tag{8}$$

for laminar interference, where the value of F is defined in Eq. (1). The preceding interference heating predictions will be compared with heating/pressure data correlations, as discussed in the following section.

It should be noted that almost all of the heating data discussed in this analysis were measured by thermocouples with thin-skin models. This indicates that the thin-skin temperatures T_W at which heat-transfer coefficients are reduced are about 70° - 80°F (slightly higher than the model initial temperatures, i.e., room temperatures).

III. Data Correlation and Discussion

A. 2-D Turbulent Oncoming Flow

The collected 2-D turbulent heating data are correlated with the computed inviscid pressure ratios P_3/P_1 as shown in

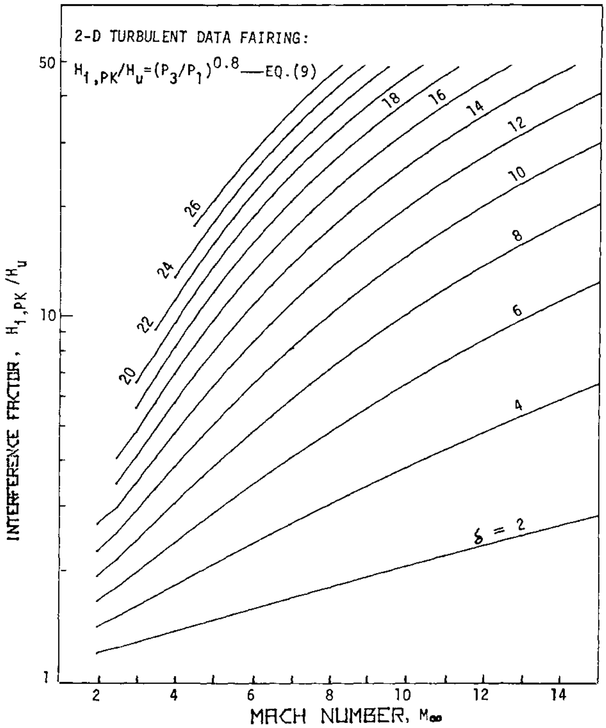

Fig. 7 Turbulent interference heating design curves.

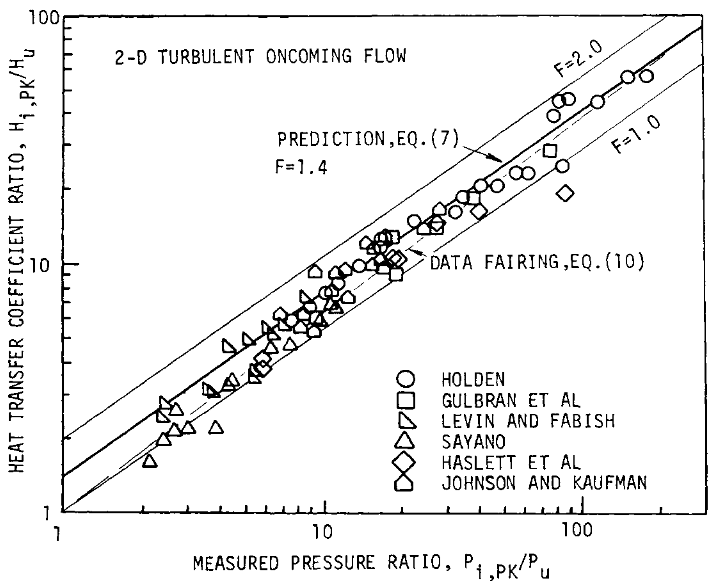

Fig. 8 2-D turbulent peak interference heating correlation with measured pressure data.

Fig. 5. The 2-D turbulent prediction, Eq. (7), also is presented for comparison purposes. The data correlation and heating prediction show good agreement. The correlated heating/pressure data also have been faired by Markarian,[17] Hung and Barnett,[18] and Popinski,[19] yielding the following result:

$$H_{i,PK}/H_u = (P_3/P_1)^{0.8} \tag{9}$$

The preceding data fairing also is presented in Fig. 5. Based on Eqs. (7) and (9), two sets of engineering design curves have been generated, as shown in Figs. 6 and 7, which provide turbulent interference heating factors once upstream Mach number and shock generator deflection angle are specified. Judgement has to be made as to which of the two sets of design curves to choose.

The interference heating data also are correlated with measured pressure data, P_{iPK}/P_u, as shown in Fig. 8, where again the data correlation shows good agreement with heating prediction. This is because the measured $P_{i,PK}/P_u$ data show good agreement with computed P_3/P_1 values, as discussed in

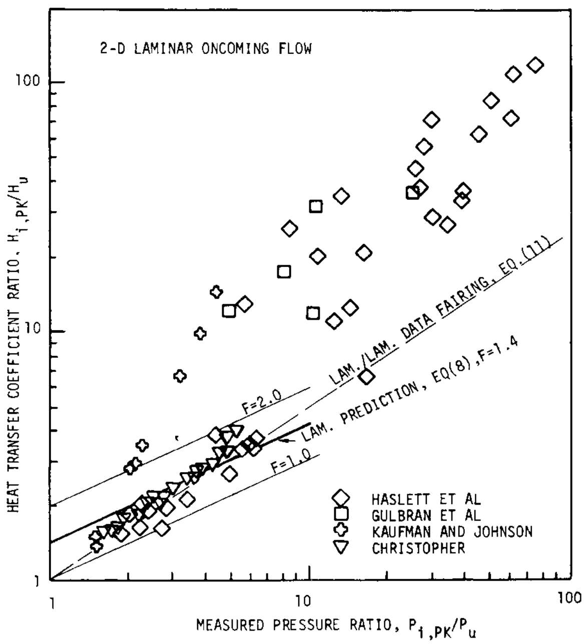

Fig. 9 2-D laminar peak interference heating correlation with measured pressure data: preliminary.

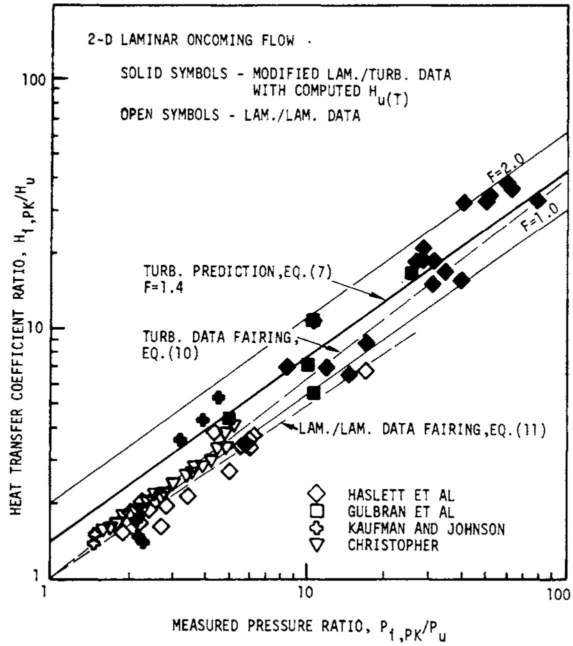

Fig. 10 2-D laminar peak interference heating correlation with measured pressure data: final.

Ref. 5. The measured heating/pressure data correlations have been faired by Haslet,[6] Sayano,[8] Holden,[15] Markarian,[17] and Neumann and Burke[20] with typical results as

$$H_{i,PK}/H_u = (P_{i,PK}/P_u)^{0.8} \tag{10}$$

which is identical to Eq. (9).

B. 2-D Laminar Oncoming Flow

The laminar interference is more complicated than the turbulent case, as the laminar oncoming flow can either remain laminar (laminar/laminar) or be tripped by the shock impingement (laminar/transitional or laminar/turbulent). This is shown in the preliminary data correlation, Fig. 9, where the high heating values and the data scattering are due to flow transition. The lower bound of the correlation should be laminar/laminar data and can be faired as

$$H_{i,PK}/H_u = (P_{i,PK}/P_u)^{0.7} \tag{11}$$

The preceding data fairing is identical to the results derived by Haslett.[6] The laminar heating prediction, Eq. (8), also is compared with laminar/laminar data, as shown in Fig. 9, with fair agreement. As discussed in Ref. 5, for laminar/turbulent flow, the extremely high $H_{i,PK(T)}/H_{u(L)}$ values include both shock impingement effect (or pressure effect) and boundary-layer transition effect. The latter must be eliminated by ratioing $H_{i,PK(T)}$ to $H_{u(T)}$ in order to derive meaningful heating factors. This is shown in the final heating/pressure correlation, Fig. 10, where solid symbols are the "modified" laminar/turbulent data, $H_{i,PK(T)}/H_{u(T)}$. The $H_{u(T)}$ values used in Fig. 10 are computed with Eckert's reference method, since no measured $H_{u(T)}$ data are available. As expected, the modified laminar/turbulent data agree with both turbulent prediction, Eq. (7), and the turbulent data fairing, Eq. (10). The data scattering is partially because measured $H_{i,PK}$ data have been ratioed to the computed $H_{u(T)}$ values. The open symbols in Fig. 10 are laminar/laminar data.

It should be noted that certain effects and experimental errors could cause data scattering shown in the laminar and

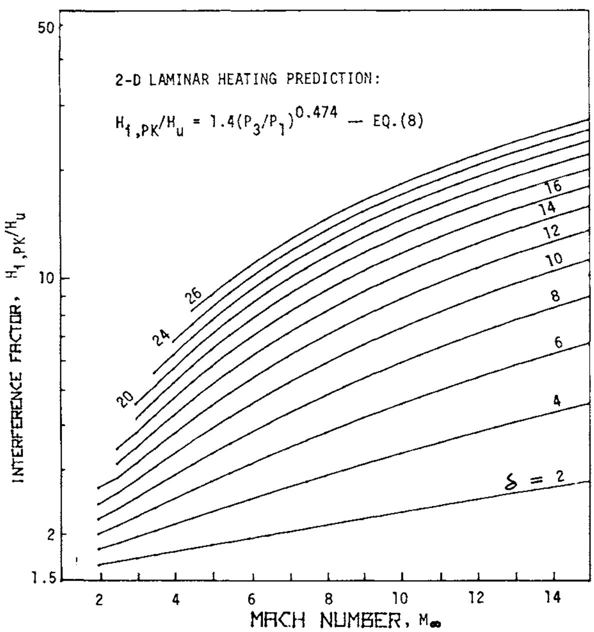

Fig. 11 Laminar/laminar interference heating design curves.

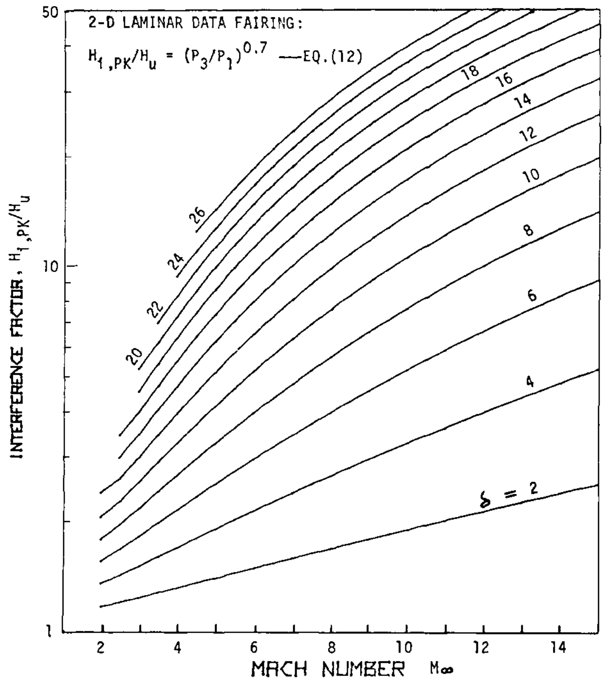

Fig. 12 Laminar/laminar interference heating design curves.

turbulent correlations. For instance, in addition to the desired shock wave generated from the wedge, the wedge/flat-plate model usually generates additional shock waves, as discussed in Ref. 6. These additional shock waves reinforce the simple wedge-generated shock wave and result in higher pressure and heating values on the plate. Besides additional shock waves, another cause of data deviation is the displacement thickness growth of the boundary layer on the wedge, as discussed in Refs. 6 and 14. The effect is not significant for the cases of large Reynolds numbers and wedge angles. However, for the cases of small Reynolds numbers and wedge angles (laminar/laminar flow), the effect must be considered, as the effective wedge angle can be significantly larger than the actual wedge angle. This results in a stronger shock wave and consequently higher interference pressure and heating on the flat plate. Also, as discussed in Ref. 10, the coalescence of the wedge trailing edge expansion fan and the shock wave impingement on the flat plate can cause lower pressure and consequently lower heating values. For the laminar/turbulent flow, the lower heating values below the correlation lines could be transitional data (i.e., laminar/transitional).

Table 2 Three-dimensional interference heating test summary

Ref.	M_∞	Shock generator (wedge or cone half-angle, deg)	Shock receiver (cone half-angle, deg)	ϕ, deg
a) Turbulent oncoming flow				
16	8	Sharp and blunt cone (7.5, 15), sphere	Sharp and blunt cone (7.5)	0, 15, 30, 45
6	8	Sharp cone (2.5, 7.5, 12.5), sphere	Flat plate	Plate ℄
12	6	Sphere	Flat plate, wind-tunnel door	Plate ℄
8	2.4, 5.0	Wedge (8, 15)	Cone-cylinder	0, 90
b) Laminar oncoming flow				
16	8	Sharp and blunt cone (7.5, 15), sphere	Sharp and blunt cone (7.5)	0, 15, 30, 45
6	8	Sharp cone (2.5, 7.5, 10, 12.5), wedge, sphere	Flat plate, cone-cylinder	Plate ℄
12	6	Sphere	Flat plate	Plate ℄

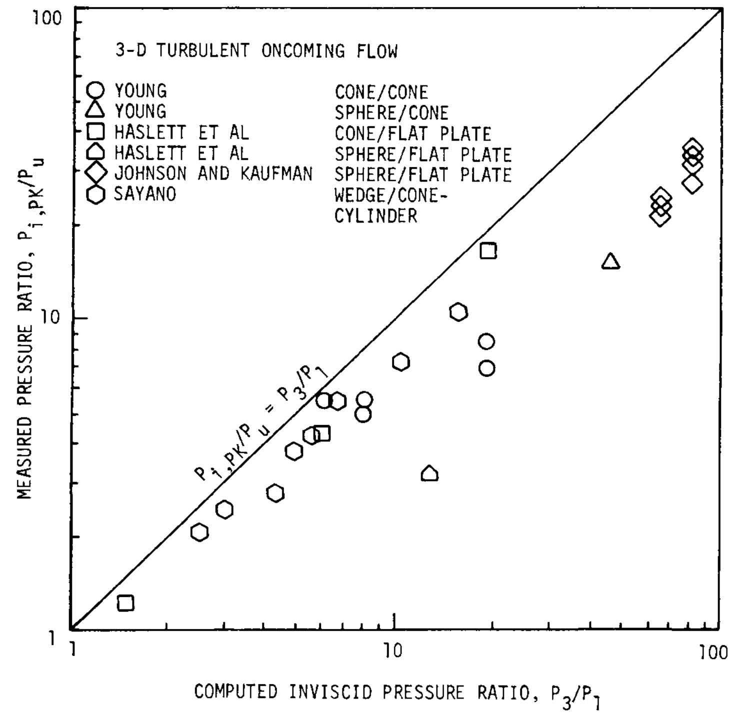

Fig. 13 Measured pressure data vs computed inviscid pressure values (3-D turbulent flow).

Some of the aforementioned effects and errors do not contribute to significant data scattering when heating data are correlated with measured pressure values, as both measurements are affected in the same manner. However, the effect on data scattering should be more significant when heating data are correlated with computed inviscid pressure values. Consequently, no attempt was made to correlate $H_{i,PK}/H_u$ with computed P_3/P_1 values for the 2-D laminar case. It is expected that, for the ideal case with no errors, laminar heating data correlation with P_3/P_1 should be identical to Eq. (11) and can be expressed as

$$H_{i,PK}/H_u = (P_3/P_1)^{0.7} \tag{12}$$

The preceding analysis can be summarized as follows. For laminar/laminar flow, the laminar prediction, Eq. (8), or laminar data fairing, Eqs. (11) and (12), should be applied to

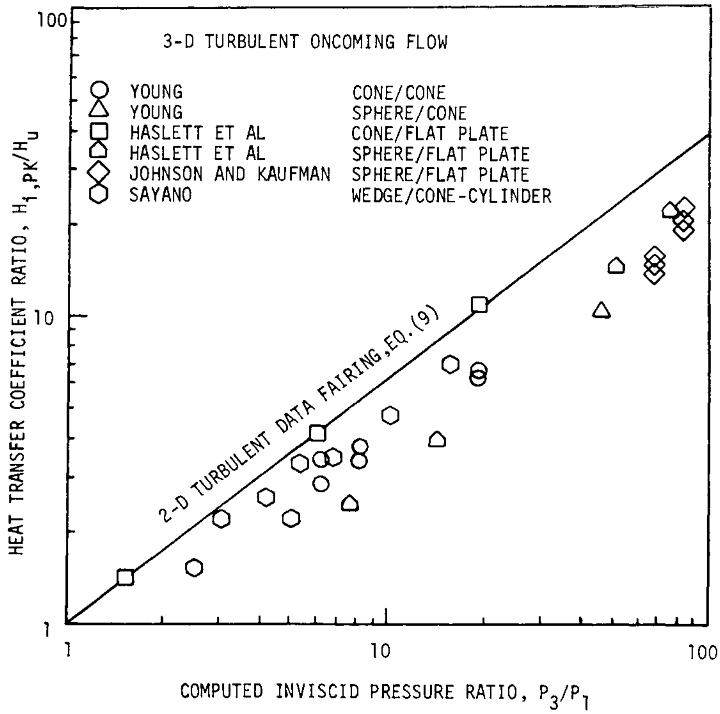

Fig. 14 3-D turbulent peak interference heating correlation with computed pressure values.

predict peak laminar interference heating. Design curves have been generated for practical purposes, as shown in Figs. 11 and 12. For laminar/turbulent flow, however, the turbulent prediction, Eq. (7), or turbulent data fairing, Eqs. (9) and (10), and design curves shown in Figs. 6 and 7 should be used to predict peak turbulent interference heating values as

$$H_{i,PK} = H_{u(L)}(H_{u(T)}/H_{u(L)})(H_{i,PK}/H_u)_{(T)} \tag{13}$$

In Eq. (13), $H_{u(L)}$ and $H_{u(T)}$ can be either measured or computed values.

C. 3-D Turbulent Oncoming Flow

Test conditions of the collected 3-D data are summarized in Table 2. The 3-D interference is more complicated than the 2-D case, as the 3-D flow always is associated with lateral pressure relief away from the peak pressure region. This makes the actual peak pressure prediction almost impossible,

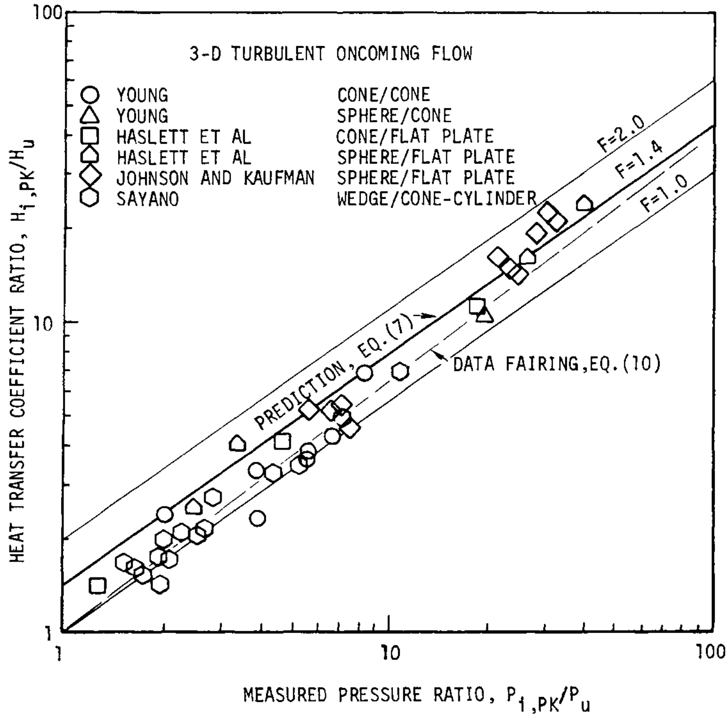

Fig. 15 3-D turbulent peak interference heating correlation with measured pressure data.

as indicated in Fig. 13, where the computed pressure values based on different approaches and assumptions, as discussed in each individual reference, are much higher than the measured data. This is reflected in the heating/pressure correlation shown in Fig. 14, where the lower heating data, in comparison with 2-D turbulent results, are due to the pressure relief, which is not included in the computed P_3/P_1 values. However, as expected, when measured pressure data are used, the 3-D turbulent correlation as shown in Fig. 15 is identical to the 2-D turbulent case.

D. 3-D Laminar Oncoming Flow

The 3-D laminar flow involves both pressure relief and boundary-layer transition. This indicates that measured pressure data have to be used and laminar/turbulent data have to be modified, as discussed in Sec. III.B. Again, the flow transition effect is shown clearly in the preliminary data correlation, Fig. 16. The modified final correlation is pre-

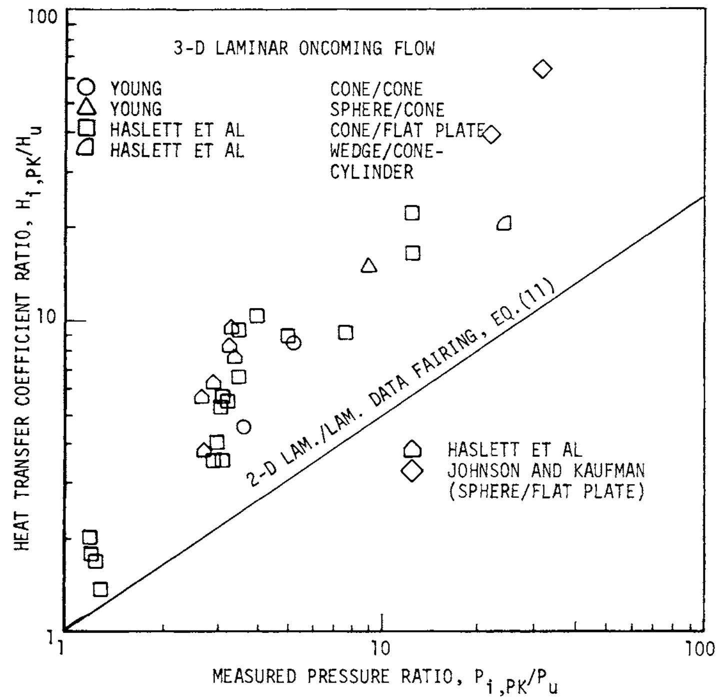

Fig. 16 3-D laminar peak interference heating correlation with measured pressure data: preliminary.

sented in Fig. 17, which indicates that again the 3-D correlation is identical to the 2-D case.

E. Space Shuttle Configuration

A series of Space Shuttle heating and pressure tests was conducted at the NASA Ames Research Center hypersonic wind tunnel. Flow conditions and model configurations of the Mach 5.3 test are summarized in Table 3. Interference data were taken with the mated configuration, Fig. 2, whereas undisturbed data were taken with Orbiter, ET, or SRB alone models. Scales of the heating and pressure models are 0.0175 and 0.010, respectively. The unit Reynolds numbers were chosen in such a way that Reynolds numbers based on Shuttle reference length in the heating and pressure tests were comparable. The undisturbed flow could be either laminar or turbulent. Boundary trips also were used for part of the tests. As indicated in the shadowgraphs, Fig. 18, the orbiter nose shock impinges on the ET upper surface and then is reflected back to the orbiter lower fuselage surface. This causes high inter-

Table 3 Space Shuttle interference heating test summary[a]

Model	Scale	Re/ft	P_0, psia	T_0, °R	Boundary-layer trip	Measurement
Mated configuration	0.0175	1.5×10^6	120	1300	Yes and no	H_i
Mated configuration	0.0175	5.0×10^6	405	1300	Yes and no	H_i
Orb., ET or SRB alone	0.0175	1.5×10^6	120	1300	Yes and no	H_u
Orb., ET or SRB alone	0.0175	5.0×10^6	405	1300	Yes and no	H_u
Mated configuration	0.010	3.06×10^6	245	1500	No	P_i
Mated configuration	0.010	3.7×10^6	300	1500	No	P_i
Orb., ET or SRB alone	0.010	3.06×10^6	245	1500	No	P_u
Orb., ET or SRB alone	0.010	3.7×10^6	300	1500	No	P_u

[a] $M = 5.3$, $\alpha = 0$ deg, $\beta = 0$ deg, NASA Ames Research Center hypersonic wind tunnel.

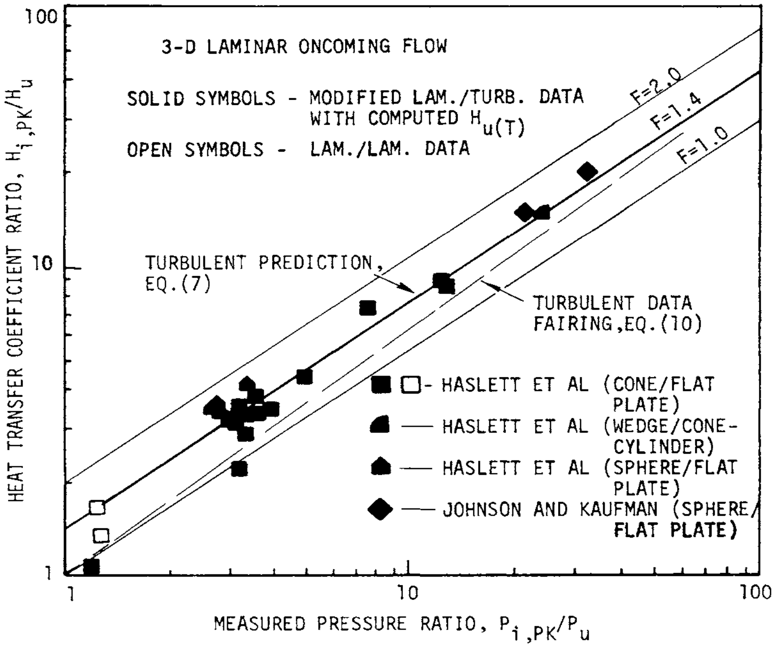

Fig. 17 3-D laminar peak interference heating correlation with measured pressure data: final.

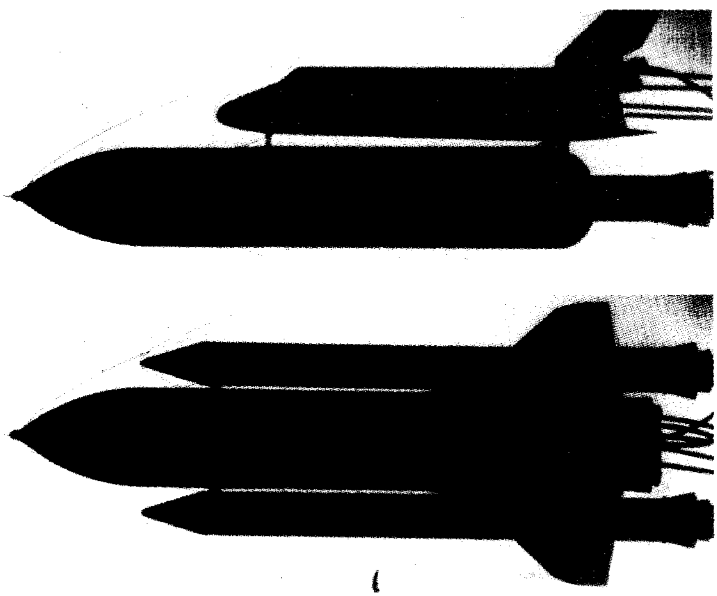

Fig. 18 Space Shuttle shadowgraphs (M_∞ = 5.0, α = 0 deg, β = 0 deg).

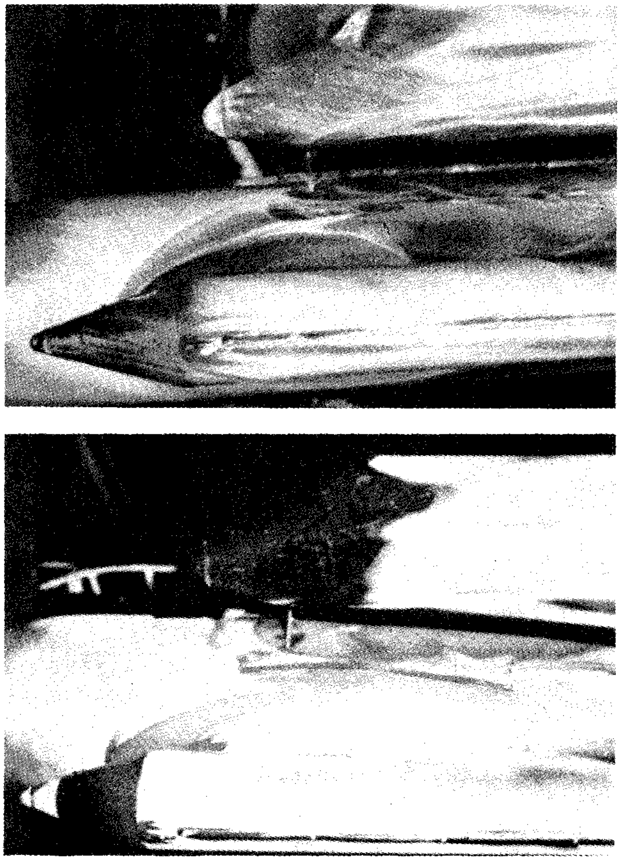

Fig. 19a Space Shuttle oil flow and thermal paint phase-change photographs (topside view).

ference heating on both surfaces, as indicated in the thermal paint test photographs, Fig. 19. The SRB nose shock interference on ET and SRB side surfaces also is shown in Figs. 18 and 19. The pressure relief as discussed previously is indicated in the oil flow photographs, Fig. 19, which show perfect agreement with thermal paint results and shadowgraphs.

Figures 20 and 21 present interference and undisturbed heating and pressure data on the ET top centerline (ϕ = 0 deg) and side surface (ϕ = 90 deg) with two different Reynolds numbers. H_{ref} in these figures is the reference heat-transfer coefficient (stagnation point value of a scaled 1-ft radius sphere computed with the Fay-Riddell equation). It is seen that, at the lower Reynolds number, Fig. 20a, natural flow transition on the undisturbed ET model (ET alone) starts at

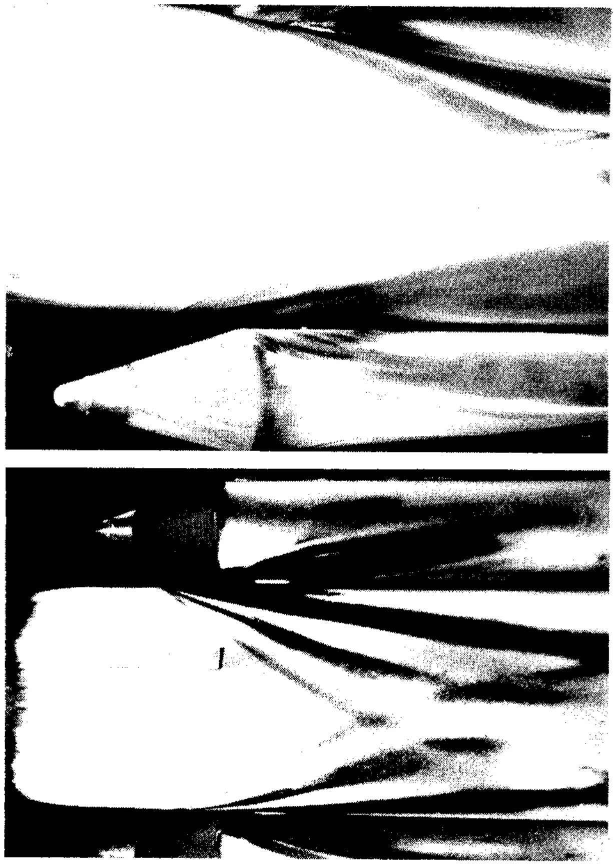

Fig. 19b Space Shuttle oil flow and thermal paint photographs (bottom view).

$\ell/L_{ET} = 0.4$, whereas boundary-layer trips cause early transition. However, it is interesting to note that boundary-layer trips have no effect on interference heating values. This indicates that the flow in and downstream of the interference region already has been tripped by the impinging shock wave even in the absence of boundary-layer trips. Thus, for the mated configuration, the effect of a boundary-layer trip is actually to change the entire flow on the ET from laminar/turbulent to turbulent/turbulent. It should be emphasized that, for the case of laminar/turbulent flow, a turbulent $H_{i,PK}$ should be ratioed to a turbulent H_u in order to derive meaningful turbulent heating factors as shown in Figs. 20 and 21. The heating factors are correlated with measured pressure data, as shown in Figs. 20 and 21. Results indicate that the

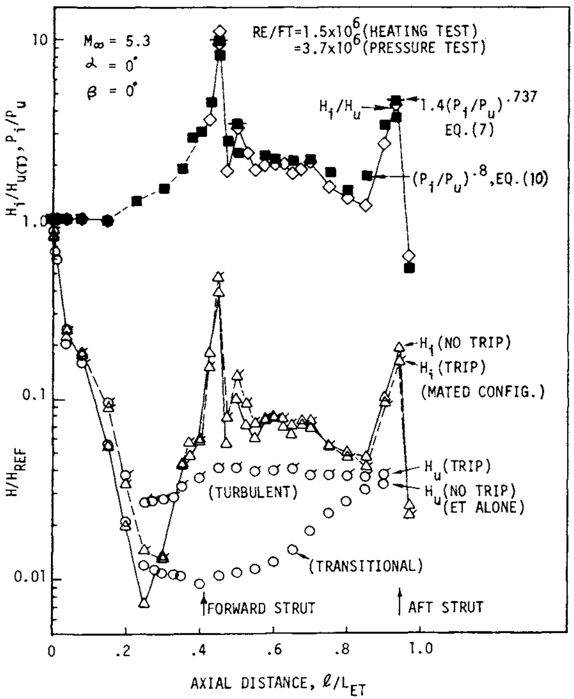

Fig. 20a Space Shuttle external tank (ET) upper centerline (ϕ = 0 deg) interference heating pressure correlation (low Reynolds number).

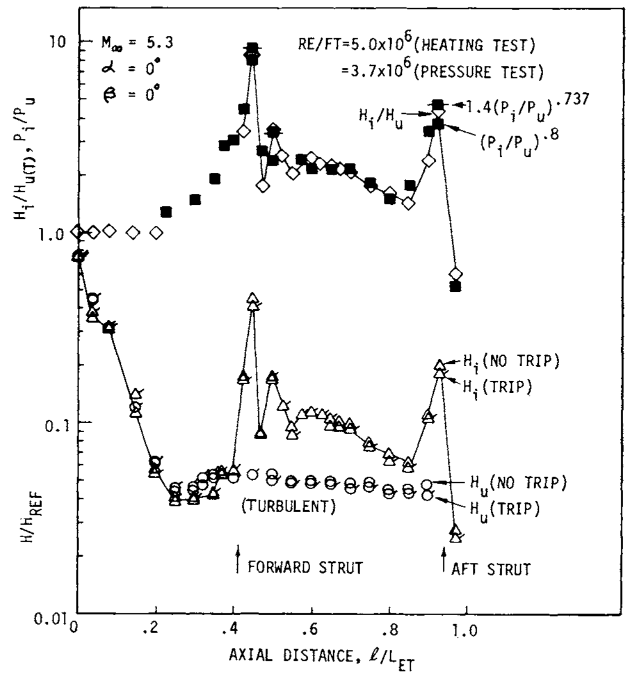

Fig. 20b Space Shuttle ET upper centerline (ϕ = 0 deg) interference heating/ pressure correlation (high Reynolds number).

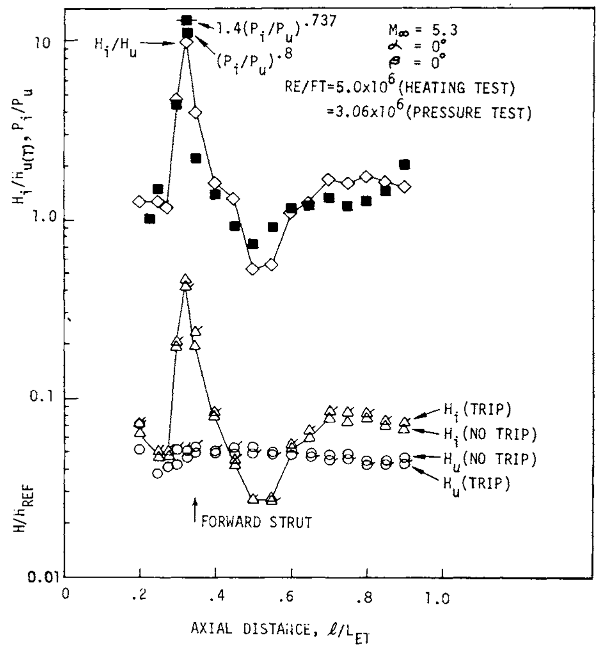

Fig. 21 Space Shuttle ET side surface (ϕ = 90 or 270 deg) interference heating/ pressure correlation.

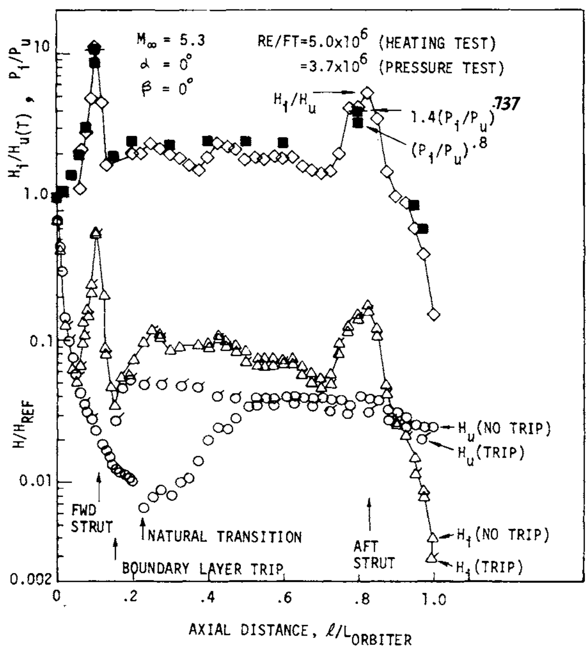

Fig. 22 Space Shuttle orbiter fuselage lower centerline interference heating/ pressure correlation.

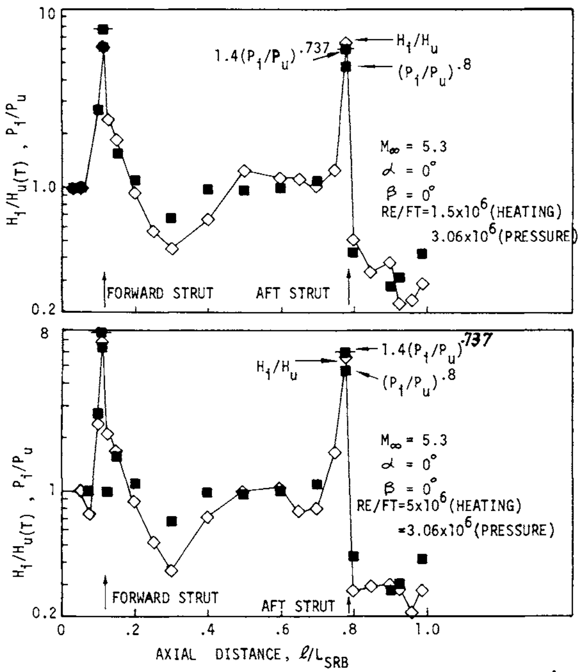

Fig. 23 Space Shuttle SRB side surface (ϕ = 90 or 270 deg) interference heating/ pressure correlation.

heating predictions and data fairings derived from simple-geometry 2-D and 3-D data can be applied directly to the more complicated Space Shuttle configuration, which involves both shock impingement and protuberance effects. The orbiter and SRB data analyses are presented in Figs. 22 and 23, with similar results.

IV. Conclusions

Interference heating, pressure, and oil flow patterns of a mated Space Shuttle configuration were investigated experimentally at a freestream Mach number of 5.3 and various Reynolds numbers. The flow upstream of the interference region could be either laminar or turbulent. Boundary-layer trips were used for part of the tests.

The existing heating and pressure data taken with simple-geometry 2-D and 3-D shock-wave interference models were analyzed, and generalized methods were developed to predict peak interference heating for both laminar and turbulent oncoming flows. Results of the analysis then were applied to the more complicated Space Shuttle configuration. The following conclusions can be drawn:

1) Eckert's reference method was applied to predict 2-D peak interference heating factors.

2) The 2-D peak interference heating data also were correlated successfully with either measured or computed pressure values.

3) The 2-D heating prediction and data correlation show good agreement.

4) For 3-D interference, heating data could be correlated only with measured pressure data. Results are identical to the 2-D cases.

5) It is interesting to note that the correlations derived from simple-geometry 2-D and 3-D data can be applied directly to the more complicated Space Shuttle configuration, which involves both multiple shock waves and protuberances.

6) Simple design curves are generated in this study to provide 2-D laminar and turbulent peak interference heating factors once upstream Mach number and shock generator deflection angle are specified. However, for 3-D cases, measured pressure data are needed for heating computations.

7) Finally, for laminar/turbulent flow, caution should be taken to exclude the effect of boundary-layer transition from the heating factor derivation.

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METHODS FOR PREDICTING RADIATION-COUPLED FLOWFIELDS ABOUT PLANETARY ENTRY PROBES

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Abstract

A method is described for efficiently obtaining numerical solutions to the radiation-coupled shock-layer equations that govern the heating events of hypervelocity entry into outer-planet atmospheres. Using a detailed radiation model, differencing and coupling procedures are developed which match the flowfield procedures to the requirements of the radiation model. Sample solutions are described for carbonaceous and reflecting heat shields in the Jovian and Saturn entry environments.

Nomenclature

C = viscous density ratio
$\bar{C}_p$ = frozen specific heat
$\mathcal{D}$ = diffusion coefficient
f = stream function
h = static enthalpy
H_T = total enthalpy
$\tilde{J}_k$ = mass flux
$\tilde{K}_k$ = mass fraction of kth element
Le = Lewis number
$\dot{m}$ = ablation rate
P = pressure
q_a = diffusive heat flux
q_r = radiative heat flux
r = local radius of body in a meridian plane
R = radius of curvature
s = distance parallel to wall

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Sc = Schmidt number
T = temperature
u = velocity component parallel to wall
v = velocity component normal to wall
x_i = mole fraction of species i
y = coordinate normal to wall
α^* = flux normalizing parameter
β = velocity gradient parameter
$\delta\eta$ = increment in surface normal coordinate
ε = density ratio across normal shock
η = Levy-Lees coordinate normal to wall
κ = 0 for plane flows, 1 for axisymmetric flows
μ = viscosity
ζ = Levy=Lees coordinate parallel to wall
ρ = density

Background

From 25% to 50% of the total weight of an outer-planet probe will be in the heat shield. To insure that the probe can carry a maximum instrument payload, the heat shield must be sized carefully, using procedures that accurately predict the heat-shield thermal response and the aerothermal environment. This study describes the development of methods for predicting the aerothermal environment. These procedures predict the important physical events, while satisfying constraints imposed by cost and reliability.

The physical events that must be considered for outer-planet probes are significantly different from these events considered in earlier analyses. For example, entry velocities of outer-planet probes are much higher (by a factor of 2 or 3) than velocities for Apollo earth entry. Thus, the probe has from four to nine times more kinetic energy to dissipate either in the atmosphere or the heat shield. Moreover, differences in the chemical composition of various model atmospheres cause significant changes in the predicted heating rates and heating mechanisms. Thus, helium-rich atmospheric models produce high shock-layer temperatures and severe shock-layer radiation. On the other hand, in helium-poor (hydrogen-rich) model atmospheres, severe convective heating pulses of extended duration (with negligible radiation) can be encountered for shallow entries.

Developing efficient, accurate methods for calculating such environments has proven to be difficult. Rapid and substantial changes in the models used to characterize the radiation field have caused many of the difficulties. Early models included 1) gray gas (optically thin), 2) nongray with the

However, these events are not within the scope of the present study.

Component Computational Requirements

The primary components of the overall calculation procedure are listed in the order of their required computational effort (greatest effort first, least effort last): 1) line radiation; 2) continuum radiation, and thermodynamic and transport properties; 3) matrix inversion (assuming inversion of a full, square matrix); and 4) evaluation of flowfield equations. Evaluating the line radiation is by far the most expensive. The matrix inversion will require only a small amount of computation, and, consequently, there is little impetus to obtain finite-difference equations that can be solved by a banded matrix, as Blottner[17] and Davis[18] have done. Other factors are much more important.

General Strategies

From the preceding discussion, it can be concluded that the most important cost benefits can be obtained by minimizing the number of iterations and nodes. To this end, the following strategies are adopted:

1) Evaluate and update all of the conservation equations and flux terms simultaneously rather than sequentially (as is normally done[17,18]).

2) Use an explicit mechanism to insure that the nodes always are distributed properly through the mixing layer.

3) Use first-order differences (rather than higher-order differences) to avoid flails.

4) Retain the nonlinear algebra in the conservative equation to minimize the number of nodes required across the layer.

Formulation

The flow in a thin shock layer of a chemically reacting, radiation-coupled gas mixture is subject to the flow conservation equations

$$(\partial \rho u r^{\kappa}/\partial s) + (\partial \rho v r^{\kappa}/\partial y) = 0 \tag{1}$$

$$\rho u\ (\partial u/\partial s) + \rho v\ (\partial u/\partial y) + \rho u v/R = - \partial P/\partial s + \partial/\partial y \left[\mu\ (\partial u/\partial y)\right] \tag{2}$$

ultraviolet (uv) included, 3) nongray with uv and atomic lines, 4) nongray, nonhydrogenic with uv and atomic lines. Current models consider nongray and nonhydrogenic effects for the uv and atomic lines of both the environmental gases plus the ablation products. With each of these increases in sophistication, greater detail has been required to describe the radiation model. In addition, costs also have increased and now can be extremely high when the radiation model is used to calculate heat fluxes as part of a coupled flowfield solution procedure.

Complicating the analytical problem, large amounts of the radiation flux can be transmitted directly from the hottest regions of the flow either to other interior regions or directly to the wall. If the flux is transmitted to the wall, it leads to massive blowing of the wall material into the shock layer. When subjected to massive blowing conditions, numerical flowfield solution procedures usually become highly unstable. Solutions are obtained only with great effort, which greatly accentuates the cost impact associated with the simultaneous evaluation of a detailed radiation flux model. Recent studies [1-7] have solved some of the problems of the blowoff instability; however, a consensus has not been reached yet on an adequate approach for handling the fully coupled problem.

Many investigators faced with the problem of providing engineering support calculations with expensive and often unstable solution procedures simplified the radiation model and (often) considered unblown flows. Thus, multiband models with from two to six bands plus (in some cases) a few select lines have been used.[8-12] Such simplifications allow solutions to be obtained and many effects to be studied. However, the approach is clearly approximate, and solution procedures of this type should be used only in coordination with experiments and detailed (benchmark) calculations.

With the multiband models, the radiation is matched to the requirements of conventional flowfield procedures. In the present study, the opposite approach is taken; a detailed radiation model is adopted, and then differencing and coupling procedures are developed which match the flowfield procedures to the requirements of the radiation model. Using this approach eliminates difficulties in maintaining high accuracy and generality in the evaluation of the radiation fluxes. In this paper, only the flowfield procedures and the coupling procedures between flowfields and radiation fields are addressed. The radiation model is discussed elsewhere,[13,14] as is the application of the coupled procedure to hypervelocity flows.[15,16]

Approach

A prediction procedure combining accuracy, flexibility, computational efficiency, and reliability is sought. To develop this optimum procedure, the important physical events and the computational requirements of various components of the procedure are considered.

Important Physical Events

A blunt probe in hypervelocity entry is shown in Fig. 1. A number of potentially important physical events are called out on the figure. The most important events include 1) detached bow shock; 2) massive blowing from the wall; 3) varying elemental composition across the shock layer; 4) a shock layer with three distinct regions: inviscid layer, mixing layer, and ablation layer; 5) very rapid changes in flowfield properties in an interior region of the flow (the mixing layer); 6) strong coupling between the position of the mixing layer and the blowing rate; 7) significant emission of radiation from the inviscid layer; 8) significant cooling of the inviscid layer by radiation losses; 9) significant absorption of radiation by the mixing and ablation layers; 10) laminar and turbulent mixing; and 11) nongray radiation. The following physical events also may be important: 1) coupling between the probe shape and the heating rate (shape change); 2) coupling between the shock shape and the blowing rates; 3) nonequilibrium regions in the flow; and 4) transition to turbulence.

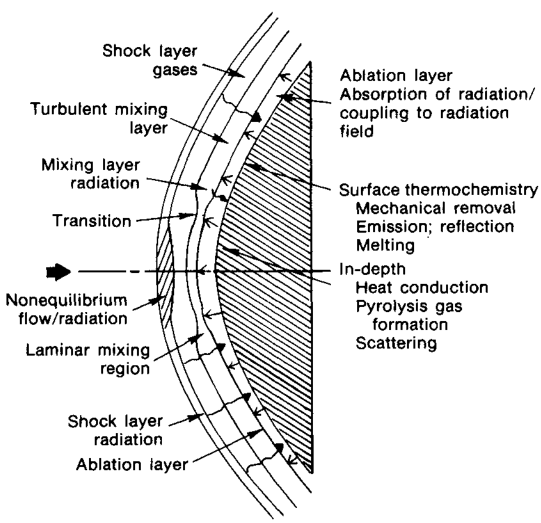

Fig. 1 Flow phenomenology.

$$\rho u\ (\partial H_T/\partial s) + \rho v\ (\partial H_T/\partial y) = -\ \partial q_a/\partial y + \partial q_r/\partial y \tag{3}$$

which govern continuity, s momentum, and energy, respectively. Moreover, the diffusion of wall elements is subject to a diffusion equation of the form

$$\rho u\ (\partial \tilde{K}_k/\partial s) + \rho v\ (\partial \tilde{K}_k/\partial y) = -\ \partial \tilde{j}_k/\partial y \tag{4}$$

when the binary diffusion approximation of Lees[19] is applicable, and when there are no sources or sinks of elements in the interior of the flow. The formulation is completed with a normal momentum equation of boundary-layer type, i.e., $\partial P/\partial y = 0$.

After introducing the Levy-Lees variables, i.e.,

$$\eta = \frac{r^{\kappa} u_e}{\sqrt{2\xi}} \int_0^y \rho\ dy\ ,\quad \xi = \int_0^s \rho_e u_e \mu_e r^{2\kappa}\ ds \tag{5}$$

and observing that the pressure gradient parameter

$$dP/ds\Big|_0 = (\partial P/\partial s) + (\rho uv/R_e) = -\left[8\rho_\infty u_\infty{}^2 s(1-\varepsilon)\right]/3R_e{}^2 \tag{6}$$

and can be approximated as being independent of y, the conservation equations for s momentum, energy, and diffusion become

$$(Cf'')' + ff'' - (2\xi/\rho u_e{}^2)\ (dP/d\xi)\Big|_0 - \beta f'^2 = 2\left[f'\ (\partial f'/\partial \ln\xi) - f''\ (\partial f/\partial \ln\xi)\right] \tag{7}$$

$$fH_T{}' + (-q_a{}^* + q_r{}^*)' = 2\left[f'\ (\partial H_T/\partial \ln\xi) - H_T{}'\ (\partial f/\partial \ln\xi)\right] \tag{8}$$

$$f\tilde{K}_k{}' + (-\tilde{j}_k{}^*)' = 2\left[f'\ (\partial \tilde{K}_k/\partial \ln\xi) - \tilde{K}_k{}'\ (\partial f/\partial \ln\xi)\right] \tag{9}$$

where the continuity equation was eliminated by using the stream function. The normalized, diffusive flux terms are related to the thermodynamic and flow variables through

$$\tilde{j}_k{}^* = -\ (C/Sc)\ \tilde{K}_K{}' \tag{10}$$

$$q_a{}^* = -\ C/Sc\ \left[h' + (1/Le - 1)\ \overline{C}_p T' + u_e{}^2 f'f''\right] \tag{11}$$

and to the physical flux terms through

$$\tilde{j}_K = \alpha^* \tilde{j}_k{}^* \tag{12}$$

$$q_a = \alpha^* q_a^* \tag{13}$$

$$q_r = \alpha^* q_r^* \tag{14}$$

where the flux normalizing parameter, viscous density ratio, Schmidt number, Lewis number, frozen specific heat, and velocity gradient parameter are given, in that order, by

$$\alpha^* = \rho_e \mu_e u_e r^k / (2\xi)^{1/2} \tag{15}$$

$$C = \rho\mu / \rho_e \mu_e \tag{16}$$

$$Sc = \mu / (\rho \mathcal{D}) \tag{17}$$

$$Le = \rho \, \overline{C}_p \mathcal{D} / k \tag{18}$$

$$\overline{C}_p = \sum_{\text{species}} C_{p_i} \, x_i \tag{19}$$

$$\beta = 2 \, d \, \ell n \, u_e / d \, \ell n \, \xi \tag{20}$$

and where the use of equal difusion coefficients is implicit in these relations, and thermal, pressure, and forced diffusion effects have not been included. The boundary conditions are the usual ones for a shock-layer problem and are written out elsewhere.[20]

Algebraic Conservation Equations

The flowfield methods adopted for this study are somewhat similar to the integral matrix method of Bartlett and Kendall.[21] The primary features of these methods are described below. Initially, the conservation equations are sampled at discrete nodes in space. Introducing finite-difference and/or interpolation relations and integrating the equations between the nodes allows the conservation equations to be reduced from differential to algebraic form. A set of iterative equations, based on the multidimensional Newton-Raphson method, is defined to solve the algebraic equations. Once converged, the solution automatically satisfies the algebraic conservation equations. A detailed description of this procedure follows.

First, consider the flow equations in the region of a given streamwise station s as being devided into N-1 strips connecting N nodal points. The nodal points are designated by η_i, where i=1 at the wall and N at the outer boundary. Let the independent variables of the flow equations be represented at points between the nodes by Taylor's series expansions.

These can be truncated to yield the familiar forward difference and spline fit relations

$$\zeta_{i-1}'' = (\zeta_i' - \zeta_{i-1}')/\delta\eta_{i-1} \tag{21}$$

where the variable ζi has been introduced to save space and considered to be any one of the usual flowfield variables (f_i, H_{T_i}, $\tilde{K}_{k_i}$). Equation (21) can be used with

$$f_i' = -f_{i-1} + 2(f_i - f_{i-1})/\delta\eta_{i-1} \tag{22}$$

to relate the higher-order variables (f_i'', ..., H_{T_i}', ..., $\tilde{K}_{k_i}'$, ...) to the primary variables (f_i', ..., H_{T_i}, ..., $\tilde{K}_{k_i}$, ...) at the wall and at interior points in the vicinity of the wall. Similarly, the Taylor series can be truncated to yield backward differences

$$\zeta_i'' = (\zeta_i' - \zeta_{i-1}')/\delta\eta_{i-1} \tag{23}$$

to relate the higher-order variables to the primary variables at the outer boundary and at interior points in the vicinity of the outer boundary. These two regions can be joined by higher-order spline fits written between one or more pairs of interior nodes, i.e.,

$$\zeta_\ell'' = -\zeta_{\ell-1}'' + 2(\zeta_\ell' - \zeta_{\ell-1})/\delta\eta_{\ell-1} \tag{24}$$

$$f_{\ell-1}' = (f_\ell - f_{\ell-1})/\delta\eta_{\ell-1} - 1/3\ (f_{\ell-1}'' + 0.5f_\ell'')\delta\eta_{\ell-1} \tag{25}$$

which completes the set of difference relations.

Equations (21-25) are particularly appropriate for massively blown flows for the following reasons:

1) The simple finite differences can turn the extremely sharp corners typically found in massively blown flows without flailing.

2) The use of joining relations in the interior of the flow, rather than at a boundary, is desirable to avoid the regions of constant shear, enthalpy, and/or composition typically present near both boundaries.

Substituting the difference relations into the conservation equations and rearranging slightly produces a set of equations for iterative solution. Because of their length, all of these equations have not been included here; they can be found elsewhere.[20] As an example, the stagnation-point energy equation assumes the form

$$\text{error}_{i-1} = (fH_T - q_a^* + q_r^*)_{i-1}^{i} - \{2f_i'H_{T_i} + f_{i-1}'H_{T_i} + f_i'H_{T_{i-1}} + 2f_{i-1}'H_{T_i}\}\,(\delta\eta_{i-1})/6 \tag{26}$$

The algebraic forms of the diffusive fluxes become

$$j_i^* = -\,(C_i/Sc_i)\,\tilde{K}_{k_i}' \tag{27}$$

$$q_{a_i}^* = -\,C_i/Sc_i\left[h_i' + (1/Le_i - 1)\,\overline{C}_{p_i}T_i'\right] \tag{28}$$

where

$$h_i' = H_{T_i}' - u_e^2 f_i' f_i'' \tag{29}$$

$$T_i' = (H_{T_i}' - u_e^2 f_i' f_i'')\,\partial T/\partial h\Big|_i + \sum_j \left(\partial T/\partial \tilde{K}_j\right)_i \tilde{K}_j' \tag{30}$$

where exact values of the partial derivatives $\partial T/\partial h$, $\partial T/\partial \tilde{K}_j$, ..., are obtained from the equilibrium thermodynamic calculation performed at each node for each iteration. The formation of the algebraic conservative relations is completed by specifying the mixing rules of Yos[22] for the diffusive properties and specifying the radiation properties and flux formation.

Having defined a suitable set of algebraic conservation equations and boundary conditions, iteration variables to be used in the numerical solution are selected. Either the nodal positions can be specified a priori and the flow variables solved for, or one of the set of flow variables can be specified and the remaining flow variables plus the nodal positions can be solved for. This latter approach insures that the nodes always will be well placed, an expecially important consideration for flows with massive blowing. Consequently, the second approach was adopted, which gave the following set of unknown variables: f_w, f_ℓ'', $\delta\eta_1$, $\delta\eta_2$, ..., $\delta\eta_{N-1}$; H_{T_ℓ}', H_{T_1}, H_{T_2}, ..., H_{T_N}; $\tilde{K}_{k_\ell}'$, $\tilde{K}_{k_1}$, $\tilde{K}_{k_2}$, ..., $\tilde{K}_{k_N}$ [a total of $3(N+1)$], to be solved at positions in space corresponding to preselected values of the velocity ratios (f_i').

There are four momentum boundary conditions; however, by specifying f_i' a priori, two conditions become redundant, ($f_i' = 0$ and $F_N' = 1$). Thus, only two meaningful momentum boundary conditions are to be retained (for f_w and f_N). Similarly, there are two boundary conditions to be retained for both the energy (${}^H T_w$ and ${}^H T_N$) and species equations ($\tilde{K}_{k_w}$ and $\tilde{K}_{k_N}$). The two boundary conditions of each type are to be solved simultaneously with N-1 conservation equations of each type [Eg. (26) plus similar equations for momentum and species], with a total of 3(N+1) equations and boundary conditions.

Iterative Equations

The algebraic conservation equations and boundary conditions are nonlinear and, in general, must be solved iteratively. A useful set of iteration equations can be obtained by extending the familiar Newton-Raphson method to multiple dimensions. Thus,

$$\{v\}_{new} = \{v\}_{old} - (\phi_{ij})^{-1} \{errors\} \tag{31}$$

where the elements in the column matrices $\{v\}$ are the unknown variables discussed earlier. The errors are obtained by evaluating the conservation equations in the following order: momentum boundary conditions (2); interior momentum equations (N-1); energy boundary conditions (2); interior energy equations (N-1); species boundary conditions (2); and interior species equations (N-1). The ϕ_{ij} are the influence coefficients, i.e., the partial derivative of the ith equation with respect to the jth unknown variable, holding the remaining unknown variables constant. With two exceptions, the development of the influence coefficients is straightforward but lengthy and will not be presented here. However, the influence coefficients involving T_i' require the second derivatives of thermodynamic variables, e.g.,

$$\frac{\partial T_i}{\partial \tilde{K}_{k_i}} = ({H_T}_i' - u_e^2 f_i' f_i'') \frac{(\partial^2 T_i)}{\partial h_i \partial \tilde{K}_{k_i}} + \sum \frac{(\partial^2 T_i)}{\partial \tilde{K}_{k_i} \partial \tilde{K}_{j_i}} \tilde{K}_{j_i}' \tag{32}$$

which are obtained using Kendall's[23] method. The influence coefficients involving the radiation flux terms $\partial q_{r_i}/\partial(\delta\eta_j)$, $\partial q_{r_i}/\partial H_T$, $\partial q_{r_i}/\partial \tilde{K}_k$ are essential to the success of the approach and allow the flux calculation to be used directly in the flowfield iteration. The radiation influence coefficients are available as part of the radiation model.[23]

Finding Solutions

Solutions are found by 1) judiciously selecting problems, 2) using constraints (damping) on the predicted corrections to the primary variables, and 3) using the mechanics of the Newton-Raphson method. A general strategy for selecting problems includes the following:

1) Start with a simple problem (usually no blowing).

2) Leave the expensive parts of the calculations turned off (radiation).

3) Assign first estimates of the primary variables.

4) Iterate to a solution.

5) Upgrade parameters defining the problem, obtaining intermediate solutions, until they define the desired problem.

6) When the parameters are correct, turn the radiation calculation back on.

Constraints (damping) on correcting the primary variables also are useful. After solving the Newton-Raphson relation, tentative values of the upgraded variables $\delta\eta_i$ and $\tilde{K}_{k_i}$ are established. If all $\delta\eta_i > 0$ and all $\tilde{K}_{k_i} \geq 0$, the new variables are kept, and the iteration proceeds. If either $\delta\eta_i < 0$ or $\tilde{K}_{k_i} < 0$, all of the corrections are divided by a number greater than 1 ($\simeq$1.5), and new tentative variables are established. This procedure is repeated until the variables satisfy both tests or until the correction becomes smaller than an arbitrary value.

The steps in the iteration procedure are as follows:

1) Estimate initial values of the primary variables.

2) Evaluate the properties: thermodynamic, transport, radiative.

3) Evaluate the errors in the conservation equations.

4) Check for convergence.

5) Evaluate the influence coefficients.

6) Invert the influence coefficient matrix, and multiply the results by the error matrix.

7) Correct the primary variables.

Sample Solutions

The computations required in a single flowfield iteration are presented in Table 1. As anticipated, the evaluation of the radiation fluxes dominates if lines are included. If the radiation fluxes are not included, the calculation is dominated by the evaluation of the real-gas effects (thermodynamic and transport properties). Thus, the actual computational efforts are consistent with those assumed in the original development of the procedures and justify the basic approach.

Because continuum-only solutions can be generated in a fraction of the time required for solutions that include lines, the continuum-only solutions are useful. They are close enough to the solution with lines to establish whether or not the procedure is stable for each particular problem. Given a continuum-only solution as a first guess, the procedure will converge swiftly to a solution with lines. Thus, costs associated with calculating a solution with lines are incurred only after all of the other nonline effects have been considered, and only under conditions where the probability for success is high.

In a typical iteration, calculations start from a known continuum-only solution and change the problem defining parameters (blowing rate, usually) in roughly 10% increments until a continuum-only solution is obtained. Then the solution with lines usually can be generated with 10 to 15 additional iterations. As an example, an iterative sequence is given in Table

Table 1 Calculation times for one iteration (19-node problem)

Operation	Central processor time for CDC 7600, sec
1) Thermodynamic and transport properties	0.71
2) Continuum radiation	0.62
3) Line radiation	14.2
4) Evaluation of flowfield equations	0.03
5) Matrix inversion	0.31
6) Total	15.87

Table 2 Relative errors for typical iterative sequences

Equation numbers[a]	Iterations 1	2	3	4	6
			Momentum equations		
(1)	0.00001	0.00036	0.00004	...	...
(10)	0.00002	0.03660	0.00262	0.00039	...
(12)	0.00011	0.00173	0.00115	0.00009	...
(15)	0.00001	0.00091	0.00015	...	...
(18)	...	...	...	...	...
			Energy equations		
(1)	0.00157	0.00317	0.00005	...	...
(10)	0.05386	0.14253	0.04091	0.00591	...
(12)	0.02075	0.26716	0.07994	0.00090	0.00001
(15)	0.00808	0.02147	0.00129	0.00045	0.00005
(18)	0.00254	0.00020	0.00001	...	...
			Species equations		
(1)	...	0.00003	...	...	...
(10)	0.00008	0.06235	0.03841	0.00465	0.00027
(12)	0.00229	0.40659	0.22130	0.06422	0.00453
(15)	0.00011	0.00506	0.00077	0.00008	0.00001
(18)	...	...	...	...	...

[a]The first two equations in each set are boundary conditions, followed by 17 interior conservation equations.

2 for a probe entering the Saturn nominal atmosphere (V_∞ = 96.8 Kft/sec, $\rho_\infty = 5.7 \times 10^{-5}$ lb/ft^3, R_N = 2.461 ft, P_s = 6.959 atm, $\dot{m}/\rho_\infty V_\infty = 0.065$). This sequence shows the changes in the relative errors in the conservation equations, from a continuum-only first-quess solution to a converged solution with lines. Thus, the number of iterations that include lines is kept to a minimum, as desired.

Additional solutions are presented in Fig. 2 to illustrate the effectiveness of the method for assigning nodes in the wall normal direction. Four solutions are presented: two for a carbon heat shield in the Jupiter nominal atmosphere, and two for a SiO_2 heat shield in the Saturn norminal atmosphere. Solutions for each condition are obtained with and without lines. It was found that 19 nodes situated at fixed values of the velocity ratios were adequate to resolve all four (significantly different) solutions.

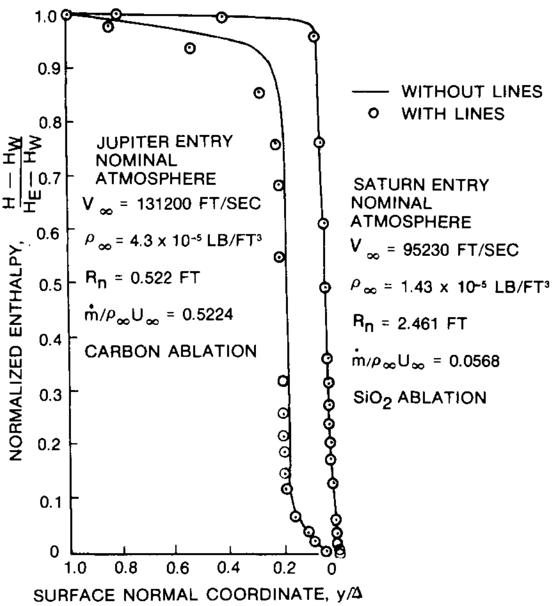

Fig. 2 Nodes across the layer.

Conclusion

A procedure has been developed to evaluate radiation-coupled flowfield problems. This procedure is structured to minimize the number of iterations and nodes while maintaining accuracy and generality in evaluating the radiation fluxes.

In this solution procedure, it has proven exceptionally useful to 1) use the multidimensional Newton-Raphson technique, which allows the radiation flux calculation to be utilized directly in the primary flowfield iteration equation; 2) select the coordinate increments as unknown variables, which insures that the nodal points always will be well situated; 3) treat the continuum and line radiation separately, which allows continuum-only solutions to be used as first guesses for complete solutions; and 4) use forward and backward finite differences combined with joining relations, which allows treatment of the sharp turns in the flow profiles without the flails that would be introduced by higher-order difference relations.

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SHOCK-TUBE STUDIES OF SILICON-COMPOUND VAPORS

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Abstract

Test gas mixtures containing SiO, SiO_2, Si_2, and SiH were produced in a 5-cm-i.d. shock tube by processing shock waves through a mixture of $SiCl_4$ + N_2O + Ar, SiH_4 + Ar, or SiH_4 + O_2 + Ar. Absorption spectra of the test gases were studied photographically in the reflected shock region using a xenon flash lamp as the light source in the range of wavelengths between 250 and 600 nm. SiO was found to be a dominant species in the vapors produced by the $SiCl_4$ + N_2O and SiH_4 + O_2 mixtures. Spontaneous combustion was observed in the SiH_4 + O_2 + Ar mixture prior to the shock arrival, and the resulting solid SiO_2 particles evaporated behind the shock wave. Spectral absorption characteristics of SiO, SiO_2, Si_2, and SiH were determined by studying the test gases.

Introduction

This paper describes the results of experiments conducted in a shock tube in which the optical characteristics of the ablation products of volume-reflecting heat shields[1] made of silica (SiO_2) were studied. The heat shields reflect away much of the radiation impinging on them and are thought to be efficient for the severe conditions expected for Jupiter entry. During an entry into the H_2/He atmosphere of the planet Jupiter, the radiative heat flux from the hot inviscid shock layer is expected to be such that a massive-blowing flowfield will be established; that is, there will be a thick, almost inviscid layer of ablation-product gases adjacent to the

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surface of the heat shield. The temperatures in the inviscid layer[2] can be as high as 18,000 K, and the temperatures in the blowing layer can range between 2500 and 9000 K. Sandwiched between the blowing layer and the hot inviscid shock layer is a viscous mixing region at temperatures near 13,000 K. Radiation from the high-temperature layer will be partly absorbed by the two layers. The performance of the heat shield therefore will depend in part on the optical absorption characteristics of the gas species present in the two layers. The absorption properties are especially important in the shorter spectral regions, because, depending on the amount of impurities present in the heat shield, silica is opaque in this region, and the impinging radiation will be taken inward as heat energy.

When silica is used as the heat-shield material, the blowing layer consists mostly of SiO, SiO_2, O, and O_2. In the mixing layer, there will be He, H, Si, Si_2, and SiH as well. Of these gas species, optical characteristics are known fairly well for atomic species and the molecule O_2. For the rest, the information is incomplete. The present work was carried out to survey the optical characteristics of these less-known molecular species in order to determine their roles in the radiation blockage.

According to a principle of radiative transfer, an incident radiation of intensity I' is attenuated, in an environment similar to the case under study, to I approximately by the relation

$$I/I' = \exp(-\sigma n L)$$

where σ, n, and L are the absorption cross section, number density of the absorbing species, and pathlength, respectively. The relationship neglects spontaneous emission because it is appropriate to do so under the conditions of interest. In order for the absorption to be significant, the product σnL must be of the order of unity or greater. According to an unpublished calculation performed by Nicolet,[2] assuming laminar flow for one point in the entry trajectory considered to be a severe case characterized by the blowing parameter value f_w of -21, the density/path-length product $\int n\,dL$ of the species SiO, SiO_2, Si_2, and SiH is 1.4×10^{18}, 7×10^{16}, 4×10^{15}, and 6×10^{14} cm^{-2}, respectively. For other molecular species, that is, OH, O_3, H_2O, HO_2, etc., the parameter is smaller. These nL values were taken as the benchmark values in the present study.

Since SiO constitutes the major species in the blowing layer, its nL parameter value will be roughly proportional to stagnation-point pressure, which is specified relatively firmly by the mission profile. For SiO_2 and SiH, however, the nL parameter could be considerably larger than the benchmark values if the assumptions of equilibrium and laminar flow are revoked. If the solid SiO_2 heat shield vaporizes predominantly into gaseous SiO_2, and if the gas-phase decomposition processes are frozen, its maximum nL value may be about the same as that for SiO. The species SiH could be produced in excess of the equilibrium laminar case if the mixing region is broadened by turbulence. Through such considerations, therefore, the regime of interest is considered to be that of nL values up to 4×10^{18}, 4×10^{18}, 10^{17}, and 10^{17} cm^{-2} for SiO, SiO_2, Si_2, and SiH, respectively.

Many band systems (i.e., at least seven) of SiO have been observed.[3] However, absolute intensity parameters (f numbers or electronic transition moments) are available for only two of these ($A^1\Pi - X^1\Sigma^+$ and $E^1\Sigma^+ - X^1\Sigma^+$). Even for the better-known $A^1\Pi - X^1\Sigma^+$ band system, however, the reported intensity parameters vary over the two orders of magnitude. For the rest of the species of interest, that is, SiO_2, Si_2, and SiH, virtually nothing is known of the absolute intensity parameters. As a first step, therefore, a survey was made in the quartz-optics spectral region between 250 and 550 nm. By studying this wavelength region with an appropriate magnitude of nL parameter, it was hoped specifically to understand the roles of 1) the $A^1\Pi - X^1\Sigma^+$ band system of SiO; 2) the $^1\Sigma - {}^3\Pi$ band system[3,4] of SiO in the 420-430 nm region; 3) the uncharacterized band of SiO_2 in the 370-450 nm region[4]; and 4) the band systems[3] of Si_2 and SiH including the $H_3\Sigma_u^- - X^3\Sigma_g^-$ (380-460 nm), $L^3\Pi_u - D^3\Pi_g$ (340-370 nm), and $K^3\Sigma_u^- - X^3\Sigma_g^-$ (300-330 nm) band systems of Si_2 and the $A^2\Delta - X^2\Pi$ (410-430 nm) and $B^2\Sigma - X^2\Pi$ (320-330 nm) band systems of SiH. In addition, an effort was made to verify whether silica evaporates predominantly into SiO (and O), as predicted by the equilibrium theory.

Experimental Methods

The test gases were produced in a 3.6-m shock tube with an i.d. of 5 cm.[5] The measurements were made in the reflected shock region near the end wall. Cold helium was used as the driver gas. A xenon flash lamp was used as the light source. Depending on the concentration of the test species, the beam from the light source was passed diametrically across the shock tube either two or four times, using front-surface mirrors. This enabled us to attain nL product values of 1.3×10^{19}, 6.6×10^{18}, 1.8×10^{17}, and 1.2×10^{17} cm^{-2} for

SiO, SiO_2, Si_2, and SiH, respectively. A Hilger and Watts E498 quartz-prism spectrograph with reciprocal dispersion varying from 8.7 A/mm at 250 nm to 105 A/mm at 600 nm, and a Spex 0.75-m Czerny-Turner spectrograph with a reciprocal dispersion of 22 A/mm were incorporated in the experimental setup. Spectra were recorded photographically over the wavelength interval between 250 and 600 nm. A set of seven-step neutral density filters was placed in front of the entrance slit of the spectrographs for the purpose of calibration of photographic plate sensitivity. One-half of the slit was covered by the filters so that comparison between the two halves of a spectral trace could give the calibration directly. The light source was flashed at varying instants past the moment of the arrival of the reflected shock wave.

Spectra of the light source and the absorption spectra of the filters were recorded on the same plate as the absorption spectra of the test gas to enable calibration of the plate sensitivity independently on each plate. The photographic plates were scanned by a Jarrell-Ash model 24-300Sp recording microphotometer to obtain a densitometric trace. The densitometric traces were digitized using a Hewlett-Packard (H.P.) 9830 computer-9864A digitizer combination, and the data were stored on H.P. magnetic tapes using integer arrays of total dimensions up to 6000. These data were analyzed by a separate code in the H.P. 9830 computer, and the results were plotted by an H.P. 9873 plotter. A three-parameter curve was fitted in the program to represent the observed sensitivity variation, that is, the "gamma" curve, for each plate at every data point.

By comparing the spectra taken at different time instants between runs, time evolutions of spectral characteristics were studied. The charging pressure varied between 25 and 60 Torr, the primary shock velocity was between 1100 and 1600 m/sec, and the driver pressure varied between 10 and 17 atm. Temperature and pressure of the test gas in the reflected shock region varied between 2100 and 5500 K and between 5 and 10 atm, respectively. The light source was found to be repeatable to within an extent indiscernible by the photographic process, that is, a repeatability of at most within ±5%, and possibly less in its intensity. The shock tube was repeatable to within about ±1% in the reflected shock temperature.

Results and Discussion

SiO and SiO_2

The test gases containing SiO and SiO_2 were produced by shock-heating a mixture of $SiCl_4$ + N_2O + Ar. Purity of

$SiCl_4$, N_2O, and Ar as certified by the supplier (Air Products and Chemicals for $SiCl_4$, Matheson for N_2O and Ar) was 99.999%, 99%, and 99.999%, respectively. Prior to the main experiments, tests were conducted with N_2O + Ar mixtures to determine the nature and concentration of impurities. Emission spectra of Na, Ca, Fe, Cr, and Cu were observed at reflected shock temperatures above 5500 K. Some of these species also absorbed in the range 4000-5000 K. The total concentration of the impurities was judged to be below 10^{-5} by volume by comparing the intensity of the impurity lines with that of a calibrated tungsten lamp. However, at temperatures below 4000 K, where the main measurements were made, impurities did not show any absorption in the 250-550 nm wavelength range. Since the concentration level of 10^{-5} is too low to affect the thermodynamic conditions, the effect of impurities was ignored in the data reduction.

Measurements were made at 0.1, 0.4, 3.5, and 4 msec after the passage of the reflected shock wave. The details of this aspect of the work will be presented in a separate report.[6] The measurements showed that steady-state conditions exist between 100 and 400 μsec after the passage of the reflected shock. These steady-state conditions were interpreted to be thermochemical equilibrium. After 1 msec in the expanding and cooling period of the shock-tube flow, the intensity of the beam emerging from the test gas became progressively dimmer. After each test, the inner walls of the shock tube were found to be covered by a thin layer of white powder. These phenomena were interpreted as follows. When an equal volume of $SiCl_4$ and N_2O is present, the predominant product species behind the shock wave are

$$SiCl_4 + N_2O \rightarrow SiO + 2\ Cl_2 + N_2$$

When N_2O concentration is n times that of $SiCl_4$, predominant product species are

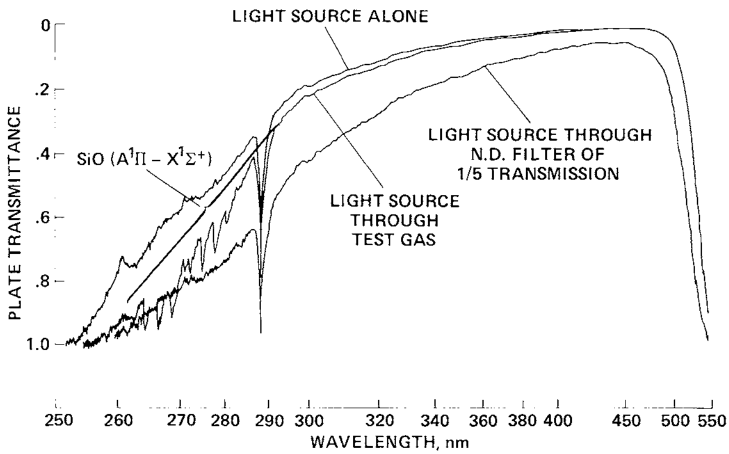
$$SiCl_4 + nN_2O \rightarrow SiO_2 + 2\ Cl_2 + 2(n-2)\ O + nN_2$$

In both cases, atomic oxygen resulting from the thermal decomposition of N_2O triggers the process. The N_2O decomposition process is very rapid (typically within 1 μsec in the present experiments) by virtue of its low binding energy (1.7 eV) and an excited complex.[7] In the present experimental conditions, temperatures were such that N_2 did not dissociate at all. In the expanding period, SiO and SiO_2 condense. When the shock tube is opened to atmosphere after a run, the solid SiO particulates convert to SiO_2 by oxidation.

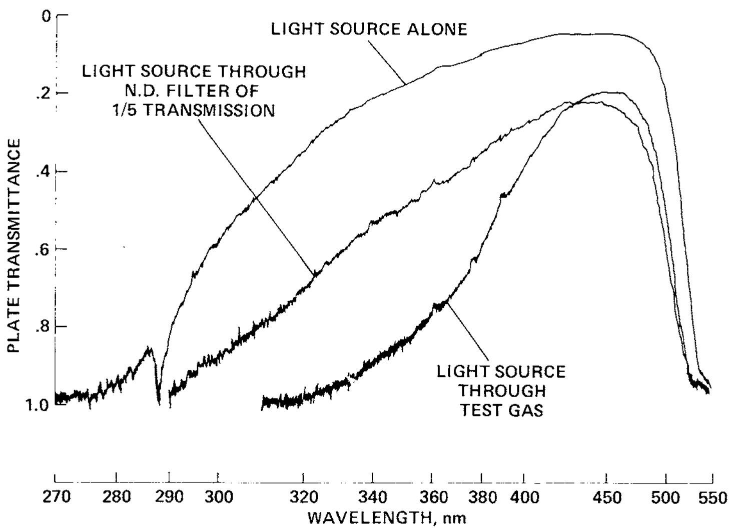
The chemical compositions, pressure, and temperature in the test gases during the nonequilibrium as well as equilibrium period were determined using a computer program in which 14 species ($SiCl_4$, $SiCl_3$, $SiCl_2$, SiCl, ClO, N_2O, O, O_2, SiO, SiO_2, Cl, Cl_2, N_2, and Ar) were accounted for. The equilibrium constants for these species were taken from the JANAF tables,[8] and the chemical reaction rate coefficients necessary for the nonequilibrium computations either were taken from the existing literature or were estimated.[6] The details of this calculation will be described in the separate paper mentioned previously.[6]

The spectral absorption data obtained from the experiment were found to be highly repeatable, that is, to within ±3% in transmittance among three sets of data.[6] The present results differ markedly from the earlier findings of Kivel and Camm,[9] who tested a $SiCl_4$ + air mixture. In their experiment, spectral intensity did not reach a steady state during the available test times, and the spectra observed were dominated by the Schumann-Runge band of O_2 and β and γ bands of NO in the region of wavelength where the SiO $A^1\Pi - X^1\Sigma^+$ band appears. Such behavior is understandable; in an $SiCl_4$ + air mixture, atomic oxygen needed in forming SiO must be produced through the thermal dissociation of O_2 at the expense of 5.1 eV/molecule, as compared to 1.7 eV for N_2O. To dissociate O_2 quickly, temperature had to be raised in the experiment of Kivel and Camm to about 7000 K. Such high temperature caused formation of NO through the reaction $O + N_2 \rightarrow NO + N$. The reaction removes O and thereby slows down the SiO-forming process. Moreover, the NO-forming reaction produced N as a byproduct, which tends to form SiN. Equilibrium in such a system is possible only when nitrogen species N and N_2 reach equilibrium, a process that takes a long time at 7000 K because of the large binding energy of N_2 (9.8 eV). One notes here that Main et al.,[10] who also conducted shock-tube investigation of SiO using solid SiO_2 particulate aerosol, erroneously attributed the lack of agreement between their results and the results of Kivel and Camm to instability of $SiCl_4$; $SiCl_4$ is always quite stable.

Figure 1a shows a densitometer trace of the spectra obtained through a test gas produced with an equal-mole mixture of N_2O and $SiCl_4$. The value for the nL parameter for SiO was 4.2×10^{18} cm^{-2} for this case, in close reproduction of the maximum condition defined in the "Introduction." As indicated in Fig. 1, absorption occurs only in the wavelength range below 350 nm. The absorption is attributed entirely to the $A^1\Pi - X^1\Sigma^+$ band of SiO. There is no sign of the $^1\Sigma - {^3\Pi}$ spin-forbidden band in the 420-430-nm region listed by Pearse and Gaydon.[4] Evidently the band is too weak to appear here.



a) Charging conditions: p = 40.5 Torr, 2.5% $SiCl_4$, 2.5% N_2O, 95% Ar, U_s = 1404 m/sec. Light source flashed at t = 100 μsec after shock arrival. Reflected shock conditions: T = 3570 K, p = 9.1 atm, [SiO] = 4.3 × 10^{17} cm^{-3}, L = 10 cm.



b) Charging conditions: p = 29.5 Torr, 4.5% $SiCl_4$, 23.2% N_2O, 72.3% Ar, U_s = 1453 m/sec. Light source flashed at t = 400 μsec after shock arrival. Reflected shock conditions: T = 2750 K, p = 10.1 atm, [SiO] = 6.7 × 10^{17}, [SiO_2] = 3.2 × 10^{17} cm^{-3}, L = 20 cm.

Fig. 1 Superposition of three densitographs: light source alone, light source through a neutral density filter of nominal transmittance 1/5, and light source through test gas.

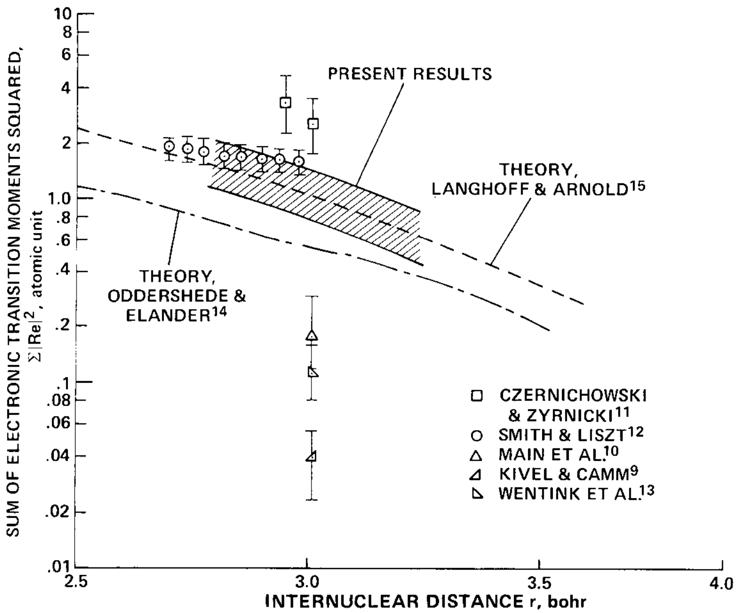


Fig. 2 Comparison of the sum of the squares of electronic transition moments $\Sigma|Re|^2$ vs internuclear distance r for the $SiO(A^1\Pi - X^1\Sigma^+)$ band systems obtained by various investigators.

In Fig. 1b, a similar densitograph is shown for a test gas rich in both SiO and SiO_2, produced by a N_2O-rich mixture, with nL parameter values of 1.3×10^{19} and 6.6×10^{18} cm^{-2}, respectively. In this case, there is a strong, near-continuum absorption over the entire observed wavelength range, attributed to particulates of SiO_2. Still there is no indication of $SiO(^1\Sigma - {}^3\Pi)$ band in the 420-430-nm range, nor the SiO_2 band in the 370-450-nm range listed by Pearse and Gaydon[4]; obviously these bands are too weak to be of interest.

The densitographs such as those shown in Fig. 1a were analyzed quantitatively to deduce the absolute intensity parameter for the $SiO(A^1\Pi - X^1\Sigma^+)$ band. Through a proper selection of the intensity parameter, theoretical synthetic spectra were able to reproduce the experimentally observed spectra quite closely. The details of this aspect of the work will be included in the separate paper mentioned earlier; only the conclusions of the analysis will be shown here. Figure 2 shows the strength of the $SiO(A^1\Pi - X^1\Sigma^+)$ band, expressed in terms of the sum of the electronic transition moments squared, $\Sigma|Re|^2$, obtained by the present work, and compares them with the available data.[9-15] As shown here, the $\Sigma|Re|^2$ values obtained are in close agreement with the experimental results of Smith and Liszt[12] and the theoretical results of Langhoff and Arnold.[15] In comparison with the works of Kivel and Camm[9] or Main et al.,[10] in which shock tubes were used, the present results are much higher. Possible causes of error in the work of Kivel and Camm were discussed earlier. The work of Main et al.[10] possibly may suffer from poor characterization of the properties of solid particles suspended in the test gas, that is, absolute concentration and size distribution of the particles, and velocity and temperature lag between the test gas and the solid particles.

Figure 3 shows the transmittance of SiO vapor in wavelengths from 160 to 320 nm in the environments expected in the blowing-layer region over a silica heat shield. The curves shown are computed taking into account the $A^1\Pi - X^1\Sigma^+$ and $E^1\Sigma - X^1\Sigma^+$ band systems of SiO. The strength of the $A^1\Pi - X^1\Sigma^+$ bands, which spread between 210 and 320 nm, is taken from the present work, whereas that of the $E^1\Sigma^+ - X^1\Sigma^+$ band in the 170–210 nm range is taken from the theoretical work of Langhoff and Arnold.[15] The theory[15] agrees closely with the experimental results of Elander and Smith[16] for the

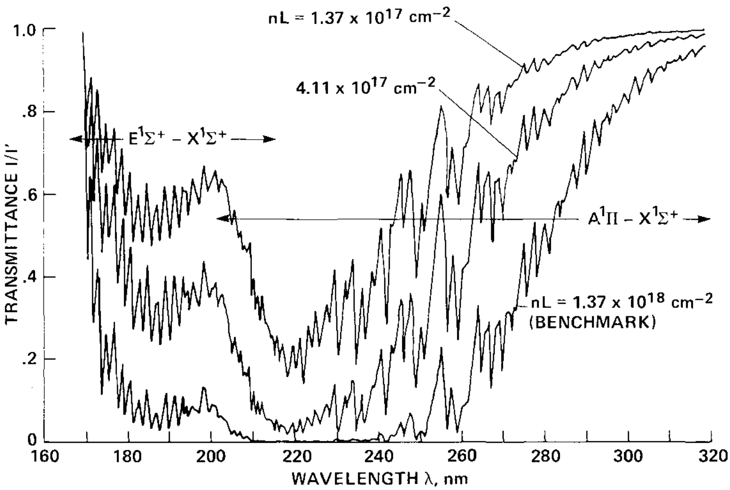


Fig. 3 Transmittance of SiO vapor calculated for conditions typical of flight environments. Assumed temperature = 4500 K; SiO number density = 10^{18}, 3×10^{18}, and 10^{19} cm^{-3}; L = 0.137 cm.

vibrational quantum number combinations for which the experimental data are available; the experimental data cover only a few vibrational quantum number combinations, whereas the theory encompasses all possible combinations. Experimental work to verify the accuracy of the theory and experimental data for the $E^1\Sigma^+ - X^1\Sigma^+$ band is in progress at NASA Ames Research Center.

The three theoretical curves shown in Fig. 3 correspond to three nL parameters ranging from 1.4×10^{17} to 1.4×10^{18} cm^{-2}, encompassing most of the regions of interest. In varying the nL parameter, the number density of SiO, n, was assumed to be varied rather than L. This procedure resulted in a small difference in the widths of individual rotational lines[6]; the difference in transmittance among the three cases shown is due partly to the difference in the line width. If L is varied rather than n, there will be less spread in the magnitude of transmittance among the three curves. In any case, the figure shows a significant absorption below about 290 nm, the extent and the absorption edge wavelength varying according to the magnitude of the nL parameter.

The role of SiO in the radiative transport through the blowing layer over a Jovian entry heat shield is illustrated in Fig. 4. Figure 4a shows the inward radiation flux calculated at the interface between the hot outer hydrogen-helium flow and the relatively colder inner blowing-layer flow, through the use of a computer program developed by Nicolet.[2] The calculation includes radiation emissions from silicon atoms that originate from the heat shield, in the photon energy range of 5.4 eV (230 nm) to 7.3 eV (170 nm). The calculation was done for the benchmark case defined in the "Introduction." That is, the Jovian atmosphere was assumed to consist of 70% hydrogen and 30% helium, and the ablation rate was assumed to be such that the blowing parameter f_w was -21.

Figure 4b shows the flux reaching the wall, calculated taking into account the absorption by SiO given by Fig. 3, and compares them with the original unpublished calculation of Nicolet. As seen here, the flux reaching the wall is much more strongly attenuated by SiO in the wavelength range of 175 to 290 nm than originally was envisioned. In Fig. 4c, the portion of the incident radiation which is absorbed by the heat shield is shown. For this calculation, the wall was assumed to reflect with a reflectance of 90% between 270 and 3000 nm, and to absorb completely elsewhere. As seen here, the portion of radiation energy absorbed by heat

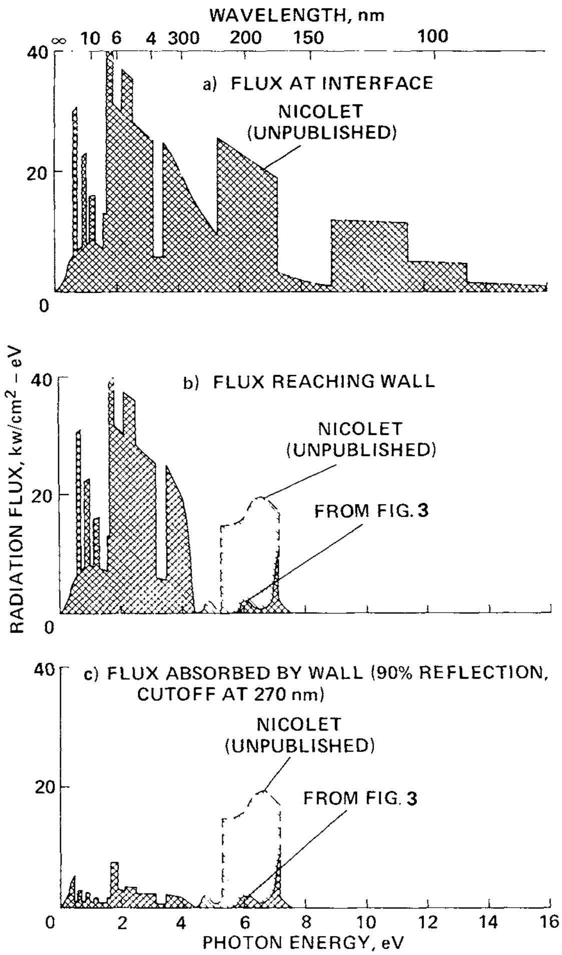
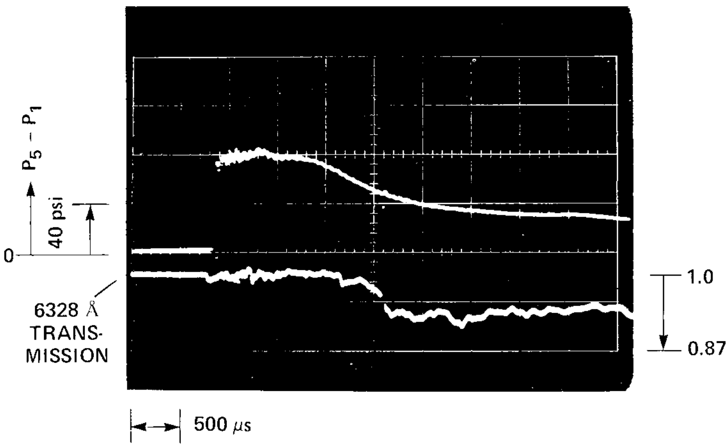


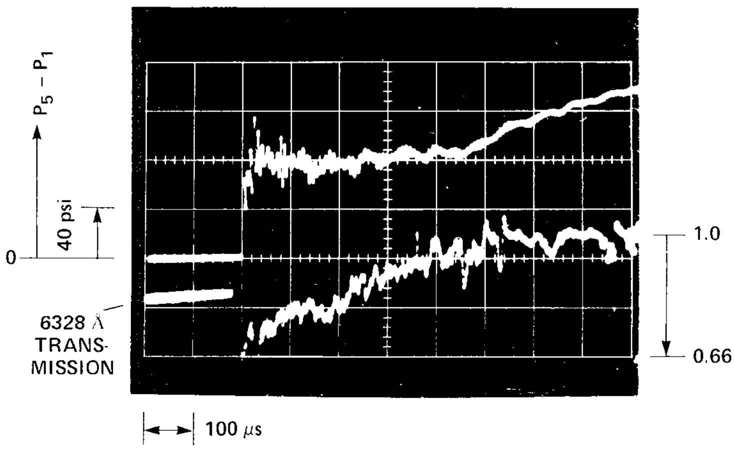
Fig. 4 Calculated radiation fluxes for volume-reflecting silica heat shield for Jovian entry (H_2/He = 70/30, f_w = -21).

shield is a small fraction of the flux prevailing at the interface, even when the cutoff wavelength of reflectance is relatively large. Since little radiation reaches the wall between 170 and 270 nm, little would be gained by improving the reflectance of the heat shield at shorter wavelengths.

It is to be noted that the absorption characteristics shown here are obtained assuming equilibrium laminar flow. In real flows, ionic nonequilibrium phenomena, and possibly



a) p = 50 Torr, 0.75% SiH_4 + 99.75% Ar, U_s = 1182 m/sec.



b) p = 30 Torr, 3% SiH_4 + 6% O_2 + 91% Ar, U_s = 1472 m/sec.

Fig. 5 Oscilloscope traces showing end-wall pressure and transmission of 6328-A He-Ne laser beam through test gas.

turbulence, will bring electrons closer to the ablating wall than predicted by the equilibrium theory. Electrons will cause broadening of individual rotational lines in the molecular bands via Stark effect, leading to a further decrease in transmittance and thereby improving the heatshield performance.

Si_2 and SiH

The test gases containing Si_2 and SiH vapors were produced by shock-heating a SiH_4 + Ar mixture. The purity of SiH_4 used for this study was 99.999%. To study the process of decomposition of SiH_4 and subsequent condensation of the silicon compound vapors produced, a beam of a He-Ne laser was passed through the test gas, and its attenuation through the test gas was monitored with a photomultiplier tube. A narrow bandpass filter and two pinhole baffles were placed between the shock tube and the photomultiplier tube in order to reject emissions from the test gas. The test result, shown in Fig. 5a, indicates virtually no attenuation of the laser beam in the first 1.5 msec, implying that only gas-phase reactions occur during this time period. After 1.5 msec, corresponding to the period of expansion and cooling, however, the light beam attenuates considerably, indicating that condensation is occurring. After each test, dark yellowish powder characteristic of metallic silicon was found deposited on the shock-tube wall. Absorption spectra, not shown here, taken at $t = 50$, 100, 200, 800, and 1.6 μsec, showed that steady-state conditions exist to about 800 μsec after the passage of the reflected shock wave. The chemical relaxation behavior behind the reflected shock was computed accounting for nine species (SiH_4, SiH_2, SiH, Si_2, H, H_2, Ar, Si, and liquid Si particulates). Most of the rate coefficients for this calculation were estimated using the formulas

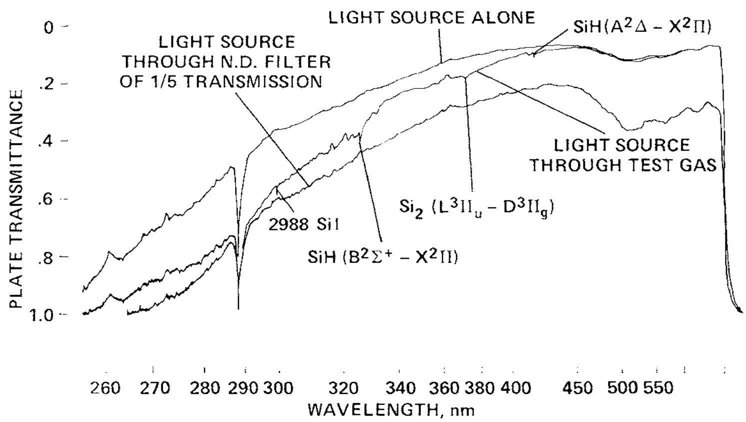


Fig. 6 Superposition of three densitographs: light source alone, light source through a neutral density filter of nominal transmittance 1/5, and light source through test gas. Charging conditions: $p = 59.8$ Torr, 0.75% SiH_4, 99.25% Ar, $U_s = 1119$ m/sec. Light source flashed at $t = 350$ μsec after shock arrival. Reflected shock conditions: $T = 2720$ K, $p = 5.8$ atm, $[Si_2] = 8.6 \times 10^{16}$, $[SiH] = 6.0 \times 10^{16}$ cm^{-3}, $L = 20$ cm.

devised by Park.[17] The calculation predicted that the original mixture will decompose and reach equilibrium within about 20 μsec after the passage of the reflected shock wave and form mostly Si, Si_2, SiH, and H.

A densitograph obtained from the test is shown in Fig. 6. As indicated, the test was made at nL values of 1.9×10^{17} and 1.2×10^{17} cm^{-2} for Si_2 and SiH, respectively, in close simulation of the maximum condition defined in the "Introduction." In the three different wavelength regions indicated in the figure, what seem to be the $Si_2(L^3\Pi g - D^3\Pi u)$, $SiH(A^2\Delta - X^2\Pi)$, and $SiH(B^2\Sigma - X^2\Pi)$ band systems are recognizable.[3] More importantly, however, one recognizes a strong continuum absorption below about 450 nm; absorption by the continuum is far greater than the sum of contributions from the three identified bands. This continuum absorption band does not belong to any known band of either Si_2 or SiH. That this is not due to particulates was deduced, aside from the tests with a laser described previously, in two different ways. First, the concentrations of Si, SiH, and Si_2 were far below the equilibrium vapor concentrations at the prevailing temperature. Secondly, in order for the particulates to cause such attenuation, it was reasoned that the total mass of the particulate matter must be far greater than the originally available amount of SiH_4.

The spectroscopic origin of this continuum is under investigation at NASA Ames Research Center. The observed intensity of absorption by this continuum system is such that, for the benchmark case involving $nL = 4 \times 10^{15}$ cm^{-2} for Si_2, the absorption by this continuum would reduce radiation flux by about 3% at 400 nm, by 10% at 280 nm, and still more at shorter wavelengths, provided that the continuum is due to Si_2. The continuum is therefore significant, although minor, in the radiative transport problem over the silica heat shield. It would be desirable to include this source of radiation absorption in the future design calculations for the silica heat shield.

Nonequilibrium Evaporation of SiO_2

In addition to the preceding experiments, all of which were conducted with equilibrium test gases, tests were conducted in a nonequilibrium mode with a test gas produced by a mixture of SiH_4 and O_2. A $SiH_4 + O_2 +$ Ar mixture was found through an experiment conducted both in a glass vessel and in the shock tube to burn spontaneously over a wide range of mixture ratios and partial pressures. As the mixture burns, a red glow appears. The combustion produces a gray-white

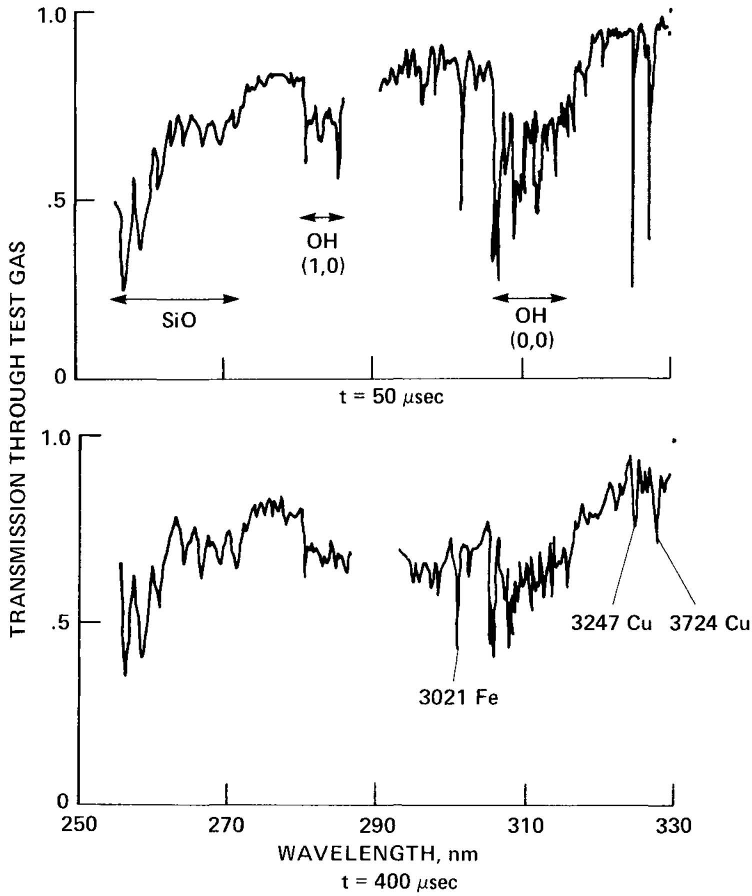


Fig. 7 Absorption characteristics of $3\%\ SiH_4 + 6\%\ O_2 + 91\%$ Ar test gas ($p = 40$ Torr, $\bar{U}_s = 1420$ m/sec). Reflected shock conditions: $3300 < T < 3700$ K, $p = 8$ atm, concentration in nonequilibrium and unknown, $L = 20$ cm.

cloud, consisting presumably of particulates of SiO_2 (when there is enough O_2) or of a mixture of SiO_2 and either Si or SiH (when there is not enough O_2). Under the stoichiometric conditions, the predominant process is considered to be

$$SiH_4 + 2\ O_2 \rightarrow SiO_2 + 2\ H_2O$$

$$SiO_2\ (gas) \rightarrow SiO_2\ (solid)$$

That the cloud consists of particulates was confirmed by passing the laser beam: as soon as the combustion commenced, the test gas started to scatter the beam so that the path of the beam became visible to the naked eye. Also, the transmission of the beam through the test gas was lowered, typically by as much as about 10%.

By shock-heating this gas-solid mixture, one should be able to study the process of vaporization of solids. Such a method would be superior to the technique used by Main et al.,[10] who injected finely ground SiO_2 powder as an aerosol; the particulates formed by in situ combustion would be finer than the mechanically ground powders. For these tests, SiH_4 + Ar and O_2 + Ar mixtures were introduced into the shock tube sequentially through a dispersing tube.[18] Combustion and ensuing condensation were confirmed by monitoring the reduction in the intensity of the laser beam received by the photomultiplier tube. The diaphragm of the shock tube then was ruptured. The oscillogram reproduced in Fig. 5b shows the change in the intensity of the beam. As seen in the picture, the beam is strongly attenuated immediately behind the shock wave; the attenuation is interpreted as being due to compression of the gas-solid mixture. As time passes, the beam intensity increases expectedly because of vaporization of the solid particles. Unfortunately, the process was found to lack reproducibility, and hence no quantitative measurement could be made in this mode.

A qualitative spectral absorption measurement was carried out while the particulates were still in the process of vaporization for the purpose of determining whether SiO exists in such a nonequilibrium condition; if the vapor is in partial equilibrium with the solid state, SiO should be a major constituent of the test gas, and hence SiO spectra should be seen. The test results are shown in Fig. 7. As seen in the figure, SiO absorption is recognizable, along with that of OH, Fe, and Cu. The OH spectra are expected because the test gas contains H_2O. There is continuous absorption throughout the entire wavelength range in the figure, underscoring the result shown in Fig. 5b. From these results, one is led to conclude that SiO is a major constituent in the vapor produced by the evaporation of solid SiO_2, as predicted by the equilibrium theory.

Conclusions

From the preceding studies, the following conclusions are drawn:

1) SiO absorbs strongly below 290 nm. As a result, a high reflectance of the volume-reflecting heat-shield

material would not be necessary below about 270 nm. Absorption in the 420-430 nm range by a spin-forbidden transition is too weak to be of interest.

2) SiO_2 does not absorb, even in the 370-450 nm region reported elsewhere, to any significant extent under the conditions of interest.

3) The mixture of Si_2 and SiH absorbs strongly below 450 nm with a hitherto unknown continuum. Their absorption characteristics are such that they could be of minor importance in the heat-shield design. It would be desirable to include this source of radiation blockage in future calculations of heat-shield performance.

4) Evidence indicates that the product of vaporization of the solid SiO_2 contains SiO as a major constituent under a nonequilibrium condition.

Acknowledgment

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MACH NUMBER AND WALL TEMPERATURE EFFECTS ON TURBULENT HEAT BLOCKAGE

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Abstract

The effect of Mach number, wall temperature, and injectant molecular weight on heat blockage resulting from mass injection has been evaluated for turbulent flow over surfaces with negligible axial pressure gradients. The evaluation resulted in semiempirical correlations that are based on a modification of a compressibility transformation theory expanded to include the latest hypersonic data up to a freestream Mach number of 8.0. The resultant blockage correlations were verified by comparison with data over a wide range of Mach number, wall temperature, and injectant molecular weight. Emphasis was placed on recently acquired data at hypersonic flow conditions on a porous sharp cone. The correlation development, based on an analytically derived compressibility transformation, employed a critical blowaway parameter concept to provide a more tractable set of correlations for engineering design.

Nomenclature

B_h = blowing parameter (λ/St)
B_u = blowing parameter ($2\lambda/C_f$)
b_{cr} = critical blowaway parameter
b_h = blowing parameter (λ/St_o)
C_f = skin-friction coefficient
C_p = specific heat
F_c = compressibility transformation function C_{f_i}/C_f

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F_s	=	viscous transformation function based on Re_s, $Re_{s,i}/Re_s$
F_θ	=	viscous transformation function based on Re_θ, $Re_{\theta,i}/Re_\theta$
H	=	total enthalpy
h	=	static enthalpy
M	=	Mach number
m	=	power law velocity exponent $2/(n+1)$
n	=	power law velocity profile exponent
$\dot{q}$	=	heat-transfer rate
Re_s	=	Reynolds number based on wetted length $(\rho_e u_e s/\mu_e)$
Re_θ	=	Reynolds number based on momentum thickness $(\rho_e u_e \theta/\mu_e)$
r	=	radius measured from axis of symmetry
s	=	wetted length along surface of body
St	=	Stanton number $\dot{q}_w/(\rho u)_e (h_r - h_w)$
T	=	absolute temperature
u	=	velocity component tangent to surface
v	=	velocity component normal to surface
y	=	distance normal to body surface
z	=	velocity ratio u/u_e
θ	=	boundary-layer momentum thickness
λ	=	blowing parameter $(\rho v)_w/(\rho u)_e$
λ_∞	=	blowing parameter $(\rho v)_w/(\rho u)_\infty$
μ	=	molecular viscosity
ρ	=	density
ω	=	viscous transformation exponent, defined in Eq. (18)

Subscripts

air	=	air
cr	=	critical value
e	=	boundary-layer edge
i	=	imcompressible flow value
inj	=	injectant
o	=	zero mass injection
r	=	recovery (adiabatic wall) condition
w	=	wall condition
∞	=	freestream condition

Superscript

$*$ = properties based on Eckert reference enthalpy defined as $h^*/h_e = 0.5\ (1 + h_w/h_e) + 0.11\ r\ (\gamma - 1)\ M_e^2$

I. Introduction

The alleviation of the high heating rates encountered by surfaces of hypersonic vehicles has been recognized as an important problem. One of the cooling methods that has shown promise is mass-transfer cooling, wherein a foreign material is transferred from the vehicle surface into the boundary layer. This has a twofold advantage in attenuating the heat-transfer problem. The transferred coolant may absorb heat from the boundary layer through a phase change (sublimation, evaporation, melting, etc.), as well as provide high thermal heat capacity. In addition, it has been shown that the introduction of a material (with its normal velocity component) at the surface acts to decelerate the flow and, consequently, to reduce the skin friction. This also implies a reduction in heat transfer at the wall, often referred to as heat blockage.

When examining the available literature concerning mass-transfer experiments, it is noted that 1) the bulk of the data was obtained under supersonic flow conditions ($M < 5$), and 2) the model geometries consisted primarily of flat plates, sharp cones, and cylinders in cross flow, precluding pressure gradient effects. Moreover, a careful inspection of engineering heat-transfer prediction techniques indicates that empirically derived correlations, developed from the transpiration experiments, have been employed to determine the level of heat attenuation at the surface of the model resulting from the injection of gases through the porous wall. However, examination of the available hypersonic turbulent boundary-layer data where mass-transfer effects are present indicates that the existing empirical correlations[1-3] for turbulent boundary layers do not characterize the heat blockage sufficiently because the effects of Mach number and wall temperature are not taken into account (see Fig. 1).

In general, most analytical investigations have centered around the flat-plate or sharp cone geometry to allow a more tractable mathematical solution. The most successful approaches involve the solution of the momentum integral equation coupled with suitable compressibility transformations and Colburn's analogy to predict skin friction and heat transfer. These approaches will be examined, modified, and expanded: first, to include the recently acquired experi-

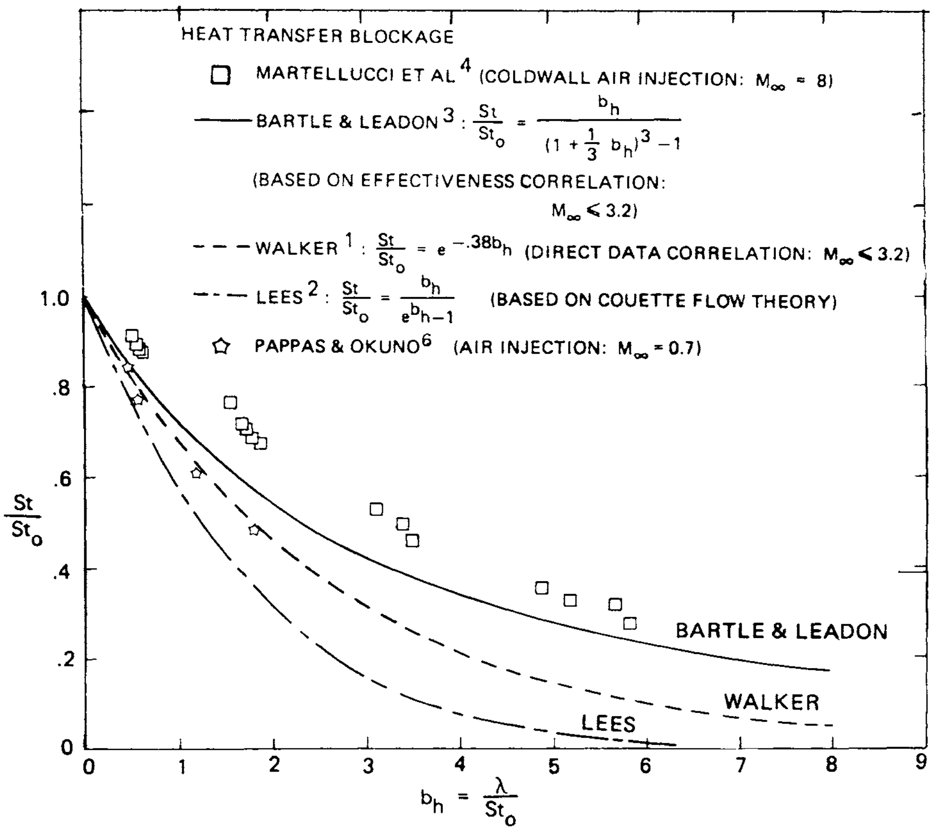

Fig. 1 Empirical heat blockage correlations of previous investigators.

mental data of Martellucci et al.[4, 5] at hypersonic (M > 5) conditions, and second, to account empirically for the effects of both Mach number and wall temperature, with the objective of providing improved engineering design relationships suitable for inclusion in existing semiempirical turbulent boundary-layer codes.

Literature Review

There have been many experiments performed on various geometric shapes in low- and high-speed tunnels to determine the effect of heat blockage on a transpiration-cooled surface. Experiments of this nature have been performed to simulate ablation effects experienced on re-entry vehicles. However, analytical predictions have weighed heavily on the Reynolds analogy to determine the attenuation of heat flux as a consequence of mass injection, thereby requiring recourse to experimental data and subsequent engineering correlations. Of the three empirical correlations alluded to previously (Fig. 1), only that of Walker[1] was a direct curve-fit of the available experimental data at that time. Lees'[2] expression was derived analytically based on a Couette-flow assumption. Bartle and Leadon[3] correlated their experimental data in terms of an effectiveness parameter. They suggested that the heat blockage could be expressed in terms of the effectiveness as shown in Fig. 1. A comparison of these correlations with the data of Martellucci et al.[4] at M = 8 and Pappas and Okuno[6] at $M \simeq 0$ shows a definite deficiency in accounting for the effect of Mach number.

Two of the most pertinent analytical investigations of turbulent boundary-layer heat and mass transfer employing compressibility transformation are those of Spalding et al.[7] and Kutateladze and Leont'ev.[8] Spalding et al. introduced compressibility transformations for skin friction, heat transfer (Colburn analogy), and Reynolds number (viscous effect). The developed technique utilized all of the available experimental data at the time (zero pressure gradient, supersonic flow, $M < 5$). Spalding et al. did not include a Mach number or wall temperature effect in their viscous transformation but recognized its potential. On the other hand, Kutateladze and Leont'ev included the effect of Mach number in their theory but neglected the viscous effect. References 3, 6 and 9-11 have reported turbulent heat blockage data obtained at supersonic conditions, whereas Refs. 4 and 5 extended this data base to the hypersonic flow regime. A detailed assessment of this data base is given in Ref. 12.

Relative to the abundant heat blockage literature, there is a paucity of applicable skin-friction experiments. References 13-17 have investigated skin-friction reduction due to mass injection at supersonic conditions. A review of these investigations is reported in Ref. 12.

Objectives

The technical objectives of the present study are to examine, modify, and expand the aforementioned heat blockage and skin-friction reduction correlations (both analytical and experimental) in order to include the recently acquired data of Martellucci et al.[4,5] and Danberg[11] obtained at hypersonic flow conditions ($M > 5$) and to account for the effects of Mach number, wall temperature, and injectant molecular weight.

II. Analysis

General Approach

Considerable success in correlating heat-transfer blockage and skin-friction reduction as a function of Mach number and wall temperature was obtained by adopting an approach similar to that of the flat-plate analysis of Spalding et al.[7] which employs compressibility transformations to the solution of the momentum integral equation. Extending Spalding's scheme to conical surfaces, the transformed momentum integral equation becomes

$$\frac{d\,(F_\theta Re_\theta \cdot R_s\, Re_s)}{d\,(F_s Re_s)} = F_s Re_s \cdot F_c \frac{C_f}{2}(1 + B_u) \frac{F_\theta}{F_c F_s} \tag{1}$$

where the transformation functions F_c, F_s, and F_θ are dependent upon Mach number, wall temperature, and blowing rate.

Spalding's basic postulate asserts that unique relations exist between $(F_c\, C_f/2)$, $(F_\theta\, Re_\theta)$, and $(F_s\, Re_s)$. This can be true only if, in Eq. (1), $(1 + B_u)\, F_\theta \,/\, (F_c\, F_s)$ is equal to a constant for all boundary layers. Consideration of the particular case of the uniform density zero mass-transfer boundary layer shows that this constant must be equal to unity. Therefore,

$$F_s = (F_\theta / F_c)\,(1 + B_u) \tag{2}$$

Thus, it is seen that F_s is not independent of the F_c and F_θ transformations. This means that only F_c and F_θ need be deduced, either analytically or empirically, to afford a solution of Eq. (1). Rewriting Eq. (1) yields

$$\frac{d\,(F_\theta Re_\theta \cdot Re_s)}{dRe_s} = \frac{F_\theta}{F_c} Re_s \left(F_c \frac{C_f}{2}\right)(1 + B_u) \tag{3}$$

The skin-friction coefficient for an incompressible flat-plate flow can be expressed in terms of the transformation functions as

$$F_c\,(C_f/2) = (E/2)\,(F_\theta\, Re_\theta)^{-m} \tag{4}$$

where E and m are constants related to the velocity power law exponent; specifically, $E/2 = 0.013$ and $m = 0.25$ for $n = 7$.

Equation (3) then becomes

$$\frac{d(F_\theta Re_\theta \cdot Re_s)}{dRe_s} = \frac{E}{2}\frac{F_\theta}{F_c}(F_\theta Re_\theta \cdot Re_s)^{-m} Re_s^{m+1} (1+B_u) \tag{5}$$

which integrates to (assuming B_u constant)

$$\frac{(F_\theta Re_\theta \cdot Re_s)^{m+1}}{m+1} = \frac{E}{2}\frac{F_\theta}{F_c}(1+B_u)\frac{Re_s^{m+2}}{m+2} \tag{6}$$

Rearranging and reintroducing Eq. (4) yields the result

$$F_c \frac{C_f}{2} = \frac{E}{2}\left[\frac{E}{2}\left(\frac{m+1}{m+2}\right)\frac{F_\theta}{F_c}(1+B_u)\, Re_s\right]^{-m/(m+1)} \tag{7}$$

For the zero blowing case ($B_u = 0$), Eq. (7) reduces to

$$F_{c_o} \frac{C_{f_o}}{2} = \frac{E}{2}\left[\frac{E}{2}\left(\frac{m+1}{m+2}\right)\frac{F_{\theta_o}}{F_{c_o}} Re_s\right]^{-m/(m+1)} \tag{8}$$

which, when combined with Eq. (7), results in the skin-friction reduction relation:

$$\left(\frac{C_f}{C_{f_o}}\right)_{Re_s} = \left(\frac{F_c}{F_{c_o}}\right)^{-\frac{1}{(m+1)}}\left(\frac{F_\theta}{F_{\theta_o}}\right)^{-\frac{m}{(m+1)}}\left(1+B_u\right)^{-\frac{m}{(m+1)}} \tag{9}$$

If one employs the use of Reynolds' analogy ($St = C_f/2$) at this point, the heat blockage relation, analogous to Eq. (9), becomes

$$\left(\frac{St}{St_o}\right)_{Re_s} = \left(\frac{F_c}{F_{c_o}}\right)^{-\frac{1}{(m+1)}}\left(\frac{F_\theta}{F_{\theta_o}}\right)^{-\frac{m}{(m+1)}}\left(1+B_h\right)^{-\frac{m}{(m+1)}} \tag{10}$$

where B_h is defined in terms of the Stanton number:

$$B_h \equiv \lambda/St \tag{11}$$

Deduction of the Transformation Functions

Spalding et al. adopted a definition for F_c which commonly has been accepted in the scientific community, namely,

$$F_c \equiv \left[\int_0^1 \left(\frac{\rho/\rho_e}{1 + B_u z}\right)^{1/2} dz\right]^{-2} \tag{12}$$

where z is the velocity ratio u/u_e.

In the foregoing, an equation of state generally is invoked, together with the boundary-layer assumption of zero pressure gradient normal to the surface. The linear Crocco relation then is applied, and numerical integration is performed to evaluate Eq. (12). In order to simplify the complexity of the preceding integral, it will be assumed that F_c can be approximated by the expression

$$F_c \cong F_c{}^* = \frac{\rho_e}{\rho^*}\left[\int_0^1 \frac{dz}{\sqrt{1 + B_u z}}\right]^{-2} \tag{13}$$

where the quantity with the asterisk is based on the Eckert reference enthalpy; then

$$F_c{}^* = (\rho_e/\rho^*)\left[(2/B_u)\left(\sqrt{1 + B_u} - 1\right)\right]^{-2} \tag{14}$$

Note that, in the limit (using L'Hôpital's rule) as $B_u \longrightarrow 0$,

$$F_{c_o}{}^* = \rho_e/\rho^* \tag{15}$$

which is the transformation function for a noninjection case.[18]

A comparison of the $F_c{}^*$ approximation with the numerically exact F_c solution from the tables of Spalding et al. is given in Fig. 2 for a realistic range of Mach numbers ($0 < M_e < 8$) and wall temperature ratios ($1 < T_w/T_e < 8$). The comparison indicates excellent agreement in the supersonic regime ($M < 5$) but a significant variation with increasing values of Mach number; however, this is to be ex-

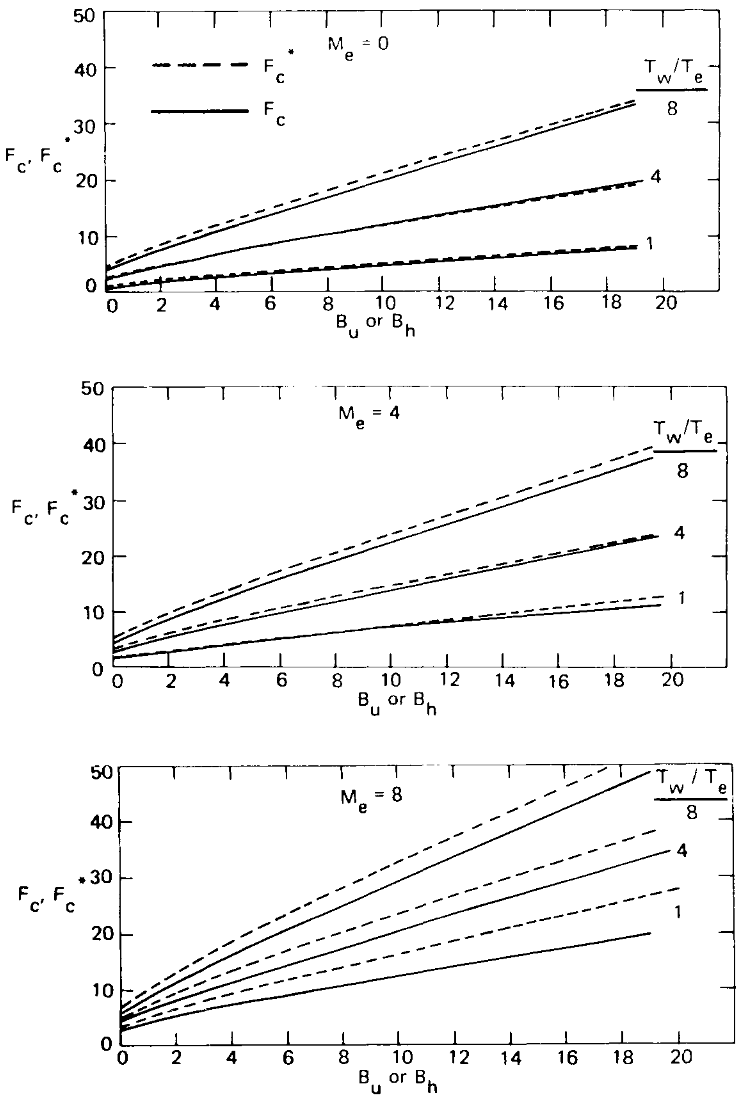

Fig. 2 Verification of compressibility transformation approximation.

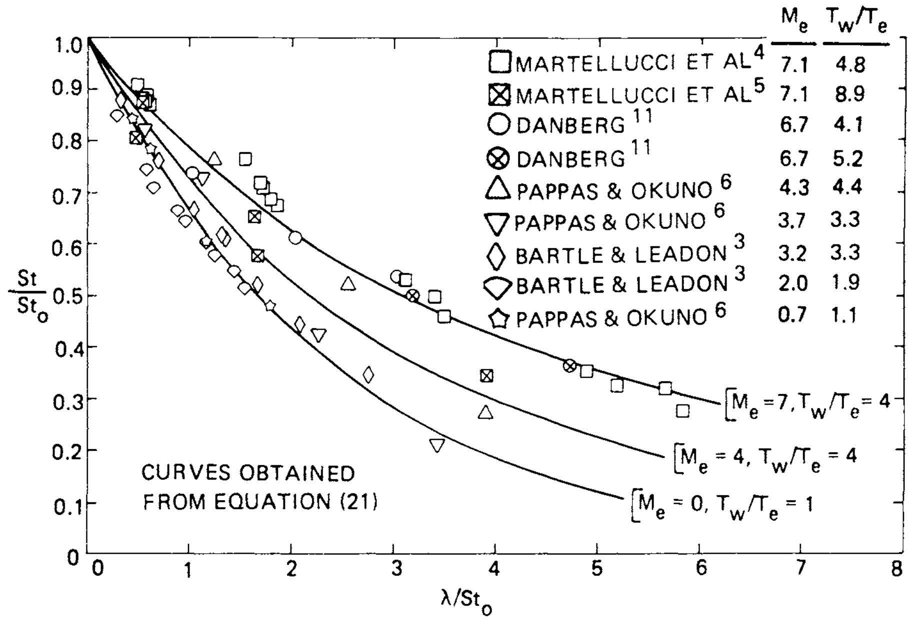

Fig. 3 Verification of the heat blockage correlation (air injection).

pected, as seen, for example, in a turbulent heating flight test analysis by Laganelli.[18] The important thing to keep in mind here is that this approximation is purely analytical, whereas the F_θ viscous transformation function (as will be seen) is deduced from experimental data. The net effect of both of these transformations will account satisfactorily for the hypersonic situation.

Although most authors agreed on the analytical form for the F_c transformation, significant disagreement prevailed in the formulation of the F_θ function. Since there existed considerable uncertainty about the nature of F_θ, Spalding et al. deduced the following empirical relation (based on all of the air injection data available at the time prior to 1962):

$$F_\theta = (\mu_e/\mu_w)\,(1 + B_u)^{-3/2} \tag{16}$$

which does not contain a Mach number effect.

Consequent modification of this relation to include the experimental air injection data of References 4-6 and 11, along with the

Eckert reference state concept, resulted in the empirical relation

$$F_\theta = (\mu_e/\mu^*)\,(1 + B_u)^{-\omega} \tag{17}$$

where

$$\omega = (T_w/T_e)^{-1/8} + (1/8)\,M_e \tag{18}$$

Eq. (18) takes into account the effects of Mach number and wall temperature through a purely empirical correlation of experimental data and judicious use of the F_c tables of Spalding et al. Note, again, that, in the limit as $B_u \longrightarrow 0$, Eq. (17) reduces to

$$F_{\theta_o} = \mu_e/\mu^* \tag{19}$$

which is the correct zero injection viscous transformation using the Eckert approach.[18]

Substitution of Eq. (14, 15, 17, and 19) into the skin-friction expression. Eq. (9), results in (for m = 1/4)

$$\left(\frac{C_f}{C_{f_o}}\right)_{Re_s} = \left[\frac{2}{B_u}\left(\sqrt{1 + B_u} - 1\right)\right]^{8/5} \left(1 + B_u\right)^{(\omega - 1)/5} \tag{20}$$

where it is noted that, in the limit as $B_u \longrightarrow 0$, C_f/C_{f_o} approaches unity, as required.

The analogous equation for Stanton number is simply (by Reynolds analogy)

$$\left(\frac{St}{St_o}\right)_{Re_s} = \left[\frac{2}{B_h}\left(\sqrt{1 + B_h} - 1\right)\right]^{8/5} \left(1 + B_h\right)^{(\omega - 1)/5} \tag{21}$$

Verification with Experimental Data

A comparison of predictions using Eq. (21) and a wide range of air injection experimental data is given in Fig. 3, where the heat

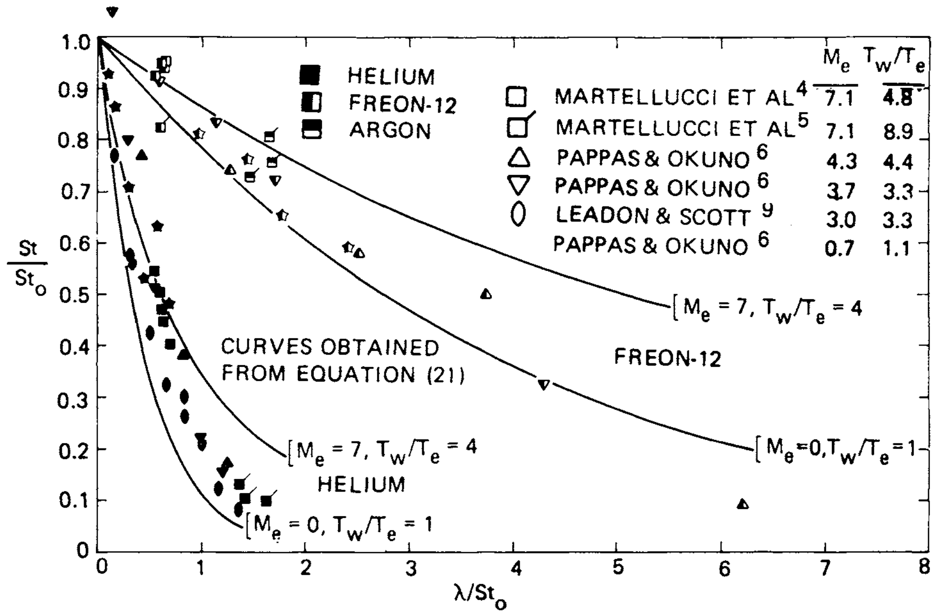

Fig. 4 Verification of the heat blockage correlation (foreign gas injection).

blockage (represented by the Stanton number ratio St/St_o) is plotted as a function of the blowing parameter λ/St_o. All of the data points of Ref. 4 were obtained at the high Reynolds number ($Re_\infty/ft = 3.7 \times 10^6$). The Danberg data points at $T_w/T_e = 4.1$ are not the exact St/St_o values reported in Ref. 11. The data points in Fig. 3 were obtained by normalizing his reported St values with a second set of St_o values which he reported at the same conditions. It was felt that the second set, which consisted of a larger number of data points, would prove to be more reliable. This conclusion seems to be borne out by the correlation curve ($M_e = 7$, $T_w/T_e = 4$) in Fig. 3. The Pappas and Okuno[6] data plotted in Fig. 3 are their "thermocouple no. 7" data, which were measured at their highest reported Reynolds numbers. Further comments concerning the choice of the data included in the foregoing will be reflected upon in Sec. III.

The wall temperature effects were taken into account [Eq. (18)] through the use of the F_c transformation function tables of Spalding et al.[7] As seen from the figure, the correlation curves bracket the data very well, verifying the approximations for F_c [Eq. (14)] and F_θ [Eq. (17)].

The influence of injectant molecular weight on heat blockage has been taken into account utilizing the foreign gas injection data of

Refs. 4-6 and 9. Correlation of heat blockage with foreign gas injection is presented in Fig. 4, where data for helium, argon, and Freon-12 injection are compared with the correlation of Eq. (21), with the blowing parameter B_h redefined as

$$B_h \equiv (C_{p_{inj}}/C_{p_{air}})\,(\lambda/St) \tag{22}$$

Considering the data scatter, Eq. (21) does an excellent job of correlating the available data.

III. An Alternate Correlation for the Results

The skin-friction reduction relation, Eq. (20), and the heat blockage expression, Eq. (21), which have been developed are expressed in the implicit parameters B_u and B_h, which contain the terms C_f and St, not C_{f_o} and St_o. As engineering design correlations, these expressions would require iterative solutions in order to obtain the desired results. Therefore, more tractable correlations for design purposes should be expressed in terms of explicit blowing parameters (i.e., terms containing the a priori known values of C_{f_o} or St_o).

This was accomplished through the use of the critical blowaway parameter, b_{cr}, introduced by Kutateladze and Leont'ev.[8] They defined b_{cr} as the limiting value of the explicit blowing parameter when C_f/C_{f_o} or St/St_o vanishes. The explicit blowing parameters (analogous to B_u and B_h) are defined as

$$b_u = 2\lambda/C_{fo} \tag{23}$$

$$b_h = \lambda/St_o \tag{24}$$

In Ref. 20, the critical blowaway parameter b_{cr} was found to be related to Mach number and wall temperature. In the present study, b_{cr} has been correlated as a function of Mach number and wall temperature through the ω parameter defined in Eq. (18). The expression that correlates b_{cr} and ω is given as

$$b_{cr} = \exp\left[1.676\,(\omega + 0.161)\right] \tag{25}$$

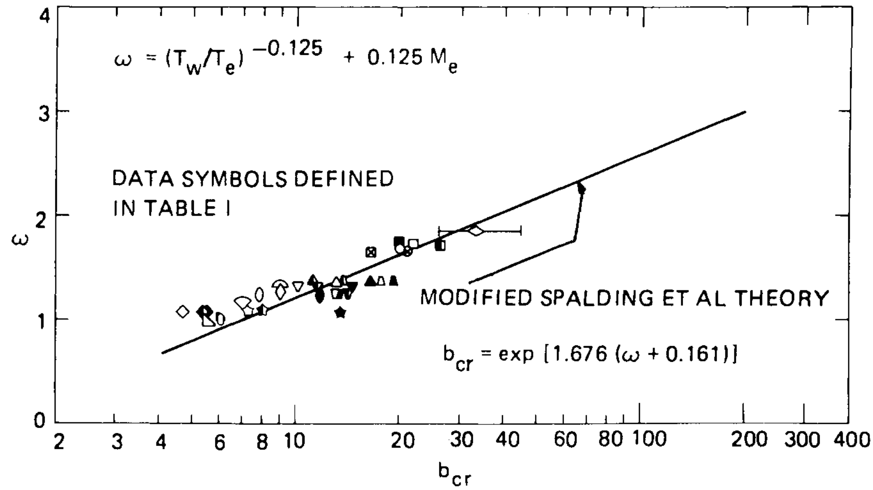

Fig. 5 Effect of Mach number and wall temperature on blowaway parameter.

Table 1 Definitions of data symbols

Symbol	Source	Ref.	M_e	T_w/T_e	Surface
	Heat transfer				
◇ (with bar)	Fogaroli and Saydah	19	8.1	3.0–5.8	Cone
□	Martellucci et al.	4	7.1	4.8	Cone
⊠	Martellucci et al.	4	7.1	8.9	Cone
○	Danberg	11	6.7	4.1	Flat plate
⊗	Pappas and Okuno	6	4.35	4.4	Cone
△	Pappas and Okuno	6	3.67	3.3	Cone
▽	Bartle and Leadon	3	3.2	3.3	Flat plate
◊	Leadon and Scott	9	3.0	3.3	Flat plate
0	Bartle and Leadon	3	2.0	1.9	Flat plate
▽ (rounded)	Pappas and Okuno	6	0.7	1.1	Cone
☆, 0	Tewfik et al.	10	0	1.0	Cylinder
	Skin friction				
⏢	Pappas and Okuno	16	4.3	(4.4)	Cone
⏷	Pappas and Okuno	16	3.21	(3.3)	Cone
⌓	Tendeland and Okuno	17	2.55	(1.0)	Cone
⬡	Dershin et al.	13	3.18	(3.3)	Flat plate
◺	Kendall et al.	14	0	(1.0)	Flat plate
D	Goodwin	15	0	(1.0)	Flat plate
◇	Pappas and Okuno	16	0.7	(1.1)	Cone

[a]Open symbols denote air injection; filled symbols denote helium injection; half-filled symbols denote freon injection.

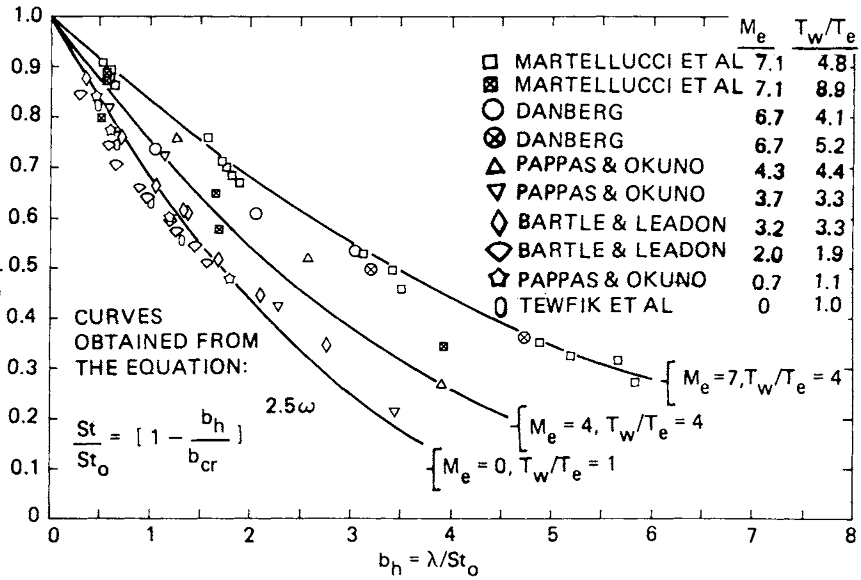

Fig. 6 Comparison of heat blackage correlation with air injection data.

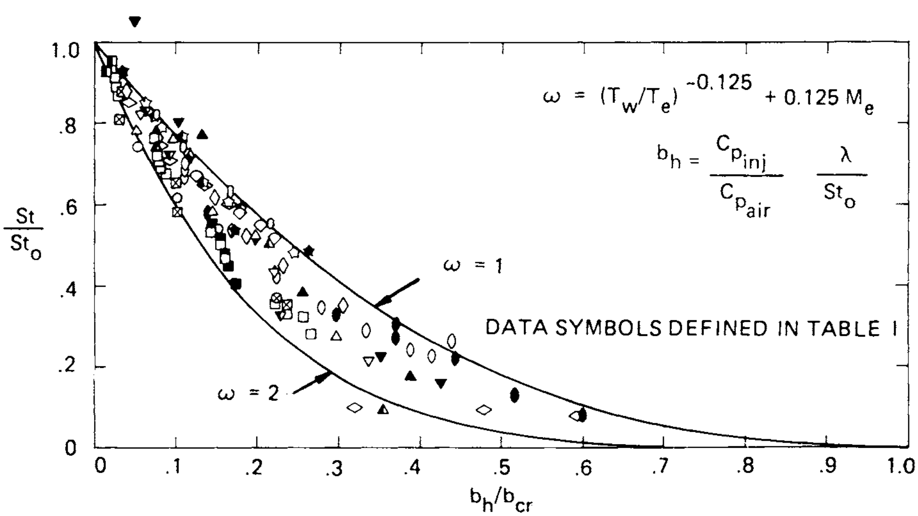

Fig. 7 Correlation of heat blockage with foreign gas injection.

This result [Eq. (25)] represents the approximation for b_{cr} which provides the most suitable agreement with the correlation equations (20) and (21) developed in Sec. II. Equation (25) is plotted in Fig. 5 and compared with all of the available heat-transfer and skin-friction mass injection data (see Table 1). Each symbol represents the average value of b_{cr} extracted from each data source. The massive blowing data of Fogaroli and Saydah[19] were included for comparison at higher values of b_{cr}. Further arguments concerning the incompressible evaluation of b_{cr} are presented in Ref. 12.

Utilizing this concept of b_{cr}, the heat blockage equation may be expressed in terms of the explicit blowing parameter b_h as follows:

$$St/St_o = [1 - (b_h/b_{cr})]^{2.5\omega} \tag{26}$$

The form of this expression was inspired by a correlation of a series of analytical solutions (for a wide range of Mach number and blowing rate) performed by Kutateladze and Leont'ev.[20]

In Fig. 6, the heat blockage correlations, Eq. (26) are compared with the air injection data reviewed previously in the discussion of Fig. 3. The curves generated from Eq. (26) provide slightly better agreement with the data than do the equivalent curves generated from Eq. (21). Overall, excellent agreement is observed for the air injection comparison over the Mach number range $0 < M_e < 7$ for three different geometrical surfaces.

The correlation for heat blockage with foreign gas injection is illustrated in Fig. 7, where St/St_o is plotted as a function of b_h/b_{cr}. [The explicit blowing parameter b_h, with provision made for foreign gas injection, is redefined as $b_h = (Cp_{inj}/Cp_{air}) \times (\lambda/St_o)$.] Most of the data for air, helium, and Freon-12 injection are bracketed by the correlation curves for $\omega = 1$ (subsonic flow) and $\omega = 2$ (hypersonic flow) which represent a realistic flight condition range. Some of the helium injection data fall outside of the subsonic limit ($\omega = 1$), but, overall, Fig. 7 verifies the success of the correlation equation derived for heat blockage [Eq. (26)]. Similar success in correlating the available supersonic skin-friction data[13-17] was reported in Ref. 12.

IV. Conclusions

Heat transfer to a transpired surface including nonair injectants and high Mach number effects can be predicted based on low-speed, constant-property air injection data using the following equations:

$$St/St_o = [(2/B_h)(\sqrt{1 + B_h} - 1)]^{8/5} \quad (1 + B_h)^{(\omega - 1)/5}$$

where

$$\omega = (T_w/T_e)^{-1/8} + (1/8)\, M_e$$

and

$$B_h = (\lambda/St)\,(C_{p_{inj}}/C_{p_{air}})$$

An alternate correlation (explicit), having the same effect, is

$$St/St_o = [\, 1 - (b_h/b_{cr})\,]^{2.5\omega}$$

where

$$b_{cr} = \exp\,[1.676\,(\omega + 0.161)]$$

and

$$b_h = (\lambda/St_o)\,(C_{p_{inj}}/C_{p_{air}})$$

The implicit formulation represents an extension of a concept proposed by Spalding et al., where viscous corrections have been added, whereas the explicit version adopted the work of Kutateladze and Leont'ev by introducing a modified value of the critical blowaway parameter. The latter concept was employed to provide a more tractable set of correlations for engineering design.

The technique was compared to the experiments of several workers over a range of Mach number ($0 < M_e < 8$) and wall temperature ratio ($1 < T_w/T_e < 8$) where it was noted that heat-transfer attenuation due to mass injection was influenced more by Mach number than by wall temperature ratio. In addition, good agreement was shown between the prediction and the mean of the data.

Acknowledgment

This work was supported by the Air Force Flight Dynamics Laboratory (AFFDL-FXG), Wright Patterson Air Force Base, Ohio.

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A COMPARISON OF CALCULATED AND MEASURED ROCKET PLUME INFRARED RADIATION

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Abstract

This paper describes a study in which exhaust plume afterburning calculations, including unburned fuel effects, were coupled with infrared radiation models to describe the infrared characteristics of a specific set of subscale hydrocarbon/O_2-fueled rocket motor exhausts. The results show that the intensity and spatial distribution of infrared radiation data from CO_2 in the 4- to 5.5-μ band are represented quite well by the nozzle/plume afterburning models employed. It also is shown that the turbulent mixing model must be chosen very carefully, since the accuracy of the predictions is extremely sensitive to the mixing parameters adopted.

I. Introduction

The objective of this study was to compare the results of a number of different exhaust plume afterburning calculations coupled with infrared (IR) radiation model predictions, with a specific set of IR data taken on a subscale hydrocarbon/O_2-fueled rocket motor at the Air Force Armament Laboratory (AFATL).[1] Calculations were made by four research organizations: AeroChem Research Laboratories, Inc.[2] (AeroChem), General Applied Science Laboratories, Inc.[3] (GASL), Lockheed/Huntsville Research and Engineering Center[4] (Lockheed), and the Rocket Propulsion Establishment, Westcott, England[5] (RPE), using different gasdynamic, chemical, and IR

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models. Three fuels were studied: RP-1 (liquid), CH_4 (gas), and C_6H_6 (liquid); all burned with gaseous oxygen. The calculations all were completed before the data were made available. A complete description of the models and computational techniques employed in the predictions is beyond the scope of this paper, as are the details of the experimental methods used by AFATL in obtaining the data; the interested reader is referred to the prior publications indicated and Refs. 6-8 for additional background information.

The discussion that follows concentrates mainly on the comparative results obtained in the RPE and AeroChem calculations. They are illustrative of the sort of agreement with experimental data that is attainable with state-of-the-art afterburning models, and serve to demonstrate the sensitivity of the models to gasdynamic and chemical kinetic variables. Similar agreement with the experimental data was obtained using the GASL model as well, -but the mixing parameters employed in the Lockheed calculations were apparently too high and gave generally much shorter plumes.

II. Model Predictions

All predictions were made in the following order: 1) calculation of combustion chamber properties for the specific fuel, 2) determination of properties at the nozzle exit plane using a nonequilibrium nozzle analysis model, 3) calculation of plume gasdynamic and chemical properties, and 4) prediction of plume IR radiation in the 4- to 5-μ band. Table 1 gives the general features of each model, all of which have been described in the open literature. Calculations were made for each of the fuels assuming 100% combustion efficiency, and, with the exception of the RPE effort, a series of calculations was included in an effort to assess the effects of unburned fuel, produced in the chamber due to the inefficient combustion, on the plume properties and IR radiation.

A. Combustion Chamber/Nozzle Calculations

In the "normal" calculations (0% unburned fuel), the combustion chamber and nozzle exit plane properties were calculated by assuming complete chemical equilibrium in the combustion chamber, followed by a nonequilibrium nozzle flow analysis. For the cases where the combustion was assumed to be incomplete, the "phenomenological model" developed by AeroChem (described in Ref. 8) was employed. Basically, this model assumes that the amount of unburned fuel (which remains inert during the nozzle expansion) is that for which the computed and measured C* efficiencies of the motor are equal.

The specific lower-order hydrocarbon used as the "pyrolysis product" or unburned fuel in the calculations is somewhat arbitrary. For this study, we chose C_2H_2 as the inert product for RP-1 and C_6H_6 fuels and CH_4 itself for CH_4 fuel. It should be emphasized that there is considerable uncertainty in the measured C* efficiencies, and the percent unburned fuel determined by the foregoing procedure is subjected to the same uncertainties.

The combustion chamber properties in all of the models are nearly identical for any of the three fuels. Nozzle exit plane temperatures are, however, slightly different from model to model, and there are some small differences in the radical concentrations. The major species are in good agreement, particularly the afterburning "fuel" species, H_2 and CO for the 100% combustion efficiency cases.

The most important effect of employing the phenomenological unburned fuel model with respect to the afterburning plume calculations is that the unburned "fuel" largely replaces CO and H_2 with a specific hydrocarbon (either C_2H_2 or CH_4), thereby changing the magnitude of the heat release potentially available from afterburning.

B. Exhaust Plume Gasdynamic/Chemical Properties

1. Initial Conditions. Initial conditions for the constant-pressure plume calculations were determined via a one-dimensional expansion from the exit plane to ambient pressure (0.186 atm), resulting in an "expanded" initial jet radius and lower initial temperatures than those computed at the exit plane. The variable pressure analyses of Refs. 3 and 5 utilized the actual exit plane properties.

An approximation of the magnitude of unburned fuel effects on afterburning plume properties may be obtained by comparing the respective initial nozzle exit compositions and temperatures, taking into account the heat of combustion of the exit plane fuel species (ΔH_c), defined as $\Delta H_c = \Sigma_i X_i \Delta H_{c_i}$, where X_i is the mole fraction of each "fuel" species, H_2, CO, and unburned hydrocarbon. It is reasonable to assume that the total ΔH_c represents the potential for afterburning in each case. For the 0% unburned fuel cases, the values of ΔH_c are almost the same, -and the AeroChem and RPE calculations give nearly identical results. Substitution of the unburned hydrocarbon for CO and H_2 substantially increases ΔH_c for both CH_4 and C_6H_6 but results in only a modest increase for RP-1. The sensitivity of the afterburning potential to the type and

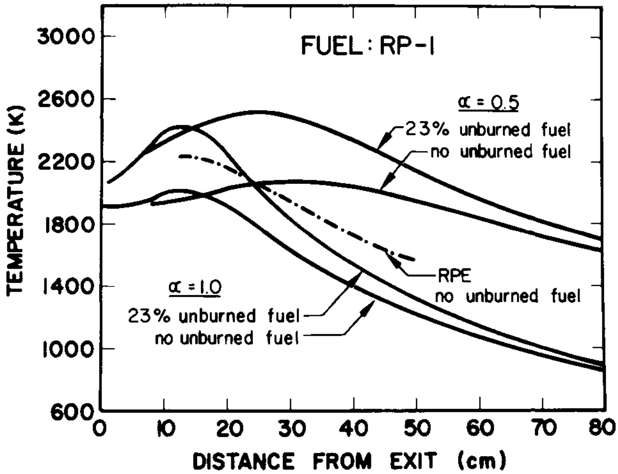

Fig. 1 Calculated centerline temperature.

amount of unburned fuel can be estimated by the calculation of the change in the available heat of combustion of the gas at the nozzle exit before and after introduction of unburned hydrocarbons.

2. Plume Temperatures and Composition. Comparison of computed centerline temperature profiles such as those given in Fig. 1 leads to the following observations:

1) With the exception of the region near the exit plane, where the RPE code accounts for plume shocks, the AeroChem and RPE codes give similar results for all three fuels (0% unburned fuel) with respect to position of peak temperatures and the rate of temperature decay. Since the overall chemistry schemes incorporated into the two models are similar (for 0% unburned fuel), the results imply that the turbulent mixing rates predicted by the models are nearly the same despite the fact that they are derived from different gasdynamic considerations (see Table 1).

2) The dependence of the temperature field on the mixing is considerable; i.e., an arbitrary decrease in the eddy viscosity by a factor of 2 (from α = 1.0 to α = 0.5 where α is an eddy viscosity adjustment parameter in the Donaldson/Gray diffusivity model[9] included for the express purpose of varying turbulent mixing rates) has a very large effect on the predictions.

3) Employing the unburned fuel model enhances the peak temperatures about 400 K for RP-1 and about 800 K for C_6H_6

Table 1 Model summary

	AeroChem	GASL	Lockheed	RPE
Combustion chamber	Equil./nonequil.	Equil./nonequil.	Equil./nonequil.	Equil.
Plume pressure[a]	Constant	Radial pressure gradients included	Constant	Radial pressure gradients included
Plume turbulent mixing	Eddy viscosity (Donaldson/Gray model)	TKE	TKE	TKE
Chemistry plume code[b]	LAPP	GASL plume code	Lockheed balanced pressure plume model	REP-3
Radiation	AFRPL code (modification of Tracy Jackson code)	Aerodyne plume radiation code (ARC 2)	NASA band model	RPE code

[a]Conditions calculated by AeroChem at nozzle exit plane and after expansion to ambient pressure were used as input to the plume calculations by AeroChem, GASL, and Lockheed.

[b]Nonequilibrium chemical schemes employed in all cases. Identical chemical reaction mechanisms and rate coefficients used by AeroChem, GASL, and Lockheed. The mechanism and rates are quite similar to those used by RPE.

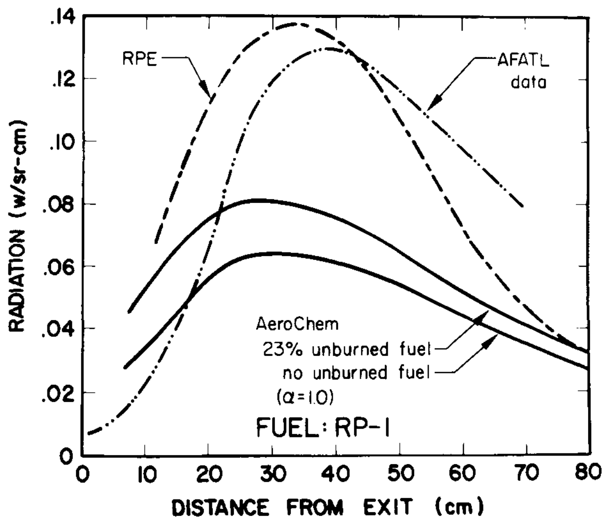

Fig. 2 Radiation as function of axial distance.

(not shown), qualitatively in accord with their relative afterburning potentials. The plume lengths, however, are about the same as in the 0% unburned fuel calculations.

The salient features of the alterations in plume chemistry induced by including unburned hydrocarbons in the exhausts are most apparent from examinations of centerline chemical composition profiles. For example, it is found that the location of peak temperature for RP-1 correlates closely with the region of maximum H_2 decay rate, similar to observations in laboratory flames. The effect of the unburned fuel is to increase very rapidly the H_2 mole fraction substantially above its exit plane value. However, the exit plane H_2 mole fraction initially has been suppressed (in comparison to the 0% unburned fuel case) by the inclusion of unburned fuel so that the absolute levels a short distance downstream of the exit are nearly the same in both cases. Comparisons between the CO_2 and N_2 (from ambient air) curves show that CO_2 production chemistry continues far downstream in the plume.

III. Comparison between Infrared Radiation Predictions and Data

In comparing the predicted station radiation from any of the models with the AFATL measurements, the calculated radiation values are too high near the exit plane; this is demonstrated in the RP-1 case of Fig. 2. However, there are some considerable uncertainties in the experimental data in this

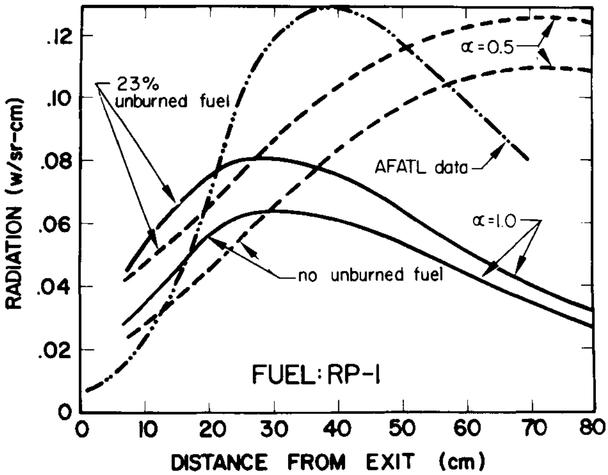

Fig. 3 Calculated effects of mixing rate and unburned fuel.

region, and so the disagreement may be less severe than it appears. Another striking general observation is that, in several cases, model calculations predicted plume lengths considerably too short, caused almost surely by mixing rates that are too fast. The general shape of the RPE predictions compares well with the data, although the location of the peak intensity is too far upstream. The AeroChem predictions are somewhat lower than those of RPE, as might be expected from comparison of the respective temperature curves, but the predicted overall plume lengths from both of these codes are in reasonable agreement with the data for all three fuels. This is illustrated by the data of Fig. 2 in which the width of the calculated radiation curves at half maximum are within ± 6 cm (± 5%) of that for the experimental curve.

Imposing an arbitrarily slower mixing rate (decreasing α from 1.0 to 0.5) in the AeroChem model gives much higher plume temperatures and correspondingly higher predicted peak radiation, as shown in Figs. 1 and 3, respectively. Although this arbitrary variation in mixing rate brings the peak radiation values into somewhat better agreement with the data, the location of the maximum moves considerably downstream of the observed peak radiation. This parametric variation demonstrates the extreme sensitivity of the models to the value of the mixing rate parameter selected.

For the unburned fuel cases, the AeroChem predicted radiation is generally in accord with what one would expect from the centerline temperature predictions. The peak values of station radiation throughout the plume are predicted quite well (i.e., to within ± 35% or less of the measured values) by the AeroChem and RPE models for the three fuels.

IV. Conclusions

Based on the results of this comparison among afterburning plume models that predict temperature, composition, and IR radiation and the comparison between these IR radiation predictions and the AFATL data, the following conclusions may be drawn:

1) A comparison among any of the predicted plume properties indicates that the results are extremely sensitive to the overall turbulent mixing rates.

2) All models predict the location of the peak station radiation to be slightly upstream of the measurements, and the two models that achieve best agreement with respect to total plume intensity also produce curves with the same general shape and length at half maximum as those of the data.

3) The maximum specific radiant intensity (W/sr-g/sec) predictions are in quite reasonable agreement ± < 40% with the data, at least in the fuel rich range examined in this work.

4) There is not enough information to assess completely the significance of the phenomenological unburned fuel model, although, in general, it appears that it produces better agreement with the data than does the same model based on 100% combustion efficiency. These effects are, however, of secondary importance to those associated with errors in turbulent mixing rates.

Acknowledgment

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COMPARISON OF THEORETICAL AND EXPERIMENTAL INFRARED RADIATION FROM A ROCKET EXHAUST

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ABSTRACT

Gasdynamics and infrared radiation computational models have been developed to predict the physical properties in hot rocket exhaust plumes. Experimental data were obtained in the exhaust plume of a small kerosene/gaseous oxygen rocket engine using a three-dimensional laser Doppler velocimeter system to measure the velocity distribution. The spatial distribution of radiated energy in the 4- to 5-μ wavelength band was measured using an infrared radiometer. Detailed comparisons between the computational and experimental results are described.

I. Introduction

This paper describes the results obtained during a unified analytical and experimental program designed to characterize the spatial distribution of infrared radiation emitted from high-temperature exhaust plumes. It is well known that exhaust plumes of high-performance turbojet aircraft and rockets are characterized by the presence of an internal shock wave structure and a highly turbulent mixing layer between the inner hot jet flow and the outer cool airstream. The presence of shock waves leads to the appearance of shock-

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heated high-temperature regions in exhaust plumes which are referred to as "hot spots." In addition, for hot flows in which the oxidizer-fuel (O/F) ratio is below the stoichiometric value, i.e., if the exhaust is fuel-rich, afterburning can occur in the turbulent mixing layers, which entrain air from the external flow around the aircraft or missile. Afterburning of fuel in the exhaust plume leads to an increase in the infrared radiation downstream of the nozzle exit plane.

Several theoretical models and computer codes have been developed which predict, in an approximate manner, the gasdynamics, chemistry, and radiation associated with high-temperature turbojet and rocket exhausts. A brief description of two of these models will be given in Sec. II.

During the course of this study, it became clear that no totally adequate theory existed for the prediction of the complex reflected shock waves and Mach disks present in the plume. In addition, there was considerable uncertainty in the ability of present turbulence models to predict the spatial distributions of velocity, temperature, and radiating species concentrations. These distributions are required to calculate the radiation emitted from the exhaust plume. An experimental measurements program was initiated to provide velocity distribution and radiation intensity data to validate the analytical models and computer codes and to provide data concerning the fundamental processes involved in the production and emission of infrared radiation from aircraft and missile exhaust plumes. The experimental equipment used to obtain the gasdynamic and radiation data is described in Sec. III. Typical experimental results are described in Sec. IV, whereas comparisons of predicted and measured results are given in Sec. V.

II. Theoretical Models

Numerous theoretical models exist for the prediction of the gasdynamics, chemistry, and radiation properties of exhaust plumes.[1-17] It is beyond the scope of this paper to describe all of these models in detail. Instead, a brief description will be given of models used by the authors to predict the plume flowfield and the emitted infrared radiation.

A. Low-Altitude Plume Program (LAPP)

One of the most versatile computer programs which predicts nonequilibrium rocket exhaust plume properties is the Low-Altitude Plume Program (LAPP) developed by AeroChem for the U.S. Air Force Rocket Propulsion Laboratory.[1] The analytical model is based on a finite-difference solution of the free shear mixing layer equations. The model assumes parallel turbulent mixing between concentric, chemically reacting streams. A mixed implicit/explicit scheme is used to eliminate instability problems. Implicit differences are used for the solution of species concentration equations, and explicit differences are used for the momentum and energy equations. Since the model treats only parallel, axisymmetric mixing, the radial momentum equation is ignored so that the pressure is assumed constant in the radial direction. It is possible to handle axial pressure gradients, but the variation of pressure with axial distance must be known a priori.

Prandtl and Schmidt numbers are assumed to be constant throughout the plume. Turbulent transport is described by an eddy viscosity model. The original computer code contained six different eddy viscosity models, including the Ting-Libby and Donaldson and Gray models.

It should be noted that LAPP has no provision for handling shock waves or pressure discontinuities and is correct only in the limit of constant pressure in the radial direction. Thus, it would be expected that the LAPP model would be invalid in the near field of a rocket exhaust containing strong shock waves. If there is a strong recirculating flow region at the base of the rocket, it is expected that LAPP will yield results that do not agree with reality because of reversed and radial flow velocities.

In spite of the approximate nature of the model, LAPP has been shown to yield results that are as close to experimental data as most of the existing exhaust plume models.[18] Rhodes[19] has shown that the LAPP code, using the Donaldson-Gray eddy viscosity model for the initial portion of the plume and then the Ting-Libby model for the remainder of the plume, yielded good results for the carbon-nitrogen ratio in the mixing layer of a rocket plume.

B. Turbulent Kinetic Energy Model

The turbulent kinetic energy (TKE) model used to calculate the properties of the exhaust plume uses an implicit numerical technique developed by Spalding and co-workers.[6,7] One of the unique features of the TKE model is use of a conservation equation for the turbulent kinetic energy and an equation for the square of the characteristic frequency of the energy-containing eddies. The turbulent kinetic energy then is related linearly to the turbulent shear stress. The TKE model of Spalding has been used successfully to predict the flowfield and radiation field from a small rocket engine exhaust plume which has multiple embedded shock waves and is afterburning.[21] The version of the TKE model used to perform the calculations described in this paper was developed and made available to MIRADCOM by the Rocket Propulsion Establishment of Great Britain.[20]

C. Infrared Radiation Models

A numerical radiation band-model was developed to predict the infrared spectral and spatial radiation intensity distributions in exhaust plumes.[22] The model essentially extends the models of Jackson[13,14] to include H_2O vapor radiation and to

Table 1 Engine design and performance parameters

Fuel	Kerosene
Oxidizer	Gaseous oxygen
Purge	Nitrogen 100 psig
Coolant	Water at 3 gal/min
Chamber pressure	50 and 148 psig
Fuel flow rate	
50 psig	1.38 g/sec
148 psig	3.80 g/sec
Oxygen flow rate	
50 psig	3.05 g/sec
148 psig	8.70 g/sec
Throat diameter	0.200 in.
Exit diameter	0.310 in.
Throat area	0.0314 $in.^2$
Exit area	0.0755 $in.^2$
Area ratio (A_E/A_t)	2.40
O/F ratio	2.25
Chamber diameter	0.780 in.
Chamber length	1.620 in.
L^*	25.20 in.
C^* at 148 psig	6045 fps

treat higher temperatures than the Jackson code. The statistical band model and spectral parameters used in the computer code were developed by NASA for application to rocket plume studies.[23,24]

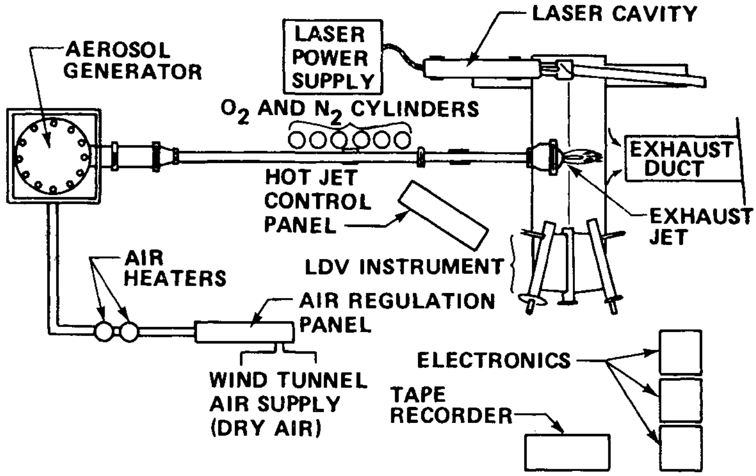

Fig. 1a LDV exhaust plume test arrangement.

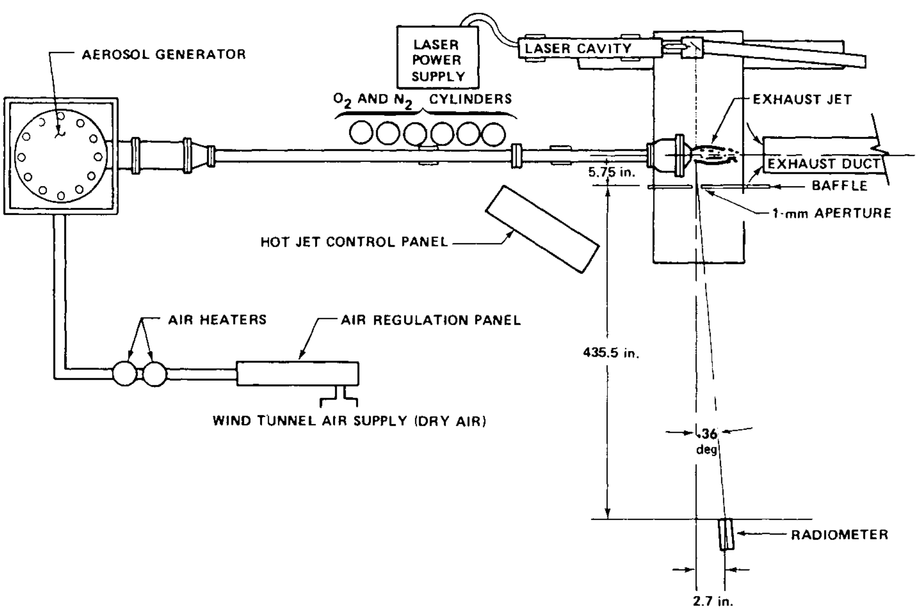

Fig. 1b Infrared radiometer test arrangement.

III. Description of Experimental Equipment

A. Small Laboratory Rocket Engine

The exhaust plumes studied in these measurements were produced by a small kerosene/gaseous oxygen rocket engine[25] operating in the NASA Laser Doppler Velocimeter Facility.[26] Engine design and performance parameters are listed in Table 1. Data were taken at a variety of oxidizer/fuel (O/F) ratios and at two chamber pressures. The experimental data used to compare with the theoretical calculations were obtained at a chamber pressure of 11.1 atm and an O/F ratio of 2.25.

B. NASA Laser Doppler Velocimeter Facility

The NASA Marshall three-dimensional laser Doppler velocimeter[26] was used to measure the velocity distribution in the exhaust plume and mixing layer produced by the inner hot rocket exhaust jet mixing with an outer subsonic airflow. A schematic diagram of the LDV experimental arrangement showing the installation of the small rocket engine is shown in Fig. 1a. As described in Ref. 25, the simultaneous measurement of the three Doppler velocity components yielded the three velocity components u,v, and w.

In order to simulate external airflow around the small rocket engine, an outer flow nozzle was used to produce an external flow velocity of 61 m/sec. A schematic diagram of the rocket engine and outer flow nozzle is presented in Fig. 2.

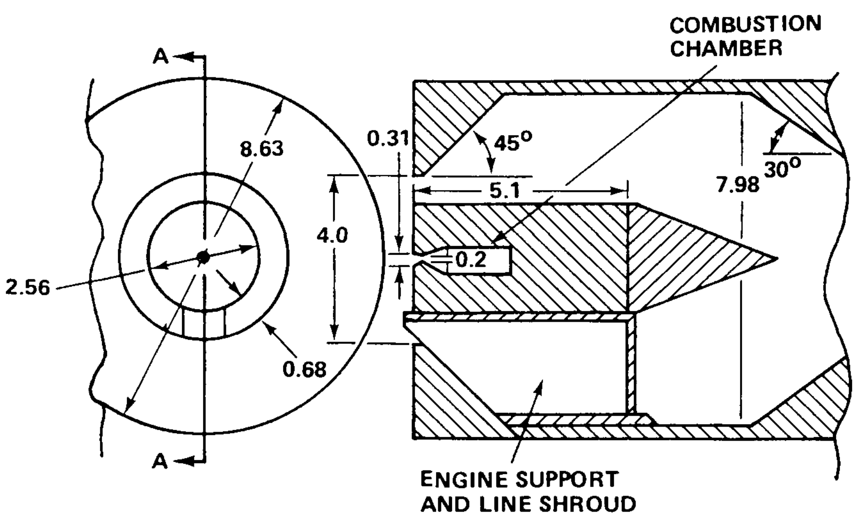

Fig. 2 Schematic of the exhaust plume and external flow apparatus.

C. Infrared Radiation Diagnostic Equipment

The primary instrument used to obtain the radiation intensity data was an Electro-Optical Industries model 470 radiometer. The data were obtained using a fixed filter, which transmitted radiation in the 4- to 5-μ band.

A large metal shield with a 1-mm aperture was moved in the axial direction to obtain the variation of radiant intensity along the exhaust plume centerline. A second set of measurements was made using a 1-mm vertical slot in the large metal shield. The slot length was greater than the diameter of the exhaust plume. The experimental setup is shown in Fig. 1b.

IV. Typical Experimental Results

A. Velocity Distribution Data

Experimental data for mean flow velocities ($\bar{u},\bar{v},\bar{w}$), turbulent intensities [$\overline{(u')^2}$, $\overline{(v')^2}$, $\overline{(w')^2}$], and turbulent velocity correlations were obtained in the radial direction at various axial locations downstream from the nozzle exit plane. Data were obtained at X/D locations of 2.4, 4.8, 8.4, 11.3, and 14.2, where X is the distance measured from the nozzle exit plane and D is the diameter of the exit nozzle. Typical

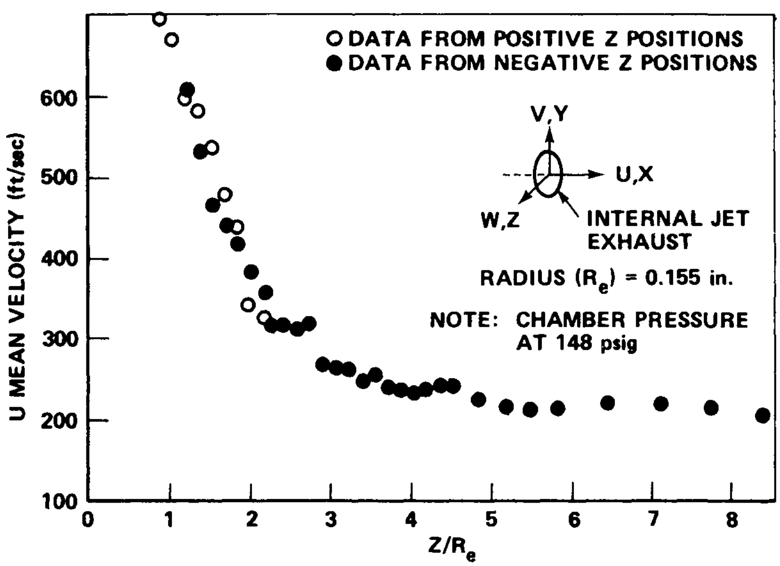

Fig. 3 Profile of u mean velocity component at X/D = 11.3.

data obtained at X/D = 11.3 for the mean axial flow velocity are presented in Fig. 3. Other data are described in Ref. 25.

B. Infrared Radiation Data

The infrared radiation data have been described previously.[25,27] Typical data obtained at an O/F ratio of 2.25 and an external flow velocity of 210 fps are presented in Figs. 4 and 5 for the 1-mm aperture and slot, respectively. The centerline radiation intensity distribution in Fig. 4 clearly shows the presence and influence of shocks and/or

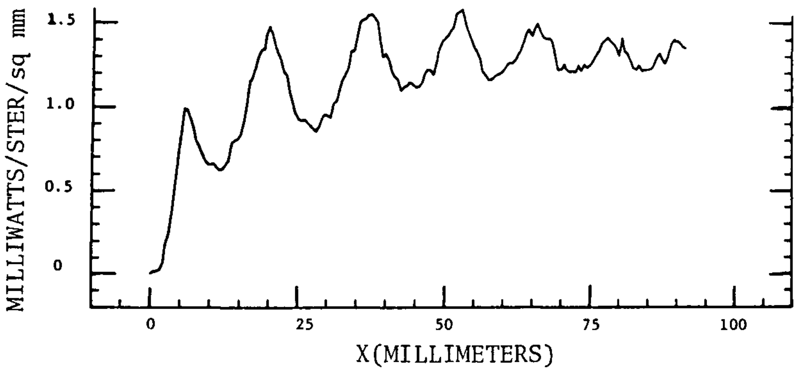

Fig. 4 Spatial distribution of radiation along axis of plume for aperture diameter = 1-mm; O/F = 2.25, velocity = 150 fps.

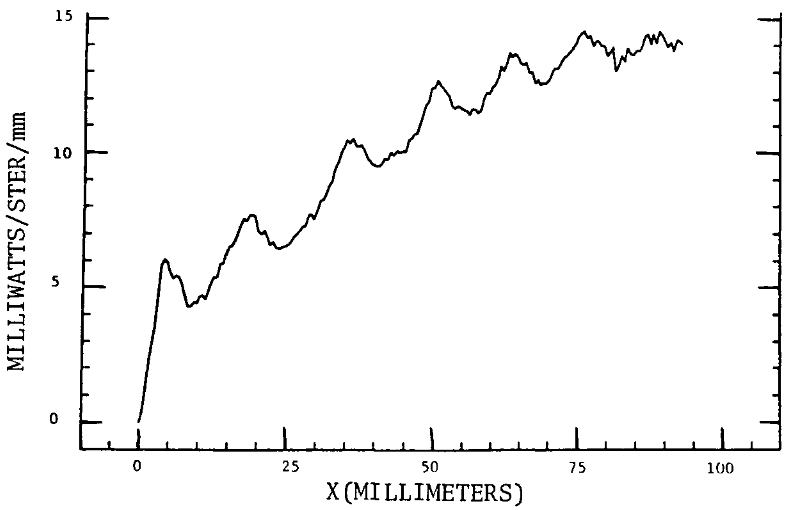

Fig. 5 Spatial distribution of radiation along axis of plume for 1-mm slot width; O/F = 2.25, velocity = 210 fps.

Mach disks in the exhaust plume. The effect of afterburning in the mixing layers on the infrared signature is shown clearly in Fig. 5 where the shock-induced radiation peaks are superimposed on an axially increasing radiation intensity. The data presented in Figs. 4 and 5 were obtained in the near field of the exhaust plume (X/D less than 12) and indicate that the peak radiation intensity has not been reached. Typically, the axial distribution of radiation is similar to that shown in Fig. 6. Even though the data presented in Fig. 6 are for a lower mass flow and chamber pressure, the axial variation is similar to that obtained for the high-chamber-pressure cases. The data presented here are only samples selected to indicate the measured trends in preparation for a discussion of the comparison between theory and experiment.

V. Comparison of Calculated and Measured Results

The LAPP and TKE computer codes were exercised for the rocket performance parameters listed in Table 1. The NASA Lewis Chemical Equilibrium Computer Program was used to calculate the exit plane gas properties. Because of the short expansion nozzle length, the chemical species were assumed to be frozen at the throat values. These exit plane conditions then were used as initial conditions for the LAPP and TKE computer codes.

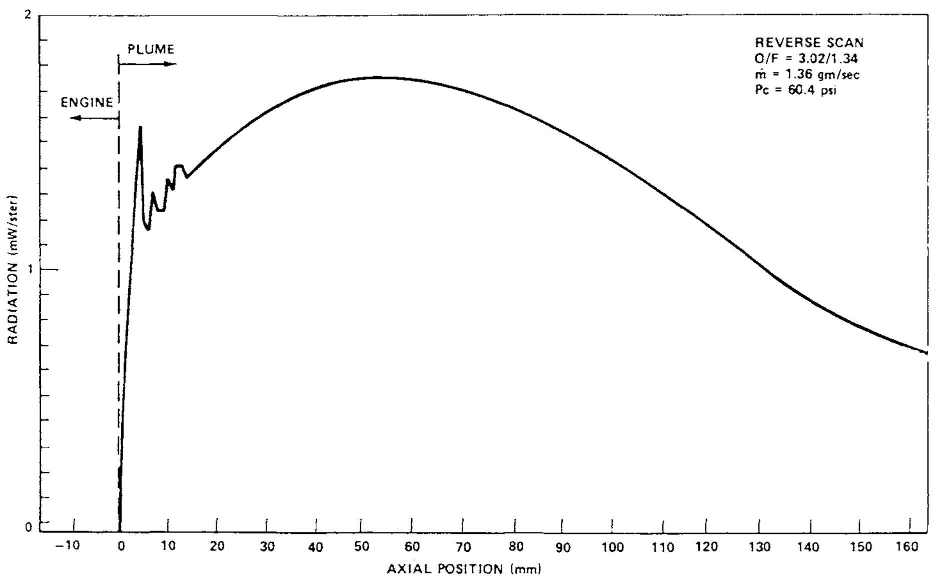

Fig. 6 Axial variation of radiation at O/F ratio of 2.27.

The results predicted by the LAPP and TKE computer codes are presented in Figs. 7-12. The variation of the centerline velocity with distance from the nozzle exit plane is presented in Fig. 7. The LAPP code predicts a constant-velocity core having a length approximately equal to X/D = 5, which is followed by an exponential velocity decay due to turbulent mixing predicted by the Donaldson-Gray viscosity model. The TKE (REP 3) code predicts a sharp decrease in velocity at approximately an X/D value of one. There are also several other oscillations in velocity, as shown by the jagged nature of the curve. A particularly large decrease in velocity occurs near an X/D value of 8. Also noted in Fig. 7 are the locations of the measured radiation peaks shown previously in Fig. 4. The X/D locations for the LDV measurements also are indicated on Fig. 7.

The predicted variations of temperature at the jet centerline with axial distance are given in Fig. 8. The LAPP code predicts a small temperature rise due to afterburning in the

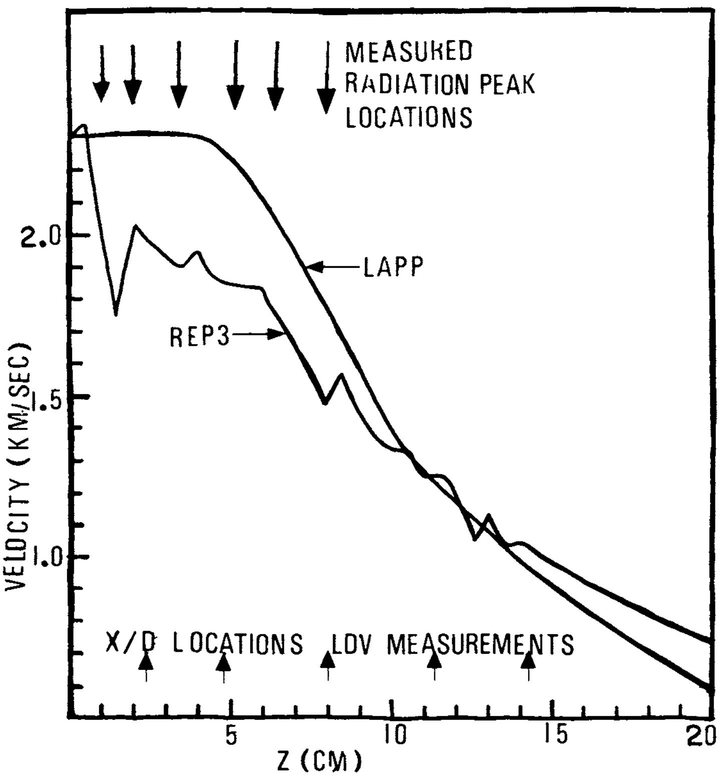

Fig. 7 Calculated axial velocity distribution.

exhaust plume. The TKE code predicts a sharp rise in temperature at an X/D value of one, which then is followed by a series of oscillations in temperature. The predicted oscillations do not agree with the measured locations of radiation peaks (denoted by arrows on the Figs.). It is quite obvious that neither of the computer codes does an adequate job of predicting the spatial variation of velocity and temperature produced by shock waves and Mach disks in the near field of the exhaust plume. This is not surprising since LAPP has no provision for treating shocks.

Figures 9-12 present the variation of the axial component of the velocity with radial distance from the jet centerline. As shown in Fig. 9 for an X/D value of 2.4, the predicted results (curves) do not agree with the experimental measurements (symbols). Because of the limited number of particles in the hot core of the jet, the NASA LDV system was not able to make measurements closer to the jet centerline than about one exit nozzle radius. The variation of the radial velocity component also is shown. The TKE code predicts an increasing value of v with radius, whereas the measured value decreases with radial distance. The external flow is an axial flow at 61 m/sec.

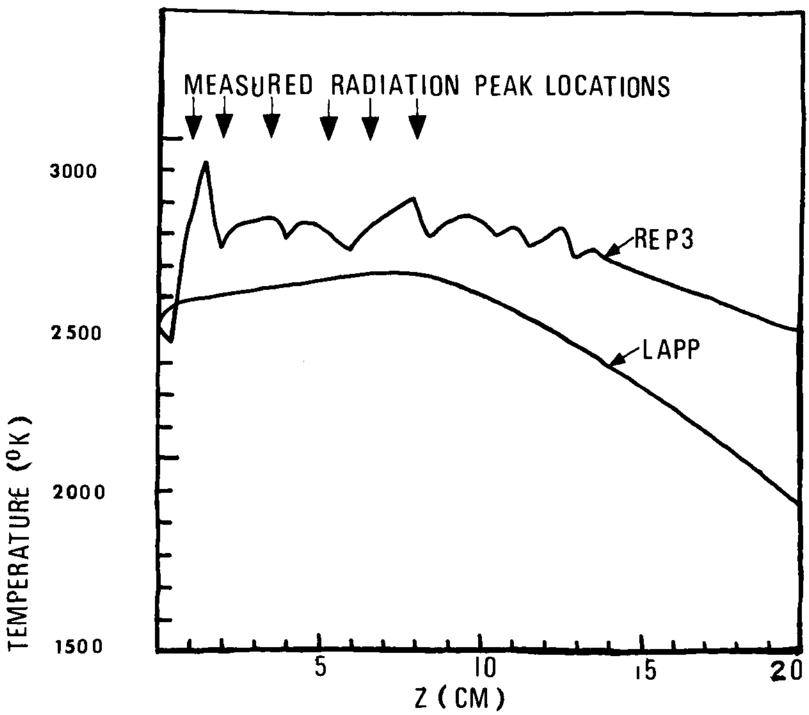

Fig. 8 Calculated temperature distribution.

The lack of agreement between theory and experiment also is indicated in Fig. 10 for the velocity profile at an X/D value of 8.0. The LAPP and TKE calculated results are similar but are in excess of the experimental values. Similar results are presented in Figs. 11 and 12 for X/D values of 11.3 and 14.2.

The LDV data indicate that the exhaust plume is much narrower than predicted by the LAPP and TKE gasdynamics models. Since both the LAPP and TKE models assumed that turbulent mixing began at the nozzle exit plane and ignored the large recirculation (base flow) region, it is expected that turbulent mixing occurs too fast compared to the physical situation where turbulent mixing begins at the end of the recirculation zone, which is several nozzle diameters downstream from the exit. It is known that the base flow region can have a pronounced effect on exhaust plume spread and initial turbulent

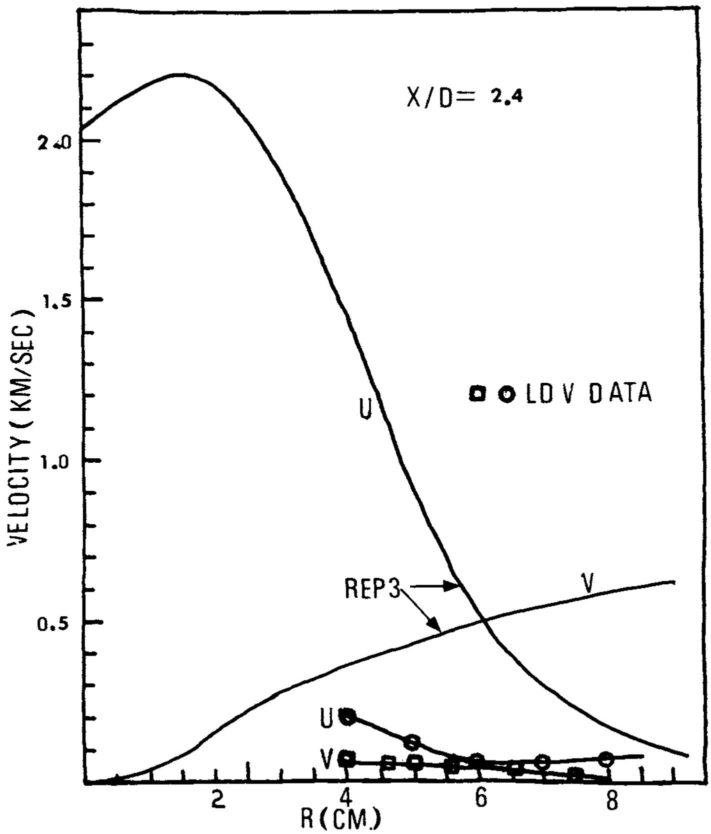

Fig. 9 Velocity profiles at X/D = 2.4.

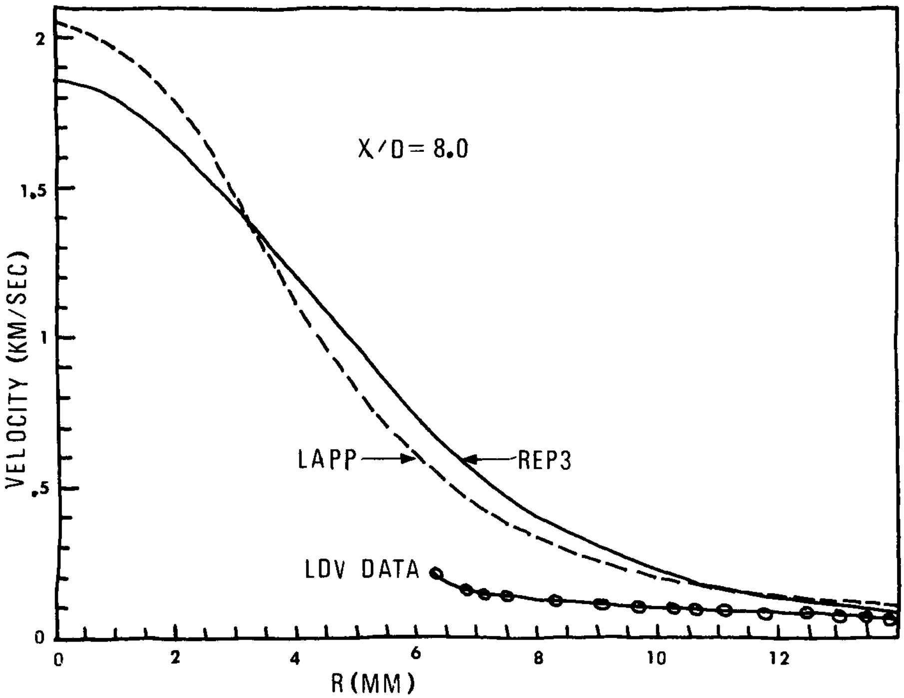

Fig. 10 Velocity profiles at X/D = 8.

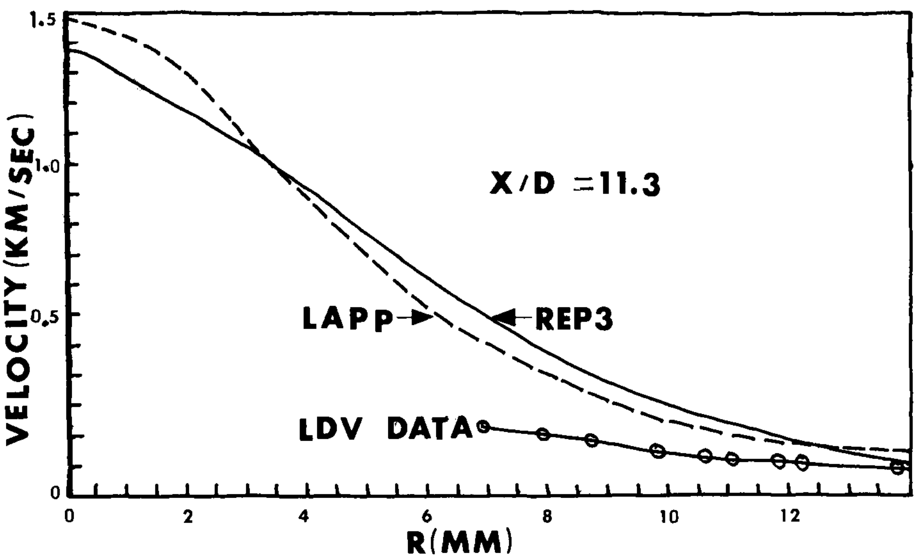

Fig. 11 Velocity profiles at X/D = 11.3.

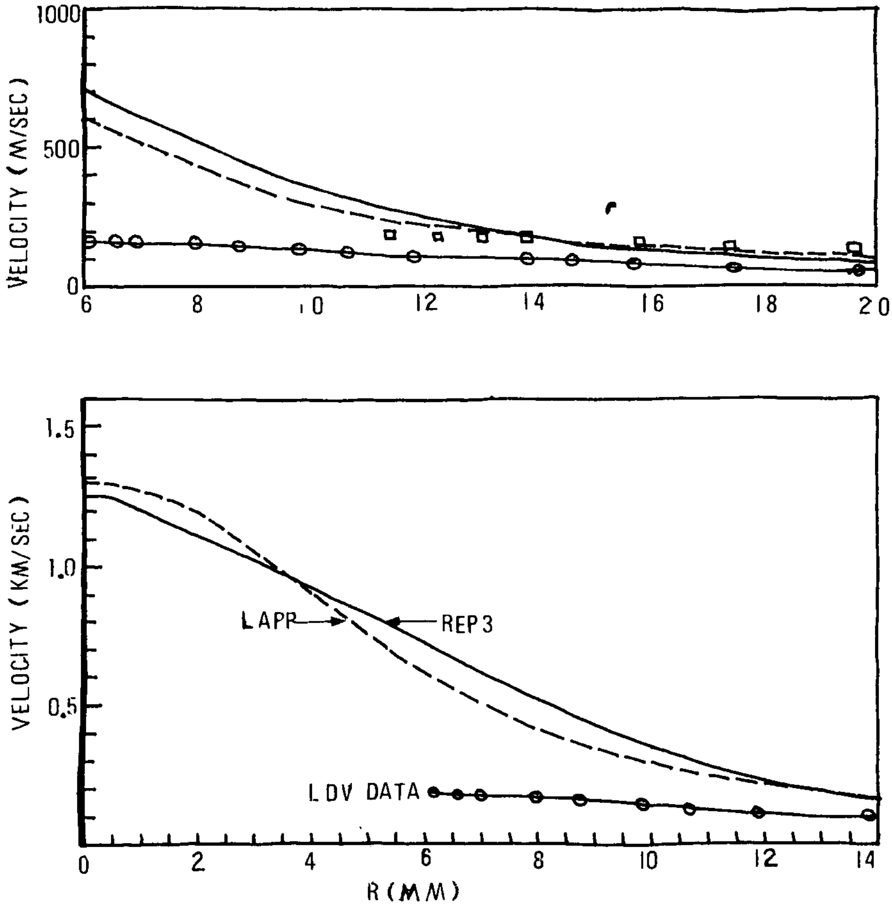

Fig. 12 Velocity profiles at X/D = 14.2.

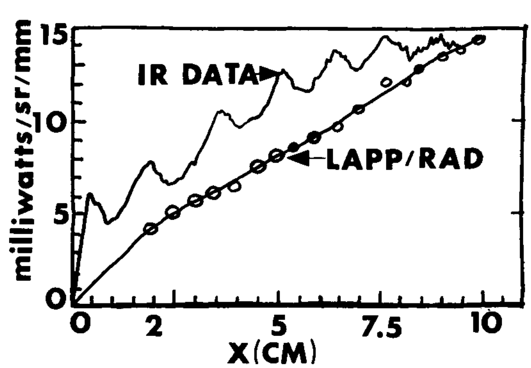

Fig. 13 Spatial distribution of radiation along axis of plume.

mixing layer growth. Because of time and budget constraints, we were unable to use a base flow model to assess its effect on the calculated temperature and velocity profiles.

The infrared radiation band model program was used to calculate the variation of infrared radiation along the exhaust plume for the temperature and species concentration profiles predicted by the LAPP and TKE gasdynamics computer codes. The computed results are shown in Fig. 13. The lack of agreement between theory and experiments was not unexpected, since the calculated temperature distributions did not yield the spatial structure due to shock waves that are evident in the measured infrared data and are visible in photographs of the exhaust plume.

VI. Concluding Remarks

In summary, the available computer models did not yield results that agreed with either LDV or infrared radiometer data. Further model development clearly is needed to predict the gasdynamic properties of a high-temperature, chemically reacting gas flow containing shock waves and turbulent mixing layers.

In addition, additional experimental measurements using a single-particle self-realization laser velocimeter to obtain the velocity distribution in the central hot core of the jet are needed. A unified analytical and experimental research program is needed to improve the state-of-the-art in exhaust plume gasdynamics prediction models.

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A BAND MODEL FOR CALCULATING RADIANCE AND TRANSMISSION OF WATER VAPOR AND CARBON DIOXIDE

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Abstract

The NASA band model formulation, originally employed for missile base heating predictions, has been modified to allow the calculation of radiance, radiant intensity, transmission, etc., from a gas. This model has been evaluated by comparing its predictions to laboratory data on H_2O, CO_2, and various mixtures containing these gases. The calculations reproduced the measured values for homogeneous and inhomogeneous samples of H_2O to within about $\pm 20\%$. The agreement with measured values for CO_2 was not as good, especially at low pressures. The theoretically derived band model parameters for CO_2 appear to be in error. For mixtures of H_2O, CO_2, and other gases (H_2,CO,H_2), the agreement was good.

I. Introduction

When calculating the mid-wavelength infrared radiation of a low-altitude rocket plume or any other body of gas at moderate temperature ($<\sim 3000$ K), the usual procedure is to approach the problem sequentially. The temperature, pressure, and composition of the gas are calculated first, and then a radiative transfer model is employed to determine the radiance, radiant intensity, transmission, etc., of the gas. When relatively little spectral structure is desired, a band model usually is employed to facilitate this radiation calculation.

There are a number of possible sources of error in this procedure, even when the uncoupling of the gasdynamics and

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Table 1 Cases considered for comparison

Subject	(Ref.)	Fig. No.	Partial pressure, mm Hg H_2O	CO_2	N_2	CO	H_2	Temp. profile[a] K	Length m
Isothermal hot H_2O	(12)	1	38.0	...	38.7	...	...	1215	0.6
Isothermal hot H_2O	(4)	2a	808	...	...	...	...	1177	0.6
		2b	243	...	...	...	...	1177	0.6
		2c	77	...	...	...	...	1177	0.6
Isothermal hot CO_2	(12)	3	...	38.0	38.0	...	...	1202	0.6
Isothermal hot CO_2	(12)	4	...	7.58	68.43	...	...	1200	0.6
Isothermal mixture	(12)	5	35.26	7.58	27.09	1.44	4.64	1201	0.6
Isothermal mixture	(12)	6	35.26	7.58	27.06	1.47	4.62	1202	0.6
Nonisothermal hot H_2O	(13)	7	692	...	...	...	...	392/513/593/671/745/ 835/918/993/1059/1152	0.6

Table 1 (continued)

Subject	(Ref.)	Fig. No.	Partial pressure, mm Hg H_2O	CO_2	N_2	CO	H_2	Temp. profile[a] K	Length m
Nonisothermal hot H_2O	(13)	8	670	...	...	...	...	382/537/723/953/1128/ 1160/990/751/558/389	0.6
Nonisothermal hot CO_2	(13)	9	...	760	...	...	...	390/464/537/641/715/ 824/914/994/1061/1153	0.6
Nonisothermal hot CO_2	(13)	10	...	760	...	...	...	386/528/719/953/1130 1160/975/737/541/387	0.6
Nonisothermal hot CO_2	(13)	11	...	760	...	...	...	518/1132/1160/781/1060 1062/777/1160/1114/536	0.6
Nonisothermal mixture		12	57	28	675	...	...	378/537/723/958/1127 1158/990/752/555/383	0.6
Isothermal cold mixture		13	8.64	88.63	0.73	...	...	301	100

[a]The temperature of the gas was either homogeneous, in which case one temperature is reported, or was inhomogeneous, in which case the temperatures are given at the center of each 6-cm-wide cell.

radiation calculations is valid. These sources include the form of the band model (regular or random, for example), the effect of nonuniformities in the optical path, and the effect of absorption of the hot-gas radiation by a cool atmosphere between the radiator and the observer. Considerable work is being done in each of these areas. Yet the question remains: how well can hot-gas radiance be calculated using available models? What is the magnitude of the error that will be introduced into the calculation in different cases? These are questions that will be addressed in this paper by comparing model predictions to laboratory data.

The band model that we used is referred to as the NASA model, originally formulated to calculate missile base heating.[1] The model recently was modified to permit the calculation of radiative transfer from the plume to an arbitrary point in space.[2,3]

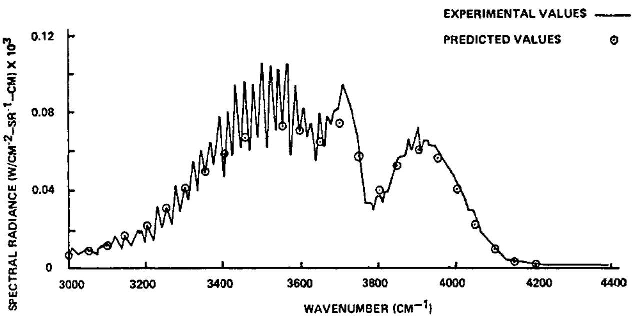

Fig. 1 Spectral radiance at 2.7 μm, isothermal water.

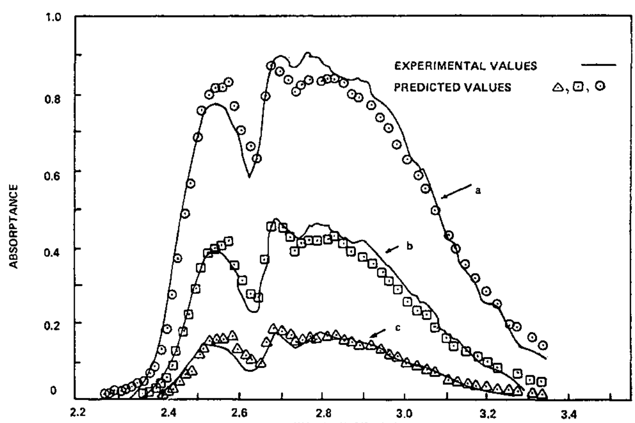

Fig. 2 Spectral absorptance, isothermal water; a=808 mmHg, b=243 mmHg, c=77 mmHg.

II. Radiative Transfer Model and Experimental Data

A. The Model

The NASA band model is well documented,[2-4] and will not be discussed in any detail here. It is a random (statistical) band model that uses band model parameters experimentally determined for H_2O and calculated for CO_2 and the diatomic molecules. These are tabulated in Ref. 4. Some recent work by Young[5,6] suggests that these parameters can be improved at the lower temperatures by judiciously combining them with parameters calculated from the line compilation of the Air Force Cambridge Research Laboratory.[7] Band model parameters for the CO_2 2.7-μm and 4.3-μm regions have been measured by Kunitomo,[8,9] but these have not been incorporated into any radiative transfer model in this country, as far as we know.

Various methods for accounting for inhomogeneities in the optical path have been developed. The Curtis-Godson approximation (Ref. 4, Chap. 4) can be applied to all of the lines in a spectral interval, as though the band parameters represented a single line, or the interval can be divided into groups of lines, each group having similar strengths, and the Curtis-Godson approximation applied to each group. Lindquist and Simmons[10] have examined an alternate formulation for inhomogeneous paths which recently was extended by Young.[11] A third possibility has been proposed by Lindquist et al.,[12] since they found that both the Curtis-Godson and Lindquist-Simmons approximations cannot accurately calculate spectra when a hot gas is radiating through a cool, intervening atmosphere.

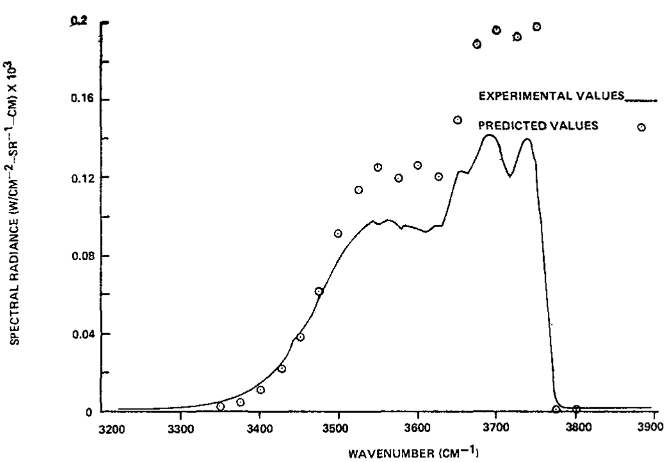

Fig. 3 Spectral radiance at 2.7 μm, isothermal CO_2.

In the calculations reported here, the simplest Curtis-Godson approximation is used. For the most part, the non-uniformities in the optical path are due solely to the temperature and pressure variations within the hot-gas emitter and not to any long atmospheric absorption path.

B. The Data

Although the NASA band model has been in existence for some time, there has been very little detailed comparison reported of its predictive ability against measured data. Ludwig et al.[4] present some comparisons, as does Young.[6] In this work, we used data from three sources.[4,12,13] These data are used to assess the ability of the model to calculate hot-gas emission from single and multicomponent isothermal and nonisothermal samples, effects of total pressure on emission, cold-gas transmission, and hot-through-cold gas emission.

The various measurements that were used in this study are listed in Table 1. They are divided into three categories: isothermal radiators, nonisothermal radiators, and isothermal absorbers. In all cases, the only infrared bands considered are the 2.7-μm band of H_2O and the 2.7- and 4.3-μm bands of CO_2. Further comparisons are given elsewhere.[7]

IV. Comparison of Measurements with Predictions

A. Isothermal Radiators

Figures 1 and 2 show measured and predicted values for the spectral radiance and absorptance, respectively, from homogene-

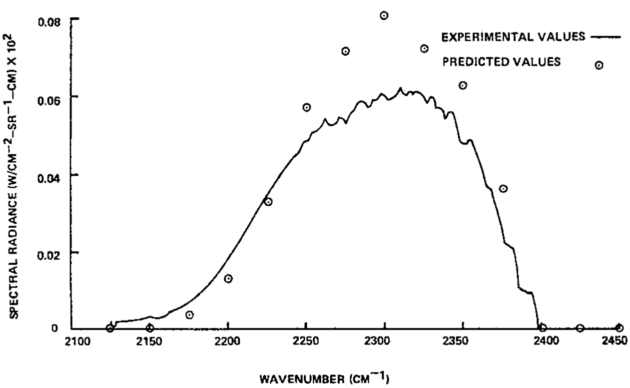

Fig. 4 Spectral radiance at 4.3 μm, isothermal CO_2.

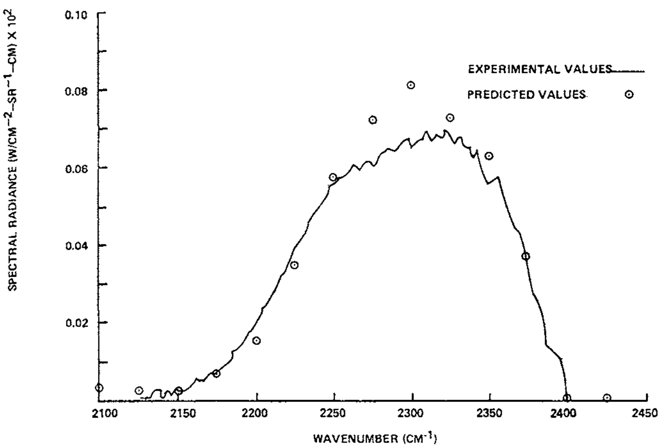

Fig. 5 Spectral radiance at 4.3 μm, isothermal mixture.

ous samples of H_2O. Near the center of the band, the predicted radiance values are about 12% less than the measurements. Overall, the agreement is excellent. Ludwig et al.[4] compare the model with other measurements in the 1.1-, 1.4-, 1.8-, 2.7-, and 6.3-μm bands. The maximum error suggested by these authors will occur for long pathlengths and should be within $\pm 20\%$. Young[6] examines some data at 1040 K. His predictions, based on the same band model parameters as used here, do not appear to agree as well as those reported here for slightly higher temperatures. Young attributes this to the extrapolation of the experimentally determined NASA parameters to lower temperatures. (In this case, the lower temperature is 1040 K; the band model parameters were derived from measurements made above 1200 K.) Young's combined band model parameters do improve the agreement, especially near the center of the band.

Although the H_2O radiance and absorptance predictions give excellent agreement with available data, such is not the case for CO_2. Comparisons of the predicted and measured radiance are shown in Figs. 3 and 4 for the 2.7- and 4.3-μm bands, respectively. Near the band centers, errors of up to 30-50% in the computed spectra are found. The integrated error over the entire band, however, would be less than this value. Ludwig et al.[4] show much better agreement between predictions and measurements in both bands at higher pressures (P_{CO_2}=1-2 atm). Young, who also shows comparisons only at high pressure, likewise finds reasonably good agreement between measurements and predictions using the NASA parameters.

The result here is somewhat disturbing. All of the available comparisons with experimental data for CO_2 emission have been at relatively high pressure and indicate reasonably good agreement. The lower-pressure calculations are significantly different from the measured data. A similar result was found by Lindquist et al.[12] They state that the reasons for this disagreement is the complete reliance on theoretical calculations to determine the CO_2 band model parameters, and the current ignorance concerning broadening parameters.

For a simulated rocket exhaust plume mixture (Fig. 5) the agreement at 4.3 μm is somewhat better than it is for a CO_2/N_2 mixture. The maximum errors at the band center are on the order of 20% for the plume mixture, as opposed to 33% for CO_2/N_2 mixture. Figure 6 illustrates the comparison between measurements and predictions for the 2.7-μm band emission from the same simulated rocket plume mixture. The agreement is again quite good in this region, with errors on the order of 15% which is about the same as for the H_2O/N_2 mixture of Fig. 1.

B. Nonisothermal Radiators

The comparisons for isothermal radiators discussed previously give an indication of how well the band model parameters used in the radiation model predict data. For nonisothermal gases, a comparison between measurements and predictions reveals the combined effects of the band model parameters, as well as the method used to treat inhomogeneous paths. In all of the calculations performed here, the Curtis-Godson approximation, in the single-line-group (SLG) mode, was employed. This is the simplest method for treating inhomogeneous paths. Further refinements in predictions, along with more difficult calculations, could be achieved by using the Curtis-Godson multiple-line-group (MLG) approximation, the

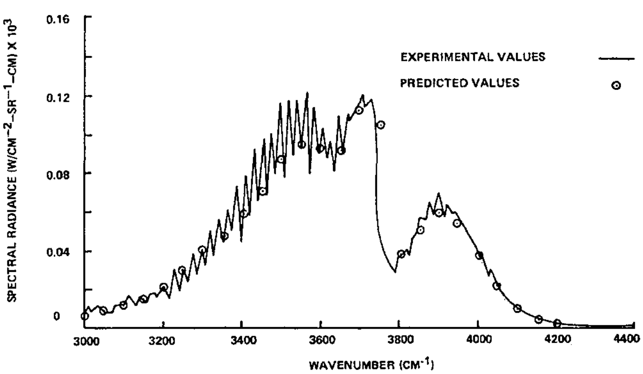

Fig. 6 Spectral radiance at 2.7 μm, isothermal mixture.

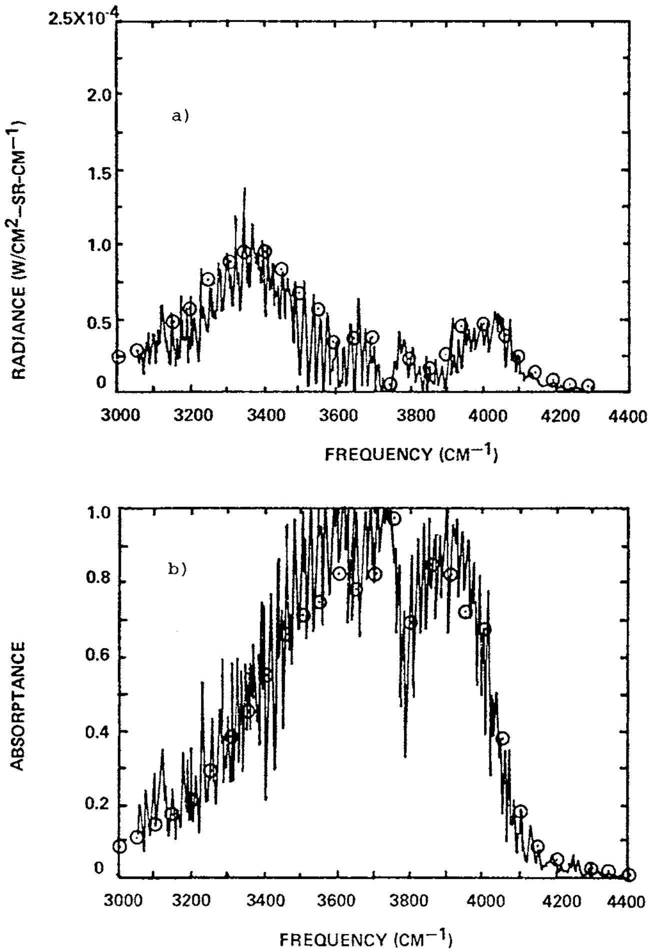

Fig. 7 a) Spectral radiance and b) absorptance at 2.7 µm, nonisothermal water (⊙, predictions; —, data).

Lindquist-Simmons[10] approximation, or the Lindquist[12] approximation. The experimental data reported here were all taken from Ref. 13 on relatively high-pressure (∿1 atm) H_2O, CO_2, and $H_2O/CO_2/N_2$ mixtures in the 2.7-µm band. Similar data at 4.3 µm were not available.

Figures 7 and 8 are comparisons of measurements and predictions for pure H_2O with two different temperature profiles. The profiles are given in Table 1. The temperature gradient for Fig. 8 is higher than that for Fig. 7. The maximum temperatures are about the same, 1160 K. Agreement in both cases is excellent, throughout the band, for both radiance and absorptance. The maximum error in either case is about 20%.

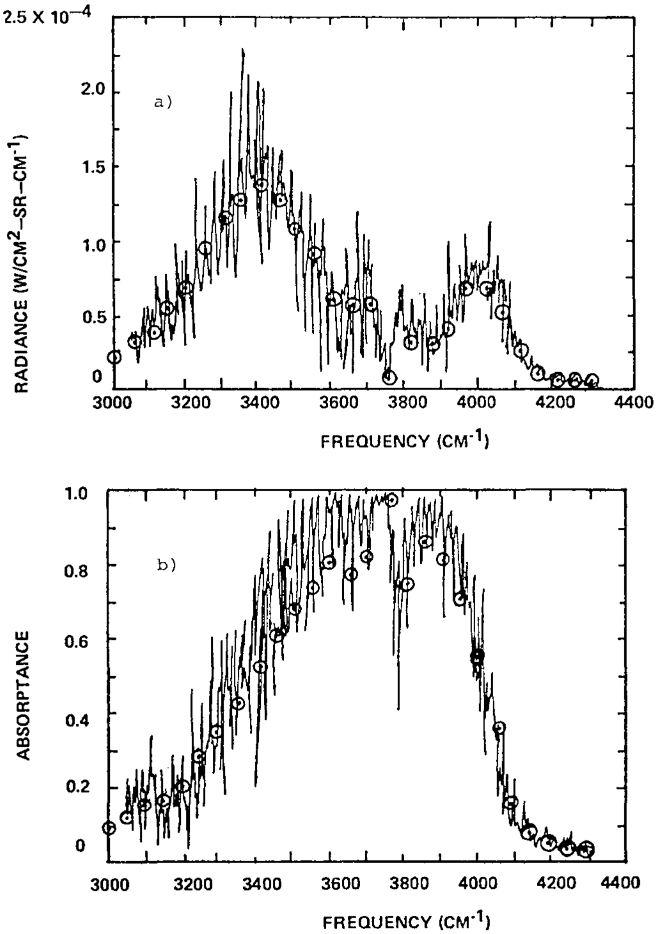

Fig. 8 a) Spectral radiance and b) absorptance at 2.7 μm, nonisothermal water (⊙, predictions; —, data).

Figures 9-11 illustrate the agreement between measurements and predictions for high-pressure, nonisothermal CO_2 in the 2.7-μm region. Except for the most severe temperature gradient of Fig. 11, which has an unrealistic shape for most physical applications, the band model does a good job in predicting the observed spectra. Since the band model parameters for CO_2 appear to be in error, it is difficult to assess which is the source for any disagreement found here: the band model parameters or the Curtis-Godson approximation. Both no doubt contribute to the error. However, for temperature profiles representative of a rocket or turbojet exhaust, such as in Fig. 10, the agreement is excellent.

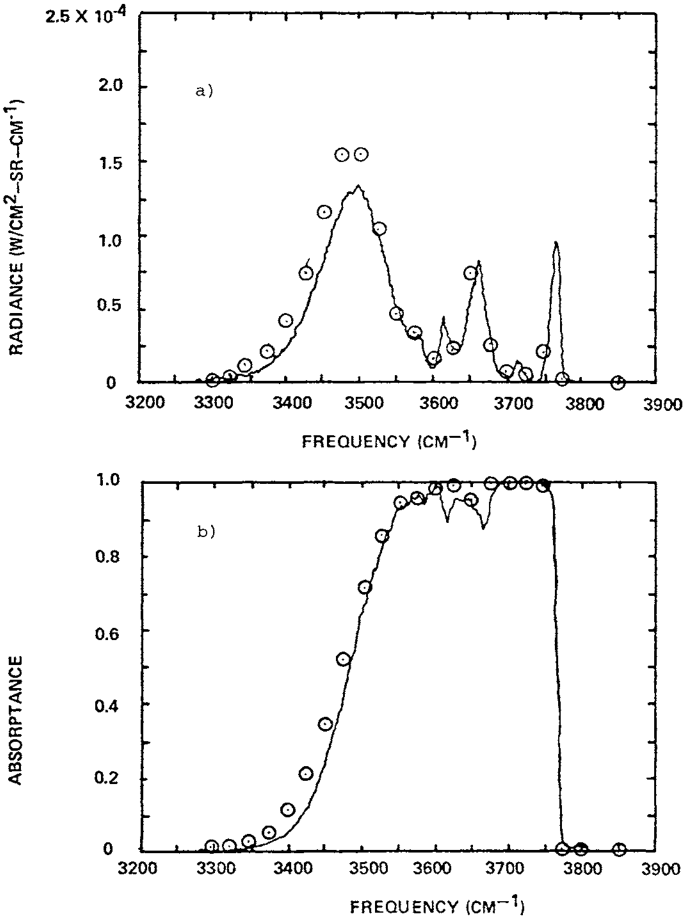

Fig. 9 a) Spectral radiance and b) absorptance at 2.7 µm, nonisothermal CO_2 (⊙ , predictions; —, data).

Excellent results likewise are achieved for the $CO_2/H_2O/N_2$ mixtures given in Fig. 12, with respect to both radiance and absorptance. The Curtis-Godson approximation appears to introduce no additional error into these calculations, since the maximum error is still on the order of only $\pm$ 20%.

C. Atmospheric Transmission

In many cases, the radiation emitted by a hot-gas sample must pass through a long atmospheric path before being received by the detector. The atmospheric transmission can give rise to two questions: knowing the source intensity, what is the observed intensity, and vice versa.

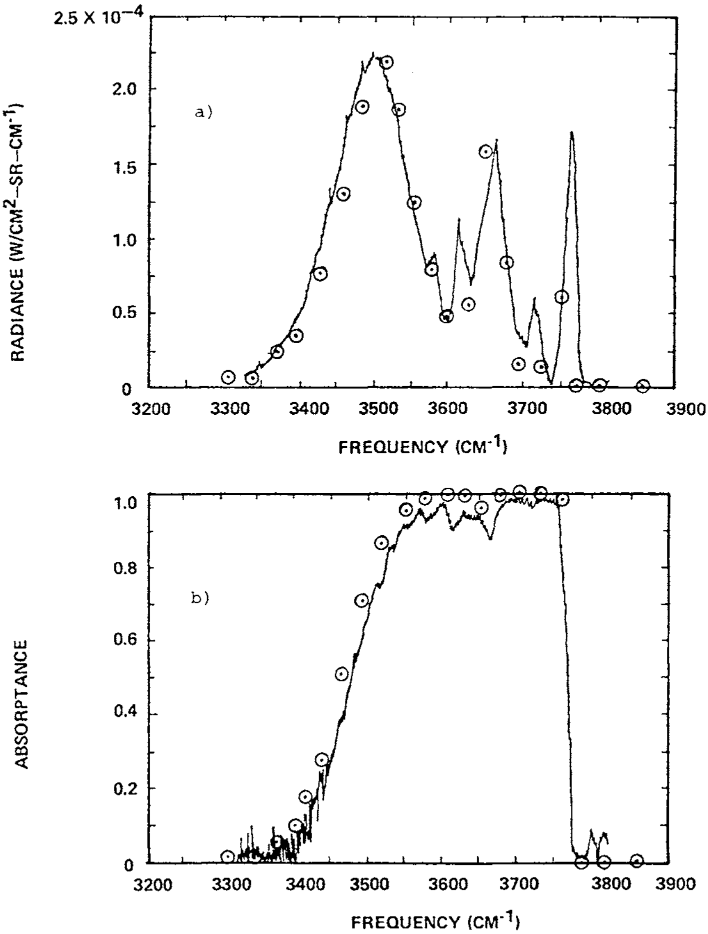

Fig. 10 a) Spectral radiance and b) absorptance at 2.7 µm, nonisothermal CO_2 (⊙, predictions; —, data).

An exact calculation of the problem, using radiation band models, requires that the atmosphere and hot-gas source be treated together. The optical path then would consist of the relatively long, slowly varying atmosphere and the short, rapidly varying hot gas. The prediction of radiation for such a problem would experience difficulties from several quarters. Band model parameters would have to be available at the low temperatures of the atmosphere. A more exact inhomogeneous path approximation than the Curtis-Godson (and perhaps even Lindquist-Simmons) would have to be employed because of the severe gradients near the boundary of the hot gas/cool atmosphere. The calculation time would increase because of the increased optical pathlength. Finally, the solution of the

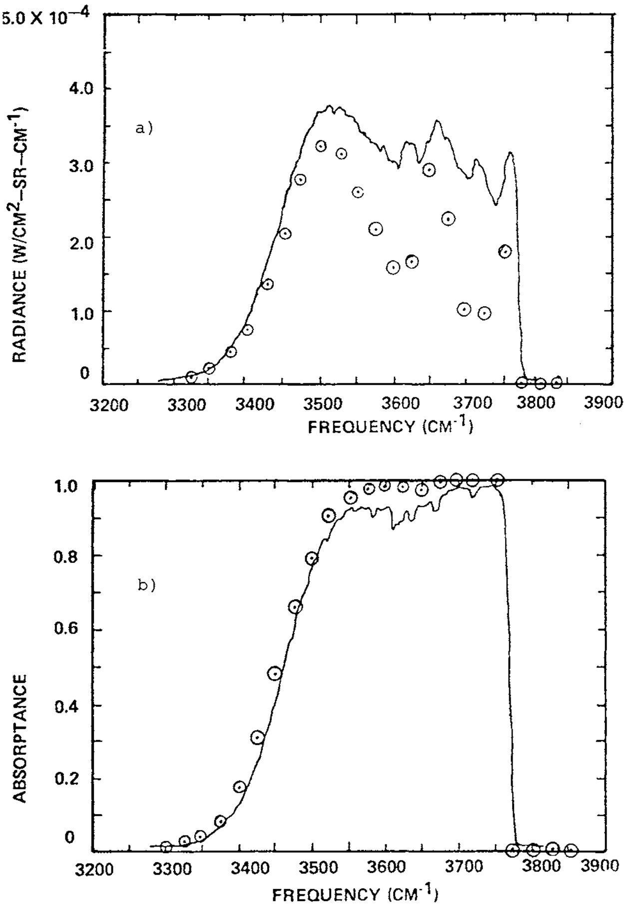

Fig. 11 a) Spectral radiance and b) absorptance at 2.7 µm, nonisothermal CO_2 (⊙, predictions; —, data).

inverse problem, viz, determining source characteristics from the observed value, would be an iterative process.

For all these reasons, the calculation of atmospheric transmission has been uncoupled. The standard procedure is to calculate the atmospheric transmission and then multiply it by the hot-gas radiance to obtain an observed radiance. Several techniques are available for calculating the transmission, and these can be divided into two classes: those accounting for the spectral structure of the absorbing gas, and those not accounting for it. The band models used for calculating hot-gas radiance fall into the latter class. Detailed line-by-line calculations are of the former class and will not be considered here.

One of the most popular of the band model formulations for the atmospheric transmission is that of Selby and McClatchey.[14] This is an empirical band model formulation, and the parameters used in this model have no relationship with those used in the NASA band model. The NASA band model and the Lowtran 2 model of Ref. 14 have been compared vis-a-vis data on the transmission characteristics of cold H_2O and CO_2 determined by Ref. 12.

Figure 13 is representative of the results. In all cases, the NASA band model, which does an admirable job in predicting hot-cell radiance, fails miserably in its ability to predict cold-cell transmission. This reflects the inaccuracy in extrapolating the experimentally derived band model parameters of H_2O to low temperatures, as well as the inaccuracies in the theoretically calculated parameters for CO_2. A similar comparison was performed by Young[6] using different data. He, too, found poor agreement between measurements and predictions based on the NASA band model parameters[4] but quite good agreement when band model parameters derived from the detailed line compilation[7] were used.

For all of these cases, the AFCRL (now AFGL) Lowtran 2 model gives much better predictions of the transmittance than does the NASA band model. Discrepancies with Lowtran 2 here are attributable to two sources, inaccuracies in the model and errors introduced in trying to determine the appropriate parameters to use in Lowtran to reproduce the properties of a laboratory cold cell. The latter is the result of the Lowtran model's being an empirical atmospheric transmittance model. The concentration of species in the model is fixed, as are the temperature and pressure of the atmosphere, depending on the model atmosphere chosen. Properties of the cold cell used to simulate an atmospheric path were quite different from an actual atmospheric path. Nonetheless, Lowtran does reasonably well in predicting the observed transmittance values.

It should be emphasized that the procedure of multiplying an atmospheric transmittance by a source radiance to obtain an observed radiance is fundamentally incorrect for correcting hot-gas radiance, although it can be used for radiators with no spectral structure. This is illustrated in Fig. 14, from Ref. 12, where the measured hot-cell radiance is multiplied by the measured cold cell transmittance and the product compared to the measured hot-through-cold-cell radiance. The measured radiance and transmittance for the hot and cold cells are given in Figs. 6 and 13, respectively. The lower curve in the figure is the observed radiance of the hot gas viewed through the cool gas. The upper curve is the best that can be

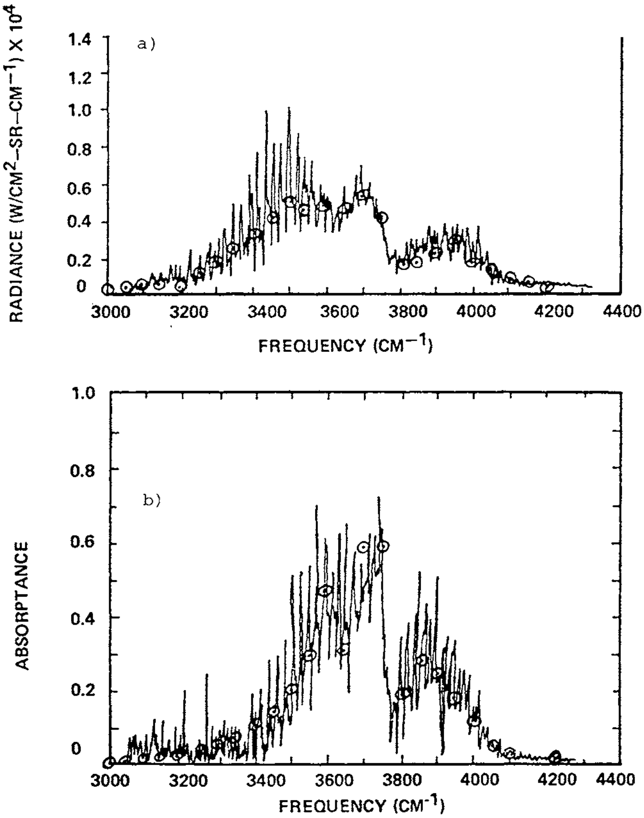

Fig. 12 a) Spectral radiance and b) absorptance at 2.7 µm, nonisothermal mixture (⊙, predictions; —, data).

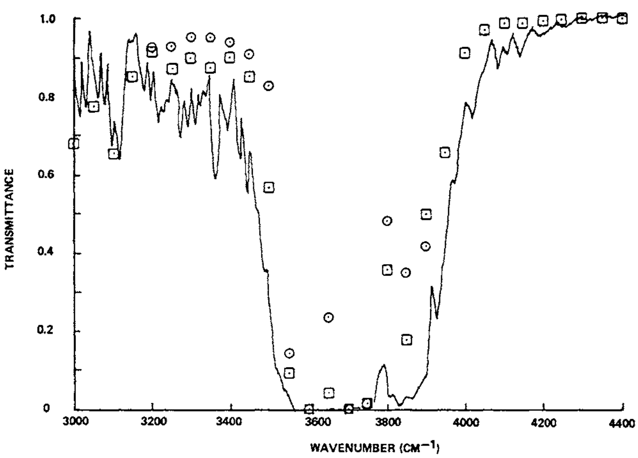

Fig. 13 Cold cell transmittance at 2.7 µm, isothermal mixture (—, data; ⊙, NASA band model; ⊡ Lowtran 2).

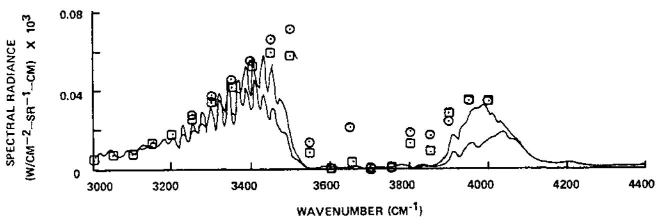

Fig. 14 Measured hot- through cold-cell radiance (—, lower) and product of measured hot-cell radiance and cold-cell transmission (—, upper) for isothermal mixture compared to product of calculated hot-cell radiance (Fig. 6) and cold-cell transmission from NASA model (⊙) and Lowtran 2 (⊡).

done in predicting the hot through cold radiance using a simple multiplicative method. It was obtained by multiplying the measured hot-cell radiance by the measured cold-cell transmittance. If our predictions of these two quantities were exact, our calculation of hot through cold radiance would match the upper curve.

Shown in Fig. 14 are predicted values of the observed radiation, calculated as the product of the predicted radiance from Fig. 6 and the predicted transmission, based on both models, of Fig. 13. Both predicted values are higher than the observed hot-through-cold radiation. Corrections to observed intensities based on values of transmission calculated in this manner introduce much larger errors than those in the calculation of hot-gas radiance. Similar results were found at 4.3 μm, although the agreement between our predictions and the data were much worse. However, at 4.3 μm the effects of line correlation are much less.

IV. Conclusions and Recommendations

In general, the ability of the NASA band model formulation to predict hot-gas radiance is quite good. The errors involved for $H_2O/CO_2/N_2$ mixtures at low pressures (∿0.1 atm) are on the order of $\pm$ 20%. The agreement for pure H_2O is much better than for pure CO_2. It seems probable that slight improvements in high-temperature H_2O predictions could be made by using the band model parameters of Young,[6] which he developed by combining the NASA parameters, obtained empirically at temperatures above 1200 K, with parameters calculated from a detailed, low-temperature (<∿300 K) line compilation. Needless to say, such a combined set should improve greatly the ability to calculate transmission in low-temperature gases, for which the NASA model does a poor job.

The predictive capability for pure CO_2 is much worse than for H_2O at low pressure, although the NASA parameters appear to provide a good representation of observed radiance and transmission at high pressures. The use of Young's[6] combined parameters would not resolve the discrepancy, since his parameters change only the low-temperature values. What is needed it appears, is a set of experimentally determined band model parameters. The errors found here are probably the result of employing the theoretically calculated parameters of Malkmus.[15,16] Kunitomo[8] compares his measured parameters with the theoretical values and concludes that the values assumed by Malkmus for γ/d, the ratio of line width to spacing, are incorrect. Therefore, it may be worthwhile to use the experimental values of Kunitomo. These values were derived for both the 2.7- and the 4.3-μm bands of CO_2 with a number of broadeners on the basis of a statistical band model with an exponential line strength distribution.

It is possible that no band model parameters could be found to improve the agreement. The error could lie in the band model formulation itself, or the errors could arise from inaccuracies in both the band model and the band model parameters. There are many possible band models. Only one has been examined here.

Finally, the effects of inhomogeneous temperature distributions on observed hot-gas radiance can be accounted for using the simple, Curtis-Godson single-line-group approximation, except for those cases when the temperature gradients are extremely severe. The utility of this approximation for calculating the observed radiance for a hot gas through a long, cool absorbing path is probably not as good as the Lindquist-Simmons or Lindquist approximations. The final answer on this, however, will have to await the incorporation into the model of an improved set of low-temperature band model parameters.

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EFFECT OF MELT REMOVAL BY AERODYNAMIC SHEAR ON MELT-THROUGH OF METAL PLATES

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Abstract

Theoretical models are constructed for melt-through of metal plates under intense one-dimensional heating and tangential airflow. The limiting cases of complete melt retention and complete melt removal are modeled by energy balances at the instant of melt-through, using the heat-balance integral method. The resulting semianalytical formulas agree well with computer solutions of one-dimensional, nonsteady heat conduction for melting and vaporizing metal plates. For partial melt removal, the effect of shear in removing melt is modeled by a liquid pool of finite depth with a shear suddenly applied at the top surface. This is incorporated in a dynamical calculation of the melting and vaporizing of the plate, using the heat-balance integral method and requiring solution of ordinary differential equations in time. Calculations show the transition from complete melt removal to complete melt retention when the heating rate is increased at fixed airflow.

Introduction

Lasers provide a means for applying high heat over small areas. Fluxes of the order of 10 to 100 kW/cm^2 can be achieved. This provides a means for constructing a weapon that can be used to burn holes in the components of aircraft,

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tanks, and other weapons systems. The efficiency of such a device depends, among other things, on the relation between the intensity of the heat flux applied and the time it takes to melt the rear surface of the material being heated (the melt-through time). This paper is concerned with modeling the melt-through time of flat metal plates subject to simultaneous laser heating and tangential airflow, as would be applicable to use of a laser to burn holes in aircraft in flight.

The melting of metal plates is a classical problem in heat-transfer theory. It is not amenable to exact analytical solution because of the nonlinearity caused by the moving phase-change boundary (or boundaries, if vaporization also is considered). Although purely numerical solutions are possible, they involve the solution of partial differential equations, which are somewhat cumbersome. Furthermore, we are interested here in adding a melt-removal mechanism to account for the effect of tangential airflow, which further complicates the problem.

The main purpose of the present work is to provide a model to use for predicting melt-through times. Its validation will rest on comparison with experiments conducted by focusing laser beams on plates in wind tunnels and measuring the time it takes for the laser beam to penetrate the plate. Such experimental results have a great deal of scatter, mostly associated with difficulty in characterizing the laser beam shape, intensity, time dependence, and repeatability. Therefore, great precision in the model is not required. The important features of the model should be ease of use, simplicity, inclusion of all important physics, and reasonable accuracy of prediction.

With those requirements in mind, it appeared that the heat-balance integral method as introduced by Goodman,[1] would provide a useful method of describing the melting and vaporizing of the plate. In fact, the method originally was introduced to provide a simple approach to phase-change problems in the days when computers were neither as widespread nor as powerful as they are today. It permits reduction of the partial differential equation of one-dimensional, nonsteady heat conduction to an ordinary differential equation. Furthermore, Goodman found it possible to integrate the ordinary differential equation analytically for many problems, sometimes under further approximations. We shall make use of the heat-balance integral method and some of Goodman's results, in obtaining the thermal response of the laser-heated plates.

The models developed here are one-dimensional, allowing a spatial variation only normal to the plate surface. The validity of this approximation clearly depends on the heat-flux profile applied by the laser beam and the thickness of the plates relative to the rate at which heat can be conducted laterally. Breaux[2] has proposed a correction to one-dimensional theory to account for two dimensional effects.

To account for removal of melt by aerodynamic shear, a one-dimensional model of rate of removal is proposed, consistent with the thermal response model. The mass rate of removal is found by considering a liquid pool of finite depth to which a shear suddenly is applied at the top surface. The thermal response model is formulated so that other models of melt removal can be used, if desired. The combination of the melt-removal and thermal response models leads to a system of ordinary differential equations in time which are solved numerically from the time the front surface reaches the melting temperature until the rear surface melts.

In the limiting cases of complete melt removal and no melt removal, the dynamical model just described can be replaced by a simpler static energy balance at the melt-through time. The energy contained in the melt is equated to the total energy absorbed. By using some of the ideas and results of Goodman,[1] algebraic formulas are obtained for the melt-through times in these two limiting cases, obviating the need for numerical integration. The results of this static model are in good agreement with numerical solutions of the one-dimensional nonsteady partial differential equation of heat conduction for both melting, and melting and vaporizing metal plates, as presented by Nash.[3] A mutual check on the static and dynamic models is provided by running the dynamic model with zero shear (zero melt removal). The results show good agreement with the static model for no melt removal.

The two models proposed here (the static energy-balance model for the limits and the dynamical model for partial melt removal) provide simple convenient tools for use in attempting to analyze and explain data from laser melt-through experiments and to predict melt-through times. A number of simplifying assumptions have been made in order to avoid undue complication in models. The first is that the properties of the material are constants in each phase, i.e., that the density, specific heat, and thermal conductivity do not change with temperature but may differ for the solid and liquid phases.

The second assumption is that the fraction of the incident heat flux absorbed is constant until the front surface reaches the melting temperature and also constant after the front is melted, although the two constants may differ. In other words, the absorptivity of the material is allowed to change suddenly when the front surface melts but at no other time. The extension to absorptivity which varies arbitrarily with time is straightforward but leads to considerable algebraic complexity.

A third assumption is that the phase changes take place at fixed temperatures, both melting and vaporization. This is strictly true for melting of only pure metals. For alloys, different constituents melt at different temperatures; for all metals, vaporization is a continuous process, although it increases exponentially with temperature, the rate of increase depending upon the pressure level at which the heating occurs. (The vaporization temperature usually quoted for a pure substance is the temperature at which its vapor pressure is 1 atm.) This continuous vaporization is probably not important from an energetic viewpoint, since the amount of energy put into the vapor below the vaporization temperature is negligible because the amount of vapor produced is very small. However, there may be other important effects of this small amount of vapor, such as enabling the ignition of plasma waves, which could affect drastically the energy deposited in the plate.

We also ignore cooling effects, i.e., radiation from the hot plate and convective cooling due to airflow, because of the high power densities of interest here. Blackbody radiation reaches 1 kW/cm^2 at 3644 K, which is above the vaporization temperature of metals of interest. Convective cooling by airflow over a 3000-K surface at Mach number unity is a few hundred watts per square centimeter. Neither of these is significant except at very low absorbed powers. Such effects should be considered below absorbed powers of 1 kW/cm^2.

As mentioned previously, we do not feel that these assumptions are significantly restrictive for application to experiments with laser heating. More sophisticated theories, with attendant increase in mathematical complication and computing time and cost, although interesting, will not at this time lead to any better understanding of the data or any more reliable predictive capability. We believe that the present simple models are quite adequate for present purposes within the scope of the physical phenomena considered.

The remainder of this paper gives the development of the static and dynamic melt-through models and results. The next section discusses the time necessary to heat the solid up to the point at which the front surface melts. Then the limiting cases of complete melt removal and complete melt retention are discussed by developing the static energy balance model. These limiting cases next are compared with the exact numerical calculations of Nash.[3] Then the dynamic partial melt-removal model is formulated for any given rate of melt removal. The following section proposes a model of the melt removal by shear, which then is used to calculate melt-through times for partial melt removal. Finally, the conclusions to be drawn are set forth in the last section.

Time to Melt the Front Surface

The first time needed is that to raise the front of the plate to the melting temperature T_m. (All temperatures are measured above the initial plate temperature.) The problem of heating a solid plate of thickness ℓ_0 with a uniform heat flux q_S at z = 0, insulated at the back face, is a classical one, whose solution is given in Eq. (1), p. 104, of Ref. 4. That result shows that the front surface z = 0 reaches T_m at time t_S given by

$$T_m = \frac{q_S \ell_0}{k_S} \left[\frac{K_S t_S}{\ell_0^2} + \frac{1}{3} - \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{\exp(-\pi^2 n^2 K_S t_S/\ell_0^2)}{n^2} \right] \quad (1)$$

where k_S is the thermal conductivity of the solid, and $K_S = k_S/\rho_S c_S$ is the thermal diffusivity for density ρ_S and specific heat c_S. The series is important only for small t_S, since its purpose is to satisfy the initial condition $T = 0$. So, for large enough t_S, i.e., if q_S is not too large, we may ignore the series and write

$$T_m = \frac{q_S \ell_0}{k_S}\left(\frac{K_S t_{SQ}}{\ell_0^2} + \frac{1}{3}\right), \quad \frac{K_S t_{SQ}}{\ell_0^2} = \frac{k_S T_m}{q_S \ell_0} - \frac{1}{3} \qquad (2, 3)$$

On the other hand, if q_S is large enough, the front will reach T_m before the heat even has reached the back surface. Then the appropriate formula is that for heating a plate of infinite thickness, also a classical result, given in Eq. (7), p. 56, of Ref. 4. It shows that T_m at $z = 0$ occurs at t_{SE}, given by

$$T_m = \frac{2}{\sqrt{\pi}} \frac{q_S \ell_0}{k_S} \sqrt{\frac{K_S t_{SE}}{\ell_0^2}}, \quad \frac{K_S t_{SE}}{\ell_0^2} = \frac{\pi}{4}\left(\frac{k_S T_m}{q_S \ell_0}\right)^2 \qquad (4, 5)$$

The two approximations (2) and (4) will be used in place of the exact expression (1). A comparison of the three expressions shows that Eq. (4) is the better approximation up to $K_S t_S/\ell_0^2 = 1/\pi$, and Eq. (2) is better after that. Unfortunately, the two expressions are never equal. But a good approximation is to use Eq. (3) for low q_S and Eq. (5) for higher q_S. In order to cover the whole range, we average the numerical factors and define the breakpoint q_S as

$$q_{St} \equiv (k_S T_m/\ell_0)/0.6441 \qquad (6)$$

so that t_S is found from

$$q_S < q_{St}: \quad \frac{K_S t_{SQ}}{\ell_0^2} = \frac{k_S T_m}{q_S \ell_0} - \frac{1}{3} \qquad (7a)$$

$$q_S > q_{St}: \quad \frac{K_S t_{SE}}{\ell_0^2} = \frac{\pi}{4}\left(\frac{k_S T_m}{q_S \ell_0}\right)^2 \qquad (7b)$$

The slight discontinuity at q_{St} from 0.3108 to 0.3258 is not serious, since t_S is only a part of the melt-through time at q_{St}.

Melt-Through with Complete Melt Removal

If the melt is removed as soon as it is formed, no energy need be put into the melt. The only energy required is that needed to raise the whole plate to T_m and then melt it. The heat input is $q_S t_S$ before front surface melting and $q_L t_L$ after front surface melting, where q_L is the heat flux after t_S, and t_L is the time from t_S to melt-through. The energy balance is then

$$q_S t_S + q_L t_L = \rho_S \ell_0 (c_S T_m + L_m) \tag{8}$$

where L_m is the heat of fusion. If we define the total time to melt-through with complete melt removal as t_{mr}, then, by solving Eq. (8) for t_L, we find

$$t_S + t_L \equiv t_{mr} = \frac{\rho_S \ell_0 (c_S T_m + L_m)}{q_L} + \left(1 - \frac{q_S}{q_L}\right) t_S \tag{9}$$

as the melt-through time. This can be written as an energy per unit volume by defining W'_{mr} as

$$W'_{mr} = \frac{q_L t_{mr}}{\ell_0} = \rho_S (c_S T_m + L_m) + \left(1 - \frac{q_S}{q_L}\right) \frac{q_L t_S}{\ell_0} \tag{10}$$

This shows that the melt-through energy is a constant if $q_S = q_L$, that is, if the absorptivity does not change at melting. In this case, it is also independent of t_S, as one might expect, since no change in heating occurs. One just must put in the required total energy.

If $q_S \neq q_L$, W'_{mr} is not constant and does depend on t_S. Substitution of Eq. (7) in Eq. (10) gives

$$q_S < q_{St}: \ W'_{mr} = \rho_S L_m + \rho_S c_S T_m \frac{q_L}{q_S} - \frac{\ell_0 q_L}{3 K_S}\left(1 - \frac{q_S}{q_L}\right) \tag{11a}$$

$$q_S > q_{St}: W'_{mr} = \rho_S (c_S T_m + L_m) + \frac{\pi}{4} \frac{\ell_0 q_L}{K_S} \left(\frac{\rho_S c_S K_S T_m}{q_S \ell_0} \right)^2 \left(1 - \frac{q_S}{q_L} \right) \tag{11b}$$

We see from Eq. (10) that, if $q_S < q_L$, that is, the absorptivity of the solid is less than that of the liquid, the t_S term increases W'_{mr} over the value it would have at q_L, because the heat input up to t_S is smaller than q_L. The reverse is true if $q_S > q_L$, as it might be for painted surfaces. It is also worth noting that the energy depends only on the combinations q_S/q_L and $\ell_0 q_L$. Thus the effects of thickness and heat input are combined. Doubling ℓ_0 and halving q_L has no effect on W'_{mr} if q_S also is halved. So, thickness variation is equivalent to q_L variation, which may be a useful property when conducting experiments.

Melt-Through with Complete Melt Retention

When the melt is retained, some of the energy put in goes to heat the melt, which effectively insulates the remaining solid to some degree, thus slowing down the melt-through process. There are two cases to be distinguished here, depending on whether or not the melt gets hot enough to vaporize. If some melt does vaporize, this represents an additional heat sink, which further retards melt-through.

No Vaporization

At melt-through there is a thickness ℓ_0 of heated melt with temperature distribution $T_L(z)$ varying from some value $T_L(0)$ at the front surface to the value $T_L(\ell_0) = T_m$ at the back surface. The energy necessary to create this state is that necessary to melt the whole thickness plus that in the melt above T_m. If we equate this to the input energy $q_S t_S + q_L t_L$ as before, solve for t_L, and add q_S, we have the time t_{mk} to melt-through with melt retained as

$$t_S + t_L = t_{mk} = t_{mr} + \frac{\rho_L c_L}{q_L} \int_0^{\ell_0} (T_L - T_m)\, dz \tag{12}$$

The second term is the energy in the retained melt. It is clearly necessary to know the T_L profile at melt-through.

Two obvious requirements on this profile are to match the rear face temperature T_m and front face heat input q_L. We shall assume a quadratic profile of T_L,[1] which has been found in the heat-balance integral method[1] to give a good representation of the integrated energy. With the two boundary conditions, a quadratic profile gives

$$T_L = T_m - \left(1 - \frac{z}{\ell_0}\right) \ell_0 \left(\frac{\partial T_L}{\partial z}\right)_{\ell_0} + \frac{1}{2}\left(1 - \frac{z}{\ell_0}\right)^2 \ell_0 \left[\frac{q_L}{k_L} + \left(\frac{\partial T_L}{\partial z}\right)_{\ell_0}\right] \tag{13}$$

The remaining parameter is the temperature gradient at the back surface of the liquid at melt-through. At first, one may feel that this should be zero if the rear surface is insulated. However, it is not the solid temperature that is involved here, but the liquid temperature. At a solid-liquid interface, the interface condition is

$$- k_L \, \partial T_L/\partial z = - k_S \, \partial T_S/\partial z + \rho_S L_m \, d S_L/dt \tag{14}$$

which expresses the fact that the heat flux out of the liquid partly goes into the solid and partly serves to melt more solid, producing a rate of movement of the location S_L (t) of the interface. At melt-through, the first term on the right of Eq. (14) vanishes, but the second one does not; the interface is moving as melt-through occurs. Thus the liquid temperature gradient is not zero. To calculate this gradient from Eq. (14) would require a dynamical calculation of S_L (t), which we wish to avoid. We therefore try to use another condition, not requiring a time integration. Such a condition has been used by Goodman in his original paper on the heat-balance integral method.[1]

We note that, at any time, the liquid temperature at S_L (t), the interface, is T_m, so that $T_L (S_L (t), t) = T_m$. The time derivation of this is

$$(\partial T_L/\partial z)\,(d\,S_L/dt) + \partial T_L/\partial t = 0 \tag{15}$$

The second term can be eliminated using the temperature equation

$$\partial T_L/\partial t = K_L\, \partial^2 T_L/\partial z^2 \tag{16}$$

whereas the $d\,S_L/dt$ term can be eliminated using Eq. (14). At any time, this would bring in $\partial T_S/\partial z$. However, at melt-through this derivative vanishes because the vanishingly thin solid has a constant temperature if its back face is insulated. Thus, the use of Eqs. (14) and (16) in (15) at melt-through, $S_L(t_m) = \ell_0$, gives

$$-\frac{k_L}{\rho_S L_m}\left(\frac{\partial T_L}{\partial z}\right)^2_{\ell_0} + K_L\left(\frac{\partial^2 T_L}{\partial z^2}\right)_{\ell_0} = 0 \tag{17}$$

This shows the essentially nonlinear nature of phase-change problems. It is an exact relation between the first and second temperature derivatives at melt-through.

If we put the derivatives of our assumed profile Eq. (13) into the condition Eq. (17), we obtain a quadratic equation for the first derivative, whose solution is

$$-\left(\frac{\partial T_L}{\partial z}\right)_{\ell_0} = \frac{\rho_S L_m}{2\ell_0 c_L \rho_L}\left[\sqrt{1 + \frac{4\,\ell_0 q_L}{\rho_S L_m K_L}} - 1\right] \tag{18}$$

This enables us to define the quadratic profile completely without any necessity for integration in time.

The use of Eqs. (13) and (18) in the energy balance expression for t_{mk}, Eq. (12), gives

$$t_{mk} = t_{mr} + \frac{\ell_0^2}{6K_L} + \frac{\rho_S L_m \ell_0}{6\,q_L}\left[\sqrt{1 + \frac{4\,\ell_0 q_L}{\rho_S L_m K_L}} - 1\right] \tag{19}$$

or, in terms of energy per unit volume,

$$\frac{q_L t_{mk}}{\ell_0} \equiv W'_{mk} = W'_{mr} + \frac{\ell_0 q_L}{6 K_L} + \frac{\rho_S L_m}{6}\left[\sqrt{1 + \frac{4 \ell_0 q_L}{\rho_S L_m K_L}} - 1\right] \tag{20}$$

We see that the additional energy depends only on material properties and the product $\ell_0 q_L$, making the $\ell_0 q_L$ scaling good in the melt-retained limit as well as the melt-removed limit of Eq. (6). For $q_S = q_L$, W'_{mr} is a constant, and W'_{mk} depends on q_L linearly through the middle term in Eq. (20) and as the square root through the last term. As q_L increases, W'_{mk} increases continually because of these terms. If $q_S \neq q_L$, as long as $q_S > q_{St}$, W'_{mk} increases with q_L, since W'_{mr} approaches a constant as q_L increases (assuming q_S/q_L fixed); see Eq. (11).

The development just given takes no account of vaporization and so is valid for $T_L < T_v$, the vaporization temperature. For large enough q_L, the front surface of the melt reaches T_v before melt-through, and vaporization must be considered. The largest value of q_L for which no vaporization occurs is that at which $T_L = T_v$ when $z = 0$ in Eq. (13). When Eq. (18) also is used, one gets a quadratic for q_L, whose solution gives the heat flux q_L for incipient vaporization, which we shall call q_v. The result is

$$q_v = \frac{\rho_S L_m K_L}{\ell_0}\left[1 + \frac{2 k_L (T_v - T_m)}{\rho_S L_m K_L} - \sqrt{1 + \frac{2 k_L (T_v - T_m)}{\rho_S L_m K_L}}\right] \tag{21}$$

For values of $q_L > q_v$, vaporization needs to be considered.

Vaporization

It is convenient to start the examination of vaporization with the limiting case of complete vaporization of the whole plate. This requires adding to the melt the energy necessary to heat to T_v, and then vaporize by adding the heat of vaporization L_v. Thus, the time to add enough energy to vaporize all of the material is

$$t_{vr} = t_{mr} + \rho_L \ell_0 \left[c_L (T_v - T_m) + L_v \right] / q_L \tag{22}$$

where the first term, from Eq. (9), is the time to melt all of the material, and the second term is the time to vaporize it starting in the melted state. In terms of energy per unit volume, this is

$$q_L t_{vr}/\ell_0 \equiv W'_{vr} = W'_{mr} + \rho_L \left[c_L (T_v - T_m) + L_v \right] \tag{23}$$

For partial vaporization, of course, we do not need all this energy. Let S_{vm} be the location of the melt-vapor interface at melt-through. Then thickness $\ell_0 - S_{vm}$ of the plate has not vaporized, and so we may subtract this much of the second term on the right of Eq. (22). However, the thickness $\ell_0 - S_{vm}$ has been heated above T_m, and so we must add the energy contained in its profile. Thus the time to melt with vaporization becomes

$$t_{mv} = t_{vr} - \frac{(\ell_0 - S_{vm})}{q_L} \rho_L \left[c_L (T_v - T_m) + L_v \right] + \frac{\rho_L c_L}{q_L} \int_{S_{vm}}^{\ell_0} (T_L - T_m)\, dz \tag{24}$$

The liquid temperature profile now extends from S_{vm} to ℓ_0. It has the temperature T_m at $z = \ell_0$ and the temperature T_v at $z = S_{vm}$. We again shall assume a quadratic profile, and, with these two conditions, it is

$$T_L = T_m - (1 - z/\ell_0)\, \ell_0 (\partial T_L/\partial z)_{\ell_0} + \frac{\left\{ T_v - T_m + \ell_0 \left[1 - (S_{vm}/\ell_0) \right] (\partial T_L/\partial z)_{\ell_0} \right\} \left[1 - (z/\ell_0) \right]^2}{(1 - S_{vm}/\ell_0)^2} \tag{25}$$

We again have used the gradient at the back face as a parameter. The condition Eq. (17) still holds for this gradient, and, with the profile Eq. (25) used, the quadratic for the gradient has the solution

$$-\left(1-\frac{S_{vm}}{\ell_0}\right)\left(\frac{\partial T_L}{\partial z}\right)_{\ell_0}=\frac{\rho_S L_m}{\ell_0 \rho_L c_L}\times\left[\sqrt{1+\frac{2\rho_L c_L (T_v - T_m)}{\rho_S L_m}}-1\right] \tag{26}$$

Substitution of the profile Eq. (25) and the gradient Eq. (26) into the energy-balance time Eq. (24) provides the expressions

$$t_{mv}=t_{vr}-\frac{\rho_L c_L \ell_0}{q_L}\left(1-\frac{S_{vm}}{\ell_0}\right)\left[\frac{L_v}{c_L}+\frac{2}{3}(T_v-T_m)-\frac{\rho_S L_m}{6\rho_L c_L}\left(\sqrt{1+\frac{2\rho_L c_L (T_v-T_m)}{\rho_S L_m}}-1\right)\right] \tag{27}$$

$$\frac{q_L t_{mv}}{\ell_0}\equiv W'_{mv}=W'_{vr}-\rho_L c_L\left(1-\frac{S_{vm}}{\ell_0}\right)\left[\frac{L_v}{c_L}+\frac{2}{3}(T_v-T_m)-\frac{\rho_S L_m}{6\rho_L c_L}\left(\sqrt{1+\frac{2\rho_L c_L (T_v-T_m)}{\rho_S L_m}}-1\right)\right] \tag{28}$$

These expressions cannot be used yet because they contain the unknown location of the melt-vapor interface at melt-through, S_{vm}. The calculation of this quantity in general would require a time integration. However, for our purposes, an approximate expression would be useful if one were available in a simple form. In fact, such an approximate expression is available, because Goodman has solved a simpler version of the present problem by integrating in time. His problem differed from the present one in that he started with the solid all at the melting temperature. The heat input then serves to provide the energy to melt, to heat the liquid, and to vaporize. For this problem, Goodman was able to integrate the differential equations of his heat-balance integral method to arrive at expressions for $S_L(t)$ and $S_v(t)$, the locations of the liquid-solid and vapor-liquid interfaces as functions of time. At melt-through, $S_L = \ell_0$, and by eliminating t, Goodman[1] arrived at a relation for S_{vm} in his Eq. (82). That relation can be written as

$$\left(\frac{\ell_0 - S_{vm}}{S_{Lv}}\right)(1 - b) + b - \frac{q_L}{q_v} = \frac{b}{a} \ln\left[1 + a - a\left(\frac{\ell_0 - S_{vm}}{S_{Lv}}\right)\right] \tag{29}$$

where S_{Lv} is the location of the liquid-solid interface when the vaporization temperature is reached at the front of the liquid, i.e., when $S_{vm} = 0$. The constants a and b are combinations of material properties, defined by

$$A = \frac{2\,\rho_L\, c_L\,(T_v - T_m)}{\rho_S\, L_m}, \quad B = \frac{2}{A}\left[1 + A - \sqrt{1 + A}\,\right] \tag{30a}$$

$$a = \left(\frac{A}{2} + 1\right)\frac{\rho_S\, L_m}{\rho_L\, L_v}, \quad b = \frac{\{[(2/3) - (B/6)]\,(A/2) + 1\}}{(A/2) + 1} \tag{30b}$$

The expression for S_{Lv} is taken from Goodman's solution for the nonvaporizing problem and is related to q_v:

$$S_{Lv} = \ell_0\, q_v / q_L \tag{31}$$

It has the correct limit $S_{Lv} = \ell_0$ when $q_L = q_v$. It also shows that Eq. (29) gives the correct limit $S_{vm} = 0$ at this value of q_L.

Equations (29 - 31) then provide the value of $\ell_0 - S_{vm}$ which enables Eq. (27) or (28) to be used to find the time or energy for melt-through in the vaporizing case. Although Eq. (29) is a transcendental equation for $\ell_0 - S_{vm}$, it is quite easy to solve either by hand or on a computer, since the right side is a fixed function of $(\ell_0 - S_{vm})/S_{Lv}$ for any material, whereas the left side is a straight line in the same variable with a slope dependent only on material but an intercept that depends on $q_L \ell_0$ as well as material properties, according to Eq. (21). Note that there is a limiting value of $\ell_0 - S_{vm}$ for very large $\ell_0 q_L$, given by the vanishing of the argument of the ln term on the right as

$$(\ell_0 - S_{vm})_{lim} = (1 + a^{-1}) S_{Lv} = (1 + a^{-1}) \ell_0 q_v / q_L \tag{32}$$

Thus, for large q_L, $\ell_0 - S_{vm}$ approaches zero, or the material is completely vaporized, which is again the correct limit.

In terms of $1 - S_{vm}/\ell_0$, as it appears in Eqs. (27) and (28), we can express Eq. (29), with Eq. (31), as

$$\frac{q_L}{q_v}\left(1 - \frac{S_{vm}}{\ell_0}\right)(1 - b) + b - \frac{q_L}{q_v}$$
$$= \frac{b}{a} \ln \left[1 + a - \frac{a\, q_L}{q_v}\left(1 - \frac{S_{vm}}{\ell_0}\right)\right] \tag{33}$$

We now have a complete solution for melt retention with vaporization, by solving Eq. (29) or (33) for $1 - S_{vm}/\ell_0$ and inserting its value in Eq. (27) or (28).

Comparison with Exact Calculations

In order to validate the use of the approximate formulas for melt retention, comparisons can be made with more exact solutions of the partial differential equation for one-dimensional heating with melting and vaporization. A convenient source of such calculations is the work of Nash,[3] who has plotted graphs of melt-through time vs heat flux for three metal alloys: Al 2024, stainless steel (SS) 304, and Ti-6Al-4V. The properties he used are given in Table 1. He worked in dimensionless variables and plotted

$$\tau_f = \frac{t_m K_S}{\ell_0^2} \quad \text{vs} \quad \alpha \hat{Q} = \frac{q\, \ell_0}{\rho_S K_S L_m}$$

where t_m is the appropriate melt-through time, and $q = q_S = q_L$, since he did the case of equal fluxes.

Table 1 Material Properties

	Stainless steel 304	Ti-6Al-4V	Al2024
T_m,[a] K	1427	1622	611
T_v,[a] K	2973	3258	2453
L_m, J/g	272	393	398
L_v, J/g	4653	8887	10,732
$\rho_S = \rho_L$, g/cm^3	8.0	4.47	2.77
$c_S = c_L$, J/g - K	0.619	0.833	0.249
k_S, W/cm - K	0.260	0.190	1.73
k_L, W/cm - K	0.260	0.190	0.866
K_S, cm^2/sec	0.0523	0.051	0.6
K_L, cm^2/sec	0.0523	0.051	0.3

[a]These are the melting and vaporization temperatures relative to an ambient temperature of 300 K.

His results are shown as the solid and dashed lines in Figs. 1 (Al 2024), 2 (SS304), and 3 (Ti). The lower dashed line is t_{mr}, and the upper one is t_{vr}, the limiting cases for complete melt removal and complete vaporization, and they, of course, agree with the present Eqs. (9) and (22) for the properties in Table 1. The solid line is the complete melt-retention case, which shows the transition, as heating is increased, from t_{mr} to t_{vr}.

Calculations with the present approximate model were performed for $\ell_0 = 0.159$ cm (1/16 in), using Eq. (19) for t_{mk} (the circles in the figures) and Eqs. (27) and (23) for t_{mv} (the X in the figures). We note that, in the dimensionless variables τ_f, $\alpha\hat{Q}$, the thickness does not enter either t_{mk} or

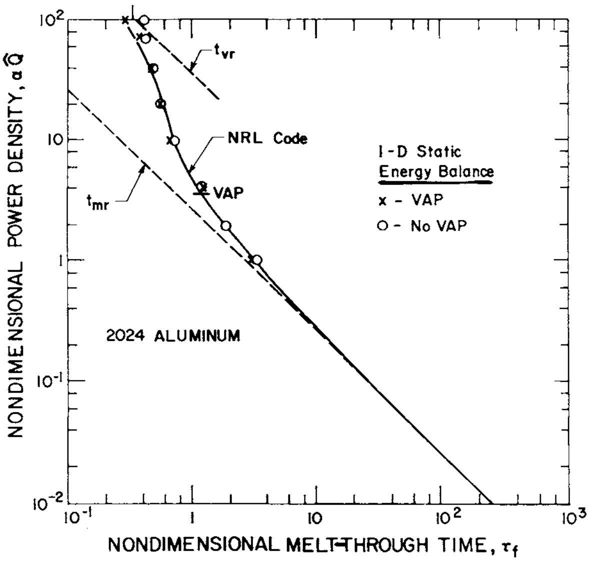

Fig. 1 Melt-through time as a function of heat flux absorbed, compared to the calculations of Nash,[3] for Al 2024.

t_{mv}. The former is obvious from Eq. (19), and the latter is clear from Eqs. (27) and (33) when we recall that $q_v \ell_0$ is independent of ℓ_0, from Eq. (21).

The figures show that the present model, as simple as it is, provides a very good approximation to the more exact calculations of Nash. The maximum error seems to be 10 to 15% for Ti in the middle of the transition range. The location of the onset of vaporization, according to Eq. (21), is shown by a mark on the curve labeled "VAP". Above that, both the vaporization and no-vaporization results are shown. The no-vaporization times stay remarkably close to the vaporization times until they finally diverge at high $q\ell_0$. The no-vaporization time asymptotes at a value $\tau_f = K_S/6K_L$, whereas the vaporization time correctly asymptotes to t_{vr}, the upper dashed line. However, for simple calculations, one could use the no-vaporization time t_{mk} of Eq. (19) up until $t_{mk} = t_{vr}$, and then use t_{vr}. This will give larger than realistic times at very high values of q, above $q\ell_0/\rho_S L_m K_S = 100$ for SS and Ti, and 70 for Al, but this may be above the interesting range of $q\ell_0$ anyway, since it is 11.4 kW/cm for SS, 8.9 kW/cm for Ti, and 66.1 kW/cm for Al.

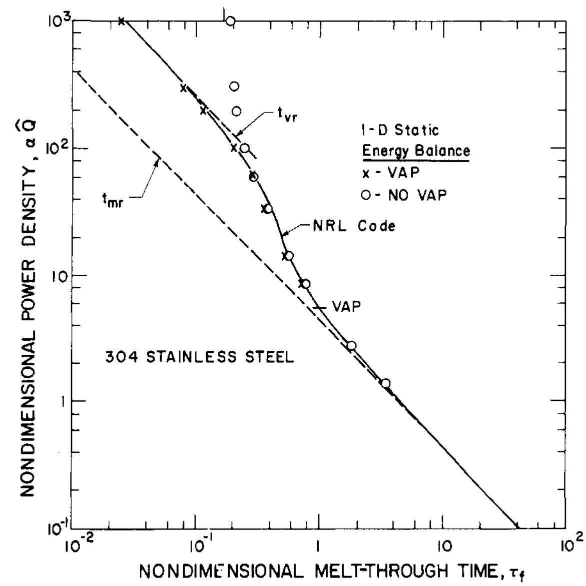

Fig. 2 Melt-through time as a function of heat flux absorbed, compared to the calculations of Nash,[3] for stainless steel 304.

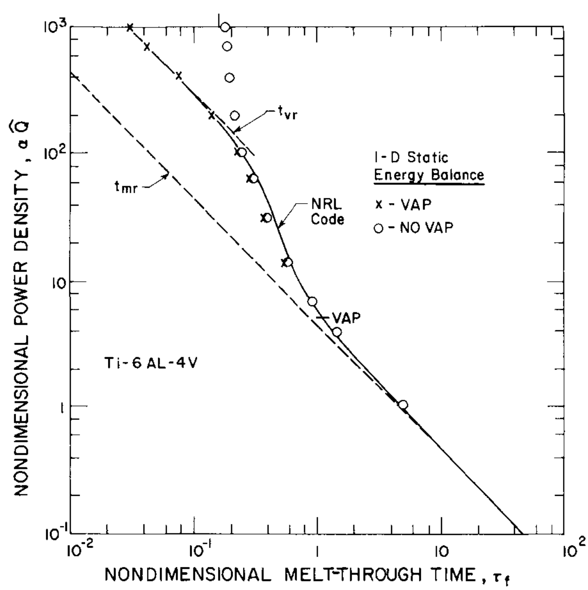

Fig. 3 Melt-through time as a function of heat flux absorbed, compared to the calculations of Nash,[3] for Ti-6Al-4V.

The comparisons of Figs. 1-3 indicate that the rather simple models proposed here, with quadratic temperature profiles, give quite accurate results, which are useful for the calculation of metal alloy melt-through times. This gives us confidence to proceed with the extension to the case of partially retained melt.

Partial Melt Removal Formulation

We shall formulate a set of differential equations (in time) for the calculation of one-dimensional melting and vaporizing with partial melt removal. It is important to realize that the melt-through time with partial melt removal can lie only between the values for complete removal and complete retention, i.e., between the solid line and lower dashed line on Figs. 1-3. Although this is a factor of 10 in τ_f at the high values of $q\ell_0$, where it is t_{mr} to t_{vr}, it is much less over the more interesting range of lower $q\ell_0$, where the range of t_m is well represented by the difference

$$t_{mk} - t_{mr} = \frac{\ell_0^2}{6K_L} + \frac{\rho_S L_m \ell_0}{6q_L}\left[\sqrt{1 + \frac{4\ell_0 q_L}{\rho_S L_m K_L}} - 1\right] \quad (34)$$

This depends on q_L, ℓ_0, and the material properties and shows a noticeable effect of the latter. Large diffusivity, as in Al, reduces the difference considerably in both terms. Thus, for Al we expect the limits between which melt removal has scope to change t_m to be much closer together than the limits for SS or Ti. It may be much more difficult to observe experimentally the effect on t_m of melt removal in Al than in other, less conducting materials. And even in SS and Ti, the range of t_m is not very large, and so only experiments in the

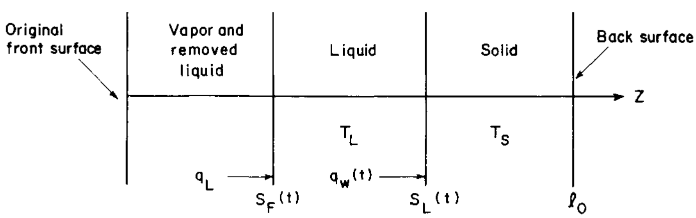

Fig. 4 Cross section of plate that is melting, vaporizing, and having liquid removed.

middle of the melt-removal region will show clear effects. For small removal rates, the results will not be distinguishable from complete melt retention, and for large removal rates, the results will not be distinguishable from complete melt removal. It is hoped that the results of the model to be described will indicate where the middle of the melt-removal range is, in terms of aerodynamic flow or shear. Notice also that thicker plates will have a larger range of t_m, making partial melt removal more observable.

Figure 4 shows a sketch of a plate of thickness ℓ_0 in the process of melting and vaporizing. The thickness coordinate z has its origin at the original surface. The liquid-solid interface is at $S_L(t)$, and the liquid surface is at $S_F(t)$. The heat flux $q_w(t)$ at S_L leaves the liquid toward the solid, where it serves both to heat the solid and to melt it. The external energy flux q_L at S_F heats and vaporizes the liquid. We shall write energy balances for both solid and liquid, using the interface conditions and assuming quadratic profiles, to derive the differential equations of the model.

<u>Solid</u>

The energy put into the solid at any time t after t_S is that necessary to bring its front surface to T_m, which is $q_S t_S$, plus that put in by $q_w(t)$, which is $\int_{t_S}^{t} q_w \, dt$. (We assume that the back face of the solid is insulated.) This energy has converted a thickness of solid S_L to liquid and put enough heat into the remaining solid to give it a temperature profile T_S. So the energy balance at any time t is

$$q_S t_S + \int_{t_S}^{t} q_w \, dt = \rho_S S_L (c_S T_m + L_m) + \rho_S c_S \int_{S_L}^{\ell_0} T_S \, dz \tag{35}$$

The interface condition at S_L is just Eq. (14), where q_w is the left side.

A quadratic profile for the solid $S_L < z < \ell_0$ is

$$T_S = T_m + \left(\frac{\partial T_S}{\partial z}\right)_{S_L} (z - S_L) \left[1 - \frac{1}{2} \frac{z - S_L}{\ell_0 - S_L}\right] \tag{36}$$

which has T_m at S_L and a zero gradient at ℓ_0. If the gradient from Eq. (14) is used with this profile in Eq. (35), the energy balance becomes

$$q_S t_S + \int_{t_S}^{t} q_w \, dt = \rho_S (c_S T_m \ell_0 + S_L L_m) - \frac{(\ell_0 - S_L)^2}{3 K_S} \left(q_w - \rho_S L_m \frac{d S_L}{dt} \right) \qquad (37)$$

This is a differential equation in time for the two variables $S_L(t)$ and $q_w(t)$, or

$$I_w(t) = \int_{t_S}^{t} q_w \, dt, \qquad \frac{d I_w}{dt} = q_w \qquad (38)$$

It must be solved with a similar equation for the liquid.

At $t = t_S+$, $S_L = 0$, and the heat going into the solid is q_S, whereas the heat input has changed to q_L, so that

$$q_L - \rho_S L_m \, d S_L/dt = q_S \qquad (39)$$

If $q_L = q_S$, the rate at which melting starts is zero. However, if $q_L \neq q_S$, this rate begins at a finite value because of the discontinuity in q. The differential equation (37) at $t = t_S +$ is

$$q_S t_S = \rho_S c_S T_m \ell_0 - q_S \ell_0^2 / 3 K_S$$

This agrees with Eq. (3) for t_{SQ}, the expression for t_S appropriate to a quadratic profile, as we might expect, since we have assumed a quadratic profile to get Eq. (37). In order to assure the validity of Eq. (37) at t_S +, we must use this expression for t_S, and so Eq. (37) is

$$\int_{t_{SQ}}^{t} q_w \, dt = \rho_S L_m S_L + \frac{\ell_0^2 q_S}{3 K_S} - \frac{(\ell_0 - S_L)^2}{3 K_S} \left(q_w - \rho_S L_m \frac{d S_L}{dt} \right) \qquad (40)$$

The formulation using Eq. (40) for the solid holds for $t > t_{SQ}$. In the numerical solution, it is convenient to start time at zero and then add t_{SQ} to refer the time back to the start of heating. Notice that, for large enough q_S, t_{SQ} will be negative, according to Eq. (7), and so the total time to melt-through will be less than the time calculated using Eq. (40). At first, this may seem puzzling. However, it becomes understandable when we realize that, for high enough q_S, the quadratic profile used for T_S has a negative energy content at the start, because it is not a physically realistic profile. (The realistic profile is an error function.) As time goes on, the energy content becomes positive, and the quadratic profile becomes realistic. The negative value of t_{SQ} compensates for the negative energy state at the start and eliminates the time necessary to bring the energy up to a positive value. So the combination of using t_{SQ} and the quadratic profile is consistent and leads to realistic results.

Liquid with No Vaporization

The energy input to the liquid up to time t is $q_L (t - t_{SQ})$ minus the integral of q_w, which goes to the solid. This energy has raised the liquid temperature above T_m. If liquid is being removed from the front by aerodynamic means at a rate of $d S_A/dt$ (where $S_F = S_A$ before vaporization begins to occur), then this carries off energy at a rate $\rho_L c_L (T_{LA} - T_m) d S_A/dt$, where T_{LA} is the liquid temperature at S_A. The energy balance is then

$$q_L(t - t_{SQ}) - \int_{t_{SQ}}^{t} q_w \, dt = \rho_L c_L \int_{S_A}^{S_L} (T_L - T_m) \, dz$$

$$+ \rho_L c_L \int_{t_{SQ}}^{t} (T_{LA} - T_m) \frac{d S_A}{dt} \, dt \qquad (41)$$

The constants in a quadratic profile for the liquid, between S_A and S_L, are determined by $T_L = T_m$ at S_L and the gradients $- q_L/k_L$ at S_A and $- q_w/k_L$ at S_A. The resulting profile is

$$T_L = T_m + \frac{q_w}{k_L}(S_L - z) + \frac{q_L - q_w}{2\,k_L}\,\frac{(S_L - z)^2}{S_L - S_A} \tag{42}$$

which gives, at $z = S_A$,

$$T_{LA} = T_m + (q_L + q_w)(S_L - S_A)/2\,k_L \tag{43}$$

Use of Eqs. (42) and (43) in Eq. (41) gives the energy balance as

$$q_L\,(t - t_{SQ}) - \int_{t_{SQ}}^{t} q_w\,dt = \frac{(S_L - S_A)^2}{6\,K_L}\,(2\,q_w + q_L)$$

$$+ \frac{1}{2\,K_L}\int_{t_{SQ}}^{t} (S_L - S_A)\,(q_L + q_w)\,\frac{d\,S_A}{dt}\,dt \tag{44}$$

This is a second differential equation for $I_w(t)$ and $S_L\,(t)$ if S_A is taken as given or determined by the other variables, so that Eq. (44) can be solved, together with Eq. (40). The initial conditions are

$$t = t_{SQ}:\ S_L = 0,\ I_w = 0 \tag{45}$$

and additional relations that hold there are

$$q_w = q_L,\ q_L - q_S = \rho_S\,L_m\,d\,S_L/dt \tag{46}$$

We may note that, if $S_A = 0$, then Eqs. (40, 44, and 45) are an integral formulation of the melting of a solid, with complete melt retention. As such, a solution until melt-through, $S_L = \ell_0$, is a dynamical calculation of t_{mk}, which could be compared with the static energy balance method previously used for the melt-retained limit. The value of q_w at melt-through could be compared with the slope Eq. (18) obtained from the use of the condition (15). In fact, it is easy to see that the only difference between the results of the two methods of finding t_{mk} is this final value of q_w. Simply add Eqs. (37) and (44) for $S_A = 0$ and evaluate at $t = t_{mk}$, where $S_L = \ell_0$. The result is exactly the expression Eq. (19)

if the q_w term is written in terms of $q_w = -k_L (\partial T_L/\partial z)_{\ell_0}$ by using Eq. (18). Such a comparison between the two methods of obtaining t_{mk} will be presented later.

Because no vaporization has been included, these equations can be used only up to a time at which $T_{LA} = T_v$. After that, vaporization must be included. This time is determined in the course of integration by the satisfaction of the condition, from Eq. (43), that

$$2 k_L (T_v - T_m) = (q_L + q_w)(S_L - S_A) \tag{47}$$

When the time at which Eq. (47) is satisfied has been reached (call it t_v), then a new set of equations must be used.

<u>Liquid with Vaporization</u>

The temperature profile is now different from Eq. (42). First, the liquid is contained between S_L and S_F, where S_F is the liquid front, now receding by the combined effects of aerodynamic removal and vaporization. Thus, we may write

$$S_F = S_v + S_A \tag{48}$$

S_v is determined by the interfacial condition at S_F:

$$q_L = -k_L \frac{\partial T_L}{\partial z} + \rho_L L_v \frac{d S_v}{dt} \tag{49}$$

Furthermore, of course, the liquid temperature at S_F is T_v.

A profile that has $T_L = T_v$ at S_F, $T_L = T_m$ at S_L, and q_w at S_L is

$$T_L = T_m + \frac{q_w}{k_L}(S_L - z) + \left[T_v - T_m - \frac{q_w}{k_L}(S_L - S_F)\right]\left(\frac{S_L - z}{S_L - S_F}\right)^2 \tag{50}$$

If the gradient of this at S_F is put into Eq. (49), a relation between q_w and $d S_v/dt$ is found:

$$q_L = -q_w + 2 k_L \frac{(T_v - T_m)}{S_L - S_F} + \rho_L L_v \frac{d S_v}{dt} \tag{51}$$

This must be satisfied for all $t > t_v$. At t_v, of course, $S_F = S_A$, and Eq. (47) then shows that $d\,S_v/dt = 0$ as well as $S_v = 0$ at t_v.

The energy balance in the liquid can be found just as we obtained Eq. (41) for $t < t_v$. We must count the energy in the liquid only after t_v, and so on the right of Eq. (41) we must subtract the energy in the liquid temperature profile at t_v. Furthermore, since $T_{LA} = T_v$, the time integral on the right of Eq. (41) can be performed, with S_A replaced by S_F to account for the energy removed by vaporization as well as aerodynamic removal. Then the energy balance becomes

$$q_L(t - t_v) - \int_{t_v}^{t} q_w\,dt = \rho_L c_L (T_v - T_m)(S_F - S_{Av}) + \rho_L L_v S_v$$
$$+ \rho_L c_L \left[\int_{S_F}^{S_L} (T_L - T_m)\,dt - \left(\int_{S_F}^{S_L} (T_L - T_m)dt \right)_v \right] \tag{52}$$

Here S_{Av} is the value of S_A at t_v, and the last term on the first line accounts for the fact that the vapor carries away its latent heat L_v as well as its sensible heat. If the profile Eq. (50) is put into this energy balance, we find, using Eq. (48),

$$q_L(t - t_v) - \int_{t_v}^{t} q_w\,dt = \rho_L c_L (T_v - T_m)(S - S_{Av})$$
$$+ \rho_L c_L (T_v - T_m + L_v/c_L) S_v + \frac{(S_L - S_F)^2}{6 K_L}\left[q_w + \frac{2 k_L (T_v - T_m)}{S_L - S_F} \right]$$
$$- \left\{ \frac{(S_L - S_F)^2}{6 K_L}\left[q_w + \frac{2 k_L (T_v - T_m)}{S_L - S_F} \right] \right\}_v \tag{53}$$

This is a differential equation for $S_L(t)$, $S_F(t) = S_v(t) + S_A(t)$, and $q_w(t)$, which replaces Eq. (44) when $t > t_v$. With the solid Eq. (40), it is two equations for S_L, S_v, and q_w (or I_w). The third equation is the interfacial condition Eq. (51). The initial conditions at t_v are the values of the variables q_w, S_L, and S_A at t_v, and $S_v = 0$. If I_w is used as the variable, then

$$\int_{t_v}^{t} q_w \, dt = \int_{t_{SQ}}^{t} q_w \, dt - \int_{t_{SQ}}^{t_v} q_w \, dt = I_w - I_{wv} \tag{54}$$

where I_{wv} is the value of I_w at t_v.

By eliminating the integral of q_w between Eqs. (35) and (52), taking $S_A = 0$, and evaluating at melt-through, $S_v = S_{vm}$, $S_L = \ell_0$, it can be shown that the present formulation with no melt removal agrees with Eq. (24).

A model of partial melt removal now has been formulated by means of energy balance, for a given one-dimensional melt-removal schedule $S_A(t)$. This schedule will depend on the mechanism of removal. In the next section, we shall propose a model for S_A appropriate for removal by aerodynamic shear which will complete the system of equations and permit solutions to be obtained for partial melt removal.

Model for Melt Removal by Shear

To be compatible with the one-dimensional energy balance, we need a melt-removal model that does not depend on distance along the plate in the flow direction, but only on time and the other parameters of the problem. We shall assume that the size of the heated spot is small compared to the distance from the edge of the plate to the spot in the flow direction. Then the change in aerodynamic shear within the spot is small compared to the value of the shear, since aerodynamic shear changes slowly with flow distance when one is not near the leading edge. So we can take the shear imposed by the flow, τ_i, to be constant in space. The effect of this shear does vary in time, however. The shear does nothing until there is liquid created. As soon as there is liquid, the shear begins to move it.

A simple picture of the effect of shear on a liquid can be obtained by imagining a pool of liquid of depth δ in the z direction, lying above a solid. A shear τ_i at the surface z = 0 is

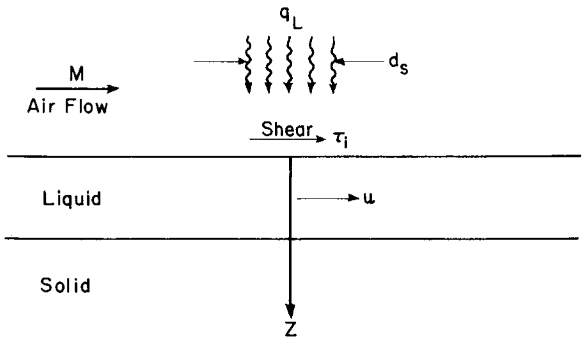

Fig. 5 Cross section of plate with aerodynamically applied shear.

applied at t = 0 (Fig. 5). This sets the liquid in motion parallel to the surface with velocity u (z, t). The equation governing this motion is

$$\frac{\partial u}{\partial t} = \nu_L \frac{\partial^2 u}{\partial z^2} \tag{55}$$

which balances the inertia force with the viscous force. The momentum diffusivity ν_L is the kinematic viscosity of the liquid, which is related to the density ρ_L and dynamic viscosity μ_L by $\nu_L = \mu_L/\rho_L$. The boundary conditions are continuity of shear at z = 0 and no movement of the liquid at z = δ, where we assume the fluid to be in contact with unmelted solid.

The solution to Eq. (55) is obtained easily from Eq. (2), p. 104 of Ref. 4 by recognizing this to be the same as the heating of a finite slab of thickness δ, with the back surface fixed at the initial temperature and the front subject to a heat flux - τ_i, where ν_L is the thermal diffusivity and μ_L the thermal conductivity. The solution is

$$u = \frac{\tau_i \delta}{\mu_L}\left[1 - \frac{z}{\delta} - \frac{8}{\pi^2}\sum_0 \cos\frac{(2n+1)\pi z}{2\delta} \times \frac{\exp[-\pi^2 (2n+1)^2 \nu_L t/4\delta^2]}{(2n+1)^2}\right] \tag{56}$$

The mass flow of liquid through any plane perpendicular to the flow direction is

$$\dot{m} = \int_0^\delta \rho_L \, u \, dz = \frac{\tau_i \delta^2}{\nu_L}$$

$$\times \left\{ \frac{1}{2} - \frac{16}{\pi^3} \sum_0 \frac{(-1)^n}{(2n+1)^3} \exp\left[\frac{-\pi^2 (2n+1)^2 \nu_L t}{4 \delta^2} \right] \right\} \qquad (57)$$

We propose to use this simple model of flow due to shear to find S_A for use in the model of plate heating which we have derived. If we take Eq. (57) as the rate at which liquid can be moved by the shear, we can equate it to the rate at which liquid disappears from the plate by motion of the liquid surface, $d\,S_A/dt$. This rate is related to the diameter d_S of the heated area by

$$\dot{m} = \rho_L \, d_S \, d\,S_A/dt$$

If we equate this to Eq. (57), we find

$$\frac{d\,S_A}{dt} = \frac{\tau_i \delta^2}{\mu_L d_S} \left\{ \frac{1}{2} - \frac{16}{\pi^3} \sum_0 \frac{(-1)^n}{(2n+1)^3} \exp\left[\frac{-\pi^2 (2n+1)^2 \nu_L t}{4 \delta^2} \right] \right\} \qquad (58)$$

and integration gives

$$S_A = \frac{\tau_i \delta^2 t}{2 \mu_L d_S} \left\{ 1 - \frac{5}{12} \frac{\delta^2}{\nu_L t} \right.$$

$$\left. + \frac{128}{\pi^5} \frac{\delta^2}{\nu_L t} \sum_0 \frac{(-1)^n}{(2n+1)^5} \exp\left[\frac{-\pi^2 (2n+1)^2 \nu_L t}{4 \delta^2} \right] \right\} \qquad (59)$$

These provide relations between S_A, δ, and t. They are to be used together with the relation

$$\delta = S_L - S_F = S_L - S_v - S_A \qquad (60)$$

which gives the depth of the liquid layer (see Fig. 4). The combination of Eqs. (59) and (60) provides a transcendental equation for δ at any given time. The integration in time is performed iteratively. To advance a step in time, the differential equations are integrated for S_L, S_v, I_w using the value of S_A from the previous time. Then Eqs. (59) and (60) are solved for δ at the new time with the value of S_L from this integration. Then the integration is repeated over the same time step using the new δ for $S_L - S_A$ in Eq. (44) or for $S_L - S_F$ in Eq. (53). When the dependent variables are the same to a prescribed tolerance in two successive iterations, the integration over that time step is considered completed, and we go on to the next time step.

In this procedure, the expressions for S_A and $d\,S_A/dt$ are used in a quasisteady manner. That is, it is assumed that the adjustment of S_A to each liquid thickness δ occurs instantaneously. In fact, this was assumed tacitly in deriving Eqs. (58) and (59), since δ was held fixed in the solution of the flow problem.

Certainly this model for melt removal is an approximate one. There is no effect of the edges of the heated area, which is beyond the scope of a one-dimensional model. The relation between $\dot{m}$ and $d\,S_A/dt$ supposes the heating to occur in a strip of width d_S but infinite depth normal to the plane of the paper in Fig. 5. And it is a quadisteady model, as just pointed out. Nevertheless, it is believed to include the correct physical phenomena of movement of the liquid by the shearing force applied by the airflow and should provide information about the amount of liquid which can be moved by the shear. It indicates, as expected, that the amount of liquid moved is proportional to the shear, which in turn is an increasing function of the flow Mach number M for any given location on the plate. The effect of the shear will be a balance between M and the heat input q_L. The larger M, the faster the shear can move the liquid. The larger q_L, the more liquid is created. For low M, we expect the shear to be ineffective in removing liquid unless the q_L is very low, and so the melt-through time will be close to t_{mk}. For high M, the shear will be very effective in removing liquid except for very large q_L, so that the melt-through time will be near t_{mr}. At intermediate shears, removal will be effective at

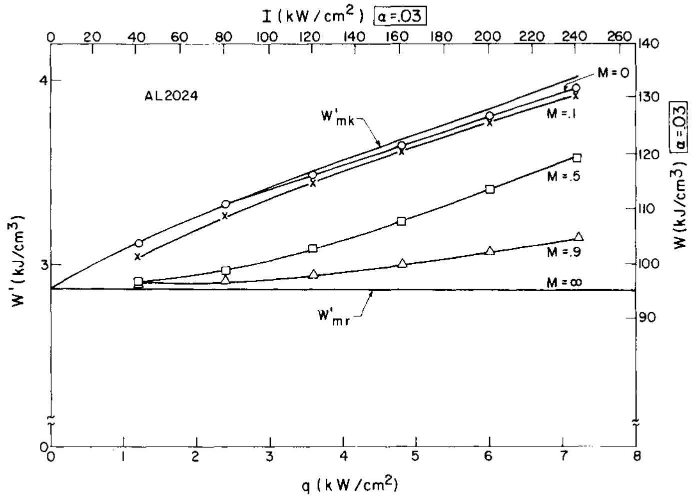

Fig. 6 Melt-through energy as a function of heat flux absorbed, including partial melt removal, for Al 2024.

low q_L, and so we shall be near t_{mr}, and its effectiveness will decrease as q_L increases, and so we shall approach t_{mk}. The results of calculations described in the next section will verify this physical intuition.

Calculations for Partial Melt Removal

Up to the time of writing, partial melt-removal calculations have been made of the melt-through times only for no vaporization of the liquid. They were made for Al, because it has a high value of heat flux at incipient vaporization, $q_v = 15.32$ kW/cm^2, and so a large range of q could be covered. The equations integrated were Eqs. (37) and (44), using Eqs. (58-60) for S_A and d S_A/dt, from t_{SQ} to $S_L = \ell_0$, which is melt-through.

The integration has to be done carefully near $t = t_{SQ}$, which is the zero of time for the integration, because Eq. (44) has a very small factor $\delta^2 = (S_L - S_A)^2$ on the right. This starts at zero and grows very slowly in time. Equation (44) is used to calculate $q_w = d\,I_w/dt$, and this quantity is 0/0 at $t = t_{SQ}$. Series expansions were used to start the integration, but, even so, difficulties were encountered in using a Runge-

Kutta scheme. The equation has a "stiff" character near t_{SQ} because of the division by δ^2, and a scheme suitable for such equations was used.

In the calculations done so far, the relation $q_L = q_S = q$ was used. Now q is related to the laser intensity I by $q = \alpha I$, where α is the absorptivity of the surface, that is, the fraction of the incident laser energy which the surface absorbs. The theory developed permits the absorptivity to change discontinuously when the surface melts, so that $\alpha_S \neq \alpha_L$, but this feature has not been implemented yet in the computer program.

The shear used was that appropriate to an experimental configuration used by the Naval Research Laboratory in heating plates with a 10.6-μm CO_2 laser. It was obtained from the designer of the wind tunnel that produced the flow.[5] The values of the heated area size d_S were obtained from Laird Towle of the Naval Research Laboratory, based on the burn patterns in plexiglas samples. Table 2 gives the parameters for the cases calculated. The shear τ_i depends only on M, and the spot size only on q. In order to relate heat input to peak laser intensity I, a value of absorptivity for Al of α = 0.03 was used. This enters in determining the spot size d_S, which was provided as a function of I.

The results of the calculations are presented in Fig. 6 as a plot of W' vs q, with scales that also give

$$I = q/\alpha, \qquad W \equiv I t_m / \ell_0 = W'/\alpha$$

for absorptivity $\alpha = 0.03$. The bottom line, W_{mr}, is also labeled $M = \infty$, since it represents complete melt removal by very high shear. It is plotted from Eq. (10) and is a constant for $q_L = q_S$. The upper curve is W'_{mk}, from Eq. (20), representing complete melt retention with the static energy balance calculation using the condition Eq. (18). The results of the dynamical integrations for each Mach number are the intermediate curves from M = 0 near the top to M = 0.9 near the bottom. (Notice that the zero of W' is suppressed on the figure.)

The M = 0 curve is actually a dynamical calculation of W'_{mk}. It agrees with the static energy balance very closely for low q but is somewhat lower at higher q, reaching 1.6% lower at $q = 7.2\ kW/cm^2$. This discrepancy is a measure of the approximation incurred by using the condition Eq. (18) instead of performing the time integration and shows this approximation to be a good one. The M = 0.1 curve is only slightly below the M = 0 one, about 1% at high q, and 3% at $q = 1.2\ kW/cm^2$. Clearly it would be impossible to distinguish between these two cases experimentally. At the lower limit, the M = 0.9 curve is 1% above W'_{mr} at low q and 10% above at high q. Again the distinction would be impossible experimentally. The M = 0.5 curve lies in the middle of the W' range and shows the greatest possibility of observing aerodynamic shear effects between the two limiting cases of complete melt retention and complete melt removal.

It must be pointed out that, for Al, over the range of q shown ($1.2 \le q \le 7.2\ kW/cm^2$, $40 \le I \le 240\ kW/cm^2$), the difference between t_{mr} and t_{mk} is never great, varying from 0.034 to 0.026 sec. This small difference is caused by the high conductivity of Al, which prevents the melt from being an effective insulator. It also prevents vaporization from occurring until $q_v = 15\ kW/cm^2$, which is $I_v = 511\ kW/cm^2$

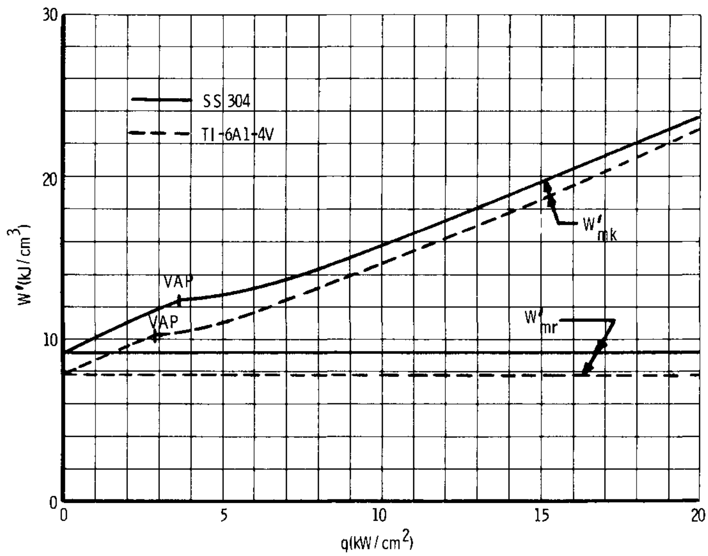

Fig. 7 Melt-through energy as a function of heat flux absorbed for complete melt removal and melt retention (stainless steel 304 and Ti-6A1-4V).

Table 2 Shear and Spot Size

M	τ_i, dynes/cm^2	I, kW/cm^2	0.03 I = q,	d_S, cm
0	0	40	1.2	3.38
0.1	55	80	2.4	2.52
0.5	441	120	3.6	2.28
0.9	1162	160	4.8	2.28
		200	6.0	2.28
		240	7.2	2.28

for $\alpha = 0.03$. This combination of high conductivity and low absorptivity means that large differences between t_{mr} and t_{mk} for Al would be expected only at such large values of I that other physical phenomena, such as plasma waves, probably will occur, and they are not included in the present model. There seems little hope of observing aerodynamic shear effects on Al without the intervention of other phenomena.

The same is not true of stainless steel (SS) or titanium (Ti). No dynamical calculations for $0 < M < \infty$ have been made yet for them, because they have such low conductivities that their values of q_v are 3.72 and 2.88 kW/cm^2, respectively. This means that the vaporization formulation of the differential equations must be used, and that has not been implemented yet on the computer. However, their low conductivity also means a larger difference between t_{mr} and t_{mk}. This can be seen in Figs. 2 and 3, but a more direct comparison can be made by looking at the W' vs q plots of Fig. 7. The spread between the curves is considerably larger for both SS and Ti than for Al of Fig. 6. The location of q_v is shown by the mark labeled "VAP" and shows a discontinuity in slope caused by the onset of vaporization.

It is clear that melt removal has much more scope in which to change W' (or t_m) for SS or Ti and this can occur at lower values of I because of the higher values of α (0.1 for SS, 0.2 for Ti). Therefore, there is a much greater chance of observing aerodynamic melt removal in these two alloys than in Al. However, it probably still is necessary to operate in the correct range of M, not near 0.1 or 0.9. This range will become clearer when dynamic calculations including vaporization are made for SS and Ti.

Conclusions

We have developed a simple one-dimensional model for melt-through of a metal plate subject to high heat flux on the front face, with insulated back face. We considered the cases of complete melt removal, complete melt retention, and partial melt removal by aerodynamic shear due to gas flow along the plate. The method of the heat-balance integral was used,[1] in which the spatial temperature profiles were taken to be quadratic with coefficients determined by appropriate boundary conditions.

For complete melt removal or retention, the melt-through time was obtained by a simple energy balance, using a condition on the time derivative of the liquid-solid interface temperature to avoid time integration. When vaporization occurs, the amount of liquid vaporized up to melt-through was determined using a solution given by Goodman[1] to a simpler vaporization problem. The results of this simple energy balance model agree very well with exact calculations of Nash[3] for the alloys Al 2024, SS 304, and Ti-6Al-4V.

For a partial melt removal by aerodynamic shear, a dynamical model was developed, requiring integration in time. The melt-removal model was obtained by using the flow induced by the sudden application of shear on the surface of a liquid pool of finite depth. Results have been calculated for Al2024, below the heat flux necessary to vaporize, for a geometry appropriate to some Naval Research Laboratory experiments. They show that, at a flow Mach number of 0.1, the shear is too small to reduce the melt-through time below the melt-retained value, whereas at M = 0.9 the shear is so high that the melt-through time is nearly as low as that of the melt-removed case. Only at M = 0.5 was the time nearly intermediate between the limiting cases.

At the time of this writing, partial melt-removal calculations have not been done for SS or Ti because they vaporize at low heat flux, and the vaporization part of the model has not been implemented yet. The case where the heat fluxes before and after front surface melting, q_S and q_L, are not equal has been formulated here, but it also awaits implementation.

The model presented here emphasizes the fact that melt removal has scope to change the melt-through time only between the complete removal and complete retention limits. The difference between these limits is small for high-conductivity metals like Al, because the retained melt is not a good insulator. It is therefore unlikely that melt removal ever can be observed for Al below the incident intensity at which plasma effects become significant. For lower-conductivity metals like SS and Ti, the limits are further apart because of the insulating properties of the melt, and so melt-removal effects may be observable. But the melt-through time will be observably different from the limits only in a limited Mach number range, which can be defined by further calculations with the present model. In any event, the detection of partial melt removal is possible only at the higher heat fluxes, since, at the lower fluxes, the complete removal and retention limits come together.

The models developed show that the melt-through time depends only on the product of the heat flux absorbed and the plate thickness, and so variation of one of these factors can substitute for variation of the other. The methods developed provide a simple and computationally fast tool for calculating melt-through times for use in prediction and data analysis. They do not include certain additional phenomena that may be important, such as combustion (especially of Ti), plasma effects, or lateral spreading of heat. However, a simple correction for the latter effect has been developed by Breaux.[2]

Acknowledgment

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ARC-HEATER CODE-VALIDATION TESTS OF HEAT-SHIELD MATERIALS

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Abstract

Candidate outer-planet entry probe heat-shield materials have been tested in two different arc-heater test facilities. Four different carbon phenolic materials were tested in air in a combined radiative and convective heating facility to evaluate the surface thermochemistry and in-depth thermal response. These materials then were compared, based on their insulative performance. Additional tests of carbon phenolic and fused silica materials were conducted under near-sublimation conditions in a hydrogen/helium environment to compare surface thermochemistry response with predictions. Results indicate that carbon phenolic surface thermochemistry is understood reasonably well. The carbon phenolic surface temperature was predicted within 7% of the measured value using a full equilibrium thermochemistry model for both air and hydrogen/helium environments. In-depth carbon phenolic model temperatures also were predicted with some accuracy. Current modeling of silica thermochemistry in hydrogen/helium is not in agreement with test

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results. The measured silica surface temperatures are 300 to 400 K lower than predicted, and the predicted silica recession rate is approximately half of the experimentally measured value. Changes to the existing silica thermochemistry model, including the possible formation of a silicon monoxide layer, should be evaluated to explain this anomaly.

Introduction

Outer-planet exploration missions to be carried out in the future will include probes that carry scientific instruments into the atmospheres of Saturn, Uranus, and Jupiter for in situ measurements. The heat-shield is one of the most important subsystems and must be effective through an extremely hostile radiative and convective entry heating environment. For these heat-shields, both the surface ablation and the in-depth thermal conduction processes are significant, since the mass of ablator required to maintain the design back face temperature dominates the total weight of the heat-shield system. It therefore is necessary to develop a sufficient data base for the in-depth thermal response of materials proposed for use on the outer-planet probes, when those materials are exposed to high combined heating environments. At the same time, an evaluation of the validity of the surface thermochemistry model for near-sublimation conditions of candidate materials in a hydrogen/helium environment is needed.

Test Facilities and Test Models

Ideally, tests should be conducted at conditions that achieve the surface temperature and surface mass loss rate in the range predicted for outer-planet entry, in a hydrogen/helium gas mixture. Because most ground-test facilities can match these conditions by testing in convective-only environments at excessively high pressures and heat-transfer coefficients, such facilities are not recommended for use. Two facilities were used for the test program. The NASA Ames Research Center's (ARC) Advanced Entry Heating Simulator (AEHS) facility was used to supply the combined convective-radiative heating environment in air, at heating rates below the peak heating rates predicted for entry. The Aerotherm Arc Plasma Generator (APG) was used to study surface thermochemistry in a hydrogen/helium environment for convective-only heating but also at heating levels below flight conditions.

The test program conducted in NASA ARC's AEHS facility consisted of both convective and combined radiative and convective heating tests of carbon phenolic models in air. The primary objective of this test program was to evaluate the in-depth thermal response of several candidate heat-shield materials when exposed to high combined heating environments. Typical stream conditions for the AEHS tests were a stagnation pressure of 2 atm, convective heat flux of 2300 W/cm^2, and a radiative heat flux of 1500 W/cm^2. Two different test models were used. The "infinite slab" model, shown in Fig. 1, consists of a fully instrumented carbon phenolic sample with in-depth thermocouples placed at various distances from the ablating surface, shown in Fig. 2. The "finite slab" model, shown in Fig. 3, is a thin ablator sample of 0.30 in. thickness and is used to obtain data on surface recession and back face temperature response. The carbon phenolic materials tested came

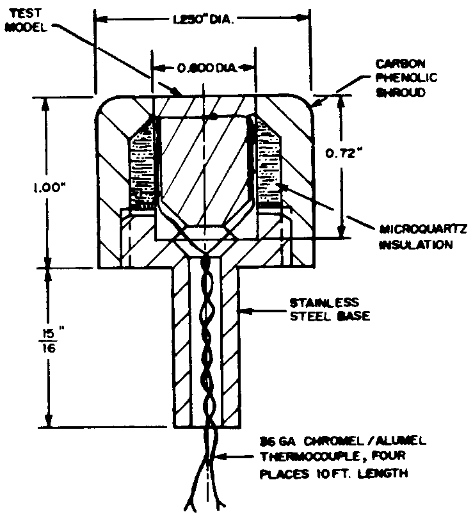
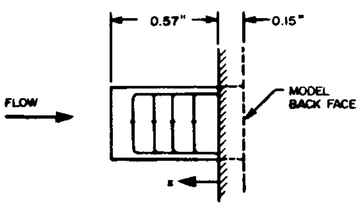


Fig. 1 Infinite slab model design.



MODEL NO.	x, THERMOCOUPLE LOCATION (from interface), in			
	1	2	3	4
CP1-1	—	—	0.235	0.10
CP1-2	0.465	0.34	0.265	0.10
CP1-3	0.465	0.335	0.250	0.10
CP1-4	0.456	0.35	0.255	0.10

Fig. 2 Thermocouple placement for infinite slab models.

from four sources: General Electric Missiles and Space Division, Martin-Marietta Corporation, McDonnell-Douglas Astronautics Company, and Hitco Corporation.

The Aerotherm APG test program consisted of convective heating tests of carbon phenolic and fused silica models in a hydrogen/helium environment. The primary objective of these tests was to validate the surface thermochemistry model of the carbon phenolic and fused silica in hydrogen/helium, in order to give greater confidence in the extrapolation of the analytical ablation model to the higher heating conditions expected in outer-planet entry. The gas composition used was 75 wt% H_2/25 wt% He, which corresponds to the Jupiter nominal model and is also in the range of interest of Saturn and Uranus mod-

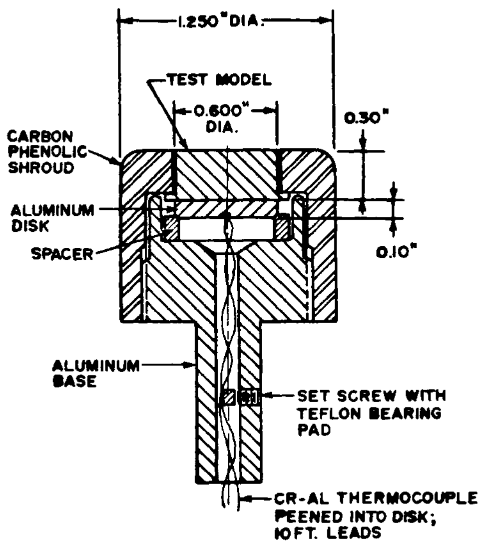


Fig. 3 Finite slab model design.

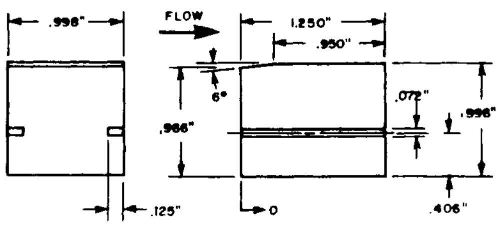


Fig. 4 Detailed dimensions of material test section "B." Note that the nozzle geometry is rectangular (0.15 x 1.0 in.) and these segments insert into the nozzle configuration.

els. The APG stream total conditions were nominally 2 atm stagnation pressure and 115 MJ/kg enthalpy. The carbon phenolic or silica test materials were machined in a two-dimensional nozzle configuration depicted in Fig. 4. The static pressure on the samples in the throat was nominally 1 atm throughout the tests.

Test Results

Test results for the four carbon phenolic materials tested in air in the AEHS, consisting of surface and in-depth temperature variations with time and ablation rates, are shown in Tables 1 and 2 and Figs. 5 - 7. Model CP1-4 had only one thermocouple (No. 4) that exhibited a reading. Therefore, no in-depth temperature traces for this model are shown in Fig. 5. These results are in reasonable agreement with each other. However, performance differences are noted, particularly in back face temperature variation with the finite slab models, as indicated in Fig. 7. The experimental normalized ablation rate was estimated from the centerline recession data, and, in general, the agreement with theoretical predictions is good. Figure 8 illustrates this agreement.

Results for the carbon phenolic and silica models tested in the APG in hydrogen/helium are shown in Table 3 and Figs. 8 and 9. The test results obtained are expressed as normalized ablation rate as a function of surface temperature. For the

Table 1 Centerline stream conditions: AEHS facility

Run no.	Data sequence[a]	P_{t_2}, atm	$\dot{q}_c$, W/cm^2 av	$\dot{q}_r$, W/cm^2 av
152	1	1.95	2300	1550
	2	...	2260	1220
153	1	1.89	2400	1575
	2	...	2320	1220
154	1	1.90	2390	1510
	2	...	...	...
155	1	1.90	2380	0
	2	1.95	2310	0

[a] 1:calibration values before model testing; 2:calibration values after model testing.

Table 2 Carbon phenolic model test results in air: AEHS facility

Model type	Model number	Material type	Run no.	Heating condition	Time exposed, sec	Peak surface temp. ε = 0.85 @ 0.8μ, °R/K	Peak back face temp., °R/K[a]	C_L recession, in.[a]
Infinite slab	CP1-1	GE 20°	152	Convective	7.5	6500/3611	...	...
	CP1-2	GE 20°	152	Combined	7.2	6900/3833	...	...
	CP1-3	GE 20°	153	Convective	7.5	6600/3667	...	...
	CP1-4	GE 20°	153	Combined	7.2	6950/3861	...	...
Finite slab	CP1-5	GE 20°	154	Combined	7.10	6900/3833	660/349[b]	0.082
	CP1-6	GE 20°	155	Convective	7.08	6650/3694	700/371[b]	0.022
	CP2-1	Martin 30°	154	Combined	7.30	6850/3806	960/516[b]	0.086
	CP2-2	Martin 30°	155	Convective	7.04	6240/3467	845/452[b]	0.023
	CP3-1	MDAC 30°	154	Combined	7.22	6685/3714	810/433[b]	0.096
	CP3-2	MDAC 30°	155	Convective	6.82	6200/3444	730/388	0.048
	CP4-1	Random fiber	152	Combined	7.04	6850/3805	852/456	0.063
	CP4-2	Random fiber	153	Convective	7.16	6555/3642	825/441	0.016

[a]Measurements not made for infinite slab models.

[b]Represents temperature at end of data collection period, not true peak back face temperature.

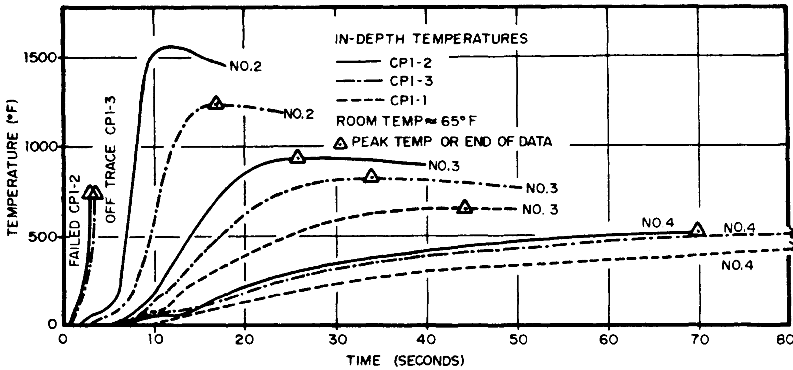


Fig. 5 In-depth temperatures of infinite slab models.

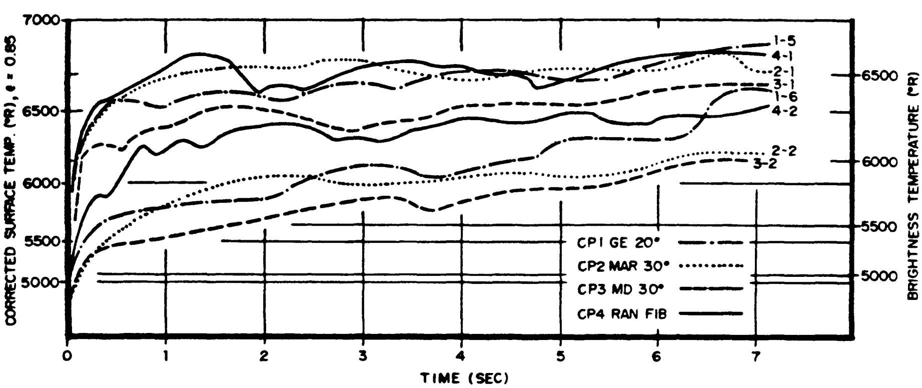


Fig. 6 Carbon phenolic surface temperatures: finite slab models.

carbon phenolic materials, good agreement with theoretical predictions is noted in Fig. 8.

Results for the fused silica materials show that the brightness temperatures are approximately 300 K below the predicted values, as shown in Fig. 9. Several reasons can be speculated for the lack of closure between predicted and measured data for silica: inaccurate modeling of silica ablation properties in hydrogen/helium, uncertainties in the measured silica surface temperatures, and variations in silica surface emissivity are all possibilities. Post-test inspection of the silica models shows large material removal, which indicates a melting mode even though the ablation data appears to fall above the silica melt temperature. Discussions with NASA Ames personnel indicate that these tests were probably in the melting mode, in agreement with data obtained recently at NASA Ames. In the NASA Ames experiments, measured surface temperatures for silica at 1-atm pressure in air were some 400 to 500 K lower than theoretical predictions. At the 1-atm pressure

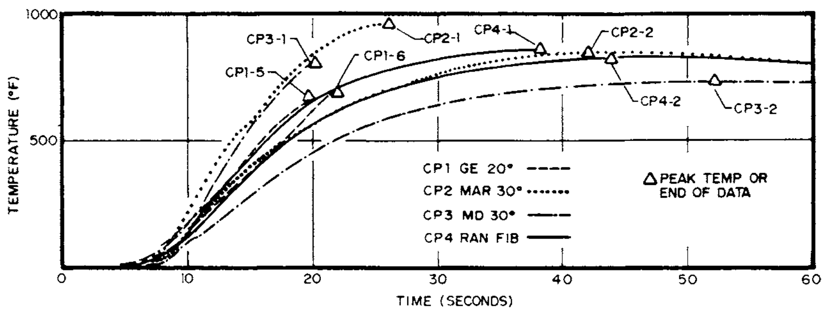


Fig. 7 Carbon phenolic model back face temperatures.

Table 3 Summary of APG test results: H_2/He

Model number	Material	P_{cham} nominal, atm	H_o, MJ/kg	$\dot{q}_{cw}$, W/cm^2	Exposure time, sec	Mass loss, g	$\dot{m}$, g/cm^2-sec
CAL	...	3.58	136.6	2310	...	...	...
002	MX4926	4.00	117.3	↓	5.0	1.215	0.030
003	Carbon phenolic	3.85	117.4	↓	9.0	1.897	0.026
004	↓	3.82	120.0	↓	11.0	1.622	0.018
005	↓	3.90	117.3	↓	13.0	1.977	0.019
006	↓	3.48	99.2	↓	18.0	2.057	0.015
011	Silica	2.45	120.6	↓	12.0	6.368	0.069
012	↓	2.30	118.1	↓	14.0	5.309	0.045
013	↓	2.66	124.1	↓	11.0	5.975	0.067
015	↓	2.57	122.3	↓	8.0	4.650	0.069
016	↓	2.47	121.3	↓	8.0	4.581	0.068

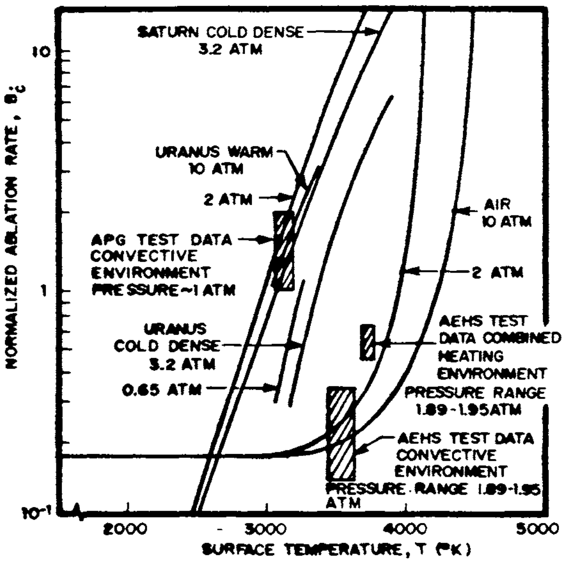


Fig. 8 Graphite surface thermo-chemistry.

test condition, it is likely that vaporization is suppressed and melt is removed mechanically from the silica surface before it can vaporize. The surface measurement technique used in the APG tests also was used at NASA Ames for 1-atm tests in air and gave satisfactory results. A more detailed study of the comparison between test results and the thermochemical model therefore is required.

Comparison of Test Results with Thermochemical Models

The material thermal response of test models CP1-2 and CP1-5 was investigated in detail. Both of these models were tested in air in the AEHS facility and were machined from GE 20° layup MX4926 carbon phenolic. Test model CP1-2 was an infinite slab model with four in-depth thermocouples, whereas test model CP1-5 was a finite slab model with one back face thermocouple. For both models, the centerline recovery enthalpy was calculated from the approximation of Fay and Riddell[1] as 12.1 MJ/kg, whereas the radiant heat flux was assumed to vary linearly from 1550 to 1220 W/cm^2 over the total test time of 7.2 sec. The heat-transfer coefficient was estimated to be 0.063 g/cm^2-sec.

The Aerotherm Charring Material Thermal Response and Ablation (CMA) code[2] was used to calculate the one-dimensional transient surface and in-depth response of model CP1-2. In addition to model dimensions and surface boundary conditions, CMA requires data on decomposition kinetics, pyrolysis gas enthalpy, heat of formation, thermal physical properties, and surface thermochemistry. The thermal physical properties used for carbon phenolic were taken from Ref.3. These properties have been used previously for the analysis of heat-shield thermal protection systems and are listed in Table 4. The comparison between predicted and measured surface and in-depth temperatures for model CP1-2 is shown in Figs. 10 and 11. The predicted surface temperatures are between 5 and 7% below measured values. The in-depth temperature response shows that the predicted temperature values always exceed the measured values; at 70 sec, the predicted temperature for thermocouple

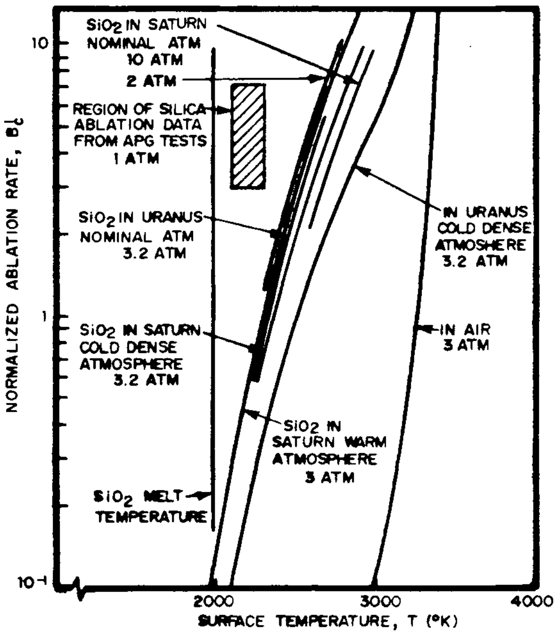


Fig. 9 SiO_2 surface thermochemistry predictions compared with APG test data.

Table 4 Properties of carbon phenolic[4]

	Virgin material			Char		
Temp., °R	Specific heat, Btu/lb-deg	Thermal conductivity, Btu/ft-sec-deg	Emis-sivity	Specific heat, Btu/lb-deg	Thermal conductivity, Btu/ft-sec-deg	Emis-sivity
530	0.21	0.0001722	0.85	0.21	0.0002268	0.85
1160	0.36	0.0002268	↓	0.43	0.000234	↓
2000	0.484	↓		0.484	0.000297	
4000	0.498			0.498	0.001269	
6000	0.500			0.50	0.002713	

No. 4 is 30% above the measured value. From a design standpoint, then, the in-depth temperature predictions are conservative. Similar results were found for test model CP1-5, a finite slab model having one back face thermocouple.

In order to compare the relative performance of the four different carbon phenolics tested, the predicted back face tem-

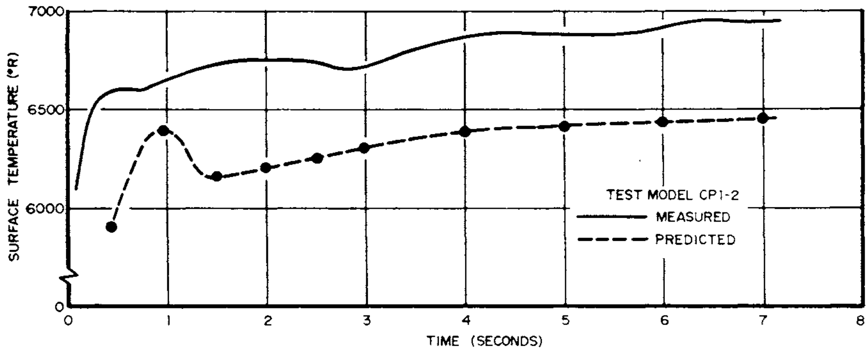


Fig. 10 Comparison of measured and predicted surface temperature response for test model CP1-2 using carbon phenolic properties from Table 4.

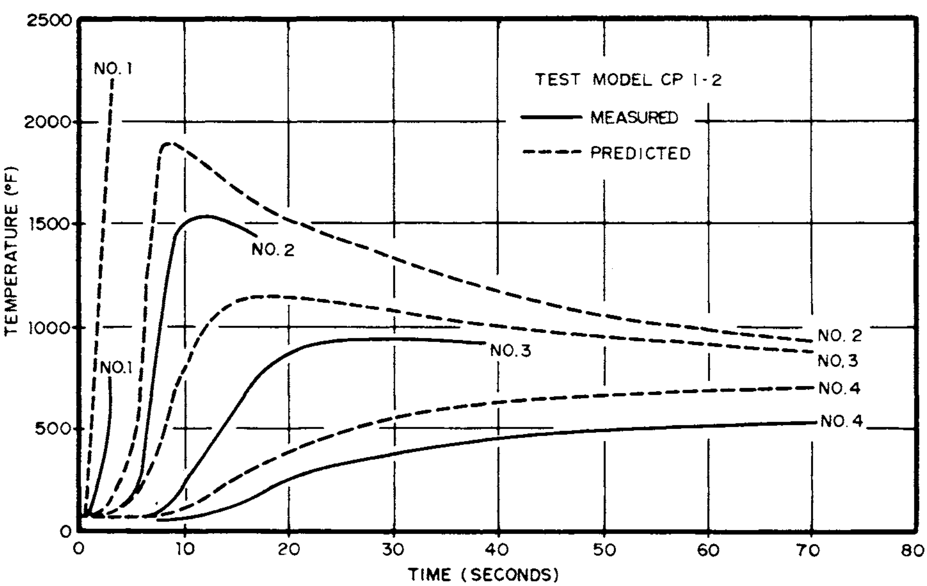


Fig. 11 Comparison of measured and predicted in-depth temperatures for test model CP1-2 using carbon phenolic properties from Table 4.

peratures for models CP1-5, CP2-1, CP3-1, and CP4-1 were compared with their experimental responses. Differences in test time and test condition were found to have a negligible effect on back face temperature measurement. Predictions for the four models were virtually identical, leading to the conclusion that differences in performance were due to differences in material thermal properties. A ranking of the four materials is given in Table 5. The models were compared for their relative insulative performance in both combined and convective heating tests. As noted in Table 5, considerable variation in relative ranking occurred between the two tests. Any additional conclusions regarding relative material performance would be strictly conjecture, because of the limited number of tests.

A further investigation of the thermochemical response of fused silica in a hydrogen/helium environment was conducted to evaluate the discrepancy between measured and predicted results shown previously in Fig. 9. Test model 016 was selected for this purpose. The observed surface temperature for this model was 2110 K. CMA was used to analyze the material thermal response of model 016 assuming an equilibrium surface thermochemistry model with SiO_2 as the condensed phase surface material. No correction of the transfer coefficients for blowing reduction was made in this solution because of the high-molecular-weight species moving into a low-molecular-weight H_2/He boundary layer. This solution corresponds to one of maximum surface temperature with minimum recession rate. The predicted mass loss rate was 0.090 g/cm^2-sec, which is in reasonable agreement with the 0.068 g/cm^2-sec measured. However, the pre-

Table 5 Relative ranking of carbon phenolic materials from AEHS tests

Ranking[a]	Combined heating test	Convective-only heating test
1	Hitco random fiber	MDAC 30°
2	GE 20°	Hitco random fiber
3	MDAC 30°	Martin 30°
4	Martin 30°	GE 20°

[a]Models of rank 1 had the best insulative performance.

dicted surface temperature was 2667 K, which is 556 K greater than measured during the test.

One explanation for the discrepancy between predicted and measured mass loss rate and surface temperature is the formation of silicon monoxide in a high-temperature hydrogen environment by

$$H_2(g) + SiO_2\ (\ell) \rightarrow SiO(\ell) + H_2O(g)$$

Currently, JANNAF supplies data for silicon monoxide vapor but not for the liquid or solid phase. An estimate was made using the "fail surface" thermochemistry model[4] to predict the effect that a SiO surface would have on temperature and mass loss rate results. The fail model assumes that all material is removed mechanically at 3780°R. The enthalpy of liquid SiO was calculated from the JANNAF value of SiO(g) at 2100 K and the heat of vaporization of SiO given in Ref. 5. The effects of blowing reduction also were included. Results give a predicted mass loss rate of 0.088 g/cm^2-sec, in good agreement with the measured value of 0.068 g/cm^2-sec.

Conclusions

For the carbon phenolic models tested in this program in both air and hydrogen/helium, good agreement of mass loss rate predictions with measured values was noted. Surface temperatures in air were predicted within 7%, and in-depth temperature predictions were shown to be conservative when compared to measured values. Design calculations based on these carbon phenolic properties should result in conservative estimates for in-depth temperature.

For the silica models tested in hydrogen/helium, questions still exist. Silicon monoxide thermochemical properties should be included in future silica thermochemistry models as a first step in explaining discrepancies.

Acknowledgement

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ABLATION PERFORMANCE OF TUNGSTEN, COPPER-INFILTRATED TUNGSTEN, AND OTHER SYSTEMS IN ARC JETS

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Abstract

The ablation performance of solid, porous, and copper-infiltrated tungstens, tantalum, a tantalum/tantalum carbide eutectic, and a niobium/niobium carbide eutectic was studied in arc-heated jets at stagnation pressures of 2.2 to 17.9 MPa and bulk gas enthalpies of 4630 to 5890 J/g. Ablation resistance was found to be inversely related to the melting point. The greatest ablation resistance was shown by the solid and copper-infiltrated tungstens. Where flaring occurred (low pressures) and the model face assumed an elliptic shape, the recession rate was related to the radius of the major axis. With solid and copper-infiltrated tungstens, a sequence of shape changes occurred (hemispheric to conic to proboscidian to right cylindrical) in the pressure range of 10 to 18 MPa, indicating that the turbulent flow was reversed to laminar flow. This conclusion was supported by a comparison of the ratio of recession rates under laminar and turbulent flow for graphites and metals. Transition appears to be dependent on the parameter of stagnation pressure/bulk gas enthalpy. The value of this parameter is similar for graphite and solid and copper-infiltrated tungstens.

Nomenclature

A_s = normalized ablation rate, $cm^{3/2}$-g^2/sec-Pa-J^2
$A_s(C)$ = normalized ablation rate for conic-shaped models
$A_s(F)$ = normalized ablation rate for flat or right-cylindrical-shaped models
$A_s(H)$ = normalized ablation rate for hemispheric-shaped models

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$A_s(P)$ = normalized ablation rate for proboscidian-shaped models
b = major elliptic axis, cm
H_o = bulk gas enthalpy, J/g
k_n = proportionality constants
P_{t_2} = stagnation-point pressure, Pa
R_{eff} = effective model radius, cm
R_n = measured nose-tip radius, cm
r = measured model radius, cm
$\dot{s}$ = recession rate, cm/sec
t = time, sec

Introduction

The ablation performance of nose tips fabricated from graphite or carbon composites, although satisfactory in clear air, is inadequate in severe particulate erosion environments containing moisture, ice, or dust. Early attempts to understand particulate erosion resistance indicated that resistance was associated with material density. This observation motivated a study, which is reported in this paper, on the ablation performance of several high-density refractory metal systems. Particular attention was given to solid and copper-infiltrated tungstens, since their performance excelled in a preliminary test program that also included porous tungsten, tantalum, a tantalum/tantalum carbide eutectic, and a niobium/niobium carbide eutectic. ATJ-S graphite was included in the test matrices for comparison.

Testing was performed in the Sandia 2-MW and the McDonnell Douglas HIP arc-heated jet facilities.[1,2] Stagnation pressure and bulk gas enthalpy ranges of 2.2 to 17.9 MPa and 4630 to 5890 J/g, respectively, were used. Results from these facilities are correlated with results obtained by Prototype Development Associates, Inc. (PDA) at the Air Force Flight Dynamics Laboratory (AFFDL) 50-MW RENT facility.[3-5] Sandia provided several tungsten and copper-infiltrated tungsten models for the PDA study.

The ablation resistance for the metal systems that constitute this study was found to be inversely related to the melting point. Flaring of the forward faces on the models (molten metal solidifying at the edges) which occurred with the higher melting materials at the lowest pressures perturbed the constant steady-state rate observed with other metal systems and ATJ-S graphite. The effect on ablation of flaring is shown in terms of a relationship that correlates recession as a function of the change in radius.

The acceptance of an erosion-resistant material should not be at the expense of reduced clear-air performance. Copper-infiltrated tungsten, which meets this requirement, showed the highest ablation resistance of the metals studied. At 2 MPa, its resistance was greater than that for ATJ-S graphite, and in the 10- to 18-MPa range, it was equivalent to this graphite.

Of particular interest was the observation that the turbulent flow condition, which exists at higher pressures for most ablators, reverted to laminar flow for the solid and copper-infiltrated tungsten models. Although the ablation mechanisms for metal systems and graphites are very different, it will be shown that values of the parameter (stagnation pressure/enthalpy) at transition are very similar for both types of ablators.

Experimental

Test Facilities

The Sandia 2-Megawatt[1] and the McDonnell Douglas HIP[2] arc heaters were used to study the ablation performance of the materials included in this report. The Sandia facility consists of a Linde N-1000 arc heater, which exhausts through a supersonic nozzle (Mach 1.7 and throat diameter of 0.70 cm), and a 1.02-cm exit into a test chamber maintained at pressures below 6500 Pa. The McDonnell Douglas HIP facility uses the MDC-200 arc heater, which has a Mach 1.7 nozzle, a 0.953-cm-diam throat, and an exit diameter of 1.14 cm which exhausts into an ambient environment. In both systems, models were injected into the heat exhaust by a Ferguson Intermitter at a fixed position of 0.25 cm from the nozzle exit. A maximum of five models were tested in a single test run. Two injector arms held a pitot probe and a null point calorimeter for pressure and heat-flux measurements. Bulk enthalpies were obtained from energy balance measurements on the arc heater. Stagnation pressures at the Sandia and McDonnell Douglas facilities were 2.2 and 2.4 MPa and 10.6 to 17.9 MPa, respectively. Bulk gas enthalpies were 5540 to 5830 and 4630 to 5890 J/g, respectively.

Materials and Test Models

The solid tungsten was a sintered material with a density of 19.1 g/cm^3 and was obtained from the Thermo Electron Corporation. The porous and copper-infiltrated tungstens were obtained from Teledyne Wah Chang Company. The porosity and density of the porous tungsten were 21% and 17.0 g/cm^3, respectively. The copper-infiltrated tungsten was prepared from

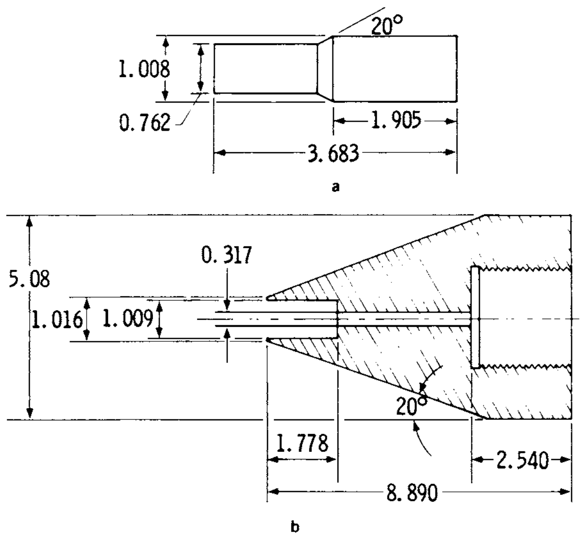

Fig. 1 Model and model holder (all dimensions in centimeters). a) Model used in the low-pressure Sandia 2-MW facility; model used in the McDonnell Douglas facility was modified to a hemisphere-cylinder of the same dimensions. b) Phenolic-Refrasil model holder.

the preceding porous tungsten and contained 18.9% copper by volume. The tantalum/tantalum carbide and niobium/niobium carbide eutectics were obtained from United Technology Corporation. The tantalum was a commercial grade whose source was unknown.

Sketches of the model and model holder are shown in Fig. 1. The right cylindrical model shape in Fig. 1a was used in the tests conducted at low pressures because this was the steady-state shape for graphite and composite materials. A hemispheric cylindrical shape of the same dimensions was used in the higher-pressure range tests conducted at McDonnell Douglas. The model holder (Fig. 1b) was fabricated from phenolic-Refrasil.

Data Reduction

Recession data and shape-change profiles were obtained from high-speed photography films. For the tests conducted at Sandia, the model position was fixed. Recession therefore was measured from the nozzle exit as a reference point or from the film-frame edge. In the tests conducted at McDonnell Douglas, a laser interferometer system maintained a constant nozzle exit to model distance. Recession therefore was measured from a reference point on the model.

Ablation performance was assessed in terms of a parameter given in the following relationship, which normalizes the ablation rate for the given test conditions:

$$\dot{s} = A_s P_{t_2} H_o^2 / r^{1/2} \qquad (1)$$

Evidence that this relationship is applicable to the metal systems considered here was obtained from the ablation data for tungsten generated at the AFFDL 50-MW arc heater.[3-5] Table 1 shows that the values of A_s are reasonably constant over stagnation pressure and gas enthalpy ranges for factors of 1800 and 8, respectively. For these calculations, recovery rather than bulk enthalpies were used.

Results and Discussion

The material loss mechanism for the metals and the metal/metal carbide eutectics is melting and for graphite it is vaporization. It is not surprising, therefore, that the ablation performances of these materials are related to their melting and sublimation points. Figure 2 shows a linear relationship between the A_s values, normalized to that for solid tungsten, and the reciprocal temperatures.

Table 1 Values of A_s for tungsten computed from test data generated at the AFFDL 50-MW arc heater[3-5]

Stagnation pressure, MPa	Recovery enthalpy, J/g	$A_s \times 10^{16}$, $cm^{3/2}$-g^2/sec-Pa-J^2
0.0070	50,200	0.962
0.0070	53,220	0.786
0.0078	65,300	0.889
0.0057	58,560	1.18
2.13	8,570	0.674
6.18	13,900	0.650
9.62	14,660	0.700
6.08	13,870	1.19
8.10	14,470	0.913
10.13	14,850	0.721

Recession was linear for tantalum, the metal/metal carbide eutectics, and ATJ-S graphite once thermal equilibrium was established. However, the solid, porous and copper-infiltrated tungstens were not linear for tests conducted at pressures of about 2 MPa. Deviations from linearity were correlated with flaring and the resultant elliptic shape. The correlation is provided in terms of Eq. (7), which is derived from the following relationships.

If it is assumed that

$$\dot{s} = k_1 q \tag{2}$$

then, from the Fay-Riddell relationship,[6]

$$\dot{s} = k_1 q = k_2 (P_{t_2}/R_{eff})^{1/2} H_o \tag{3}$$

When P_{t_2} and H_o are constant,

$$\dot{s} = k_3/R_{eff}^{1/2} \tag{4}$$

where $k_3 = k_2 P_{t_2}^{1/2} H_o$. For an elliptically shaped model, McBride[7] has shown that R_{eff}/b can be related to R_n/b. Furthermore, R_{eff}/b is essentially constant for values of R_n/b greater than about 50 when the Mach numbers are in the range of 1.5 to 2.0. Thus, the relationship $R_{eff} = k_4 b$ is applicable to the results of this study, and

$$\dot{s} = k_3/k_4 b^{1/2} = k_5/b^{1/2} \tag{5}$$

To integrate Eq. (5), it is necessary to relate b to t. It was ascertained that

$$b^{1/2} = k_6 t^{1/2} \tag{6}$$

A representative plot (Fig. 3) shows that the relationship is linear once thermal equilibrium is established and during the time interval that b increases. When Eq. (5) and (6) are combined, integrated, and the boundary conditions (s = 0 when t = 0) eliminated, then

$$\dot{s} = k_7 t^{1/2} \tag{7}$$

where $k_7 = (2k_5/k_6)$. The adequacy of Eq. (7) is shown in Fig. 4 for solid tungsten. For comparison, s vs t also is plotted.

The sequence of shape changes observed for solid and copper-infiltrated tungstens in the tests conducted at stagnation pressures of 10.6 to 17.9 MPa was of particular interest. Both indicated that an initial turbulent flow condition reverted to a steady-state condition of laminar flow. The shape change sequence for solid tungsten was hemispheric to conic to proboscidian at 12.8 MPa for one model, and hemispheric to conic to proboscidian to conic for another model. This fluctuation suggests that the turbulent flow field was not stable.

At 17.9 MPa the final shape, subsequent to the proboscidian, approximated a right cylinder. For copper-infiltrated tungsten, the sequence was hemispheric to conic to right cylindrical. These sequences are the reverse of those observed for graphite and carbon composites, during which laminar flow undergoes transition to turbulent flow. Figure 5 shows representative sequences of shape changes for these metal systems.

Evidence for reverse transition also is found in a comparison of the ablation rates for the metals and graphites prevailing during each shape and the ratios of these rates.

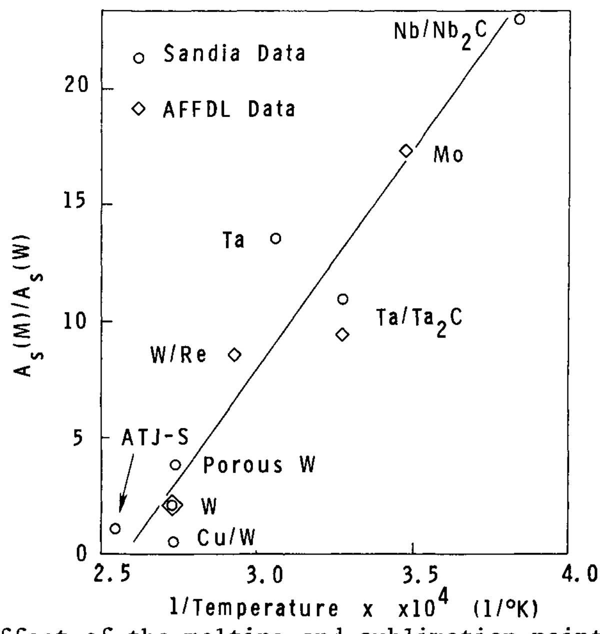

Fig. 2 Effect of the melting and sublimation points on the ablation rate. The A_s values for the various materials (M) have been normalized to that for tungsten (W). Nb-Nb_2C: niobium/niobium carbide eutectic; Mo: molybdenum; Ta-Ta_2C: tantalum/tantalum carbide eutectic; W-R_e: tungsten rhenium; Cu-W: copper-infiltrated tungsten.

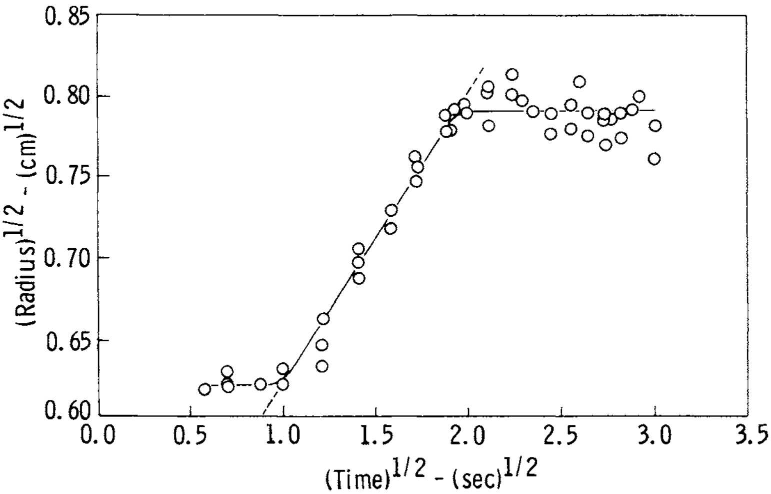

Fig. 3 Linear relationship between the major elliptic axis (b) and time for solid tungsten ablation at a stagnation pressure of 23.9 MPa and bulk gas enthalpy of 5560 J/g.

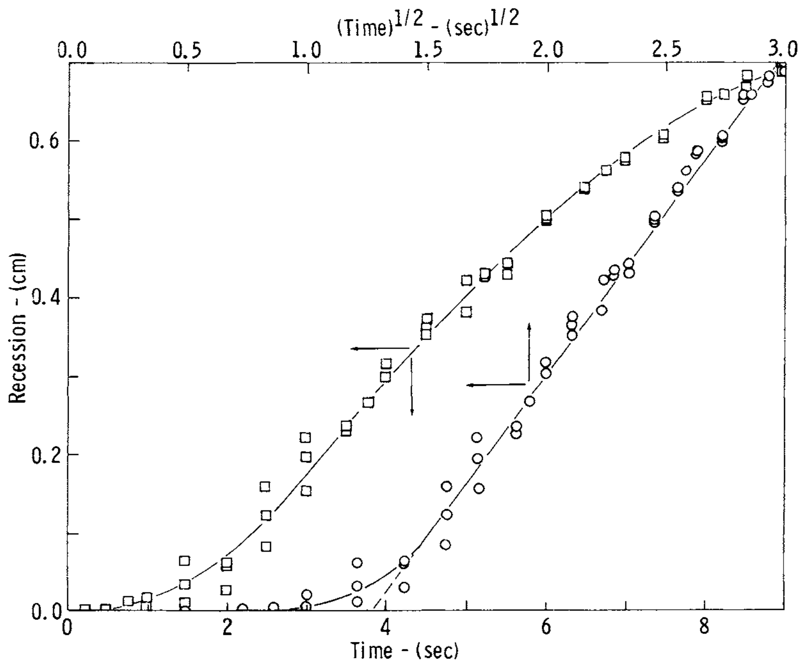

Fig. 4 Linear relationship between recession and (time)$^{1/2}$ provides evidence for Eq. (7). For comparison, a plot of recession vs time is shown for the same plot. Test material and conditions are the same as those for Fig. 3.

Figure 6 shows representative plots of recession vs time for solid and copper-infiltrated tungstens. The times when shape changes were observed are also noted. It is evident that the plots are segmented and that discontinuities occur at the same times that shape changes take place when the time period for a particular shape is sufficiently long.

The plot for copper-infiltrated tungsten illustrates the decrease in recession rate which occurs when the shape changes from conic to flat-faced. The time period for the hemispheric shape was too short and too close to initial conditions to show measurable rate changes.

The plot for solid tungsten does not come from the same model whose shape change sequence is illustrated in Fig. 5. Instead, it represents a model whose shape vacillates between conic and proboscidian configurations. The period up to 0.65 sec encompasses a time when the shape changes several times. The line, therefore, represents an average value. Beyond 0.65 sec, the model is conic and the linear plot indicates an increased constant rate.

Table 2 lists the ratios of the normalized ablation rates for several graphites under initial laminar conditions, $A_s(H)$, while the model faces were still hemispheric, and under subsequent turbulent conditions, $A_s(C)$, when the model faces had become conic. The corresponding value of the ratio $A_s(C)/A_s(F)$ for copper-infiltrated tungsten which became right cylindrical

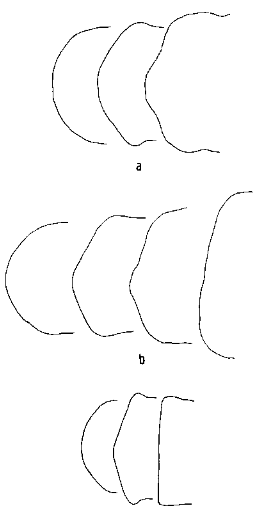

Fig. 5 Shape change sequence. a) solid tungsten, stagnation pressure 12.7 MPa, bulk gas enthalpy 4650 J/g; b) solid tungsten stagnation pressure 17.9 MPa, bulk gas enthalpy 5720 J/g; c) copper-tungsten, stagnation pressure 12.7 MPa, bulk gas enthalpy 4650 J/g.

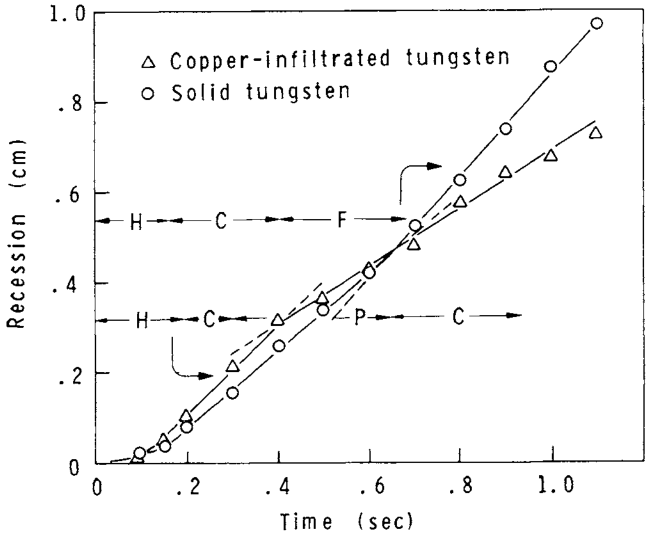

Fig. 6 Effects of solid and copper-infiltrated tungsten shapes on ablation rate. H: hemispheric cylinder; C: conic; F: flat or right cylinder. Models were tested at a stagnation pressure of 12.7 MPa and bulk gas enthalpy of 4650 J/g.

Table 2 Comparison of the normalized ablation rates under turbulent and laminar flow

Material	Number of models tested	Mean $A_s(C)/A_s$ (H, F, or P)	Standard deviation
ATJ-S graphite	8	2.02	0.45
IP graphite	12	2.11	0.49
Molded graphite	6	2.20	0.61
Copper-infiltrated tungsten	8	2.14	0.44
Solid tungsten	1	1.36	...

or flat-faced falls within the range of values provided by the graphites, thereby providing evidence that reverse transition does indeed take place.

The steady-state shape for solid tungsten is proboscidian at a stagnation pressure of 12.8 MPa which is an intermediate

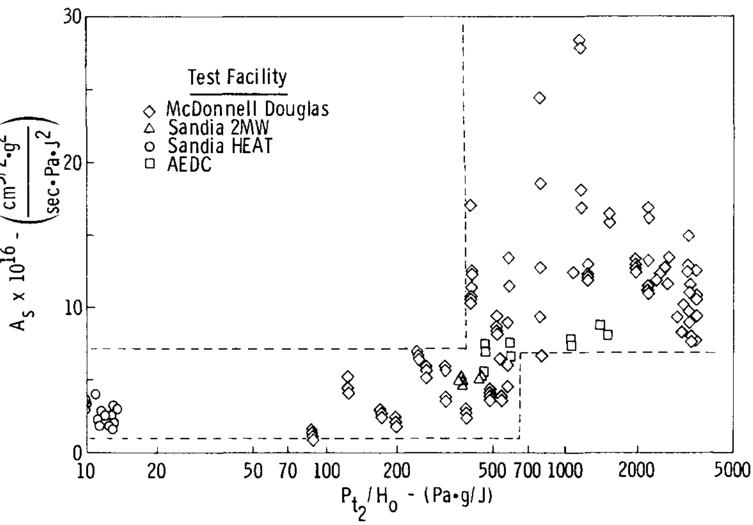

Fig. 7 Effect of the parameter P_{t2}/H_0 on the normalized ablation rate (A_s) for ATJ-S graphite.

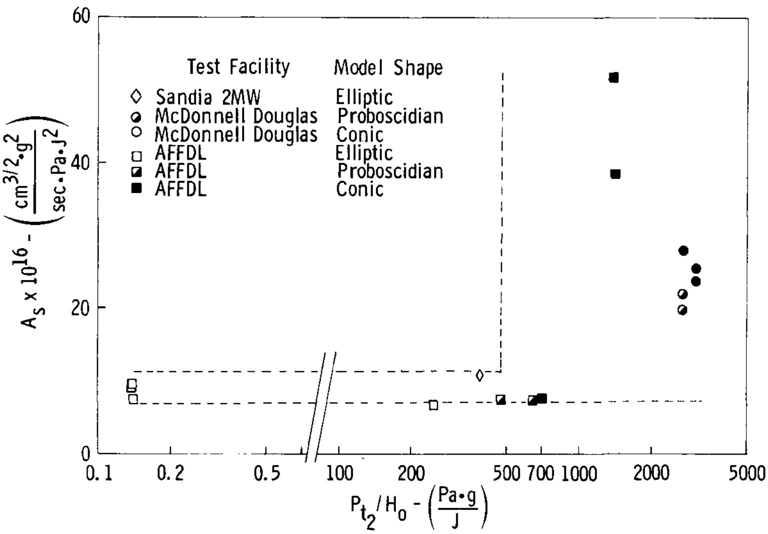

Fig. 8 Effect of the parameter P_{t2}/H_0 on the normalized ablation rate (A_s) for solid tungsten.

shape during transition. It is not surprising, therefore, that the value of the ratio $A_s(C)/A_s(P)$ is intermediate to the value for no transition (1) and transition (~2). Although solid tungsten models were right cylindrical at the end of the tests conducted at a stagnation pressure of 17.9 MPa, sufficient recession data were not available to calculate rates.

Table 3 Comparative ablation rates for tungsten, copper-infiltrated tungsten, and ATJ-S graphite

Material	Number of models tested	P_{t_2}	H_o	$A_s(P)$, $cm^{3/2}$-g^2/sec-Pa-J^2	$\hat{s}^a$	$A_s(C)$, $cm^{3/2}$-g^2/sec-Pa-J^2	$\hat{s}$	$A_s(F)$, $cm^{3/2}$-g^2/sec-Pa-J^2	$\hat{s}$
Solid tungsten	3	2.4	5540	...	...	...	...	0.983×10^{-15}	0.028
	2	12.8	4530	2.04×10^{-15}	0.15×10^{-15}	2.77×10^{-15}	...	...	...
	2	17.9	5720	...	...	2.42	0.12×10^{-15}	...	...
Copper-infiltrated tungsten	3	2.2	5670	...	...	...	...	0.336	0.033
	3	10.6	5410	...	...	2.59	0.17	1.35	0.09
	3	12.8	4530	...	...	2.36	0.035	1.36	0.09
	3	17.9	5720	...	...	2.24	0.14	0.811	0.09
ATJ-S graphite	9	2.3	5830	...	...	...	...	0.490	0.025
	3	10.6	5360	0.804	0.059	1.30	0.04	...	...
	3	12.7	5000	...	...	1.25	0.09	...	...
	4	16.5	5390	...	...	0.966	0.121	...	...
	3	20.6	5890	...	...	0.899	0.126	...	...

[a] $\hat{s}$ = standard deviation.

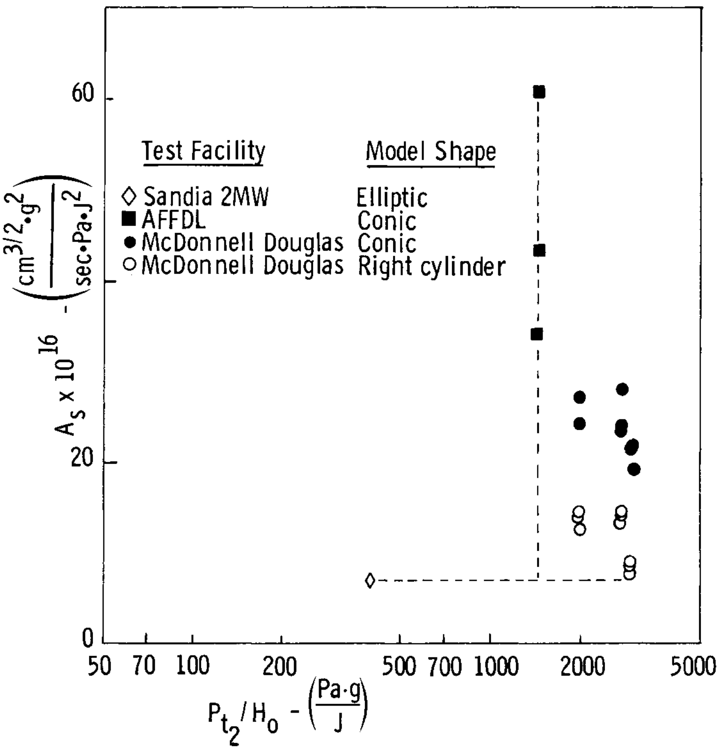

Fig. 9 Effect of the parameter P_{t_2}/H_0 on normalized ablation rate (A_s) and shape for copper-infiltrated tungsten.

The existence of laminar flow under these high-stagnation pressure and moderate bulk-gas enthalpy conditions is unusual. Since transition is a function of the roughness of the ablator surface as well as the environment, it is assumed that the smoother surface of molten metal, in contrast with that of graphite, is the critical factor for the existence of laminar flow in the later stages of ablation. It also is assumed that the metal surfaces are sufficiently rough to initiate transition during the early ablation stages. The low melting point of copper in the copper-infiltrated tungsten provided a more fluid surface than that for solid tungsten; hence, the complete sequence, conic to right cylindrical, could occur at the lower pressure of 10.6 MPa. For tungsten, the complete sequence did not occur except at the highest heat-flux condition exiting at 17.9 MPa.

It is of interest that the conditions under which transition occurs for the metals early in the tests are similar if not the same as for graphite. Figure 7 is a plot of A_s vs

P_{t2}/H_o for steady-state ablation of ATJ-S graphite. It is obvious that a much higher range of A_s values occurs above a value of 400 Pa-g/J. Associated with this and higher values, proboscidian and conic shapes become prevalent. Beyond the value of 600 Pa-g/J, the right cylinder shape associated with laminar flow no longer is observed.

Solid and copper-infiltrated tungstens indicate a similar dependence on P_{t2}/H_o. Figure 8 shows that the threshold value for solid tungsten is also about 400 Pa-g/J. The limited data available for copper-infiltrated tungsten (Fig. 9) indicate that they are not inconsistent with the results for graphite and solid tungsten.

This study was initiated to determine the ablation resistance as well as the ablation characteristics of several metal systems in clear air which have the potential for particulate erosion-resistant nose tips. Table 3 summarizes the ablation rates observed in this study for tungsten and copper-infiltrated tungsten in the pressure range of 2.2 to 20.6 MPa. ATJ-S is included for comparison. When comparisons are made for relative performance, the corresponding steady-state shapes should be considered. At 2 MPa, they would be right cylindrical for ATJ-S and the metals; in the 10.6 to 20.6 MPa range, the corresponding steady-state shapes would be conic for ATJ-S, proboscidian for solid tungsten, and right cylindrical for copper-infiltrated tungsten. It is evident that the ablation resistance of copper-infiltrated tungsten is better than that for ATJ-S graphite at 2 MPa and equivalent to that for ATJ-S graphite in the 10.6- to 17.9-MPa range. Solid tungsten is less resistant than ATJ-S graphite in both pressure ranges. The better performance of copper-infiltrated tungsten over solid tungsten is due to the improved thermal conductivity and transpiration cooling effect imparted by the copper. The overall performance of copper-infiltrated tungsten suggests that it would be suitable for both clear air and particulate erosion environments.

Conclusions

1) The ablation resistance performances of the metal systems studied are proportional to their melting points.

2) Where flaring occurs and the model face assumes an elliptic shape, the recession rate can be related to the radius of the major axis, as well as the environmental conditions.

3) Solid and copper-infiltrated tungstens instigate reverse transition (turbulent to laminar flow) in the stagnation pressure range of 10 to 18 MPa.

4) Transition for the preceding metals and graphite appears to be dependent on P_{t2}/H_o.

5) Copper-infiltrated tungsten shows greater resistance to ablation than ATJ-S graphite at 2 MPa and is equivalent to ATJ-S graphite in the pressure range of 10 to 18 MPa.

Acknowledgment

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IN-DEPTH OXIDATION DISTRIBUTION OF REINFORCED CARBON-CARBON MATERIAL

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Abstract

A method was developed for predicting oxidation distribution through the thickness of reinforced carbon-carbon material in an Earth atmospheric entry environment. A mathematical model was developed which describes oxygen diffusion and reaction rates within material pores. Pertinent rate constants were derived, and material mass loss was computed for a range of temperatures and pressures. Results indicate that both temperature and pressure have an important effect on mass loss distribution. Analytical results were quite consistent with results of ground oxidation tests.

Nomenclature

A, B	=	pre-exponential constants for CO and CO_2 product reaction, lb/ft^2-sec
A_e, A_f	=	area, external surface, fissure flow, ft^2
C	=	oxygen concentration, dimensionless
C_p	=	specific heat, Btu/lb-^{o}F
D	=	diffusion coefficient, ft^2/sec
E	=	defined constant[9]
E_w, E_{CO}, E_{CO_2}	=	activation energies for product transition and CO and CO_2 product reaction, kcal/mole
f(E)	=	defined constant[9]
f_o	=	fraction of molecules undergoing diffuse reflection, dimensionless
f(P)	=	function defining pressure dependency of K', dimensionless
K'	=	reaction rate constant for oxygen consumption, lb/ft^2-sec

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K = thermal conductivity, Btu/ft-sec-°F
K_1 = shape factor, dimensionless
L = pore length, ft
L_e = Lewis number, dimensionless
m = hydraulic radius, ft
$\dot{M}$ = mass loss rate per unit of external surface area, lb/ft^2-sec
$\dot{M}'$ = mass loss rate per unit of pore wall surface area, lb/ft^2-sec
MW = molecular weight, dimensionless
n = reaction order, dimensionless
n_p = number of pore mouth openings at substrate coating interface per unit of external surface area, ft^{-2}
P = total pressure of air, atm
q = tortuosity, dimensionless
r = pore radius, ft
R = gas constant, kcal/mole-°R
R_{ox} = ratio of oxygen to carbon consumed, dimensionless
T = temperature, °R
$\overline{V}$ = average molecular speed, fps
V = pre-exponential constant for product transition, dimensionless
W = fraction of reaction area forming CO, dimensionless
X = displacement along pore path, ft
X_o = value of X at which C = 0 (oxygen penetration depth), ft
y = straight-line depth into substrate, ft
y_o = depth to end of pore, ft
δ^o = $(2-f_o)/f_o$
δ^o_s, δ_c = substrate and coating thickness, respectively, ft
ΔM = cumulative mass loss, psf
$\Delta\rho$ = density reduction, lb/ft^3
ε = porosity, dimensionless
ρ = gas density, lb/ft^3
ρ_S = theoretical density of nonporous substrate, lb/ft^3

Subscripts

c = carbon, coating, continuum
e = freestream
f = fissure
i = segment number
I = initial value for unoxidized substrate
L = value at end of pore

0	= value at coating-substrate interface
ox	= oxygen
p	= pore
s	= substrate

Introduction

The Space Shuttle leading-edge thermal protective system is constructed of reinforced carbon-carbon (RCC) material developed by Vought Corporation. This material consists of a laminated carbon-carbon substrate with a silicon carbide oxidation inhibiting coating diffused into all surfaces. Plasma arc tests[1] have shown that oxidation rate of this material is orders of magnitude below that for uncoated RCC. Nevertheless, coated RCC is susceptible to moderate oxidation. The objective of this study was to develop an analytical method, supported by test data, for predicting oxidation distribution through the substrate thickness of coated carbon-carbon-type materials. The analytical and test data presented herein apply to a specific developmental stage of RCC, which included certain minor imperfections. This material is similar, but not identical, to that for the Space Shuttle.

Previous studies of RCC oxidation[2] were concerned with overall mass loss rate, without regard to mass loss distribution within the substrate. This distribution has an important effect upon material integrity. If the mass loss is distributed relatively uniformly, then the effect will primarily be a reduction in mechanical load-carrying capability of the substrate. If the mass loss is concentrated near the coating-substrate interface, then the primary effect will be a reduction in strength of the coating-substrate bond.

It is extremely difficult, if not impossible, to measure mass loss distribution directly within oxidized test specimens and to correlate it with oxidation test conditions in such a manner as to permit entry mission predictions. A mathematical model for computing substrate mass loss distribution, therefore, was developed. This model was used to correlate overall mass loss rates from oxidation tests on RCC in order to infer required rate constants as a function of temperature and pressure. Predictions then were made of mass loss distribution for a range of temperatures and pressures and were compared with post-test specimen examinations.

Oxidation Model

Oxygen diffuses across the coating and into the substrate through pores and minute fissures in the coating and pores in the substrate, as shown in Fig. 1. Oxidation occurs on the substrate pore walls, and the oxidation products diffuse back through the pores and fissures to the freestream. At a given temperature and pressure, oxidation rate of the substrate is determined by the oxygen concentration profile along the pores. This is, in turn, controlled by a balance between oxygen diffusing down the pore and oxygen consumed by substrate reaction. The problem is complicated by radius growth and increase of diffusion and reaction areas as the pore wall oxidizes. Closed-form solutions were obtained for substrate mass loss distribution for a constant pore radius. A finite-difference solution then was developed for progressive reaction area and was programmed for computer solution.

Several simplifying assumptions were made. Oxygen was assumed to be the only reactant at temperatures of interest to the Space Shuttle (2700°F maximum), nitrogen being considered inert. The coating was assumed to be inert at temperatures up to 1800°F. At higher temperatures, the coating reacts with oxygen to form a product film that partially blocks oxygen diffusion across the coating. This blockage was taken into account. Oxygen consumption due to reaction with the coating was assumed to be negligible.[2]

Mass Loss Rate for Constant Pore Size

Cross-sectional photomicrographs of RCC reveal that individual pores take torturous paths through the substrate.

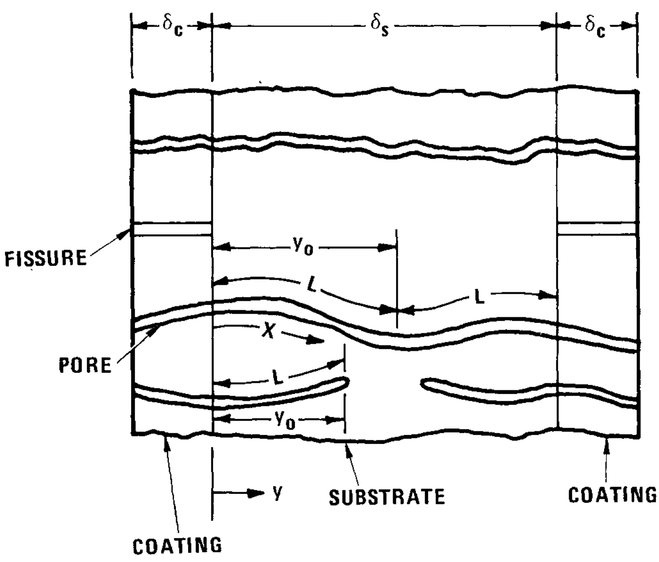

Fig. 1 Oxidation model.

Oxygen diffusion and reaction were analyzed for a single pore of constant diameter, and total oxidation rate was determined from the product of rate per pore and number of pores. Oxygen penetration distance and reaction rate profile were analyzed as a function of displacement along the pore length. This approach is similar to that of Blyholder and Eyring,[3,4] although the present solutions are more general. Mass loss as a function of depth normal to the surface is related to $\dot{M}(x)$ through a tortuosity constant $q = dx/dy$.

Oxygen concentration profile was determined from a mass balance between oxygen diffusing down the pore and oxygen consumed by reaction on the carbon pore wall:

$$\pi r^2 \overline{\rho D} d^2C/dx^2 = 2\pi r K' C^n \tag{1}$$

The boundary conditions are

$$C = C_o \text{ at } X = 0 \tag{2}$$

$$dC/dx = 0 \text{ at } X = L \tag{3}$$

Two possibilities must be considered: the oxygen may be consumed completely before reaching the end of the pore ($C_L = 0$), or it may not be consumed completely ($C_L < 0$).

Mass Loss Rate for $C_L = 0$

The solution to Eq. (1) for this case is

$$C = \left\{ C_o^{(1-n)/2} - x(1-n)\left(\frac{K'}{(n+1)r\overline{\rho D}}\right)^{1/2} \right\}^{2/(1-n)} \tag{4}$$

Total substrate mass loss rate per unit of external surface area is

$$\dot{M}_s = \frac{2\pi r n_p C_o^{(n+1)/2}}{R_{ox}} \left(\frac{r\overline{\rho D}K'}{n+1}\right)^{1/2} \tag{5}$$

This equation indicates that pore diffusion has a strong effect on substrate oxidation kinetics. The pressure sensitivity of mass loss rate is a function of $\overline{\rho D}$ as well as of K'.[3,4]

Oxygen penetration depth is determined by solving Eq.

(4) for X with C = 0:

$$X_o = \frac{C_o^{(1-n)/2}}{(1-n)} \left[\frac{(n+1)r\rho D}{K'}\right]^{1/2} \tag{6}$$

Penetration depth is seen to be a function of the ratio of oxygen diffusion rate to reaction rate $\overline{\rho D}/K'$. Diffusion rate is relatively insensitive to temperature, whereas K' increases strongly with temperature. Both $\overline{\rho D}$ and K' increase with pressure, but $\overline{\rho D}$ increases faster, as will be shown. Hence, we expect that at low temperature and high pressure, oxygen will penetrate relatively deeply into the substrate; whereas at high temperature and low pressure, the penetration will be shallow.

There is an additional effect of temperature upon X_o because of its effect upon oxygen concentration at the coating-substrate interface. This concentration is obtained by a balance of oxygen diffusing across the coating and that consumed by reaction:

$$C_o = C_e - \dot{M}_s \delta_c R_{ox} / (\overline{\rho D}\ A_{flow}/A_e) \tag{7}$$

where A_{flow}/A_e is the ratio of flow area across the coating to external surface area.

At low temperatures, K' is low, $\dot{M}_s$ is low, and C_o approaches C_e. In this reaction rate control regime, $\dot{M}_s$ is controlled by oxidation kinetics. As temperature increases, K' and $\dot{M}_s$ increase, and C_o decreases. In this transition regime, $\dot{M}_s$ is controlled by both oxidation kinetics and oxygen diffusion rate across the coating. At high temperature, K' and $\dot{M}_s$ are large, and C_o becomes small compared to C_e. In this diffusion control regime, $\dot{M}_s$ is controlled by diffusion rate. This reduction in C_o with increasing temperature will lead to a reduction in X_o, as shown by Eq. (6).

A review of carbon oxidation literature reveals values of n from 0 to 1. For temperatures and pressures of interest here, experimental data [5-8] indicate that n is near 1/2. For this value of n, substrate reaction rate profile is given by

$$\dot{M}_s' / (\dot{M}_s')_{x=0} = (1 - X/X_o)^2 \tag{8}$$

where

$$(\dot{M}_s')_{x=0} = C_o^{1/2} K'/R_{ox} \tag{9}$$

At a depth into the pore of $(1/2)X_o$, $\dot{M}_s'$ is only 25% of that at the interface.

Mass Loss Rate for $C_L>0$

An approximate solution to Eq. (1) was derived[9] for this case. Oxygen concentration and substrate reaction rate profile are given by

$$(C/C_o)^{1/2} = \dot{M}_s'/(\dot{M}_s')_{x=0} = 1-Ef(E)X/L+3\left[f(E)-1\right](X/L)^2 \quad (10)$$

where E and f(E) are functions of the ratio of reaction rate to diffusion rate.[9] This relation is shown in Fig. 2. Profiles are presented for three values of C_L corresponding to different ratios of reaction rate to diffusion rate. As reaction rate increases, the mass loss profile changes from a uniform loss for $C_L = C_o$ to a steep reduction in mass loss with increasing depth for $C_L = 0$. Comparative results from a finite-difference solution, presented in the next section, show good agreement with Eq. (10) for E = 0.832.

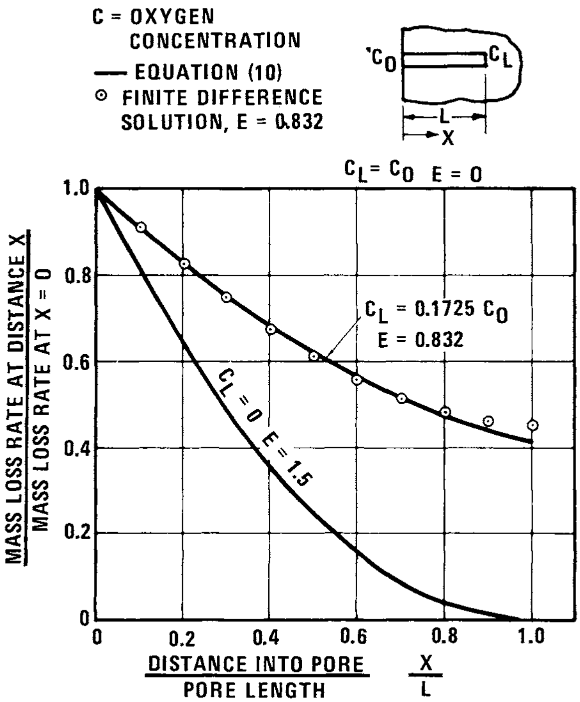

Fig. 2 Mass loss distribution, constant pore size ($C_L > 0$).

Progressive Reaction Surface

Thus far, the analysis has considered a pore radius of some specified value. As the pore wall oxidizes, its radius will increase, increasing the reaction surface area and diffusion area. Hence, for the reaction-controlled and transition oxidation regimes, substrate oxidation rate will increase with time. This is important for predicting in-depth oxidation and for inferring rate constants from test data.

Radius growth rate generally will vary from a maximum at the pore mouth to a minimum at the pore end. Hence, even an initially uniform radius pore eventually will become nonuniform. This was analyzed by dividing the pore length into finite segments and computing the growth of each segment, concentration profiles, and mass loss rate using the method of finite differences. Concentration profile at zero exposure time was computed using equations from the constant pore size solutions. An iterative procedure was used to define concentration profile and mass loss rate at each of a series of finite time steps using finite-difference equations derived from Eqs. (1-3 and 7). Radius growth for each pore segment and time step was computed from

$$dr_i/dt = (C_i)^{1/2} K' / R_{ox} \rho_s \tag{11}$$

The reduction in apparent or bulk density was evaluated from

$$\Delta\rho_i = \pi n_p \rho_s q \left(r_i^{\,2} - r_I^{\,2}\right) \tag{12}$$

A digital computer routine was used to perform numerical calculations.

Evaluation of Rate and Pore Structure Constants

Theoretical (nonporous) substrate density was estimated from bulk density and porosity to be 103.4 lb/ft^3. Initial pore radius was taken from porosimeter measurements, which gave an average value of 2 X 10^{-5} in. Coating thickness is 0.025 in.

Oxidation Product

The ratio of oxygen to carbon consumed by substrate oxidation is a function of the relative amounts of CO and

CO_2 product[9]:

$$R_{ox} = 4/3 \left[(2+CO/CO_2) \ / \ (1+CO/CO_2) \right] \tag{13}$$

The ratio CO.

The ratio CO/CO_2 was defined[9] by use of experimental data[5] and theoretical considerations[10, 11]:

$$CO/CO_2 = 5.834 \times 10^{16} \ e^{-62,510/T} \tag{14}$$

At temperatures above 1400°F, Eq. (14) gives CO as the primary product, and R_{ox} approaches 4/3. At temperatures below 1000°F, CO_2 is indicated to be the primary product, and R_{ox} approaches 8/3. Hence, the transition between CO and CO_2 product occurs between 1000° and 1400°F.

Substrate Pore Tortuosity

Pore path tortuosity is related to porosity, pore radius and numbers of pores by[9]

$$q = \varepsilon \ / \ 2\pi r^2 n_p (y_o/\delta_s) \tag{15}$$

Substrate porosity has been measured as 0.16, and average pore radius is 2 X 10^{-5} inch. For pores which extend completely across the substrate, $y_o/\delta_s = 1/2$, and it was assumed to have the same value for other pores. The number of pores was inferred from mass loss data for bare uncoated RCC, since the coating evidently blocks some pores to oxygen flow. This yields a value of 71 x 10^8 pores/ft^2. Using these values, Eq. (15) gives a tortuosity of 2.583. Carman[12] reported values of tortuosity for consolidated media, most of which fell between 1.41 and 3.54; hence the inferred value is reasonable.

Oxygen Diffusion in Substrate

The product of density and diffusion coefficient was represented by

$$\overline{\rho D} = \left[(L_e K/C_p)^{-1} + (2\rho r \overline{V}/3)^{-1} \right]^{-1} \tag{16}$$

where $L_e K/C_p$ is the coefficient for the continuum regime, and $2\rho r\overline{V}/3$ is that for the free molecule regime.

The values of $\overline{\rho D}$ given by Dq. (16)

The values of $\overline{\rho D}$ given by Eq. (16) are only weakly temperature dependent; however, they increase significantly with pressure. This is shown in Fig. 5, for a temperature of 1200°F and pore radius of 2. X 10^{-5} in.

Oxidation Tests

Results of air oxidation tests (Table 1) were used in evaluating the other rate and pore structure constants. Plasma arc specimens were disks retained in graphite holders with air flow normal to the specimen face. Radiant and thermal cycle specimens were bars and disks, radiantly heated on one surface, and with air flowing over all surfaces. Furnace specimens were bars and disks, heated and oxidized on all surfaces. Specimens were exposed to constant temperature and pressure for 30 to 60 min/exposure cycle and 5 to 20 total cycles. Mass loss was computed from weight before and after each cycle for 84 specimens at 40 test conditions.

At 800 to 1600°F, RCC mass loss rate increased with increasing temperature.[2] The rate of increase was steep from 800° to 1200°F, but reduced from 1200° to 1600°F. This was interpreted as being reaction rate control and transition oxidation regimes. Mass loss rates increased with increasing exposure, and the mass loss histories were used to infer K' and n_p values.

At 1600° to 1800°F, $\dot{M}$ was essentially constant with temperature[2] and increased only slightly with increasing exposure, approaching steady values near the end of exposure. The final M values were interpreted as being diffusion-controlled and were used to infer $\overline{\rho D}\ A_{flow}/A_e$ values.

Above 1800°F, $\dot{M}$ decreased with increasing temperature.[2] Substrate oxidation is diffusion-controlled, but coating oxidation forms a product film which partially blocks coating pores and fissures. Differential thermal expansion between the coating and substrate further reduces fissure flow area with increasing temperature. Final mass loss rates were correlated by the empirical relation:

$$\dot{M}_D = P^{0.8}\Big[0.245 \times 10^{-4} - 0.185 \times 10^{-4} \times (1 + 1.28 \times 10^{-26} e^{156,800/T})^{-1}\Big] \tag{17}$$

This relation is based upon data from 37 specimens and 28 test conditions. The diffusion-controlled mass loss rates M_D were used to infer $\overline{\rho D}\ A_{flow}/A_e$ values for T > 1800°F.

Table 1 Oxidation test conditions

Test facility	Test specimen	Temperature, °F	Pressure, atm
Rockwell International radiant	0.8- x 4.8-in. bars 2.8-in.-diam. disks	1200-2600	0.01-0.132
NASA Johnson Spacecraft Center plasma arc	2.8-in.-diam. disks	2040-2960	0.015-0.045
NASA Ames Research Center plasma arc	2.8-in.-diam. disks	1500-3010	0.021-0.10
NASA Langley Research Center plasma arc	2.8-in.-diam. disks	1925-2490	0.011-0.077
Vought Corporation furnace	2.8- and 3/4-in.-diam. disks 1.0- x 5.5-in. bars	800-1600	1.0
Vought Corporation thermal cycle	1.73- x 6.75-in. bars	1000-1800	0.05-1.0

Mass loss rates in the plasma arc tests were generally higher than those in the radiant tests. The rates used to establish $\overline{\rho D}\, A_{flow}/A_e$ values were intermediate between plasma arc and radiant test measurements. Plasma arc test mass loss rates were nominally 20% higher than those used to compute $\overline{\rho D}\, A_{flow}/A_e$ at 1600°F and 25% higher at 2100°F. The rates used to establish K' and n_p values were based on radiant tests only, since the temperatures at which plasma arc tests were conducted were too high for inference of those properties.

Oxygen Diffusion in Coating

The model for oxygen diffusion across the coating was derived in similar fashion to that for the substrate. Diffusion down both fissures and pores, however, was included. Expressions for fissure and pore diffusion coefficients were taken from Carman.[12] The resulting expression for the coating diffusion parameter is

$$\overline{\rho D}\,\frac{A_{flow}}{A_e} = \frac{P\ MW}{RT}\left\{\left(\frac{A_f / A_e}{(PC_p MW/L_e KRT) + (3K_1/4\delta_o \overline{MV})}\right)_f + \left(\frac{\varepsilon/q}{(PC_p MW/L_e KRT) + (3K_1 q/4\delta \overline{MV})}\right)_p\right\} \tag{18}$$

Coating properties were evaluated by correlating Eq. (18) with values of $\overline{\rho D}\, A_{flow}/A_e$ inferred from RCC oxidation test data, using the following relation:

$$\overline{\rho D}\, A_{flow}/A_e = \dot{M}_D \delta_C R_{ox}/C_e \tag{19}$$

This was derived from Eq. (7) with $C_o = 0$. The inferred values of $\overline{\rho D}\, A_{flow}/A_e$ are presented in Fig. 3. The following constants were selected so as to fit Eq. (18) to the data:

$$A_f/A_e = 0.0012 \qquad (K_1/\delta_o M)_f = 0.1224 \times 10^6\ ft^{-1}$$

$$\varepsilon/q = 0.011 \qquad (K_1 q/\delta m)_p = 5.176 \times 10^6\ ft^{-1}$$

Separate contributions of pores and fissures to oxygen diffusion are shown in Fig. 3.

The inferred value of A_f/A_e corresponds to a temperature of 1600°F. The coating fissures result from differential ex-

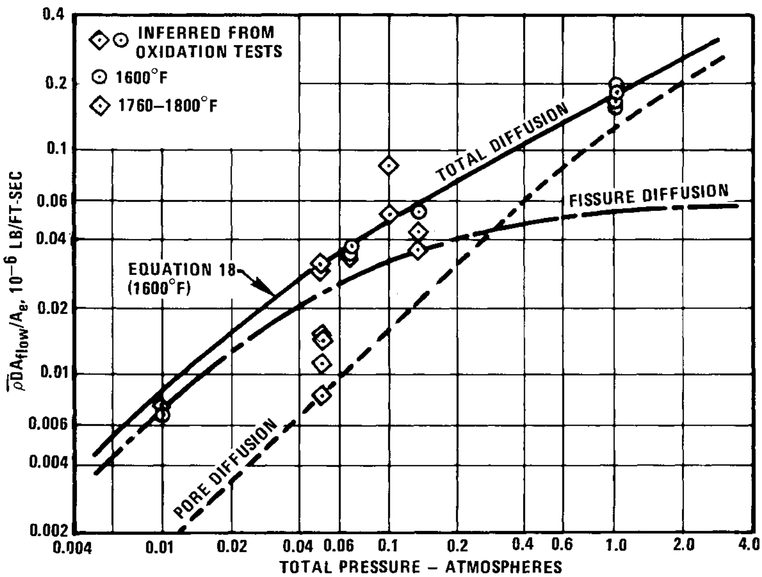

Fig. 3 Correlation of oxygen diffusion parameter for coating with pressure.

pansion between the coating and substrate. Previous analyses of expansion effects[2] resulted in $A_f/A_e = 0.0015$ at 1600°F. Hence, the inferred value is quite reasonable. The expansion analysis results were used to correct $\overline{\rho D}\ A_{flow}/A_e$ values for Eq. (19) for fissure expansion/contraction with temperature change. Similar examination of the other three parameters indicated that the inferred values are reasonable.

Equation (18) applies at T < 1800°F. Equations (17) and (19) were used to compute $\overline{\rho D}\ A_{flow}/A_e$ for T > 1800°F.

Pore Number and Reaction Rate Constant

These constants were inferred from test data in the reaction control and transition regimes using the progressive reaction surface computer routine. Mass loss rate for a given pore radius is a function of both n_p and K'. The rate of increase of r with time and hence the progressiveness of $\dot{M}$, is a function of K' alone see Eq. (11) . Values of both n_p and K' were inferred, therefore, by matching analytical to experimental results for both initial mass loss rate and increase of $\dot{M}$ with time. A comparison of computed and measured mass loss rate is presented in Fig. 4 for RCC at 1400°F and 0.1 atm. pressure. A good match was obtained with $K' = 0.0404 \times 10^{-6}$ lb/ft^2-sec and $n_p = 27.62 \times 10^8$ pores/ft^2.

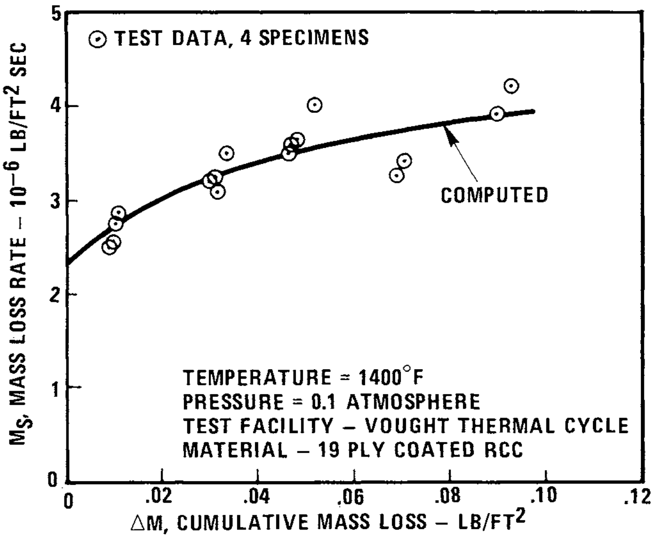

Fig. 4 Mass loss rate correlation.

A summary of inferred values of K' and n_p for other tests is presented in Table 2. A total of 174 mass loss measurements on 35 specimens were used to infer these values. In the following mass loss predictions, $n_p = 27.62 \times 10^8$ was used for 19 ply RCC and $n_p = 50.96 \times 10^8$ was used for 35 ply RCC. The inferred values of K' increased with both increasing temperature and pressure, as expected.

Correlation of Reaction Rate Constants

The reaction rate constants inferred previously were correlated with temperature and pressure for use in Earth entry mission analysis. Data in the literature [5-8] indicate a reaction order n = 1/2 for carbon oxidation. This implies both that reaction rate is proportional to $C^{1/2}$, as assumed in the mathematical model, and that K' is proportional to $P^{1/2}$. The inferred K' values for 1200° and 1400°F were plotted against pressure as a check of reaction order, as shown by Fig. 5. A correlation line of n = 1/2 does provide a reasonable fit to the data, and this further justifies use of a 1/2 reaction order in the mathematical model.

An improved correlation of the data is provided by the relation

$$K'/K'_{P=1} = f(P) = (0.9272 + 0.0728/P)^{-1} \qquad (20)$$

Table 2 Inferred rate constant and pore number

Test facility	Temperature, °F	Pressure, atm	Material[a]	K' rate constant, 10^{-6} lb/ft^2-sec	n_p pore number 10^8 pores/ft^2
Thermal cycle	1000	1.	19-ply	0.0384	27.62
	1000	1.	35-ply	0.0384	50.96
	1400	0.10	19-ply	0.0404	27.62
	1400	0.10	35-ply	0.0404	50.96
Furnace	800	1.	19-ply	0.0008843	27.62
	800	1.	35-ply	0.0008843	50.96
	1100	1.	35-ply	0.0604	27.62
	1200	1.	35-ply	0.0526	27.62
	1200	1.	15-ply	0.0526	27.62
	1400	1.	35-ply	0.0595	27.62
	1000	1.	35-ply (bare)	0.0384	71.00
RI radiant	1200	0.132	19-ply	0.03	27.62
	1200	0.066	19-ply	0.027	27.62
	1200	0.01	19-ply	0.00605	27.62

[a] Coated except where noted

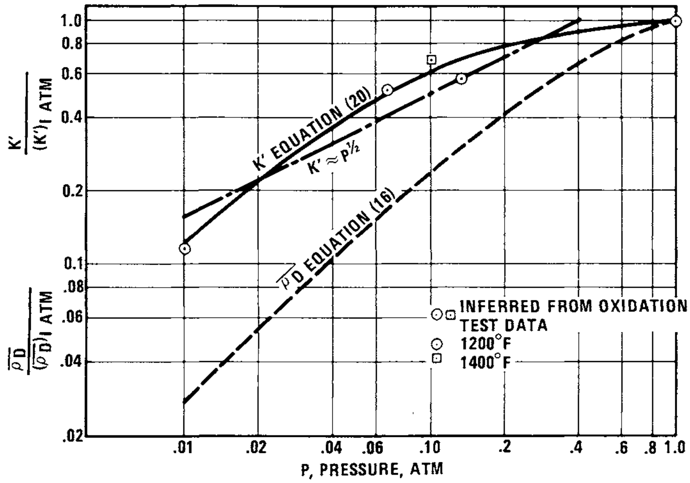

Fig. 5 Variation of reaction rate and diffusion rate constants with pressure.

Maahs[13] presented a relation for carbon oxidation which is similar, with respect to pressure dependency of reaction rate, to Eq. 20. It was based upon an adsorption-desoprtion mechanism, where n = 0 for desorption and n = 1 for adsorption. The data in Fig. 5 tend to indicate, therefore, that adsorption effects predominated at low pressure and desorption predominated at high pressure.

The relative pressure sensitivities of K' and $\overline{\rho D}$ determine the pressure sensitivity of X_o see Eq. (6) . Comparison of K' and $\overline{\rho D}$ values in Fig. 5 shows that $\overline{\rho D}$ varies more strongly with pressure than K'.

For the correlation of rate constant with temperature, K' values were divided by f(P) and R_{ox} to normalize out pressure effects and to convert to rate constant for carbon consumption, as shown in Fig. 6. For a single reaction, a linear correlation would be expected. In fact, there is a slope change at 1000°F, which is the temperature at which a transition from CO_2 to CO product begins. The slope change, therefore, was interpreted as representing a difference in reaction rates for CO vs. CO_2 product.

A similar slope change was observed by Ong[10] for graphite oxidation, although at higher temperature. Ong developed an equation for correlation of the graphite data which accounts for the difference in reaction rates for CO and CO_2 product.

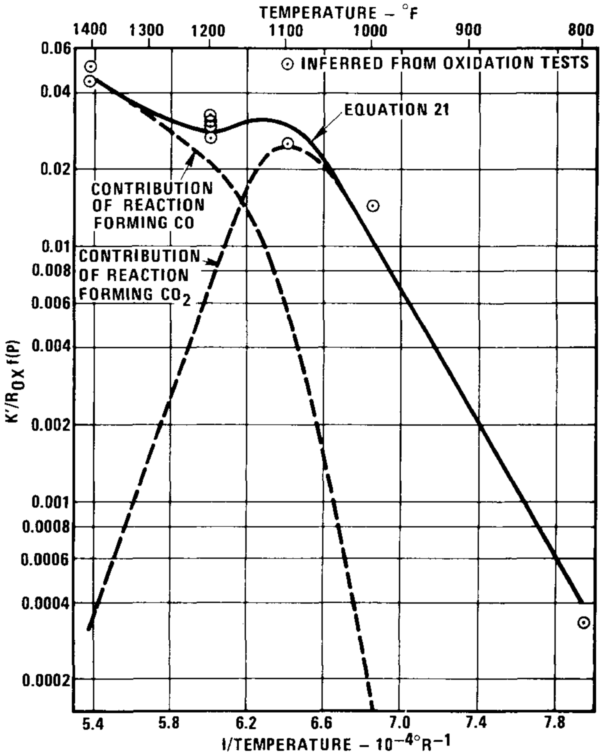

Fig. 6 Correlation of rate constant with temperature.

His equation exhibits a slope change corresponding to the point of product transition for his graphite data. A simplified version of Ong's relation was used to correlate the data in Fig. 6:

$$K'/R_{ox}\ f(P) = WAe^{-E_{co}/RT} + (1-W)Be^{-E_{co_2}/RT} \tag{21}$$

In order to satisfy Eq. (14), Eqs. (21) and (14) were combined, yielding the following relation:

$$W = \left\{1 + \frac{A}{5.834 \times 10^{16} B} \exp\left(\frac{62{,}510 + E_{co_2} - E_{co}}{RT}\right)\right\}^{-1} \tag{22}$$

The following constants were selected to fit Eqs. (21) and (22) to the data in Fig. 6:

$$A = 29.20 \times 10^{-6}\ \text{lb/ft}^2\text{-sec} \qquad E_{co}/R = 11{,}960^{\circ}R$$

$$B = 12.54 \text{ lb/ft}^2\text{-sec} \qquad E_{co_2}/R = 30,480^\circ R$$

The separate contributions of the two reactions are shown in Fig. 6.

Mass Loss Distribution Predictions

The progressive reaction surface computer routine was used to predict density reduction distributions for a range of constant temperature and pressure conditions. Three of these conditions corresponded to tests for which specimens were subjected to post-test examinations, including photomicrographs of cross-sectioned specimens and coating adherence evaluation. Comparisons are made between examination results and predictions.

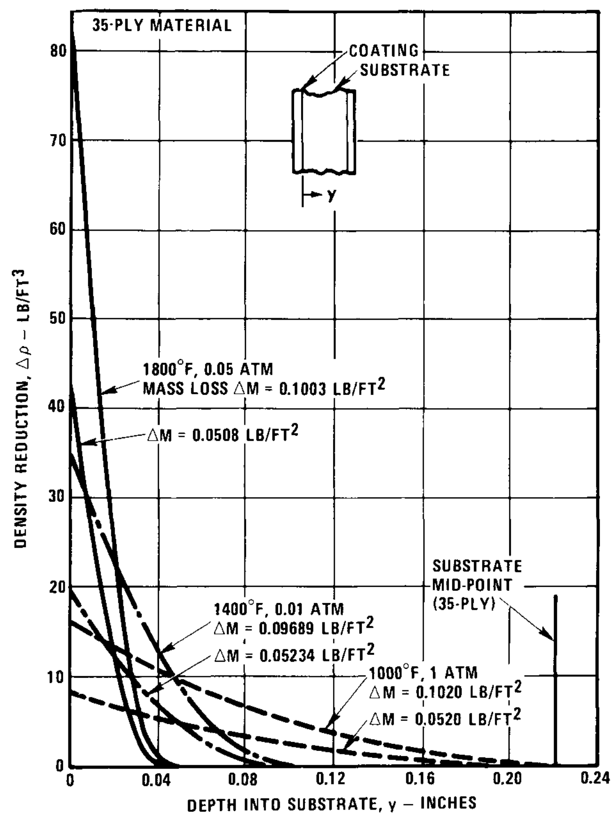

Fig. 7 RCC density reduction distribution.

Comparison of Predictions with Test Results

Predicted density reduction is presented in Fig. 7 as a function of depth into the substrate for three oxidation conditions. Results are presented for 35-ply RCC at exposure times corresponding to 0.05 and 0.10 psf nominal cumulative mass loss.

At 1800°F and 0.05 atm, there is a very steep drop in predicted density reduction from the value at the coating interface to that in depth. At a mass loss of 0.10 psf, the interface density reduction of 82 lb/ft^3 represents almost total loss of substrate-coating bond. At 0.05 psf mass loss, the interface density reduction is also high, 43 lb/ft^3. The predictions are corroborated by the fact that test specimens exposed to these conditions had little or no coating adherence to the substrate. Photomicrographs revealed heavy loss of substrate near the interface but little loss apparent in depth.

At 1400°F, 0.10 atm, predicted interface density reduction is less, 34 lb/ft^3 at 0.10 psf mass loss, but there is greater in-depth density reduction than at 1800°F. Coating adherence of test specimens at this condition was poor at ΔM = 0.10 psf but good at ΔM = 0.06. Photomicrographs revealed substantial loss of substrate near the interface but considerably less than at 1800°F. In-depth oxidation attack was apparent.

At 1000°F, 1 atm, predicted density reduction is distributed much more uniformly than at the higher temperatures and lower pressures. Coating adherence at this condition was good at mass loss of both 0.05 and 0.10 psf. Photomicrographs revealed much less severe substrate loss near the coating interface than at the other conditions but greater in-depth attack.

In summary, mass loss distribution predictions are consistent with post-test examinations of test specimens. Although the test data serve to substantiate the prediction method, it should be noted that recent coating improvements for the Space Shuttle material preclude the high density reductions discussed.

Parametric Data

Correlations of predicted density reduction at the coating-substrate interface $\Delta\rho_o$ with cumulative mass loss ΔM showed that the results could be approximated reasonably by a linear

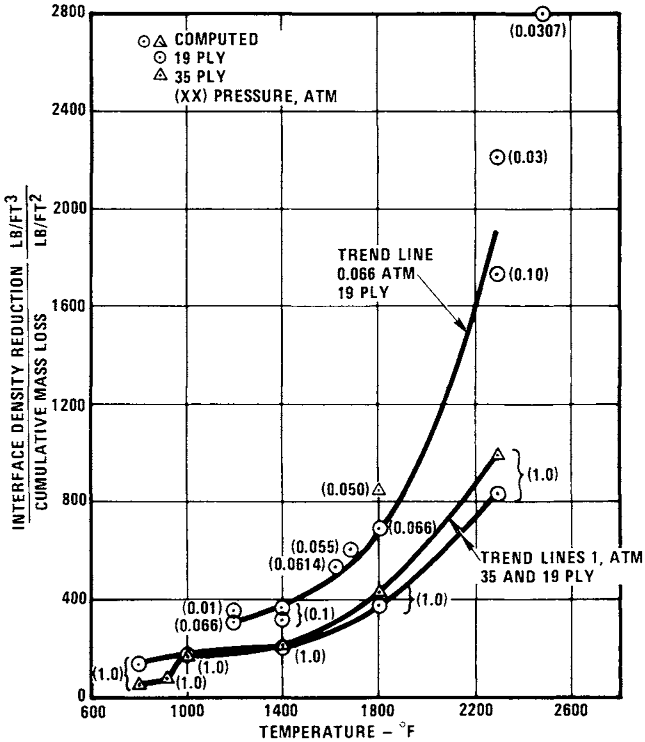

Fig. 8 Density reduction at coating: substrate interface vs temperature.

relation of slope $\Delta\rho_o/\Delta M$. This slope characterizes the mass loss distribution quite well. High values of $\Delta\rho_o/\Delta M$ on the order of 800 ft^{-1} indicate a steep mass loss gradient across the coating, with mass loss concentrated near the coating-substrate interface. Low values of $\Delta\rho_o/\Delta M$ on the order of 100 ft^{-1} indicate a relatively uniform mass loss distribution.

Values of $\Delta\rho_o/\Delta M$ determined from the computer predictions are presented in Figs. 8 and 9 as a function of temperature and pressure for thick and thin material. These results indicate that interface mass loss increases relative to total mass loss with both increasing temperature and decreasing pressure. This is consistent with previous predictions that oxygen penetration depth decreases with increasing temperature and decreasing pressure.

Conclusions

The mathematical model presented herein provides predictions of mass loss rate distribution through the thickness

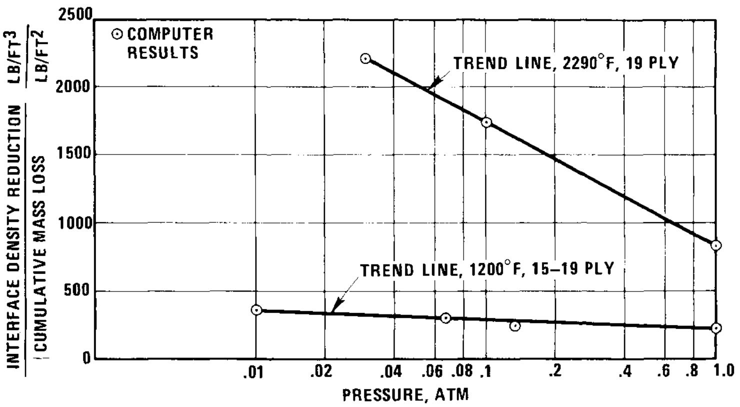

Fig. 9 Density reduction at coating: substrate interface vs pressure.

of carbon-carbon material which are reasonably consistent with photomicrographs and coating adherence for oxidized test specimens. The model also properly characterizes the increase in overall mass loss rate with exposure time for temperature up to 1800°F. Substrate mass loss distribution is relatively uniform at low temperatures and high pressures but becomes increasingly concentrated at the coating-substrate interface as temperature increases and pressure decreases.

Acknowledgment

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ASYMMETRIC NOSE-TIP SHAPE CHANGE DURING ATMOSPHERIC ENTRY

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Abstract

A method is presented to assess the probability of asymmetric nose-tip shape change during atmospheric entry. The method uses the variability of nose-tip material surface roughness to determine random, circumferentially nonuniform boundary-layer transition front behavior. Nose-tip shape histories are characterized by several key parameters and are suitable for inclusion in a Monte Carlo analysis of vehicle trim angle-of-attack history. The effects of surface roughness characteristics, re-entry trajectory, and vehicle configuration on expected shape development are examined.

Nomenclature

A = constant in Eq. (2)
B = constant in Eq. (2)
C = constant in Eq. (3)
C_{l_n} = nose-tip rolling moment coefficient
C_{M_α} = pitching moment coefficient derivative with respect to angle of attack
C_{N_n} = nose-tip normal force coefficient
ΔC_{N_n} = increment in nose-tip normal force coefficient
D = constant in Eq. (4)
F = normal force
k = roughness height
L_v = re-entry vehicle reference length
M = rolling moment
q = dynamic pressure
$\bar{r}$ = laminar region centroid distance from centerline
r_c = radial distance to gouge edge

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R_B = re-entry vehicle base diameter
R_n = nose-tip initial nose radius
$\dot{s}$ = nose-tip recession rate
s_t = running length from stagnation point to transition location
t = time
V = velocity
x_{cg} = axial distance from nose to vehicle center of gravity
α_T = trim angle of attack
γ = re-entry trajectory inclination angle
Δ = laminar region radial offset from vehicle centerline
ε = nose-tip eccentricity
θ = normalized nose-tip recession
θ_c = turbulent gouge angle
σ_k = standard deviation of roughness height distribution
ϕ = body-fixed meridional angle
$\Delta\phi$ = turbulent gouge angular width
ψ = nose-tip cant angle

Subscripts

e = entry conditions
i = i^{th} nose-tip segment
m = mean value
w = wind vector
∞ = freestream conditions

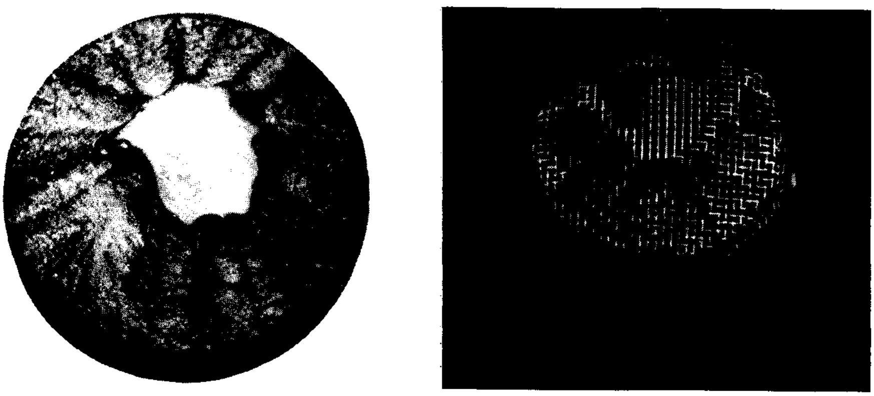
I. Introduction

Graphitic materials including bulk graphites and carbon-carbon composites commonly are used for re-entry vehicle nose tips because of their excellent high-temperature ablation resistance and, in the case of carbon-carbon composites, excellent resistance to thermal strain failure. However, re-entry vehicle aerodynamic performance is sensitive to relatively small asymmetries that develop on the nose tip during entry as a result of spotty or asymmetrical boundary-layer transition. In the past, these nose-tip asymmetries have been ignored in re-entry vehicle nose-tip shape change calculations. Although this approach allows a relatively detailed procedure to be employed in the calculation scheme and has yielded fairly good prediction of nose-tip recession during entry, no information about the inherent randomness of the nose-tip shape change process is obtained. In particular, nose-tip asymmetries that induce body-fixed normal forces and nonzero trim angle of attack are not addressed.

For a spinning, statically stable re-entry vehicle, changes in the lift vector (or roll rate) cause impact point dispersion due to the nonaveraging of the lift vector as the vehicle spins.[1] In addition, circumferential drift or jump in the lift vector orientation relative to the body-fixed axis also induces precession rate change and in some cases precession rate stoppage (lift vector fixed in inertial space) for short periods. Either roll through zero (vehicle roll rate equals zero) or precession stoppage causes relatively large dispersion as the lateral velocity jump incurred propagates to impact. Analysis of these sources of dispersion requires a technique for estimating the probability of asymmetric nose-tip shape change and the time history of the shape development. This paper presents a technique for characterizing the development of asymmetric nose-tip shapes which is the required input to determining vehicle aerodynamic performance. The technique is founded on basic material parameters and therefore yields information on potential material improvements that may reduce re-entry vehicle dispersion.

II. Boundary-Layer Transition

The primary mechanism leading to the development of asymmetric nose-tip shapes for ballistic re-entry vehicles is increased local ablation rate in regions of turbulent boundary-layer flow. Since the ratio of turbulent to laminar convective heating is larger than unity except at momentum thickness Reynold numbers less than about 40, transition to turbulent flow results in enhanced local recession rate and the development of turbulent gouges on the nose tip. This phenomenon has been observed in numerous ground and flight tests for both graphitic materials and low-temperature ablators such as camphor.



a) RECOVERED FLIGHT TEST NOSETIP b) 50-MW TEST MODEL

Fig. 1 Typical asymmetric nose-tip shapes.

Figure 1 shows the recovered bulk graphite NRV re-entry vehicle nose tip[2] and a carbon-carbon nose-tip model tested in the Air Force Flight Dynamics 50MW RENT arcjet facility. The qualitative similarity between the two nose tips is striking. Both materials exhibit longitudinal gouges attributed to turbulent vortices. The central laminar stagnation region is irregular in shape and offset from the nose-tip centerline. It is postulated that circumferentially nonuniform transition typically leads to the development of these shapes. For graphitic materials, the occurrence of boundary-layer transition on the nose tip is related directly to the surface roughness developed by the nose-tip material during ablation.

Typical results relating the location of boundary-layer transition to surface roughness are shown in Fig. 2. Surface roughness is seen to cause boundary-layer transition at substantially reduced values of momentum thickness Reynolds number compared to those obtained for smooth surfaces. The theoretical result shown is based on a vortex shedding criteria as discussed in Ref. 3 and depends on the ratio of both roughness height to momentum thickness and roughness height to nose-tip radius of curvature. The last effect accounts for the increased stability of laminar boundary layers in accelerating flow on convex bodies.[4]

Although the data shown in Fig. 2 were obtained for uniform roughness, it is well known that the size of roughness elements developed during ablation of graphitic materials is

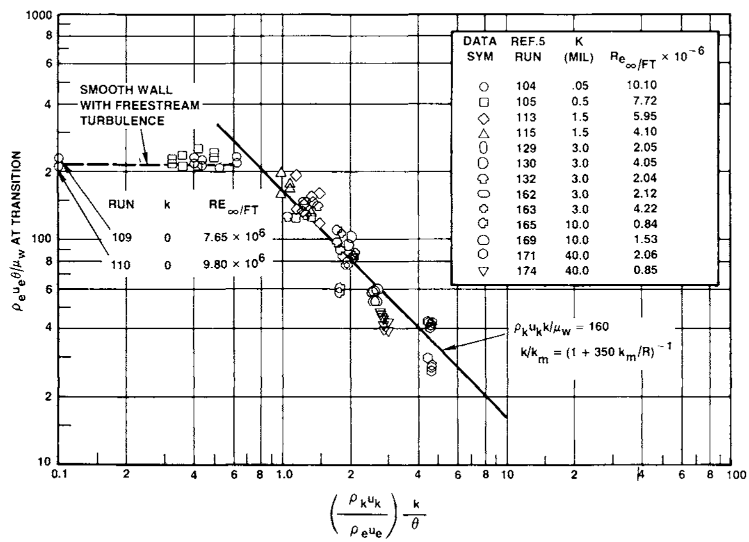


Fig. 2 Transition criteria for rough surfaces.

not constant over the surface at any instant in time.[6,7] Optical measurements on post-test ablation models which have been sectioned and carefully prepared using metallurgical techniques have indicated that a relatively wide range of element heights is present during ablation (see Fig. 3). This variability in roughness element size induces a pseudorandom variation in transition location.

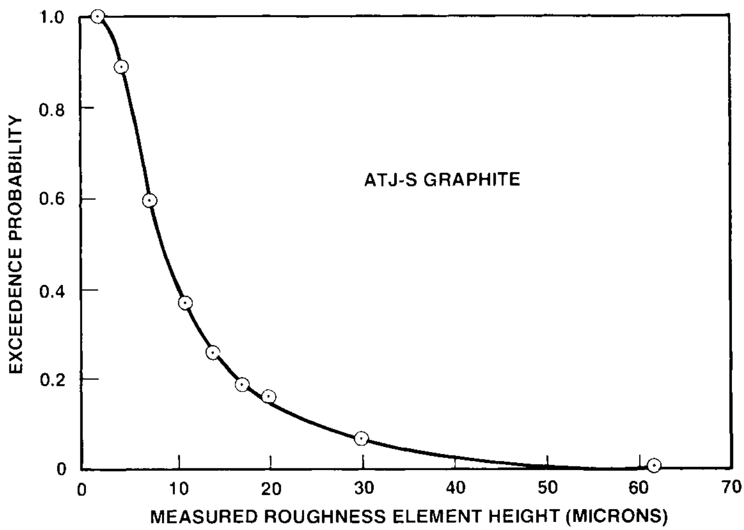


Fig. 3 Roughness character of ablated graphite surface.

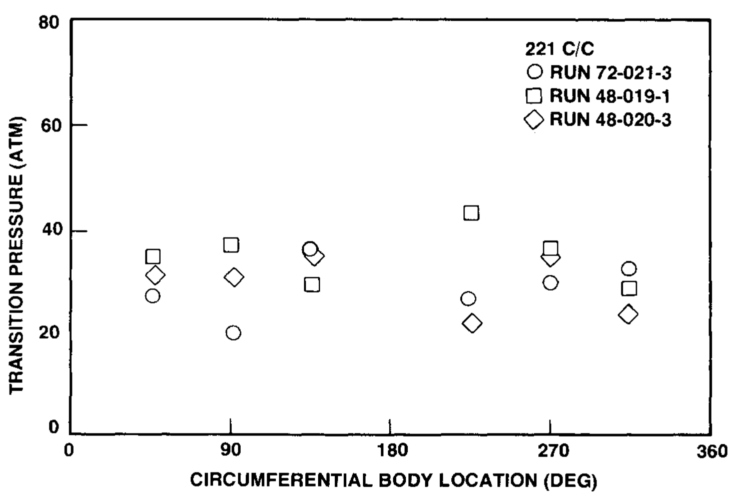


Fig. 4 Transition onset variability.

For instance, Fig. 4 shows typical data for transition onset pressure as a function of circumferential nose-tip location as determined from the point at which turbulent gouging of the nose-tip surface just begins.[8] In these tests, nose-tip stagnation pressure is increased linearly with time in an arcjet ablation environment. These data and others from arcjet and ballistic range ablation tests of various graphitic nose-tip materials were analyzed in Ref. 9, and the direct correlation between transition onset variability and surface roughness nonuniformity was established. It is this variability in roughness element size coupled with the re-entry environment which dictates the degree of asymmetry developed during flight.

III. Asymmetric Shape Change Modeling

Based on the preceding observations, the following model for asymmetric nose-tip shape change is developed. The re-entry vehicle nose tip is modeled as 20 segments, each initially covering 18 deg of the circumference and each having a randomly assigned constant roughness height based on the material's roughness element height distribution. The number of segments is chosen to match previously published empirical and theoretical estimates of the spreading angle of turbulent wedges in subsonic flow[10] and also correlates well with the number and size of gouges typically observed, as seen in Fig. 1. During early re-entry, as nose-tip stagnation pressure increases, each segment experiences transition onset and movement according to the roughness height assigned to that segment and the transition law of Ref. 3.

The asymmetric transition history thus determined provides the driving function for the simplified asymmetric shape change illustrated schematically in Fig. 5. In addition to the asymmetric transition front locations s_{ti} and the turbulent wedge widths $\Delta\phi_i$ the radial offset of the centroid of the laminar stagnation region from the nose-tip centerline Δ and the cant angle ψ are used to specify the asymmetric shape.

Simplified equations relate these parameters to the nominal recession history:

$$\theta = \int \dot{s}\,\frac{dt}{R_n} \tag{1}$$

$$\frac{d\Delta}{d\theta} = \left(\frac{R_n}{1+A\theta}\right) h(\bar{r}-\Delta) + \psi R_n - B\Delta \tag{2}$$

$$\frac{d\psi}{d\theta} = C(\alpha_T - \psi) \tag{3}$$

$$\frac{d\Delta\phi_i}{d\theta} = D \tag{4}$$

where $\dot{s}$ is calculated using a symmetric shape change program with the mean nose-tip surface roughness, and A, B, C, and D are constants.

The preceding equations give a close representation of observed shape change behavior at angle of attack for values of the constants A through D equal to 3.0, 0.2, 6.28, 1.41 derived considering geometrical aspects of Fig. 5 and the difference in heating and recession rates on the windward and leeward rays. The first term in Eq. (2) represents the driving potential due to asymmetrical transition, where $\bar{r}$ is the centroid of the laminar region and h is the unit step function. That is, the offset tends to move toward the centroid of the laminar region as the transition front progresses around the nose and the nose tip recedes. Equation (2) also indicates that the offset will increase for positive values of cant angle but tend to decrease as larger values of offset are obtained because of the higher recession rate on the leeward side. Equation (3) specified the approach of the nose-tip shape to symmetry with the wind vector as the nose tip recedes. Experimental shape change histories

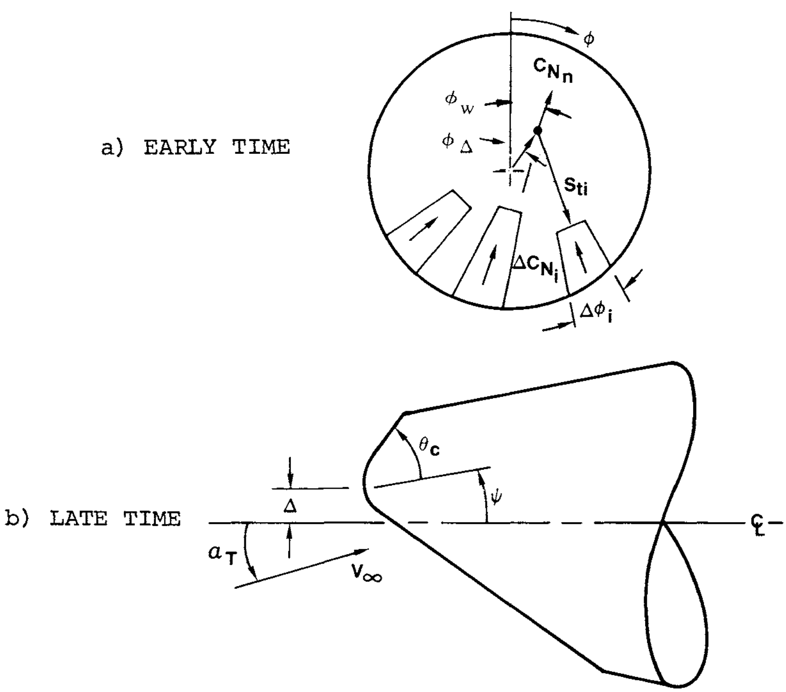


Fig. 5 Asymmetric shape change schematic.

for both low-temperature ablators[11] and bulk graphite[12] illustrate these trends. In addition, as nose-tip recession increases, a turbulent gouge will spread laterally as given by Eq. (4) because of the higher turbulent ablation rate in the gouge. This effect tends to produce a slow coalescence of the gouges separated by a low roughness laminar region.

IV. Nose-Tip Normal Force and Roll Torque

Solution of the shape change equations coupled with the asymmetric transition behavior discussed previously requires specification of the vehicle angle of attack. Consider the schematic shown in Fig. 5. The asymmetric nose tip generates a nose normal force coefficient C_{N_n} and rolling moment coefficient C_{ℓ_n}, defined by

$$C_{N_n} = F_n/q_\infty \pi R_n^2 \tag{5}$$

$$C_{\ell_n} = M_n/2q_\infty \pi R_n^3 \tag{6}$$

Nose normal force coefficient is derived assuming a hypersonic Newtonian pressure distribution. The excess force relative to an ellipsodial nose of eccentricity ε on a plane cut section is shown in Fig. 6 as a function of cut location r_c.

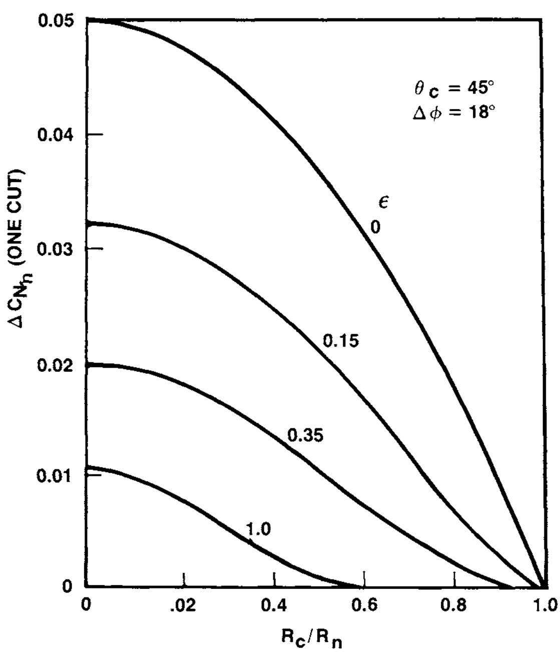


Fig. 6 Normal force increment for wedge cut.

The normal force coefficient increment ΔC_{N_n} is seen to increase rapidly with decreasing ε and r_c. The total nose normal force coefficient is the vector sum of these increments plus the contribution of nose-tip cant relative to the wind vector:

$$C_{N_n} = \sum_i \Delta C_{N_{ni}} + \psi + \alpha_T \tag{7}$$

The roll torque produced also is determined by the C_{N_n} force vector and its offset from the vehicle centerline:

$$C_{\ell_n} = (\Delta/2R_n)\, C_{N_n} \sin(\phi_w - \phi_\Delta) \tag{8}$$

where ϕ_w is the normal force vector angle, and ϕ_Δ is the laminar region offset angle (see Fig. 5).

In order to complete the system of equations describing asymmetric nose-tip shape change, it remains to relate nose-tip normal force to vehicle angle of attack. The effect of an asymmetric nose tip on the vehicle's frusta pressure distribution is described in Ref. 13, where it is shown that substantial modification of the vehicle's pitching moment characteristics can occur compared to a symmetric vehicle. An accurate assessment of nose-tip shape change history therefore requires coupling of both nose-tip shape change and the attendant changes in vehicle aerodynamics. However, in order to demonstrate the predominant effects of asymmetric nose-tip shape change, a first-order moment balance in the pitch plane yields the following equation:

$$\alpha_T = \frac{C_{N_n}}{-C_{M_\alpha}} \frac{x_{cg}}{L_v} \left(\frac{R_n}{R_B} \right)^2 \tag{9}$$

where C_{M_α} is the derivative of the symmetric vehicle pitching moment about the center of gravity at zero angle of attack.

Table 1 Typical re-entry parameters

Case	V_e, kft/sec	$-\gamma_e$, deg	R_n/R_B	x_{cg}/L_v
1	23.0	27.5	0.08	0.65
2	21.5	21.0	0.08	0.65
3	25.5	21.0	0.08	0.65
4	26.5	30.	0.08	0.65
5	23.0	27.5	0.18	0.66

Reference 13 indicates that this approach underpredicts the trim angle at bluntness ratios greater than about 10% while overpredicting α_T for more slender vehicles.

V. Nose-Tip Performance

The foregoing analysis has been programmed for computer solution using a Monte Carlo approach to define initial nose-tip roughness characteristics. Nose-tip material roughness height distribution is characterized by a gaussian distribution having a mean of k_m and a standard deviation σ_k. Initial

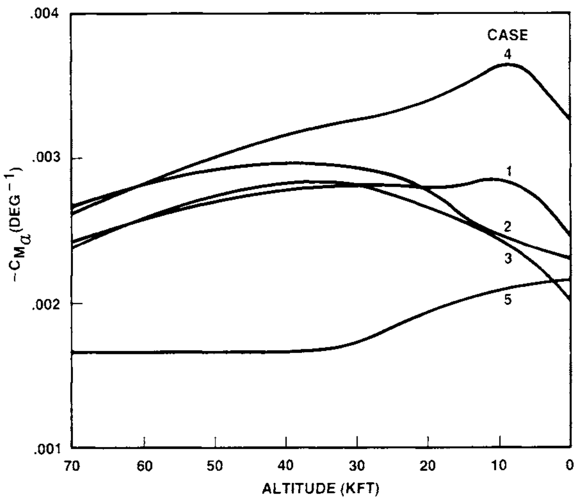


Fig. 7 Stability derivatives.

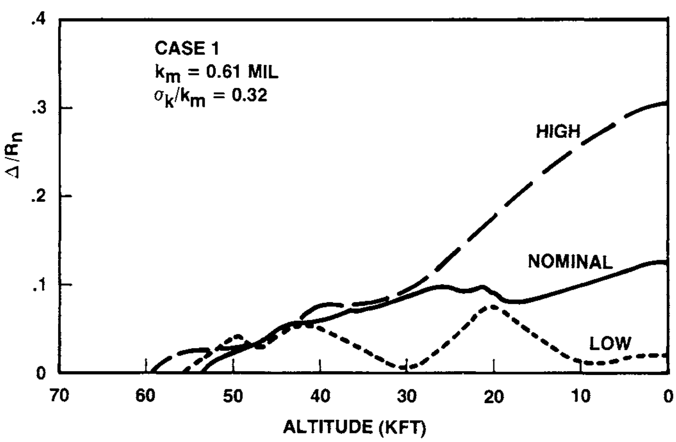


Fig. 8 Laminar region offset development.

values of the roughness height k_i are chosen randomly from the distribution and assigned to the 20 nose-tip segments for each trial run. This approach allows generation of a sample of nose-tip shape change histories suitable for use in dispersion analyses.

Application of the method for typical re-entry conditions is illustrated by the following examples. Table 1 and Fig. 7 show the values of pertinent parameters for the calculations. The stability derivatives were calculated for the symmetric ablated vehicle configuration using the general aerodynamics computer code described in Ref. 13 and roughness parameters typical of a fine-weave carbon-carbon material[9] (k_m = 0.61 mils, σ_k/k_m = 0.32). In order to illustrate the expected variability in the asymmetric shape change, 20 Monte Carlo trials were computed for each case. Figures 8-12 present results for case 1 from three trials selected to illustrate the nominal and extremes of asymmetric shape development. Figures 8 and 9 show the asymmetric shape parameter (Δ and ψ) histories as a function of altitude. Both parameters are seen to follow similar histories, with the cant angle responding slower to angle-of-attack changes than the offset responds to turbulent gouge development. Both of these parameters increase initially as transition begins on the nose at high roughness locations and then tend to stabilize for the majority of trials, as indicated by the nominal results. However, in the extreme cases, completion of turbulent gouging is delayed to lower altitudes, and stabilization is not evident. Figure 10 shows the corresponding normal force vector angle for each of the three trials. For these data, the 0-deg position is taken as opposite the location of the first turbulent gouge. It is observed that the force vector angle is not constant and may even rotate 360 deg

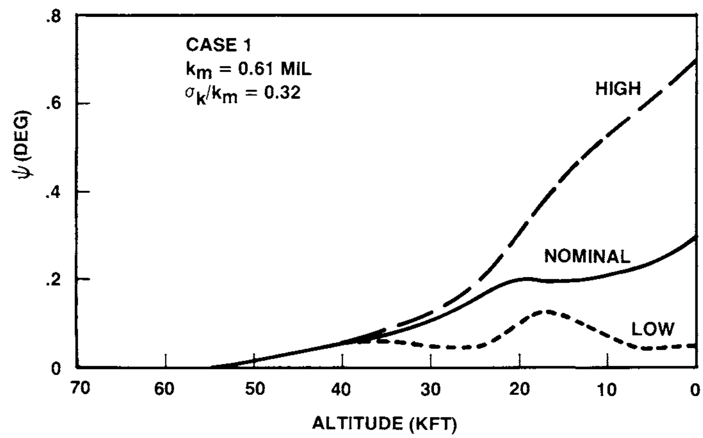


Fig. 9 Nose-tip cant angle development.

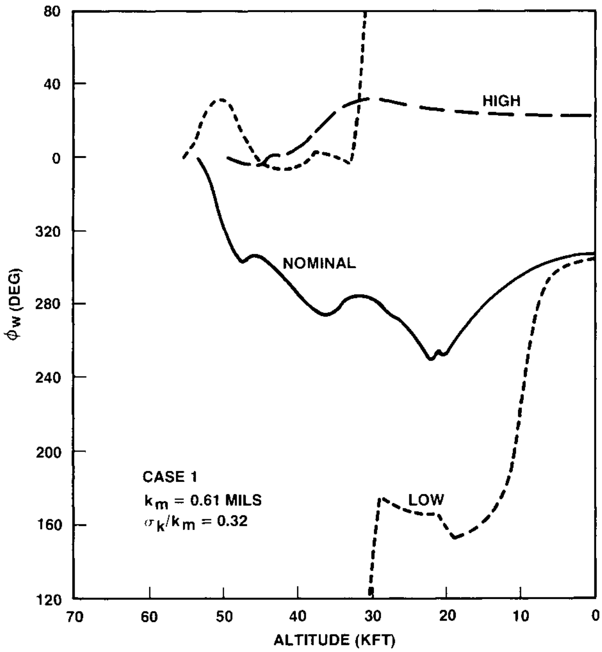


Fig. 10 Normal force vector angle variability.

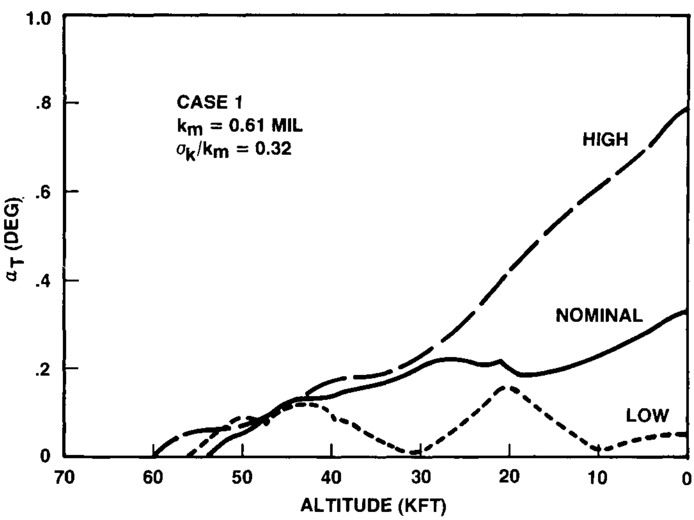


Fig. 11 Trim angle history variability.

during the re-entry, as seen in the "low" trial. However, the largest nose-tip asymmetry and trim is obtained when the force vector maintains a near-constant position. Trim angle of

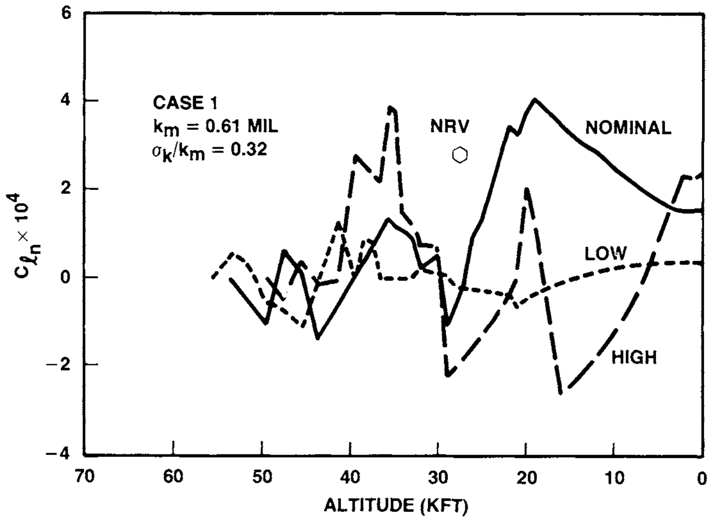


Fig. 12 Nose-tip rolling moment history variability.

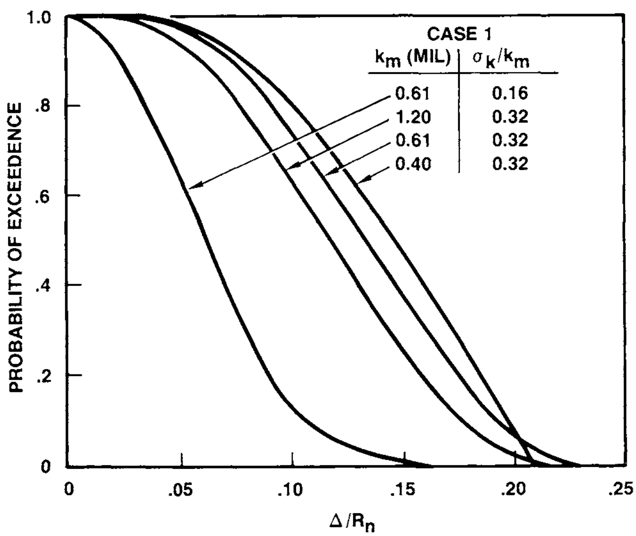


Fig. 13 Laminar region offset sensitivity.

attack and nose-tip rolling moment coefficient are presented in Fig. 11 and 12. The trim angle clearly follows the laminar region offset and indicates the dominant effect of the parameter. Predicted roll torque coefficient histories are highly erratic and have a low value relative to total vehicle roll torque coefficients deduced from flight-test data. However, the calculated results are consistent with wind-tunnel measurements[14] of roll torque on a model of the NRV nose tip, as shown in Fig. 12.

As indicated previously, the key asymmetric shape parameter is the laminar region radial offset. Figure 13 shows the offset distribution function at the completion of transition as a function of the roughness distribution parameters for case 1. The offset distribution functions indicate that a substantial variation in offset can occur for a given nose-tip material and trajectory. For instance, for the nominal values of the roughness parameters, the mean offset is 13% of the nose radius. However, the probability that Δ/R_n exceeds 0.19 or is less than 0.07 is 10%. Figure 13 further indicates that the offset distribution mean is sensitive primarily to the standard deviation of the roughness distribution and only slightly dependent on the mean roughness height. As might be expected, larger roughness materials exhibit a slightly reduced tendency to form asymmetric shapes because the transition process occurs at higher altitudes where both the ratio of turbulent to laminar recession rates is lower and the increase in nose-tip stagnation pressure occurs more rapidly. It also should be noted that, although lower mean roughness height slightly increases the mean offset, it also reduces the probability of the largest values of offset occurring. Analysis of the detailed results indicates that the higher values of offset occur when one to

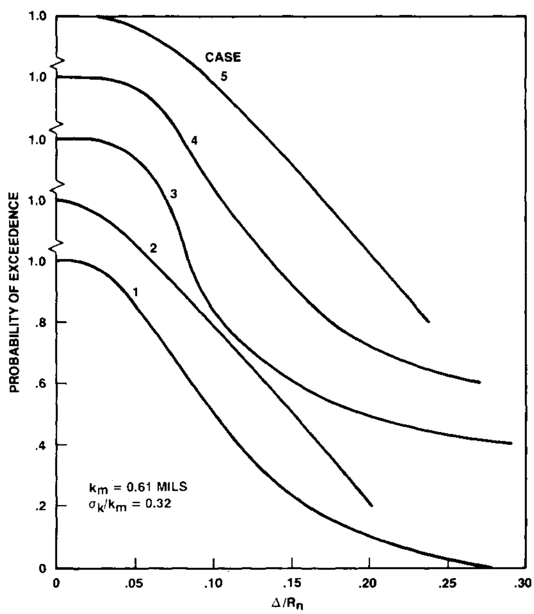


Fig. 14 Offset sensitivity to trajectory and nose radius.

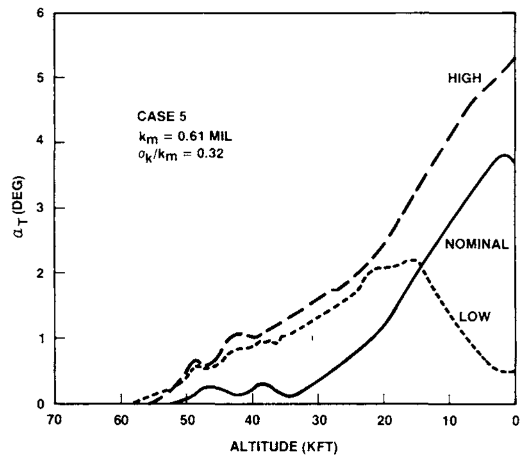


Fig. 15 Trim angle variability for blunt vehicle.

two nose-tip segments fail to experience boundary-layer transition due to low roughness values. For the lower mean roughness, however, more segments fail to transition, and, since these segments are distributed randomly circumferentially, they tend to compensate for one another, thus reducing the offset.

Figure 14 shows the sensitivity of the laminar region offset at impact to trajectory and vehicle parameters for the typical roughness values. Little trajectory sensitivity is evidenced, with the exception of a drop in the probability of occurrence of the larger offsets for the slowest trajectory (case 2). The results for this case are similar to the preceding results for 0.4 mil mean roughness, again because transition is not completed on this mild trajectory. The effect of larger nose-tip radius is indicated by comparing case 5 with case 1. A shift in mean offset to a higher value and the absence of the low-probability largest offset values is evident for the larger nose radius. In this case, the reduced nose-tip recession and transition altitude again produces incomplete nose-tip transition. Further results for the blunt vehicle are shown in Fig. 15. Trim angles of attack substantially larger than those for the sharper vehicle are produced because of the greater relative influence of the nose tip on vehicle aerodynamics.

VI. Conclusions

A method has been developed to assess the probability of the development of asymmetric nose-tip shapes during atmospheric re-entry. The method uses the variability of nose-tip surface roughness to determine the asymmetrical boundary-layer transition front behavior. Nose-tip shapes are characterized by several key parameters rather than by employing a full three-dimensional shape change analysis. The results obtained using this technique indicate that asymmetrical shape development is influenced primarily by the variability of surface roughness developed during ablation and, to a lesser extent, by trajectory and vehicle parameters and nose-tip material mean roughness height. The key nose-tip shape feature determining nose-tip normal force and hence vehicle trim angle of attack is the radial offset of the laminar stagnation region from the vehicle centerline. The magnitude of the trim angles obtained is primarily sensitive to vehicle bluntness ratio and somewhat dependent on trajectory parameters. Nose-tip-induced roll torques are predicted to be low and agree well with those measured in wind-tunnel tests of a recovered flight nose tip. Apparently the much larger random excursions in roll torque observed in flight tests are not the result of nose-tip asymmetries, as previously thought.

Sensitivity studies on the role of material properties indicate that reduction in the variability of roughness element sizes will reduce almost proportionally the degree of nose-tip asymmetry developed, whereas increases in mean roughness height to force higher altitude transition will yield only slight improvements. However, extrapolation of the results for asymmetric shape development and trim histories to expected vehicle dispersion is not feasible without considering detailed trajectory analysis. Low-altitude roll-trim dispersion is sensitive to both the rate of change of trim angle over a revolution cycle and the magnitude of the trim angle. In addition, shifts in the normal force vector angle may have the same influence as roll through zero in that, if the shift occurs in the opposite direction to the roll rate and at the same rate, the lift vector may be fixed momentarily in inertial space and yield a large lateral velocity jump and high dispersion.

Acknowledgment

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ANALYSIS OF A FILM-COOLED NOSE TIP IN A PARTICLE EROSION ENVIRONMENT

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Abstract

A method of analysis is described for predicting heat-transfer, coolant mass loss, thermal response, and surface erosion characteristics of a liquid film-cooled nose tip in a particle erosion environment. The analysis includes the effects of particle impingement heating, convective heating augmentation, and coolant film boiling. The mechanisms contributing to coolant mass loss are vaporization, erosion, and droplet entrainment. Calculated results of film-cooled nose tip performance are presented for a particular configuration subjected to particle erosion testing in rocket motor, arc heater, and ballistic range facilities. The surface temperature is shown to be the significant parameter for evaluating nose tip performance.

Nomenclature

B_o = blowing parameter
C_H = Stanton number
C_N = erosion coefficient
C_P = specific heat
D_P, D_C = particle and crater diameters
E_P = particle kinetic energy flux
g = acceleration due to gravity or aerodynamic forces
G = erosion mass flow ratio
h = film thickness
h_c = convective heat-transfer coefficient

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h_{fg} = heat of vaporization
H = enthalpy
k = thermal conductivity
K = surface roughness height
K_s = equivalent sand grain roughness height
K_{aug} = heating augmentation factor
L_o = reference length
$\dot{m}_e$ = surface mass loss rate
M = Mach number
$\dot{M}$ = coolant mass flow rate
$\dot{M}_{vap}$, $\dot{M}_{eros}$, $\dot{M}_{ent}$ = coolant mass loss rates by vaporization, erosion, and entrainment, respectively
P = pressure
Pr = Prandtl number
$\dot{Q}_B$ = boiling heat flux
$\dot{Q}_{conv}$ = convective heat flux
r = radial coordinate
S = surface distance
t = time
T = temperature
U = velocity
X_e = entrainment parameter
Y = distance normal to surface
α = thermal diffusivity
γ = specific gravity
δ^* = displacement thickness
δ_T = thermal penetration depth
μ = viscosity
ρ = density
σ = surface tension
τ = shear stress
θ = body slope

Subscripts

b = bulk liquid
e = edge of boundary layer
g = gas value at gas-liquid interface
i = initial value
ℓ = liquid
m = melt
o = nonblowing
p = particle
r = recovery
s = stagnation point
v = vapor
w = wall
∞ = freestream

I. Introduction

Nose tips for strategic re-entry vehicle systems must be capable of withstanding severe aerothermal loads as well as particle impingement effects resulting from flight through hydrometeor or dust environments. Passive nose tips generally experience high ablation and erosion rates in these environments, and the resulting shape change may have undesirable effects on re-entry vehicle performance. Therefore, consideration has been given to actively cooled nose tips utilizing transpiration or film-cooled injection schemes. The performance of these nose tips in severe thermal environments has been evaluated for the case of clear-air re-entry flight,[1,2] but only minor consideration has been given to the evaluation of active nose tips in re-entry flight through hydrometeor or dust environments. This problem therefore is addressed in the current paper, with results provided for a new film-cooled nose tip concept specifically designed for high thermal and erosion resistance.

II. General Problems

Active cooling concepts generally employ metallic nose tips that are either sintered, drilled, or slotted to provide flow passages for the liquid or gaseous coolant. The metal surfaces are designed to remain relatively cool by flowing sufficient coolant to block and absorb the incident convective heat fluxes adequately. If properly designed, the metal surfaces will not experience any melting or vaporization that could result in nose tip shape change. This already has been demonstrated in clear-air re-entry flight tests and can be accomplished with a moderate supply of stored coolant. However, in a hydrometeor or dust environment, the following thermal problems are encountered:

1) Particle impingement heating. When liquid droplets or dust particles impact metallic surfaces, approximately 70% of their kinetic energy component normal to the surface is transformed into heat energy.[3] This heating can result in a significant temperature rise of the surface unless the heat is transferred effectively to the coolant. However, the transfer of heat to the coolant internally could result in two-phase flow, which alters the coolant flow rate.

2) Convective heating augmentation. It has been shown[3] that impinging particles and erosion debris can interact with the flowfield to cause a distortion of the bow shock. The resulting vorticity amplification causes a significant increase in the laminar heating rates. For flow conditions that are already turbulent, the bow shock distortion caused by

the particles does not appear to augment significantly the turbulent heating rates. However, the particle impingement causes a roughening of the liquid film surface, which does augment significantly the convective heating rates.

3) Coolant film boiling. For liquid film-cooled nose tips, there is a problem that occurs when the surface temperature exceeds the coolant vaporization temperature. This condition results in film boiling at the solid-liquid interface, in addition to the ordinary vaporization that occurs at the gas-liquid interface. The mass flow of the liquid coolant is decreased by the surface boiling, and there is a reduction in cooling effectiveness due to the formation of a vapor film between the liquid coolant and solid surface.

4) Erosion damage. As particles impinge the surface, a portion of its kinetic energy is used to create high pressures within the material. These pressures cause local failures and subsequent removal (erosion) of surface material. As the metallic surface heats up, its erosion resistance is reduced, and the rate of mass removal is increased. The resulting surface recession may be sufficient to cause a total loss of the nose tip. This is a critical design problem for porous wall transpiration-cooled nose tips that generally utilize thin walls (less than 0.2 in). In addition to the recession problem, the surface tends to become quite rough, with a charac-

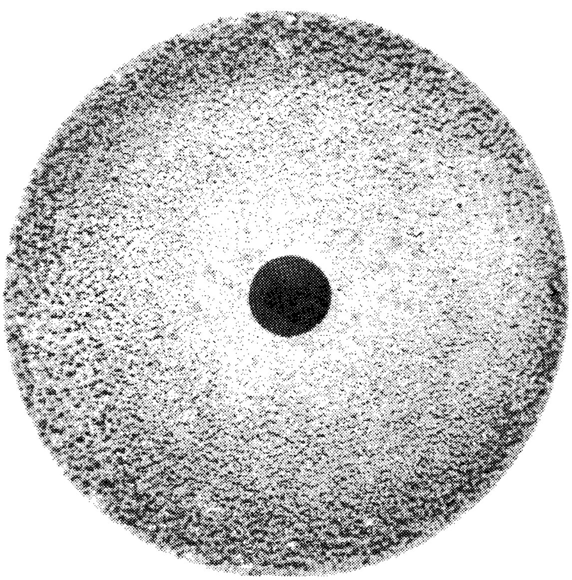

Fig. 1 Photograph of active nose tip after particle impacts.

teristic pebbly appearance, as shown in Fig. 1. Since the characteristic roughness height often exceeds the liquid film thickness, there is a potential for gross distortion of the coolant flow.

III. Analysis

To evaluate the performance of a liquid film-cooled nose tip in a particle erosion environment, one must perform coupled analyses of the coolant flow, the gaseous flow, the particle field, and the nose tip material thermal and erosion response. Figure 2 shows schematically the heat- and mass-transfer modeling that is used for these analyses. It is indicated that the liquid film loses mass by vaporization and entrainment, and, at some point, the film is lost completely. In designing the nose tip and determining the required coolant flow rates, this point generally is planned to be downstream of the high heating region. Beyond this point, there is some gas-vapor cooling, although it is not nearly as effective as liquid film-cooling. The following paragraphs describe the methods used for evaluating the heating rates, liquid mass losses, and material thermal/erosion response.

A. Heating Rates

The convective heating rates from the gas boundary layer to the surface of the liquid film are computed for both laminar and turbulent conditions. Using a conventional smooth surface, nonblowing Stanton number (C_{H_o}), the relation is

$$\dot{Q}_{conv} = \rho_e U_e C_{H_o} (H_r - H_g)(C_H/C_{H_o}) K_{aug}$$

where $\rho_e U_e$ is the mass flow at the boundary-layer edge, H_r is the recovery enthalpy, and H_g is the gas enthalpy at the gas-liquid interface. The ratio of blowing to nonblowing Stan-

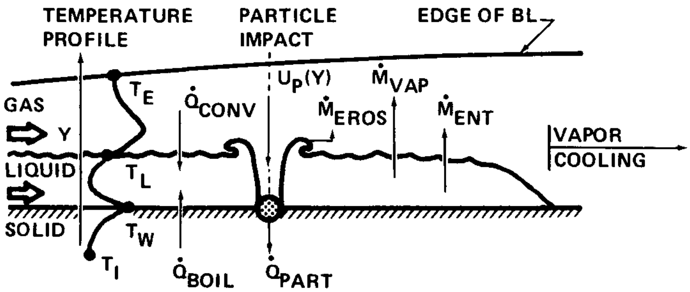

Fig. 2 Heat- and mass-transfer model for liquid film cooling in a particle erosion enviroment.

ton number (C_H/C_{H_o}) is computed in terms of a blowing parameter ($B_o = \dot{M}_{vap}/\rho_e U_e C_{H_o}$) by the correlations of Arne[4]:

$$(C_H/C_{H_o})_{lam} = 1. - (C_{P_V}/C_{P_e})^{0.4}(0.68\ B_o - 0.08\ B_o^2)$$

$$\left(\frac{C_H}{C_{H_o}}\right)_{turb} = \frac{(C_{P_V}/C_{P_e})^{0.8}\ B_o}{\left[1 + 0.25\ (C_{P_V}/C_{P_e})^{0.8}\ B_o\right]^4 - 1}$$

The quantity K_{aug} is the heating augmentation factor due to particle flowfield interactions or surface roughness. For laminar flows in a particle erosion environment, the method of Dunbar et al.[3] is utilized:

$$K_{aug} = 0.098\ \frac{\rho_\infty U_\infty}{\rho_e U_e C_{H_o}} \left[\frac{\rho_P U_P}{\rho_\infty U_\infty}\ (1+G)\right]^{0.317} \sin^2\theta$$

where $\rho_P U_P$ and $\rho_\infty U_\infty$ are the mass flows of particles and free-stream gas, respectively, and G is the ratio of eroded liquid mass flow to the particle mass flow. The effect of surface slope (θ) was not included in Ref. 3 but has been reported by the same authors in unpublished work.

For turbulent flows, the heating augmentation is attributed to the rough surface of the liquid film. In a particle erosion environment, the liquid surface has a cratered appearance, with a characteristic roughness that is similar to scallop patterns on ablating nose tips. Using the correlation of Grabow and White,[5] the heating augmentation factor is:

$$K_{aug} = \exp\left\{0.2497 + 0.211\ \ln\left(\frac{K_s}{\delta^*}\right) + 0.01025 \left[\ln \frac{K_s}{\delta^*}\right]^2 + 0.001675 \left[\ln\left(\frac{K_s}{\delta^*}\right)\right]^3\right\}$$

where δ^* is the smooth-wall nonblowing turbulent boundary-layer displacement thickness, and K_s is the equivalent sand grain roughness obtained from the relations

$$K_S \begin{cases} = 0.0164\ K\ (D_c/K)^{3.78} & \text{for } D_c/K \leq 4.93 \\ = 139.K\ (D_c/K)^{-1.9} & \text{for } D_c/K > 4.93 \end{cases}$$

where K is the peak-to-valley roughness (assumed to be the film thickness), and D_c is the crater diameter.

In addition to convective heat transfer from the boundary layer, the liquid film also may experience boiling heat transfer from the hot surface. This occurs when the surface temperature (T_W) exceeds the coolant vaporization temperature (T_V). Initially, the process is characterized by forced convection nucleate boiling, and the heating rate in this regime is computed by the method of Rohsenow[6]:

$$(\dot{Q}_B)_{nuc} = \mu_\ell h_{fg} \left[\frac{C_{P\ell}\ (T_W - T_V)}{0.013\ h_{fg}\ P_{r\ell}^{\ 1.7}} \right]^3 \sqrt{\frac{(\rho_\ell - \rho_V) g}{\sigma_\ell}} + h_c\ (T_W - T_B)$$

where μ_ℓ, ρ_ℓ, C_{P_ℓ}, σ_ℓ, and Pr_ℓ are liquid viscosity, density, specific heat, surface tension, and Prandtl number, respectively. The heat of vaporization is h_{fg}, and g is the acceleration of the film due to gravity and/or aerodynamic forces. The second term in the equation represents the forced convection heat transfer from the wall temperature (T_W) to the bulk temperature (T_B) of the flowing liquid. The Rohsenow equation is used until the nucleate boiling heating rate reaches a maximum value, which, according to Gambill,[7] is given by

$$(\dot{Q}_B)_{max} = 0.145\ h_{fg} \rho_V \left[\frac{\sigma_\ell g (\rho_\ell - \rho_V)}{\rho_V^{\ 2}} \right]^{1/4} \left[1 + \left(\frac{\rho_\ell}{\rho_V} \right)^{0.923} \frac{C_{P_\ell}(T_V - T_B)}{25\ h_{fg}} \right]$$

As the surface temperature is increased further, a stable film boiling condition is reached. The heating rate in this regime is predicted by the forced convection film boiling equation of Bromley[8] for cross flow over a cylinder:

$$(\dot{Q}_B)_{film} = \left\{ (7.3/D)\ U_\ell k_V \rho_V (T_W - T_B) \left[h_{fg} + 0.4 C_{P_V} (T_W - T_B) \right] \right\}^{1/2}$$

where ρ_V, C_{P_V}, and k_V are the coolant vapor density, specific heat, and conductivity, respectively. U_ℓ is the liquid velocity, and D is the cylinder diameter (nose tip diameter in current application). This equation is valid for liquid velocities, $U_\ell > 2\sqrt{gD}$, which is the case of interest for film-cooled nose tips in re-entry flight or ground-test environments.

The preceding equations permit calculations of the rate of heat transfer to the liquid film. This energy is used to heat the liquid to its boiling point and then vaporize it at a temperature that is a function of the local pressure. If the liquid film remains intact, there should not be any transfer of the boundary-layer convective heating to the surface. However, in a particle erosion environment, the entire surface is heated by the conversion of particle kinetic energy into thermal energy. This type of energy flux cannot be blocked or absorbed by the liquid film. Assuming 70% kinetic energy accommodation,[3] the particle impingement heat flux is computed to be

$$\dot{Q}_{part} = 0.7 \left[1/2 \ \rho_P U_P^{\ 3} \right] \sin^2\theta$$

where ρ_P is the particle field density and U_P is the particle impact velocity. For impacts normal to the surface, slowdown in the liquid film is computed by the method of Waldman and Reinecke[11]:

$$U_P = U_{P_\infty} \ \exp \left[-0.75 (\gamma_\ell/\gamma_P)(h/D_P) \right]$$

where h is the film thickness in the direction of particle travel, D_P is the particle diameter, and γ is the specific gravity of the liquid (ℓ) and particle (P).

B. Liquid Mass Loss

The mass loss due to vaporization is the result of convective and boiling heat transfer to the liquid. The integrated vaporization rate over a surface element (ΔS) at a radial distance (r) from the axis of symmetry is

$$\dot{M}_{vap} = \frac{\dot{Q}_{conv} + \dot{Q}_B}{C_{P_\ell}(T_V - T_i) + h_{fg}} \ 2\pi r \ \Delta S$$

The next liquid mass loss mechanism to be considered is the result of particle erosion. As particles impinge on the thin liquid film, they displace some of the liquid and thereby reduce the total coolant mass flow. To estimate the magnitude of this loss, it is assumed that each impinging particle removes a column of water having the diameter of the impact crater (D_c) and the thickness of the liquid film (h). The integrated erosion mass loss over a surface element ΔS is

$$\dot{M}_{eros} = \rho_P U_P \ (\gamma_\ell/\gamma_P)(3h/2D_P)(D_c/D_P)^2 \ 2\pi r \ \Delta S$$

where the crater diameter ratio is obtained from the empirical correlation of Kinslow[12] for thin targets:

$$D_C/D_P = 0.9 + 0.137(U_P/1000)(h/D_P)^{2/3} \quad (U_P \sim \text{fps})$$

The final mass loss mechanism is the mass entrainment of liquid droplets, which is the dominant effect in high-pressure environments. The liquid mass loss due to entrainment is computed in a stepwise manner using the method of Gater and L'Ecuyer.[13] By this approach, the mass entrained over a surface element is a function of the total mass flow ($\dot{M}$) entering that element, according to the relation

$$\dot{M}_{ent} = \dot{M}\left\{1 - \exp\left[-5 \times 10^{-5}\ (X_e - 1000)\ (\Delta S/L_o)^{0.9}\right]\right\}$$

where the entrainment parameter is

$$X_e = \sqrt{\rho_e U_e^2}\ (T_e/T_V)^{0.25}\ \sigma_\ell \sim lb_f^{-1/2}$$

and ρ_e, U_e, T_e, are the boundary-layer edge density, velocity, and temperature, respectively. The quantity L_o is a reference length of 10 in., based on the experiments of Ref. 13.

The total mass flow for the next section is

$$\dot{M}_{r+\Delta r} = \dot{M}\ (r/r+\Delta r) - \dot{M}_{vap} - \dot{M}_{eros} - \dot{M}_{ent}$$

where the first term represents the total mass flow from the current element modified for the radial spreading of the film between the current and successive elements. The film thickness is determined from the local total mass flow ($\dot{M}$) and gaseous shear stress (τ). Assuming a linear velocity profile in the liquid film, the thickness is given by

$$h = (\dot{M}_{\mu_\ell}/\pi r \rho_\ell \tau)^{1/2}$$

In applying this relation, it is assumed that the shear stress is augmented by particle and roughness effects in the same manner as the convective heating rate. Therefore, the thickness of the liquid film is predicted to decrease in a particle erosion environment.

C. Nose Tip Thermal/Erosion Performance

The coupled analysis of the gas-liquid-particle flow yields a net heat-transfer rate to or away from the nose tip surface. For metallic nose tips in a particle erosion environment, there is always net heating to the surface. This heat is conducted to the interior portion of the nose tip, resulting in temperature gradients within the material. Predictions of the internal temperature distributions are required for the determination of material thermal strains and the potential for internal crack formation.

In the present analysis of material thermal response, the temperature distributions are computed approximately using one-dimensional, semi-infinite slab solutions. For a specified net heating rate history and constant thermal properties, Goodman[14] gives the following solution for the surface temperature history:

$$T_W = T_i + \frac{1}{K}\left(\frac{4}{3}\alpha\dot{Q}_{net}\int_0^t \dot{Q}_{net}\,dt\right)^{1/2}$$

where α is the thermal diffusivity, and k is the material conductivity. This solution is based on an assumed in-depth temperature distribution,

$$T(Y) = T_i + (T_W - T_i)\left[1 - Y/\delta_T\right]^3$$

where δ_T is the thermal penetration depth, obtained from the relation $\delta_T = 3.5\sqrt{\alpha t}$. The main requirement in utilizing this semi-infinite slab solution is that the material thickness exceeds the thermal penetration depth.

The computation of surface temperature is required primarily to determine if melting of the nose tip material will occur. If excessive surface temperatures are predicted as a result of loss of the coolant film, then the solution may be simply to increase the coolant flow rate. However, if the excessive temperatures are caused by particle impingement heating, then an increased coolant flow will provide only a minor benefit through the effect of particle slowdown. The more effective solution may be the replacement of the nose tip material with one that has either a higher melt temperature or higher conductivity.

Another important reason for computing the surface temperature is to evaluate the reduction in material erosion resistance as the surface heats up. This variation is modeled in the form of an erosion coefficient (C_N) vs the ratio of surface temperature to melt temperature (T_W/T_m), where

$$C_N = \frac{\text{(particle kinetic energy flux)}}{\text{(surface mass loss rate)}} = \frac{E_P}{\dot{m}_e}$$

For refractory metals commonly used as active nose tip materials, C_N tends to remain constant at a high value

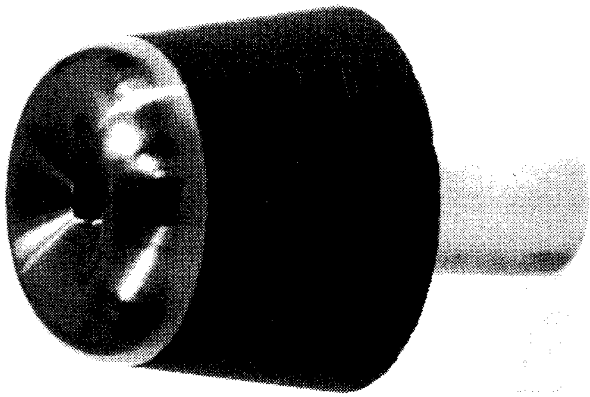

Fig. 3 Photograph of swirling cooling active nose tip (SCAN).

($\simeq 10^4$ J/g) for $T_w/T_m < 0.3$. At higher temperature ratios, C_N is reduced drastically as a result of the degraded erosion resistance. Close to the melt condition, C_N may be reduced by as much as a factor of 30 relative to its room-temperature value. The actual variations with temperature ratio used in the present analysis are based on a limited amount of experimental data.

IV. Results

A. Configuration

The methodology discussed in the preceding section has been incorporated into a generalized computer program for analyzing film-cooled nose tips of arbitrary shape and mode of coolant injection. The majority of the analyses have been performed for a film-cooling concept known as the "swirling coolant active nose tip" (SCAN). A photograph of this configuration is presented in Fig. 3, which shows the tantalum -10% tungsten (Ta-10W) tip and phenolic carbon heat shield.

The schematic drawing of the SCAN nose tip in Fig. 4 shows how the coolant is ejected with a swirling motion from a single central orifice. The swirl is created by tangential injection of the liquid into the swirl chamber at two or more locations. The chamber necks down to the injection orifice, and the coolant exits with a radial velocity that is large relative to its axial velocity. The advantages of swirling the coolant flow are as follows: 1) the coolant is injected nearly tangential to the surface, which maximizes the cooling effectiveness; 2) the coolant is distributed uniformly over the nose tip, thereby reducing the potential for local hot spots; 3) swirling eliminates the problem of "jetting" which would result in bow

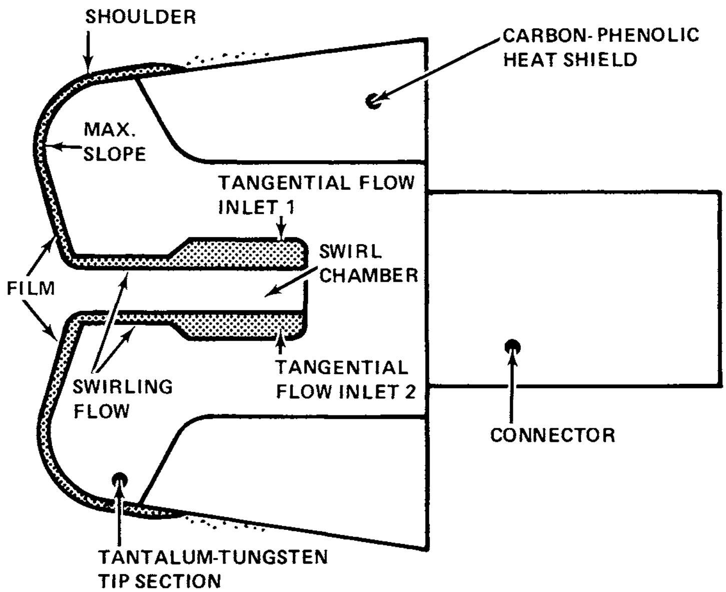

Fig. 4 Schematic of swirling cooling active nose tip (SCAN).

shock perturbations and loss of cooling effectiveness; and 4) the radial momentum of the coolant will provide flow over the entire nose tip, even at large angles of attack.

As indicated in Fig. 4, the nose tip has a blunt concave shape with the forward surface nearly tangent to the coolant flow for optimum cooling effectiveness. The thick shock layer associated with the blunt shape contributes to slowdown of incident particles, thereby reducing the erosion damage. An important feature of the blunt concave shape is that it experiences maximum heating and erosion at the shoulder, which is located sufficiently far from the coolant supply area.

SCAN already has been subjected to particle erosion testing in the Aeronutronic METS Rocket Motor Facility[15] and the AVCO 10 MW Arc Heater Facility.[16] Plans have been made for testing in the Arnold Engineering Development Center (AEDC) Ballistic Range Facility. The following paragraphs present the results of analyses of SCAN performance in each of these ground-test environments. In all cases, the liquid coolant is water, which is injected from a 0.16-in.-diam central orifice.

B. METS Results

The analyses were performed for the following conditions: 140-μ-diam graphite particles (γ_p = 1.73); particle velocity

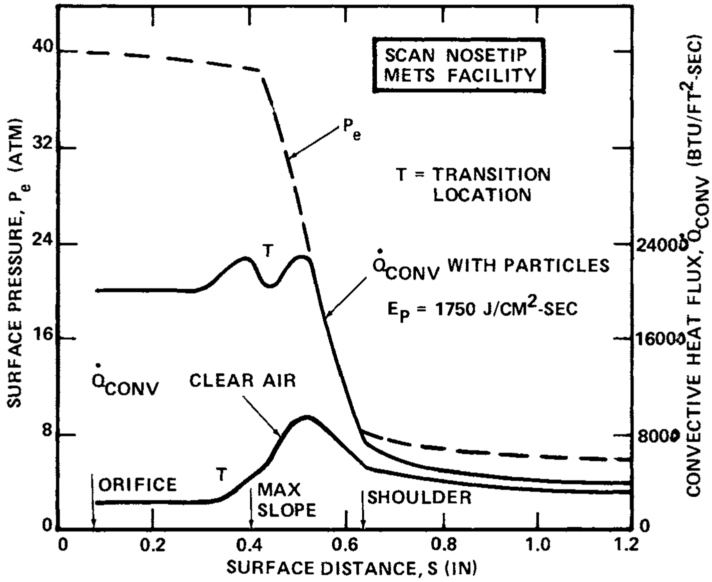

Fig. 5 Pressure and heating distributions for SCAN test in METS Rocket Motor Facility.

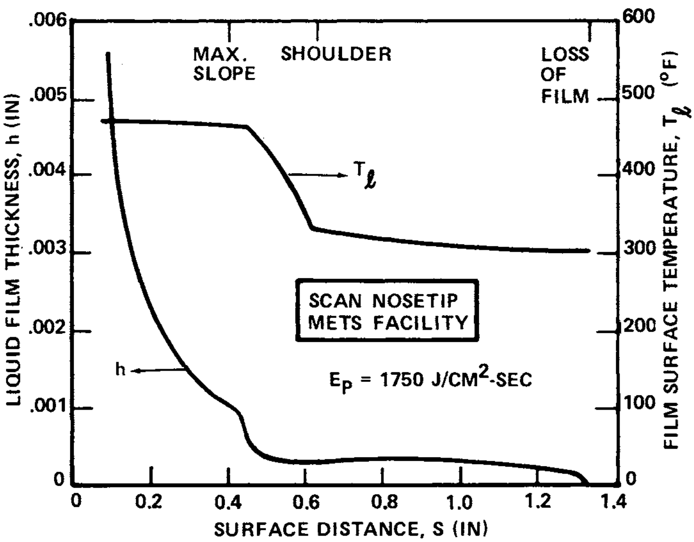

Fig. 6 Liquid film thickness and film temperature distributions in SCAN/METS test.

U_p = 6500 fps; particle field density $\rho_p = 2.83 \times 10^{-4}$ lb/ft^3; particle exposure time t_p = 1.75 sec; gas velocity U_∞ = 9700 fps; stagnation pressure P_S = 40 atm; and stagnation temperature

T_S = 6300 °F. The maximum particle kinetic energy flux was 1750 J/cm^2-sec.

Figure 5 presents the computed nose tip surface pressure and non-blowing convective heating distributions with and without the particle field. The heating rates, expressed as cold-wall values (H_g = 0), clearly show the large augmentation that occurs with particle impingement. The location of boundary-layer transition is difficult to predict for film-cooled nose tips, but as a conservative design procedure is selected to insure that the greater of laminar or turbulent heating rates prevails. It is noted that the heating rates provided in Fig. 5 do not include the effects of blowing.

The predicted liquid film thickness and temperature distributions are presented in Fig. 6 for a coolant flow rate of 0.19 lb/sec. The film thickness is shown to decrease rapidly between the orifice and maximum slope location. This is attributed to the combined efforts of radial spreading, liquid mass loss, and boundary-layer shear stress variations. The film surface temperatures are identical to the vapor temperatures for water and therefore vary in the same manner as the surface pressure (Fig. 5). The liquid mass loss distributions are presented in Fig. 7, where the losses due to vaporization and entrainment are shown to be of the same magnitude, whereas the erosion mass loss is much smaller. The film is predicted to be totally lost at a surface distance of 1.33 in., which is actually on the heat shield, since the metallic tip section ends at a distance of 0.85 in. To provide adequate protection of the tip only, a coolant flow rate of 0.14 lb/sec would be required.

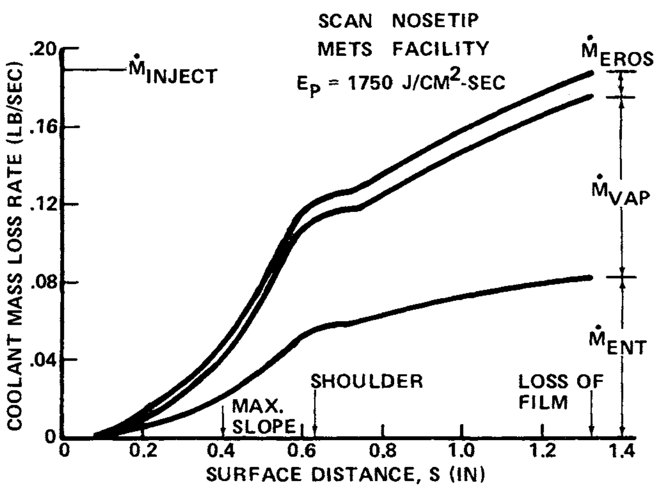

Fig. 7 Coolant mass loss distributions in SCAN/METS test.

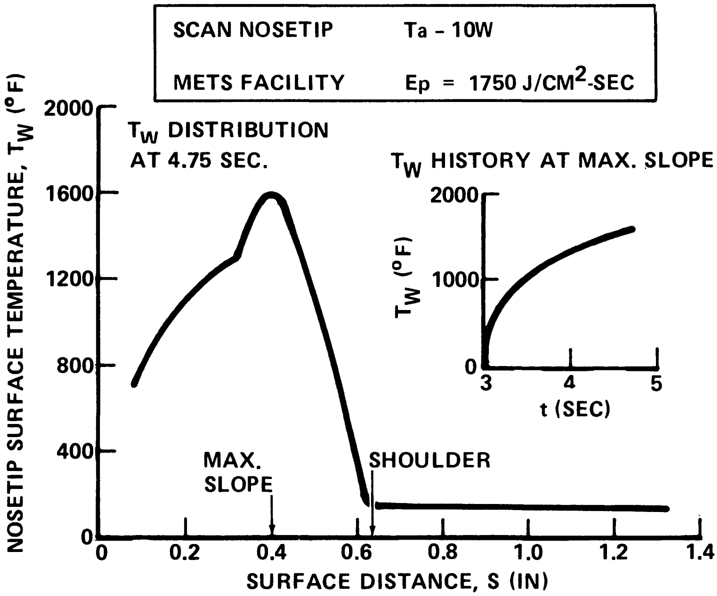

Fig. 8 Nose tip surface temperature response in SCAN/METS test.

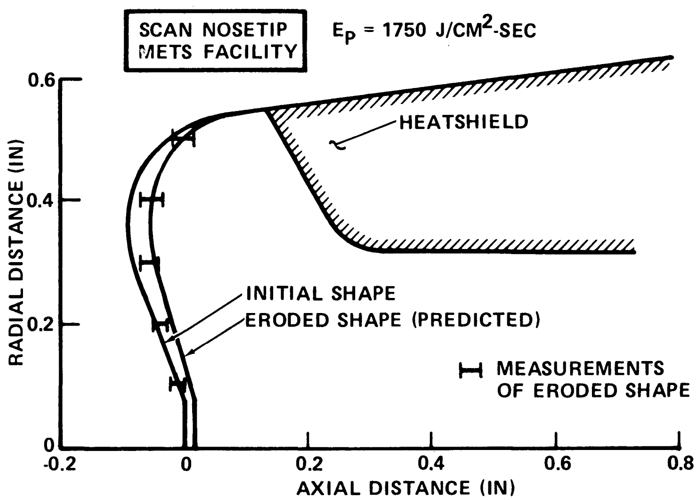

Fig. 9 Predicted and measured nose tip erosion profiles in SCAN/METS test.

The predicted Ta-10W surface temperature distribution at the end of the particle exposure period and the maximum surface temperature history are shown in Fig. 8. Comparing these surface temperatures to the liquid film temperatures of Fig. 6 reveals that surface boiling prevails over the forward portion of the nose tip (to a surface distance of 0.6 in.). A maximum surface temperature of 1600°F is predicted to occur at the maximum slope location. This is well below the melt temperature of any tantalum oxides that may form on the surface (e.g., the melt temperature for Ta_2O_5 is in the range of 3200°-3500°F). For the Ta-10W virgin material melt temperature of 5500°F, the temperature ratio T_w/T_m equals 0.35, which is sufficient to cause some degradation of erosion resistance.

The predicted nose tip recession profiles are presented in Fig. 9 and compared to the measured recession data at several locations. The data bands represent the circumferential recession variations, indicating some degree of asymmetric erosion. The predictions (with a maximum recession of 0.04 in.) agree fairly well with the data in the vicinity of the maximum slope and shoulder locations but significantly overestimate the recession close to the injection orifice. This may be attributed to improper modeling of either the erosion coefficient variation with surface temperature or the film thickness variation along the nose tip and its particle slowdown capability.

C. 10-MW Arc Results

The analyses were performed for a SCAN nose tip slightly smaller than the one tested in METS and exposed to the following conditions: 90-μ-diam graphite particles (γ_P = 1.73); particle velocity U_P = 5570 fps; particle field density ρ_P =

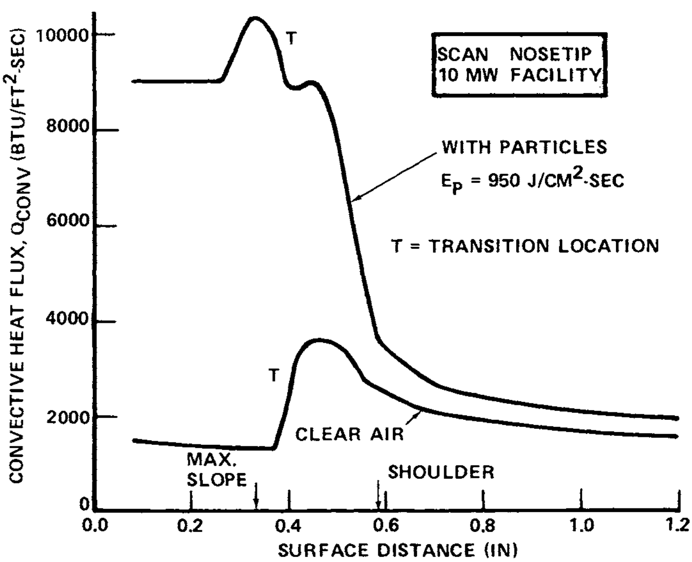

Fig. 10 Heating distributions for SCAN test in 10-MW Arc Heater Facility.

2.42×10^{-4} lb/ft^3; particle exposure time t_P = 4.0 sec; gas velocity U_∞ = 9720 fps; stagnation pressure P_S = 6.6 atm; and stagnation temperature T_S = 10,300°F. The maximum particle kinetic energy flux was 950 J/cm^2/sec.

Figure 10 presents the non-blowing convective heating rate predictions for this test condition with and without particles. As already indicated for the METS test, a significant heating augmentation occurs as a result of particle impingement.

For these tests, it was desired to use coolant flow rates low enough to approach a condition of incipient failure. To establish this condition, analyses of the SCAN nose tip thermal response were performed for a range of coolant flow rates. Figure 11 presents the predicted maximum surface temperatures as a function of water flow rate with and without particles. The surface temperatures are shown to be influenced only slightly by water flow rate for rates greater than 0.02 lb/sec. For

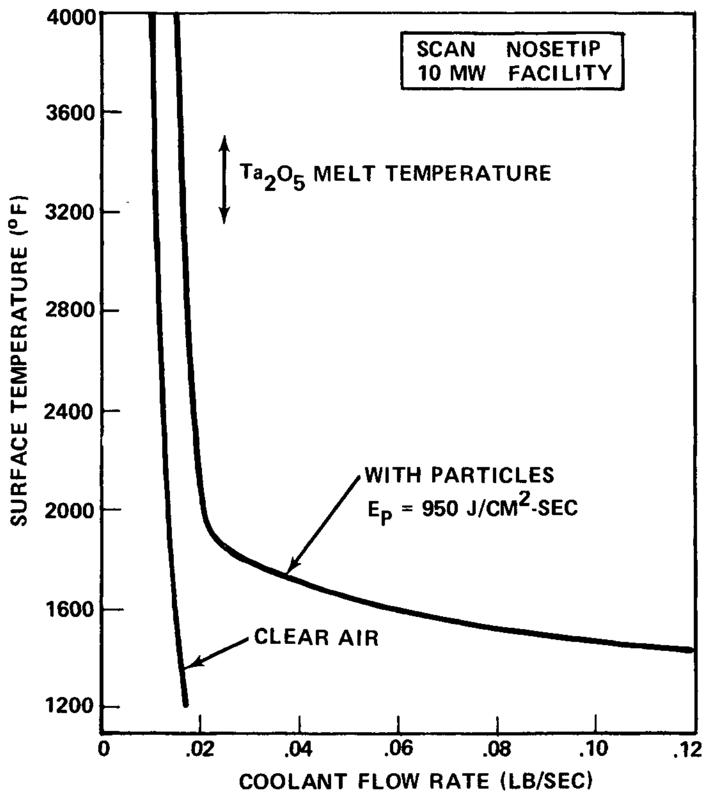

Fig. 11 Maximum nose tip surface temperature as a function of coolant flow rate in SCAN/10-MW test.

lower water flow rates, the surface temperatures increase sharply as a result of loss of the liquid film prior to the shoulder.

Unfortunately, the predicted incipient failure condition could not be verified experimentally because of facility expulsion system problems at low coolant flow rates. The lowest flow rate at which SCAN could be tested with no expulsion system problems was 0.06 lb/sec. The computed maximum surface temperature for this flow rate is 1600°F, similar to the METS condition. The predicted recession profile also is close to METS, since the time-integrated particle kinetic energy fluxes are nearly the same. The prediction is shown in Fig. 12 and compared to the experimental data at several surface locations.

The measured recession is again shown to be much lower than the prediction. This disparity possibly may be attributed to particle ablation and breakup prior to impacting the model. Another possible explanation stems from the observation that the liquid film was considerably thicker than predicted, which could reduce the particle impact velocity. However, since it was difficult to measure liquid film thickness from motion picture coverage, this explanation could not be supported.

D. Ballistic Range Results

The analyses were performed for the following conditions: 325-μ-diam glass particles (γ_p = 2.84); average particle density $U_p = U_\infty$ = 15,500 fps; particle field density ρ_p = 1.06 x 10^{-4} lb/ft^3; particle exposure time t_p = 0.039 sec (first 600 ft of range); stagnation pressure P_S = 42 and 145 atm; and

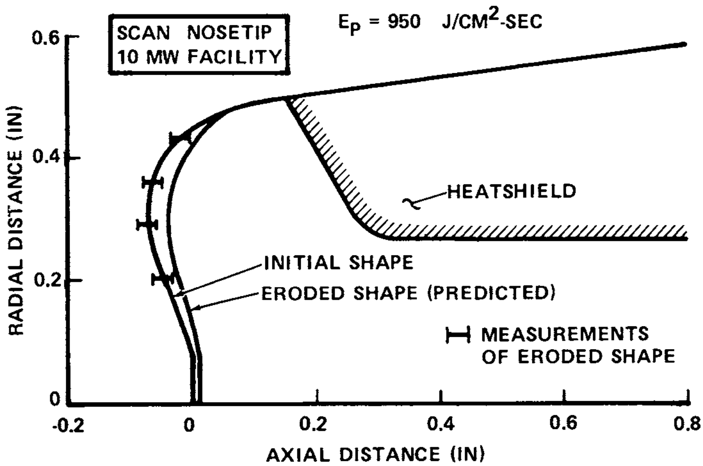

Fig. 12 Predicted and measured nose tip erosion profiles for SCAN/10-MW test.

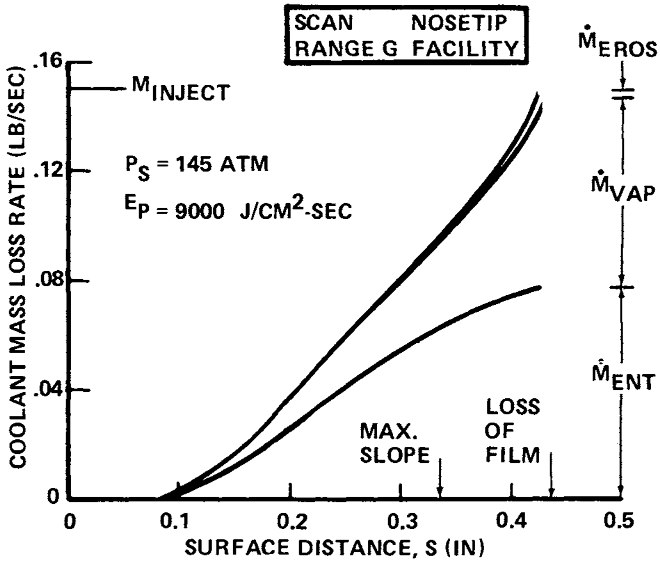

Fig. 13 Coolant mass loss distributions in SCAN/Range G Ballistic Range test.

stagnation temperature T_S = 10,700°F. The maximum particle kinetic energy flux was 9000 J/cm^2-sec.

The analyses for the high-pressure condition (P_S = 145 atm) reveal some interesting features of SCAN performance. For a water flow rate of 0.15 lb/sec, the coolant film is predicted to be lost just aft of the maximum slope location. As indicated by the mass loss curves of Fig. 13, the loss by entrainment is the dominant effect, especially in the area forward of the maximum slope. The vaporization losses are next in severity, with negligible water losses due to erosion. This is attributed to the very thin liquid film at high pressures and the relatively large particle size. This behavior is believed to be characteristic of film-cooled nose tips in re-entry flight environments.

Figure 14 presents the predicted surface temperature distribution at the end of the erosion period for the high-pressure case. The temperatures are shown to increase rapidly downstream of the location where the film is lost. A comparison of the maximum surface temperature histories is shown in Fig. 15 for the low- and high-pressure cases. Based on these results, no problems are expected at the low-pressure condition, but, at the high-pressure condition, the nose tip temperatures rapidly exceed the Ta_2O_5 melt temperature and become dangerously close to the Ta-10W melt temperature. Therefore,

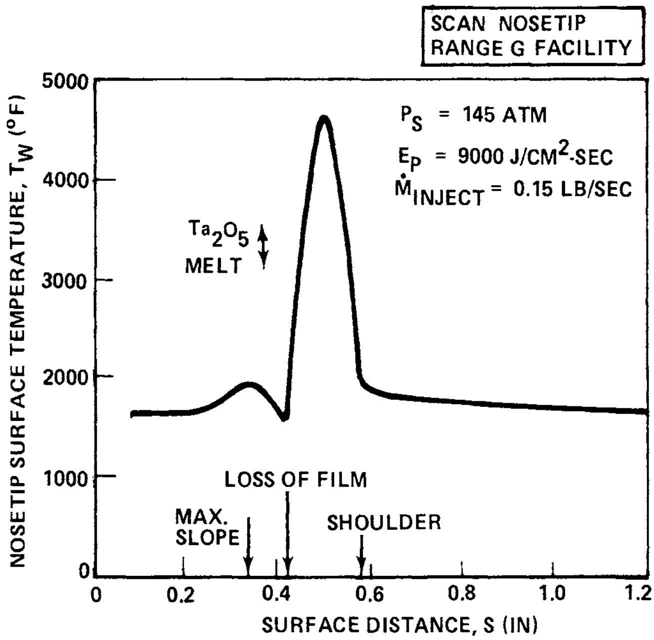

Fig. 14 Nose tip surface temperature distribution in SCAN/Range G Ballistic Range test.

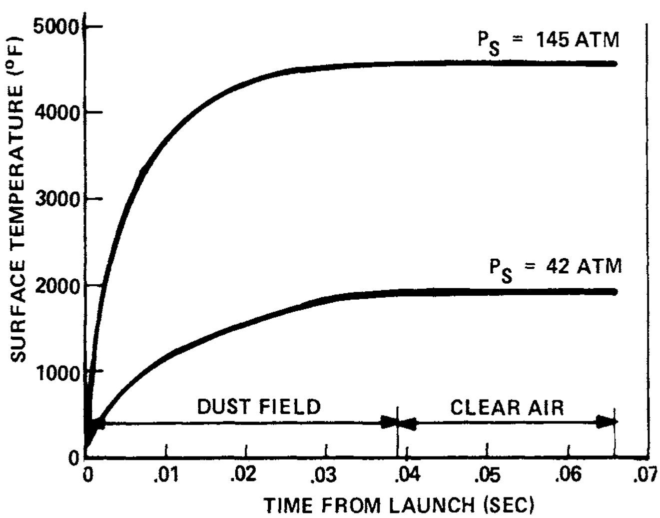

Fig. 15 Maximum nose tip surface temperature history in SCAN/ Range G test.

a severe degradation of erosion resistance is expected for the coolant flow rate of 0.16 lb/sec.

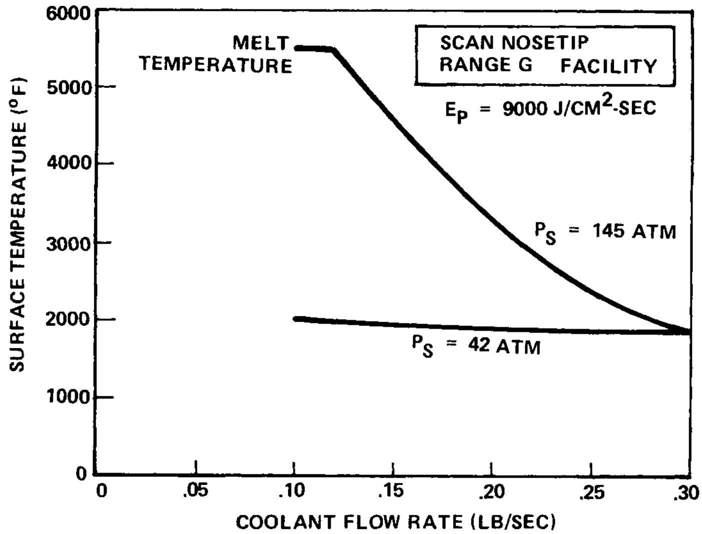

Fig. 16 Maximum nose tip surface temperature as a function of coolant flow rate in SCAN/Range G test.

The problem at the high-pressure condition can be eliminated by increasing the coolant flow rate to 0.25 lb/sec, as shown in Fig. 16. The results for the low-pressure condition are shown to be relatively insensitive to coolant flow rate, since the film is always intact (for flow rates greater than 0.1 lb/sec), and particle impingement heating is the only mechanism causing the temperature rise. In the test planning for this facility, it is recommended that external pyrometers be used to measure surface temperature at various stations along the range. These data would be extremely valuable in evaluating the nose tip performance and in assessing the validity of the present analytical techniques. This type of data was lacking in the METS and 10-MW Arc tests, which severely limited the usefulness of the test results.

V. Conclusions

1) The main thermal problems associated with liquid film-cooled nose tips in a particle environment are particle impingement heating, convective heating augmentation, film boiling, and erosion damage.

2) The mechanisms contributing to mass loss of the liquid coolant are vaporization, erosion, and droplet entrainment. In high-pressure environments, the entrainment mechanism is dominant.

3) The surface temperature is a critical parameter in evaluating active nose tip performance, since it is linked directly to the melting and erosion characteristics of the nose tip material.

4) The analysis of the SCAN nose tip in the METS facility predicts a maximum surface recession of 0.04 in. The measured recessions are close to the predictions over most of the nose tip, but less than predicted close to the injection orifice.

5) The predicted SCAN surface temperatures and recessions in the 10-MW Arc facility are nearly the same as for METS. However, the measured recession is much less; this disparity has not been resolved adequately.

6) The SCAN nose tip is predicted to survive the AEDC Ballistic Range environment at a stagnation pressure of 42 atm, with a coolant flow rate of 0.1 lb/sec. At a stagnation pressure of 145 atm, where droplet entrainment is the major mass loss mechanism, a coolant flow rate of 0.25 lb/sec is required.

Acknowledgment

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HIGH-PRESSURE ARC TEST PERFORMANCE OF CARBON-CARBON NOSE TIPS

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Abstract

Results are presented of recent high-pressure arc tests of three-dimensional woven carbon-carbon nose tips. A steady-state energy balance is used to compute the degree of heating augmentation caused by surface roughness that is compatible with the measured recession rate and surface temperature. Surface roughness deduced from augmented heating correlations is compared with values obtained from analysis of photomicrographs of post-test sectioned models. Consistent results are obtained for GE 2-2-3 carbon-carbon over the tested pressure range of 80 to 168 atmospheres. Boundary-layer transition inferred from shape change profiles during ramp tests is shown to be non-axisymmetric. In addition, two distinct types of nose sharpening obtained during ramp tests are identified, and a criterion is developed for this behavior.

Nomenclature

B' = blowing parameter, $B' = \dot{m}_W/C_h$
C_h = heat-transfer coefficient
D^* = nozzle throat diameter
D_e = nozzle exit diameter
H_{CL} = centerline enthalpy
H_W = wall enthalpy
K_s = equivalent sand grain roughness height
M_∞ = freestream Mach number
$\dot{m}_W$ = mass loss rate at wall
p_c = arc chamber pressure
p_s = stagnation pressure on test model
p_{TR} = value of p_s at which transition occurs in ramp test
$\dot{q}$ = heat flux

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R_C = corner radius
R_N = nose radius
Re = Reynolds number
$\dot{s}$ = steady-state recession rate
T = temperature
U_τ = shear velocity
δ^* = boundary-layer displacement thickness
θ = boundary-layer momentum thickness
θ_c = cone half-angle
ν = kinematic viscosity

Subscripts

o = smooth surface value
W = wall value
c = convective
RR = reradiative
∞ = freestream

Introduction

It has become current practice to test advanced carbon-carbon nose-tip materials in high-pressure arcs to determine their relative ablative performance prior to flight-testing selected candidate materials. Two types of tests commonly are conducted: 1) steady-state (Fig. 1), in which a constant environment is imposed on the model; and 2) ramp (Fig. 2), in which the pressure is increased steadily by advancing the model toward the nozzle exit.

The parameter normally used to compare material performance in steady-state tests is the steady-state recession rate. A material with a high steady-state recession rate would be expected to ablate more rapidly in re-entry than one with a lower rate. This parameter is reported routinely for all tests.

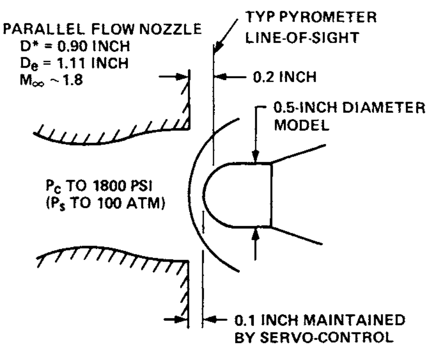


Fig. 1 Steady-state test technique (50-MW arc).

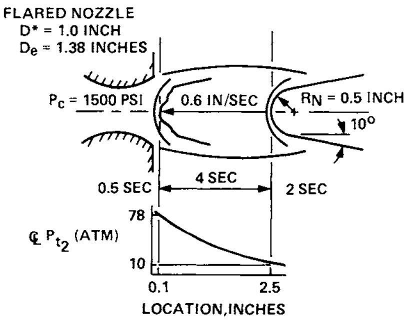


Fig. 2 Ramp test technique (50-MW arc).

The parameter normally used to compare material performance in ramp tests is transition pressure, since it is a measure of the ability of a material to withstand breakdown to turbulent flow. A material with a high transition pressure will maintain its blunt laminar shape to a lower altitude during re-entry than a material with a low transition pressure, with subsequent lower total recession due to delay of the period of high turbulent heating rates. This parameter is reported routinely for all tests and is thus a convenient and meaningful yard-stick for performance comparisons.

Several publications have appeared in the past on high-pressure arc steady-state test results [1-5] and ramp test results.[6] These references consider various facets of ablative performance of graphitic materials. The objective of the present paper is to report results of recent high-pressure arc tests of GE 2-2-3 carbon-carbon nose-tip models which illustrate unique aspects of performance which hitherto have not been widely publicized. These results originally were documented in Ref. 7 and 8.

The General Electric Company's GE 2-2-3 C-C material is a three-dimensional orthogonally reinforced, carbon-carbon composite with an x:y:z direction graphite fiber reinforcement ratio of 2:2:3. This material is manufactured by weaving the reinforcing yarns into the desired geometry and then densifying this woven preform using processing techniques to infiltrate the woven structure with a graphitic matrix.

Table 1 Nominal test conditions

Facility	Pressure, atm	Enthalpy, Btu/lb	Model diameter, in.
50-MW arc	80	5000	0.5
HIP	124	3000	0.3
HIP	168	3800	0.3

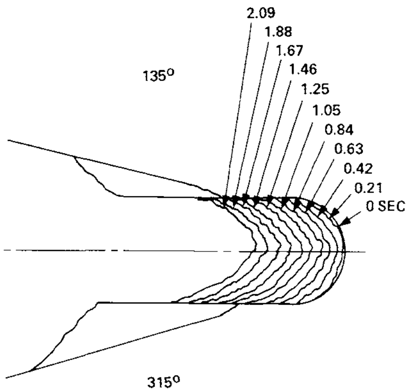


Fig. 3 Typical model shape change history (50-MW arc steady-state test).

Test Facilities

The two facilities most frequently used for steady-state tests are the 50-MW arc of the Air Force Flight Dynamics Laboratory and the HIP facility of the MacDonnell-Douglas Research Laboratory. In the 50-MW arc, initially spherical-nose models sharpen to a biconic shape (Fig. 3) with a quasi-steady-state turbulent recession rate. In the HIP facility, the models are initially biconic and retain their turbulent shape during exposure. Nominal test conditions for steady-state tests in these facilities are given in Table 1.

The enthalpy values quoted in the table are flow centerline values that are inferred from calorimeter heat-flux measurements as required to match theory, and tacitly assume no heat-transfer augmentation due to freestream turbulence in the arc flow. In the case of the 50-MW arc, this assumption has been the subject of considerable controversy. On the one hand, the study of Hoshisaki et al[9] suggests that the discrepancy between bulk enthalpy obtained from an arc chamber heat balance and that computed from stagnation-point heat-flux calorimetry and theory can be caused by a moderate turbulence level, such as exists in arc flow. On the other hand, the study of Brown-Edwards[10] concludes that the freestream turbulence in the arc is unlikely to cause heat-flux enhancement because the main frequencies are too low.

The ramp test technique has become a standard test procedure in the 50-MW arc and also has been used at the HIP facility. Typical determination of transition pressure is shown in Fig. 4. By using three cameras (as shown in Fig. 5), transition can be observed on six rays for each model, as inferred from the onset of gouges in the profile.

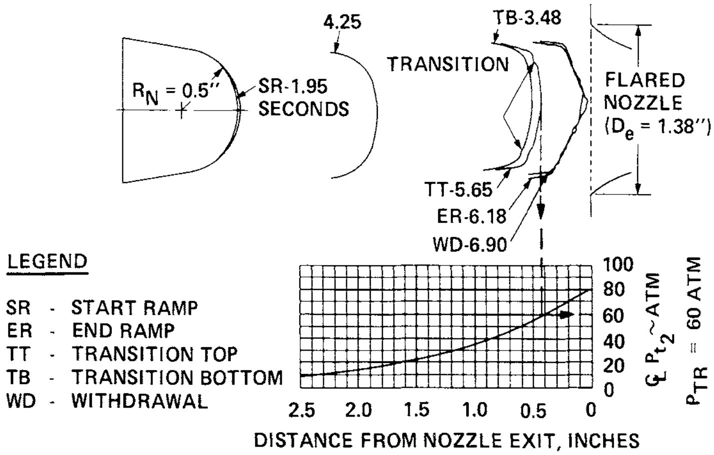


Fig. 4 Typical determination of transition pressure for 50-MW arc ramp test.

Steady-State Test Evaluation

Steady-state energy balances were performed for each steady-state test condition for GE 2-2-3 carbon-carbon to compute the convective heat flux to the conical surface of the model, as described in detail in Ref. 5 and 7. Assuming negligible thermomechanical loss of material, the steady-state energy balance equation becomes

$$\dot{q}_C = \dot{q}_{RR} + \dot{m}_W H_W$$

The reradiated heat flux was based on pyrometer measurements of surface temperature. The wall enthalpy was determined from the JANAF (Aerotherm) thermochemical response model for graphite, using the test pressure and measured mass loss rate.

The test conditions selected for the 50-MW arc are representative average values of many runs, whereas specific runs

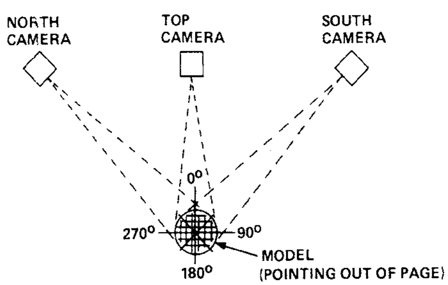


Fig. 5 Ramp test camera positions (50-MW arc).

Table 2 Augmented heating ratios calculated from steady-state recession rates

Facility	Run	p_s, atm	H_{CL}, Btu/lb	$\dot{s}$, in./sec	θ_c, deg	T_w, °R	← Calculated: C_{h_o}	C_h	C_h/C_{h_o} →
50-MW	Typical	80	5000	0.25	45	7200	3.15	4.40	1.4
HIP	1070	124	2980	0.19	57	6700	4.56	5.95	1.3
HIP	1064	124	2980	0.20	57	7500	4.56	6.40	1.4
HIP	1063	124	3520	0.26	57	8100	4.93	7.40	1.5
HIP	1071	168	3460	0.31	57	7500	5.86	8.80	1.5

Table 3 Summary of roughness parameters for surface shown in Fig. 8

Sample	Length, in.	Rms height, mils	Average peak height, mils	Average distance between peaks, mils	Dirling sand grain roughness, mils	White sand grain roughness, mils
1	0.101	1.00	0.820	1.37	2.27	3.15
2	0.158	2.40	1.72	2.79	5.64	7.36
3	0.068	1.39	1.22	1.73	4.95	5.95
$\bar{x}$	...	1.76	1.34	2.13	4.46	5.77

were selected for HIP tests. The test conditions and calculated values of Ch_o, Ch, and Ch/Ch_o are shown in Table 2, in which Ch is the heat-transfer coefficient based on the measured mass loss rate, whereas Ch_o is the theoretical smooth-surface heat-transfer coefficient for the test conditions and ablated shapes.

The values of augmented heating ratio Ch/Ch_o were used to compute the equivalent sand grain roughness K_s from two different correlations of the PANT wind-tunnel data: the latest PANT correlation (Fig. 6a, from Ref. 11) and the Grabow-White correlation (Fig. 6b, from Ref. 12). These correlations were evaluated for typical 50-MW and HIP test conditions to generate the curves of Ch/Ch_o vs K_s shown in Fig. 7. Although the two correlations give divergent predictions for higher values of K_s, they are in reasonable agreement for the range of augmented heating values (Ch/Ch_o) inferred from the steady-state energy balances.

The values of Ch/Ch_o calculated in Table 2 then were entered into Fig. 7 at each test condition to obtain the values of equivalent sand grain roughness required to cause the augmented heating. These values also are included in Fig. 7 and exhibit a range of $0.4 < K_s < 0.8$ mils, depending on test condition and augmented heating correlation.

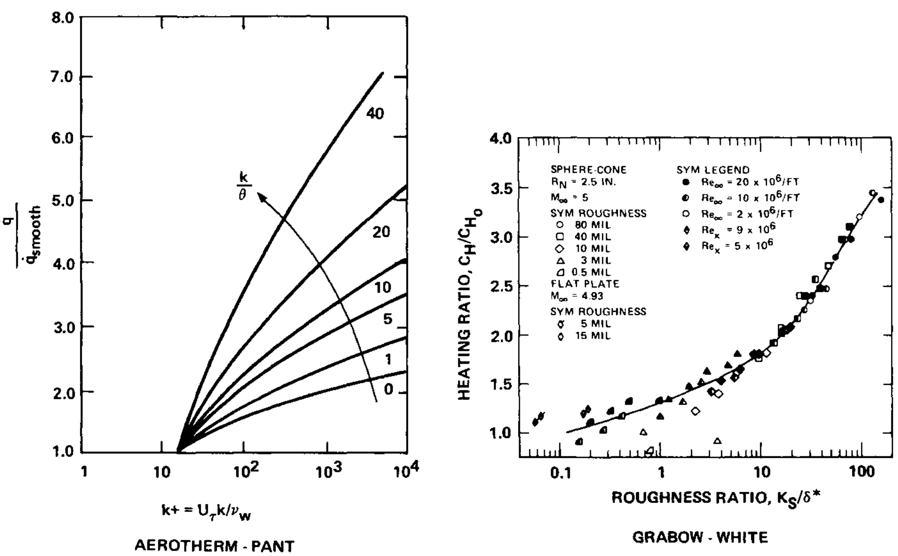
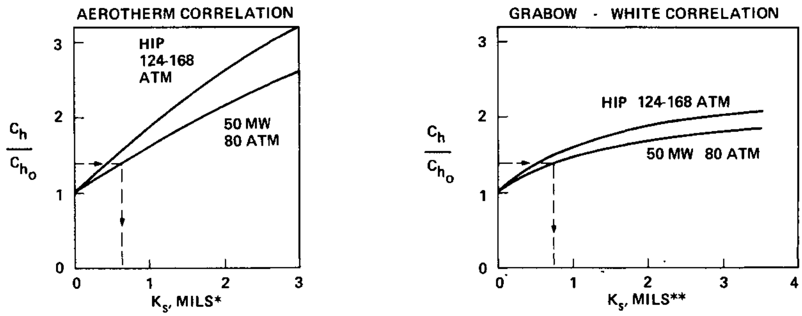


Fig. 6 Augmented turbulent heat transfer due to surface roughness.



FACILITY	p_s (ATM)	MAT'L	K_s MILS*	K_s MILS**
50 MW	80	GE 2-2-3	0.7	0.8
HIP	124	GE 2-2-3	0.4	0.6
↓	168	GE 2-2-3	0.6	0.8

Fig. 7 Equivalent sand grain roughness corresponding to augmented heating ratios.

The values of C_h/C_{h_o} and K_s just inferred should be regarded as first approximations, because of the inherent uncertainties in test conditions and correlative techniques. The effects of relaxing some of the assumptions made in the analysis are summarized below:

1) No thermomechanical loss. Finite thermomechanical loss yields lower C_h/C_{h_o}, K_s.

2) "Standard" H_{CL}. Lower H_{CL} yields higher C_h/C_{h_o}, K_s; e.g., H_{CL} = 3500 Btu/lb yields C_h/C_{h_o} = 1.9, instead of 1.4 for the 50-MW arc.

3) No blowing effect on θ, δ^*. Increase in θ or δ^* due to blowing yields higher K_s for given C_h/C_{h_o}.

It is concluded from these evaluations that the steady-state recession rate of GE 2-2-3 is consistent over the tested pressure range of 80 to 168 atm in that it follows thermochemical response theory based on a consistent degree of augmented heating. However, the absolute values of equivalent sand grain roughness just deduced are not directly applicable to re-entry conditions, because of differences in the blowing parameter B'. As pointed out in Ref. 13, values of B' are lower in arc tests than in re-entry, because of the lower stagnation enthalpy of the arcs. Studies of the coupled effects of mass addition and surface roughness are needed in order to adapt arc test values of equivalent sand grain roughness to re-entry conditions.

Fig. 8 Photomicrograph of post-test sectioned ablation model (GE 2-2-3 model, 168-atm HIP test).

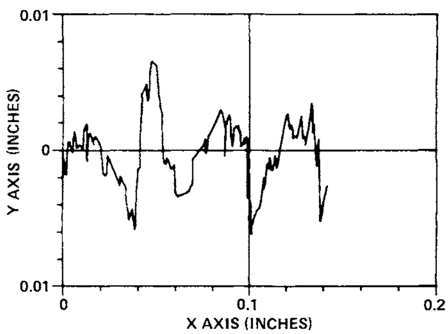


Fig. 9 Post-test surface variation of model shown in Fig. 8.

In a study of post-test surface roughness, a model that had been tested at 168 atm in the HIP facility was sectioned, polished, and photographed (Fig. 8). The surface roughness of this sectioned profile was characterized by "digitizing" the photomicrograph using computer code processing of the surface coordinates of a large number of points.

A typical computer graph of the surface variation relative to a flattened reference line determined by least-squares regression is shown in Fig. 9. A summary of roughness parameters for the model is given in Table 3. The model was divided into three sections; the results for each of these sections were combined by weighting according to the respective lengths of the sections. The "Dirling sand grain roughness" was computed by the method of Ref. 14, whereas the "White sand grain roughness" was computed by the method of

Ref. 12. In each case, the roughness elements were assumed to be conical for purposes of computing shape factors and drag coefficients.

Comparison of the various values of roughness in Table 3 with the calculated values of Fig. 7 suggests that the effective surface roughness for augmented heating in an ablating environment is closer to the post-test average peak height, rather than the equivalent sand grain roughness computed from the surface profile. This conclusion appears to be true even when blowing effects on boundary-layer momentum thickness are included. The dimensionless blowing parameter B' for the test conditions of Table 2 is on the order of 0.3; for this value of B', the correlations of Laganelli et al[15] indicate an increase in boundary-layer thickness of less than 20%.

A similar conclusion regarding the inadequacy of the use of equivalent sand grain roughness in predicting augmented turbulent heat transfer was obtained by Denman[16] in a study of roughened calorimeter response in the 50-MW arc. In the same study, Denman also matched the steady-state recession rate of graphite specimens in the 50-MW arc by computing an augmenta-

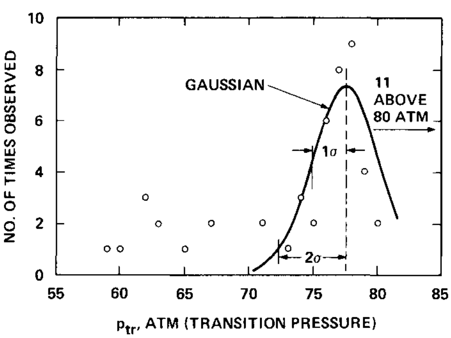


Fig. 10a GE 2-2-3 transition pressure distribution, all cameras.

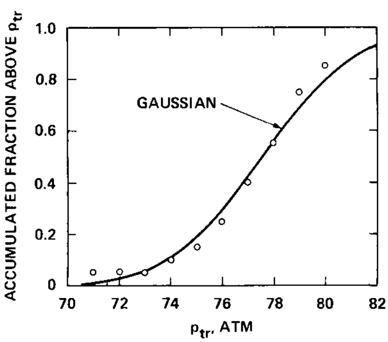


Fig. 10b GE 2-2-3 transition pressures, top camera along weave axis.

tion factor of 1.45, based on measured surface roughness and the augmented heating correlations of Nestler[17] and Powars[18].

Ramp Test Evaluation

A frequency distribution plot of transition pressures for a set of GE 2-2-3 carbon-carbon models is shown in Fig. 10. The distribution is approximately gaussian, except for a group of lower values which occurred exclusively on rays that were 45 deg from the orthogonal weave directions. A study of transition gouge occurrence from post-test examination of a group of models, e.g., Fig. 11, showed a consistent bias for transition to occur more frequently on 45 deg rays (Fig. 12), in agreement with previous results of Heinonen and Jumper.[19] This nonaxisymmetric character of transition is due to the three-dimensional weave geometry of carbon-carbon.

The ramp test results just discussed were for the "peaked enthalpy" test mode. In order to promote transition at lower pressures, a "flat enthalpy" test mode also is employed, in which higher Reynolds numbers are produced due to the lower enthalpy. Because of the longer test time following transition onset, more sharpening occurs for ramp models tested in the flat enthalpy mode. Two generic types of behavior were noted, as shown in Fig. 13: 1) maintenance of a stable shape in the 0.5-sec dwell period at the end of the ramp, and 2) rapid sharpening in the dwell period.

It was noted that models that had achieved a relatively small laminar region at the end of the ramp had a greater propensity to sharpen during dwell. The size of the laminar region in turn was noted to be dependent upon the transition

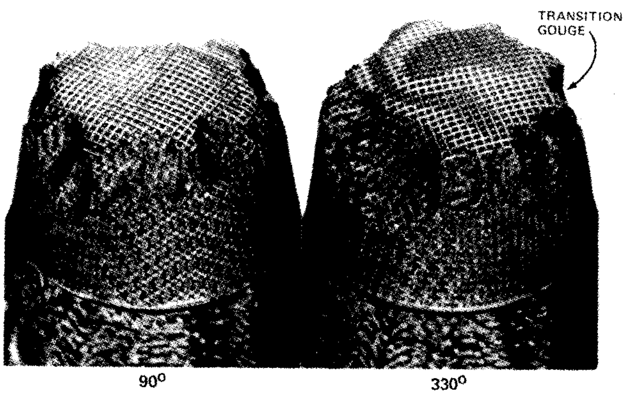


Fig. 11 Nonaxisymmetric transition gouges (50-MW arc ramp test).

pressure; models that had relatively low values of transition pressure had smaller laminar regions at the end of ramp. Hence, the two quantities that appear to determine whether or not sharpening will occur during dwell are the laminar region corner radius R_C at the end of ramp and the transition pressure p_{TR}.

Values of R_C and p_{TR} were tabulated for models tested in the flat enthalpy mode from several test series, along with a statement of whether significant sharpening occurred during dwell. A plot then was made of R_C vs p_{TR} (Fig. 14), using solid symbols for sharpening and open symbols for no sharpening. The symbol key also identifies the type of carbon-carbon and whether implants were present.

Two things are readily apparent from Fig. 14: 1) the final corner radius at the end of ramp correlates with transition pressure; and 2) a threshold region of p_{TR} of 24 atm can be defined below which sharpening always occurred during dwell, and a value of p_{TR} of 29 atm above which sharpening never occurred during dwell. In the region of $24<p_{TR}<29$, each type of behavior was observed. The sharpening criterion suggested in Fig. 14 is independent of material for those tested, and whether or not implants were present.

The physical reason for the existence of a threshold value of transition pressure below which sharpening occurs during dwell is not clear at this time. Several possibilities can be suggested, among them the following: 1) increased laminar heating associated with smaller values of R_C causes increased surface roughness in the laminar region, promoting transition

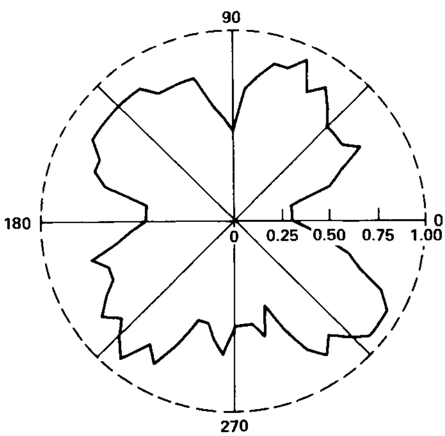


Fig. 12 Frequency of transition vs clock angle.

close to the tip; and 2) a flow instability is induced by the coupling of turbulence feeding forward into the concave streamline region near the tip. The latter effect would be more pronounced for the smaller laminar regions associated with lower value of transition pressure. Kuethe[20] observed an unstable flow region at Reynolds numbers well below the critical value given by theory and speculated that the source of instability was the stretching of vortex filaments in the region of diverging flow near the stagnation point. Kuethe states that the disturbances "were probably produced by surface roughness." A similar mechanism is suspected in the present case.

The potential effect of such a nose-sharpening mechanism in re-entry is of obvious importance. Further work is needed

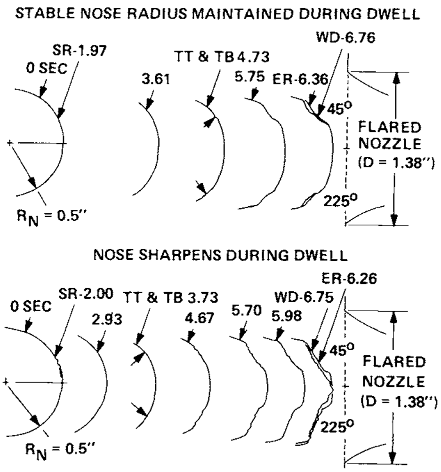


Fig. 13 Typical shape changes in 50-MW flat enthalpy ramp tests.

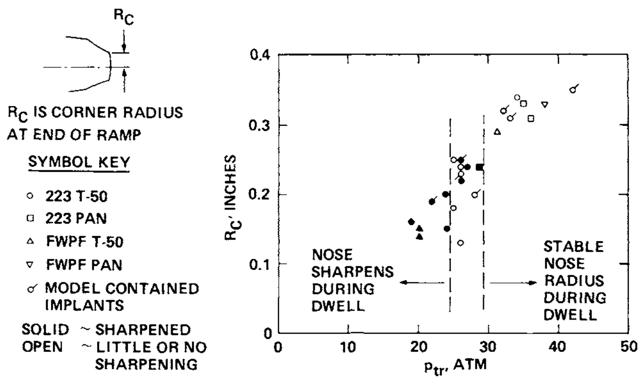


Fig. 14 Sharpening criterion for 50-MW flat enthalpy ramp tests.

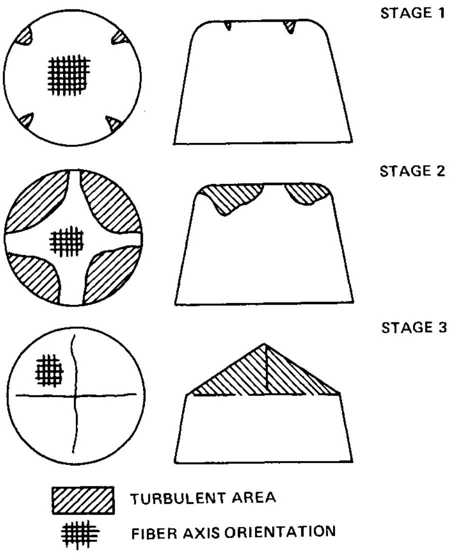


Fig. 15 Off fiber axis turbulent area progression.

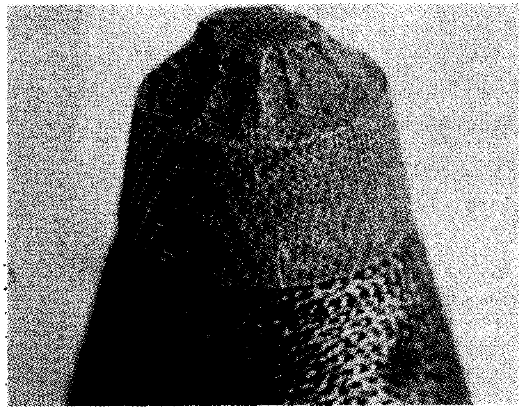


Fig. 16 Typical flat enthalpy ramp test model (50-MW arc).

to determine the cause of the sharpening, as well as its scaling laws, in order to assess the significance to re-entry nose-tip performance.

The ramp test results from flat and peaked enthalpy test modes together suggest a sharpening behavior, as shown in Fig. 15. The preferential off-axis transition leads to a nonsymmetric shape that is approximately a four-sided pyramid in the fully sharpened state. Deep grooves occur between ridges, because of the nonuniform transition behavior (Fig. 16). Increased heating will develop on such configurations due to crossflow effects associated with longitudinal vortices. Further study of the augmented heating to transitional shapes such as shown in Fig. 16 is needed to provide improved modeling of re-entry nose-tip shape change, including asymmetric shape development.

Conclusions

The principal conclusions of this study are summarized below:

1) The steady-state recession rate of GE 2-2-3 carbon-carbon is consistent over the tested pressure range of 80 to 168 atm.

2) The equivalent sand grain roughness based on post-test surface evaluation is significantly larger than the effective roughness deduced from thermochemical theory with roughness-augmented heating. The use of average peak height yields better agreement.

3) Boundary-layer transition on carbon-carbon nose-tip models is not axisymmetric, because of the three-dimensional weave geometry. This nonaxisymmetric transition behavior leads to ablated shapes having ridges, grooves, and local gouges, with concomitant increased heating. Similar behavior is expected during re-entry.

4) Two types of nose-tip shape change can occur during the dwell period following a ramp test: stable shape or rapid sharpening. A criterion has been developed for defining the transition pressure below which sharpening will occur.

Acknowledgments

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ABLATION MEASUREMENT ON THE GASJET NOSE TIP

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and

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Abstract

The Gasjet nose tip is proposed to reduce the erosion at the apex of a missile nose flying at hypersonic speeds through a rain storm as it reenters the atmosphere. A forward-facing sonic jet is directed through the tip, introducing a secondary counterflow, which displaces the bow shock and blankets the tip with a protective layer of relatively cool gas. Wind-tunnel experiments are described which proved the validity of measuring nose recession in flight by recording the pressure in the blast tube supplying the Gasjet.

Nomenclature

a = recessed length of nose, in.
D_i = inside diameter of tip discharge, in.
$\dot{m}$ = mass flow ratio $(\rho V)_{jet}/(\rho V)_\infty$
$\dot{m}^*$ = mass flow ratio at which sonic flow begins
P_{I_1} = tip internal static pressure, psia
P_{I_2} = blast tube pressure, psia
P_{sup} = supply pressure, psia
P_{T_2} = total pressure behind normal shock, psia
V = jet or tunnel freestream velocity, fps
W_a = weight flow rate, lb/sec
W_a^* = weight flow rate at which sonic flow begins, lb/sec

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α = angle of attack, deg
Δ = bow shock standoff distance, in.
θ = half-angle of tapered supply pipe to tip, 1.5 deg
ρ = gas density

Introduction

As part of the proposed Gasjet development sponsored by the Defense Nuclear Agency, a wind-tunnel test program was conducted in the 20-in. Mach 6 tunnel and the Continuous Flow Hypersonic Tunnel at the NASA Langley Research Center. The objectives of these tests were to 1) verify a technique for assessing nose recession in flight by measuring pressures in the blast tube of the Gasjet, and 2) determine the range of mass flow ratios ($\dot{m}$) required to sustain choked flow at the tip.

The test of the Gasjet nose tip on the material screening vehicle (MSV) required a technique for measuring nose-tip recession during flight which could be telemetered, since the test items were not recovered. A means of measuring recession of the Gasjet nose tip by employing a double choked flow coolant gas system consisting of a gas generator, gas generator throat area, blast tube, and tapered jet exit passage has been developed. Recession of the tip results in an increased jet exit flow area and reduces the total pressure of the gas in the blast tube. Measurement of the blast tube pressure and gas generator pressure will produce an effective measurement of tip recession as long as choked flow (i.e., sonic velocity) is maintained in both the exit area and the gas generator throat area. To assess the effects of hypersonic external flow conditions and simulated tip recession on these pressure measurements over a range of collant gas pressures, models simulating recessed nose tips were tested over a range of flow conditions.

The wind-tunnel model was a full-scale representation of the Gasjet tip mounted to a 6.3-deg half-angle cone simulating the forward 19 in. of the MSV with a Gasjet tip. The coolant flow was simulated by room-temperature air introduced at the station corresponding to the gas generator location. The coolant passageway was tapered near the Gasjet orifice so that recession of the tip results in an increase in the orifice exit area, with a decrease in the pressure in the blast tube. Tests were conducted with the initial shape and with two foreshortened tips to simulate receded configurations. As in flight, pressures were measured in the blast tube and the gas supply chamber to substantiate and calibrate the sensitivity of internal pressure to recession of the tip. A range of mass flow ratios which brackets full-scale conditions were used. The range of mass flow ratios required to sustain choked flow at the tip was determined from internal pressure measurements made at the tip and in the blast tube during the tests. The experiments proved the validity of measuring nose recession on the Gasjet by recording blast tube pressures during flight.

Apparatus

The two facilities used for these experiments were the NASA Langley Research Center Continuous Flow Hypersonic and the 20-in. Mach 6 Wind Tunnels. Both the freestream and internal gas flow medium were air.

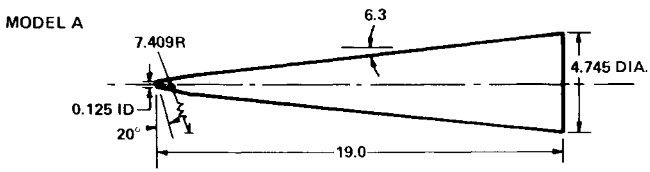


Fig. 1 Full-scale Gasjet nose tip.

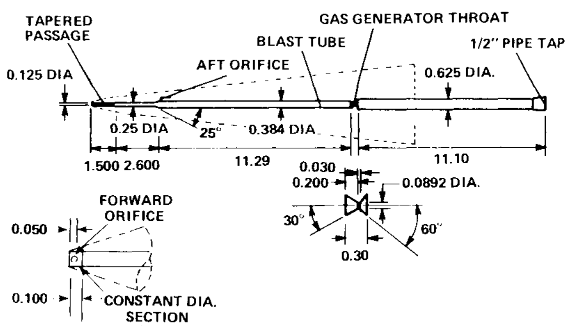


Fig. 2 Pressure model internal gas supply plumbing.

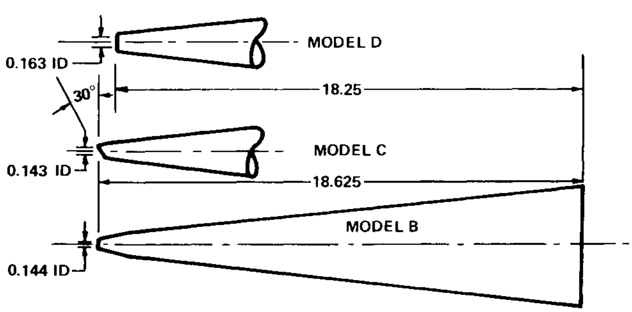


Fig. 3 Full-scale Gasjet nose tips.

Fig. 1 shows the basic Gasjet nose-tip model (model A) as mounted on a 6.3-deg half-angle cone. Fig. 2 gives the internal dimensions of the pressure model plumbing showing the gas generator throat, the blast tube, and 1.5 deg half-angle tapered jet exit passage. Room-temperature air was fed into the model sting through the 0.0892-in.-diam nozzle, simulating the end of the gas generator, and then into the tapered tip pipe, where it was discharged through a 0.125-in-diam hole in the nose tip. Internal model instrumentation consisted of a pressure tap located 0.050 in. from the model nose and a second located 4.1 in. aft of the nose. The forward orifice has a 0.020-in.-diam hole in the tip pipe and is vented through 0.030-in. slots machined into the nose prior to electro-forming the nickel shell skin. The slots were connected to 0.042-i.d. steel tubing that was enlarged to 0.071-i.d. tubing at model station 12, and this tubing was connected to the facility transducers.

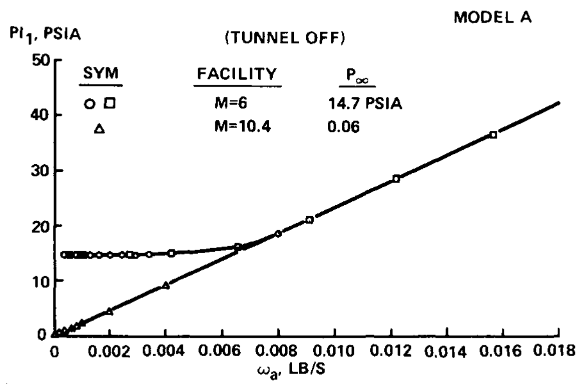


Fig. 4 Ambient discharge pressure at Gasjet tip.

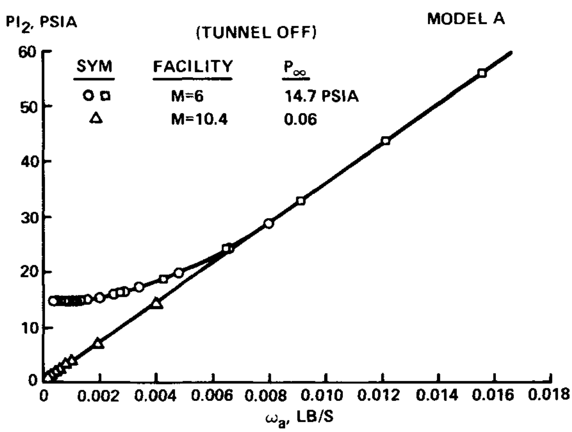


Fig. 5 Ambient discharge pressure in blast tube.

To determine the change in blast tube exit pressure with a nose tip that erodes, three recessed nose configurations were tested, as shown in Fig. 3. Normal cuts were made in the nose at station 0.375 (model B) and at station 0.75 (model D). In addition, an oblique cut was made at nose station 0.375 (model C). The internal blast tube plumbing is the same as that on the basic Gasjet nose tip (model A), but the 1.5 deg tapered tip tube results in an enlarged tip hole for the recessed noses, as enumerated on Fig. 3. No attempt was made to match expected recession shapes; only large changes in nose geometry were made for this parametric study. The foreshortened nose configurations have the forward internal pressure orifice located 0.040 in. aft of the nose.

Results

As shown in Fig. 2, the tunnel mount for the experiment used internal plumbing that provided a 0.0892-in.-diam orifice that separated the supply gas from the model simulated blast tube. This orifice was in the same axial location from the tip as the orifice of the gas generator for the full-scale MSV. At the forward end of the blast tube, a pressure tap was used to measure static pressure (P_{I2}). A second internal tap, located as far forward in the Gasjet tip as could be fabricated, measured tip internal static pressure (P_{I1}). The supply pressure was recorded several feet aft of the model base and is designated P_{sup}.

Ambient Discharge

A calibration of the jet into quiescent air was made in both wind tunnels. These results are given in Figs. 4 through 6. When discharged into a near vacuum, both the Gasjet and simulated gas generator orifices become sonic almost instantaneously, and a linear variation of pressure with weight flow results. If the measured pressure in the blast tube (P_{I2}) is considered to be the total pressure at the tip exit and the measured pressure 0.050 in. upstream of the exit (P_{I1}) is considered to be the static pressure, then the ratio of P_{I1}/P_{I2} gives a Mach number of 0.8. Since sonic flow must exist at the Gasjet tip, the measured values of P_{I1} and P_{I2} are not the exact measure of total and/or static pressure precisely at the sonic point.

Subsonic flow exists up to a weight flow of approximately 0.008 lb/sec for the Gasjet tip discharging into the atmosphere, as shown in Figs. 4 and 5. The supply pressure

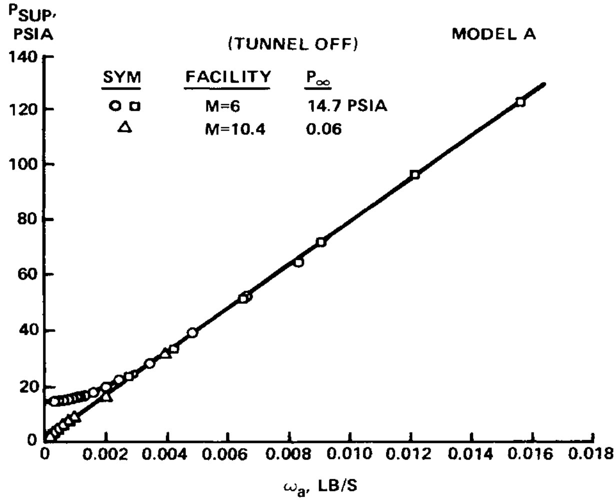


Fig. 6 Ambient discharge supply pressure.

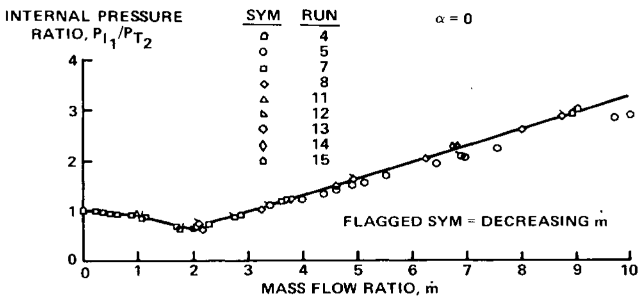


Fig. 7a Pressure in Gasjet tip, Mach 6, model A.

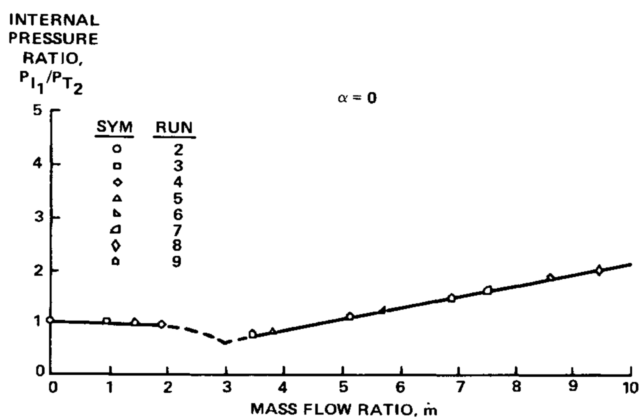


Fig. 7b Pressure in Gasjet tip, Mach 10.4, model A.

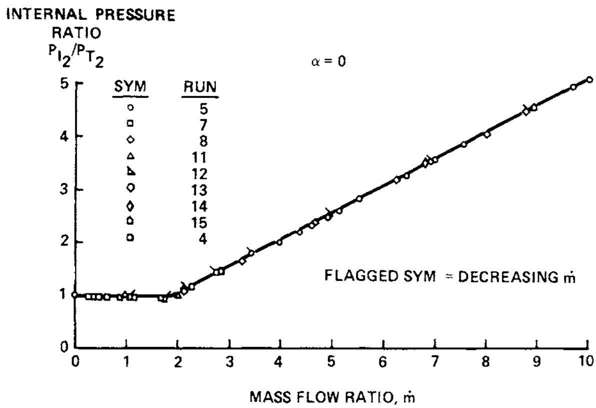


Fig. 8 Pressure in blast tube, Mach 6, model A.

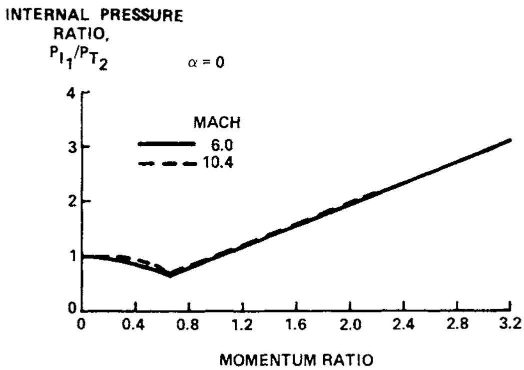


Fig. 9 Pressure in Gasjet tip, model A.

variation (Fig. 6) shows the upstream orifice is sonic at a weight flow of approximately 0.004 lb/sec for a room-pressure discharge. This orifice has 51% of the area of the tip orifice for configuration A, and the calibration appears reasonable.

Discharge Against Freestream with Simulated Tip Recession

Operation of the Gasjet against the tunnel freestream resulted in internal pressure ratios as shown in Fig. 7 for configuration A. As mass flow ratio increased, the measured static pressure at the Gasjet decreased to a ratio of P_{I1}/P_{T2} of 0.65 when sonic flow was established at the tip. Taking the ratio of P_{I1}/P_{I2} corresponding to the measured minimum pressure ratio (P_{I1}/P_{T2} = 0.65) gives an indicated exit Mach number of 0.8, matching the value observed in the ambient discharge calibration. If model measurements

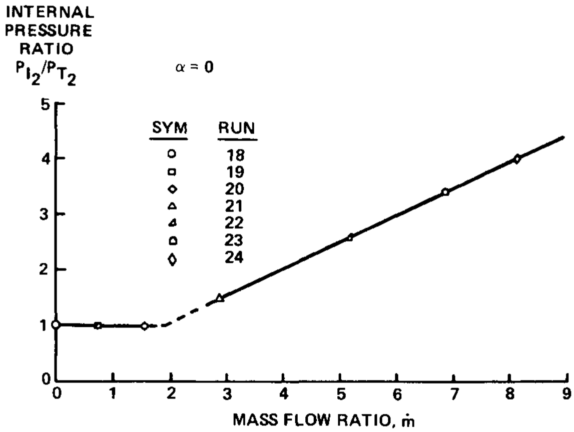


Fig. 10a Pressure in blast tube, Mach 6, model B.

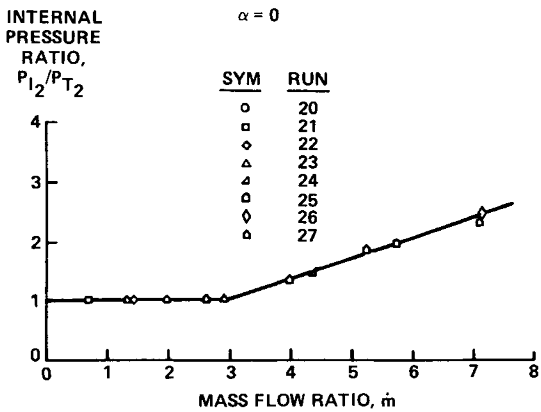


Fig. 10b Pressure in blast tube, Mach 10.4, model B.

could be located at the precise point of sonic flow, it then would be expected that the theoretical value of P_{I1}/P_{T2} of 0.53 would be reached.

As shown in Fig. 8, operation of the Gasjet subsonically against tunnel flow is different from the ambient calibration case inasmuch as the freestream acts like an ejector, reducing the static pressure at the tip while the blast tube pressure remains at the value of tunnel stagnation pressure (P_{T2}). Similar results were observed in Ref. 1.

Converting mass flow ratio to momentum ratio $(\rho V^2)_{jet}/(\rho V^2)_\infty$ for each Mach number and replotting the internal pressure ratios as shown in Fig. 9, it is observed that the process can be correlated with momentum ratio. Internal

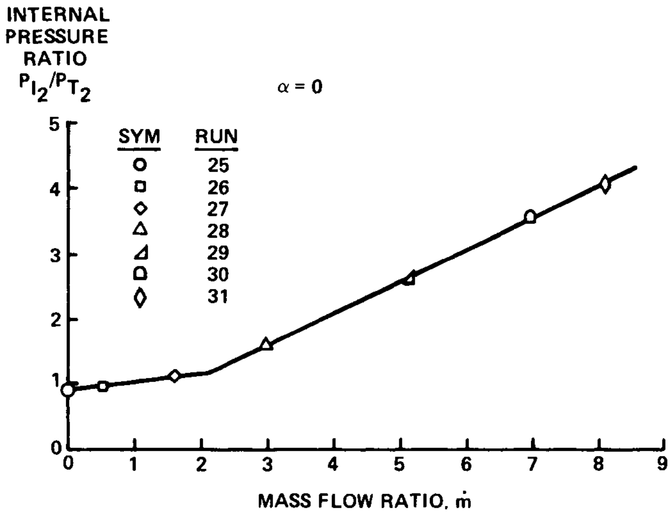


Fig. 11a Pressure in blast tube, Mach 6, model C.

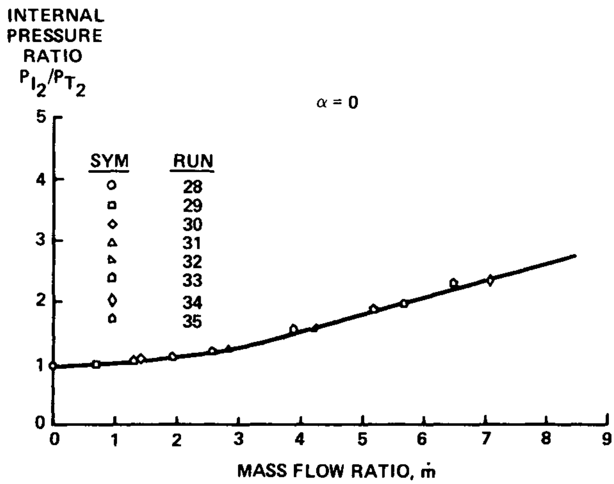


Fig. 11b Pressure in blast tube, Mach 10.4, model C.

pressure data for models B - D, which simulated recessed nose tips, are shown in Figs. 10 through 12. The mass flow ratio at which the recessed tip becomes sonic is constant with each Mach number. Fig. 13 shows that the measured weight flow ratio $wa^*/wa^*_{initial}$ is in good agreement with the jet discharge area ratio. Fig. 14 summarizes the sensitivity of the measured blast tube pressure, P_{I2}, to jet tip diameter ratio. The theory is based on the exit area change of the jet for both sonic Gasjet and supply orifice

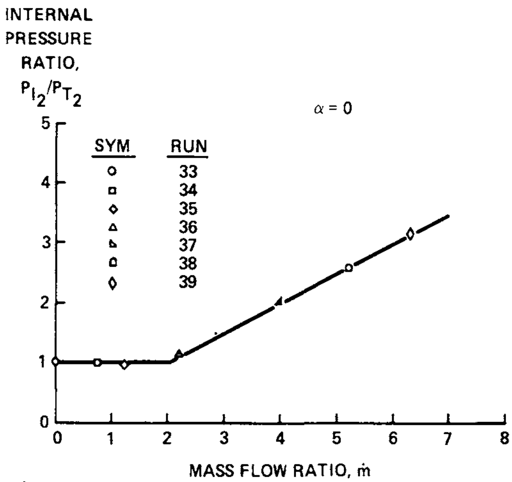


Fig. 12a Pressure in blast tube, Mach 6, model D.

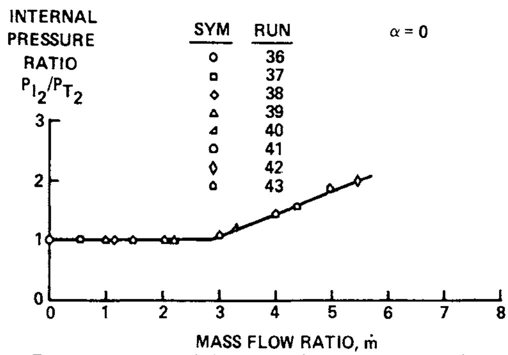


Fig. 12b Pressure in blast tube, Mach 10.4, model D.

(double choked flow) as follows:

$$\left(\frac{P_{I_2}}{P_{sup}}\right)\Bigg/\left(\frac{P_{I_2}}{P_{sup}}\right)_{\text{initial}} = \left(1 + \frac{4a \tan\theta}{D_i} + \frac{4a^2 \tan^2\theta}{D_i^2}\right)^{-1}$$

where θ is 1.5 deg, a is recessed length in inches, and D_i is 0.125 in. If the supply pressure is the same, the left side of the preceding equation reduces to

$$P_{I_2} \Big/ P_{I_2\ \text{initial}}$$

This agreement is excellent and confirms the technique of measuring tip recession with telemetered blast tube pressures

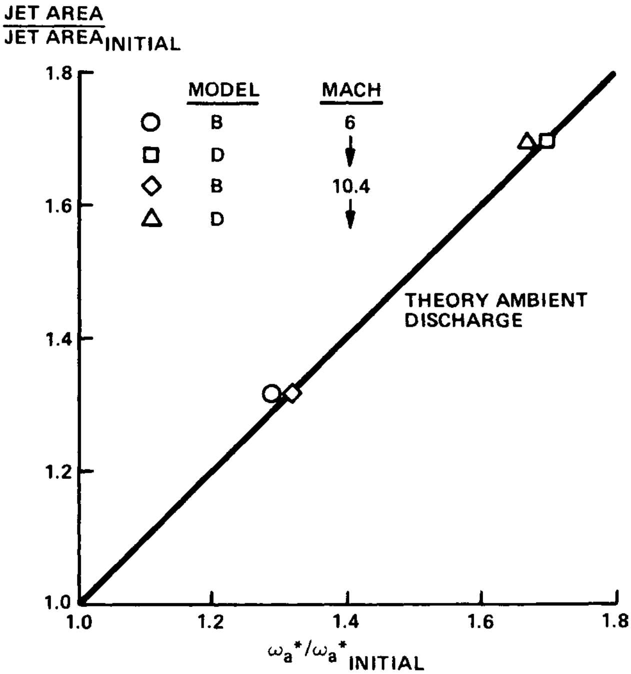


Fig. 13 Weight flow ratio for sonic flow vs jet area ratio.

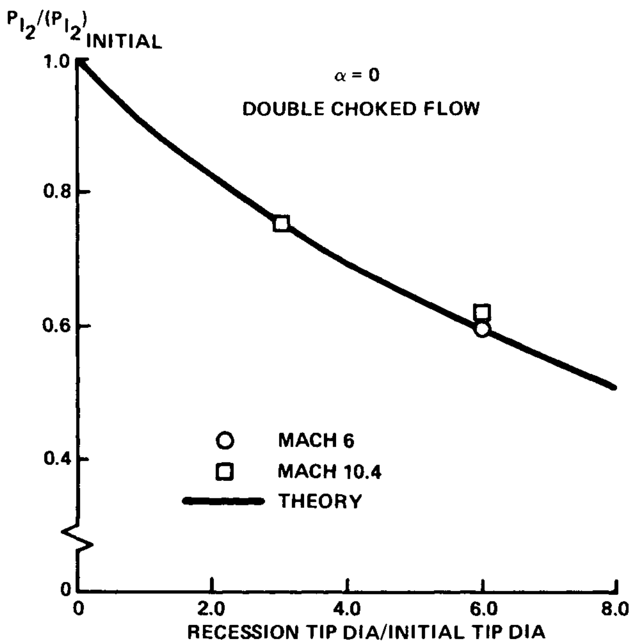


Fig. 14 Sensitivity of blast tube pressure to recession.

Fig. 15 Subsonic Gasjet shadowgraph: Mach 6, α = 0, $\dot{m}$ = 0.96, run 11.

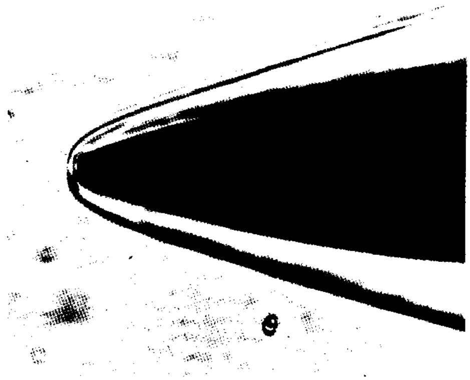


Fig. 16 Sonic Gasjet shadowgraph: Mach 6, α = 0, $\dot{m}$ = 3.7, run 13.

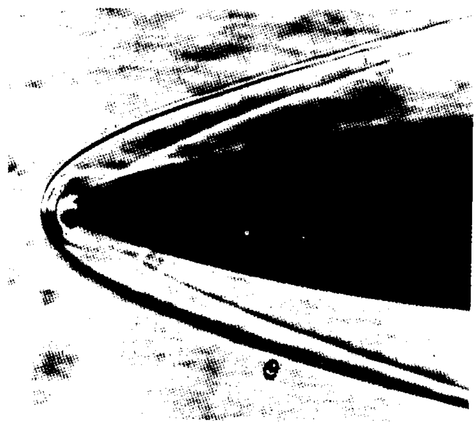
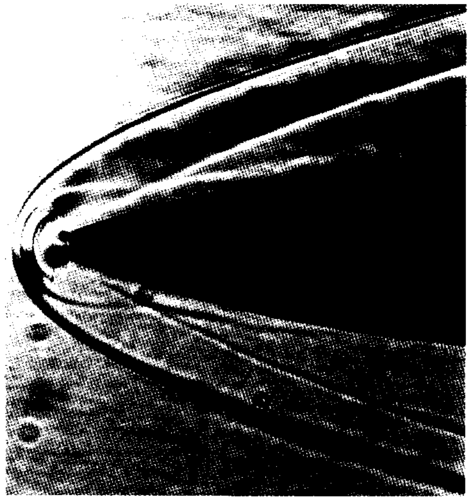


Fig. 17 Sonic Gasjet shadowgraph: Mach 6, α = 0, $\dot{m}$ = 8.9, run 15.



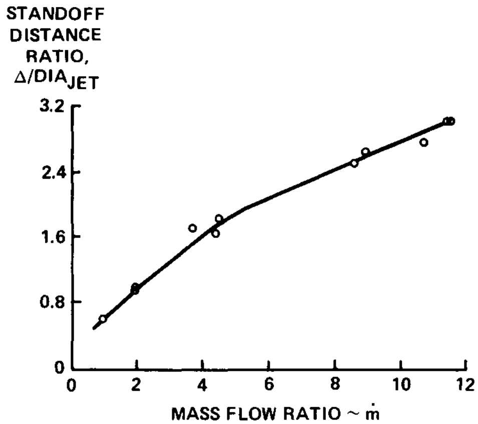


Fig. 18 Shock standoff distance, Mach 6, model A.

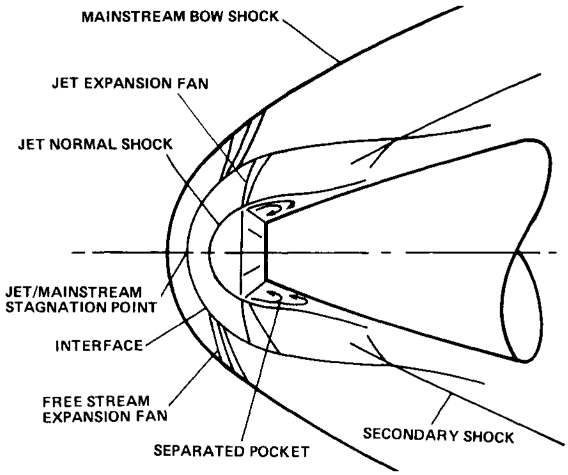


Fig. 19 Sonic jet description when directed against tunnel flow.

Shadowgraphs

Flow visualization was possible only in the Mach 6 tunnel. Figs. 15 through 17 show shadowgraphs at Mach 6 of the Gasjet flow. The mass flow ratio at which the jet goes sonic occurred at a value of 2 at this tunnel free-stream condition. Fig. 15, therefore, shows Gasjet for subsonic flow, whereas Figs. 16 and 17 show the characteristic supersonic bubble established after sonic flow occurs. As the mass flow ratio increases, the bow shock is displaced forward, and this has been plotted as a standoff distance ratio (Δ/jet exit diam) in Fig. 18.

Fig. 19 shows the flow model of the sonic jet using comments from Refs. 2-4. The description can be compared with flow visualization pictures given in Figs. 16 and 17 for the sonic jet. Flow from the sonic Gasjet expands to a supersonic Mach number in the bubble to a condition on the centerline where the stagnation pressure across the jet normal shock matches the stagnation pressure across the freestream bow shock. Both the bubble size and bow shock displacement will vary until this condition is matched.

Conclusions

Wind-tunnel tests have been completed at Mach numbers 6 and 10.4, which have 1) verified the nose recession measurement technique of telemetering the blast tube pressures in the MSV with Gasjet and 2) established the mass flow ratios ($\mathring{m}$) for sonic flow at the tip. The technique for determining nose-tip recession in flight by measuring pressures in the blast tube of the Gasjet agrees very well with theoretical predictions.

Acknowledgment

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SEGMENTED TUNGSTEN NOSE TIPS FOR HIGH-PERFORMANCE FLIGHT VEHICLES

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Abstract

A segmented tungsten nose-tip concept has been developed for use on high-performance flight vehicles in erosive weather environments. Early designs that used a conventional monolithic construction approach experienced critical thermostructural problems. The segmented concept avoids these problems by fabricating the nose tip from a number of individual segments. Two methods for fabricating the tungsten segments were investigated, and both were found to produce very high-strength material. Ground-test programs were conducted to validate the ablation and thermostructural performance of the segmented designs. The test results indicate that the segmented concept is now a viable design for use on high-performance all-weather flight vehicles.

Introduction

A problem that has received considerable attention in recent years is the development of nose tips to provide an all-weather flight capability for re-entry and interceptor vehicles. These erosion-resistant nose tips (ERN) utilize two different components to provide optimum performance in either a clear-air or an erosive weather environment. A graphitic (carbon-carbon or bulk graphite) external, or primary, nose tip is used to assure minimum recession and shape change during normal (i.e., clear-air) flight. For flight through weather environments, the primary nose tip is removed by particle erosion a short time after weather encounter. For these cases, a backup nose tip, or subtip, with high resistance to particle erosion is

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provided to assure vehicle survival with minimal degradation of accuracy. After extensive materials screening test programs, tungsten was selected as the optimum state-of-the-art subtip material. The general validity of the ERN concept then was demonstrated in several ground and flight-test programs.

The tungsten subtips in the early designs were monolithic configurations that were machined from large billets of extruded or swaged material. Several critical problems were encountered which complicated the design of flight nose tips with these materials. First, it was found that large-size extrusions and swagings (diameters greater than 2 in.) have unsuitably low-strength properties in the transverse plane due to grain orientation characteristics and flaws. Thermostructural failures of several of these designs were experienced during ground tests in rocket motor facilities in which the thermal stress environment was representative of high-performance re-entry flight conditions. These failures were a result of the poor transverse strength of the material, coupled with a large volume of critically stressed material at temperatures below the ductile-brittle transition temperature (DBTT). The fractures in these monolithic subtips were observed either to propagate to the surface, producing an immediate catastrophic failure, or to propagate to the ductile-brittle material interface. In the latter case, the crack subsequently became catastrophic if the ablation front progressed to the fracture location.

A second problem associated with the design of tungsten subtips for flight is the fact that the material can be manufactured in a maximum billet diameter of only about 3 in. (Metal-infiltrated tungsten is available in larger billet sizes, but arcjet tests have shown that it provides less than satisfactory ablation performance in severe environments.) This size limitation precluded application of the ERN concept to vehicles with initial nose radii much larger than about 1.25 in.

The segmented nose-tip concept was conceived as an approach to solving both the thermostructural and the material size limitation problems associated with the design of tungsten subtips. With this technique, the subtip is manufactured in the same external shape as the monolithic designs but is constructed from a number of small components, as illustrated in Fig. 1. The subtip consists of a number of forward-facing conical segments with a tapered center retention stud. Segmentation of the subtip results in lower thermal stress levels in the segments compared to the monolithic designs. The

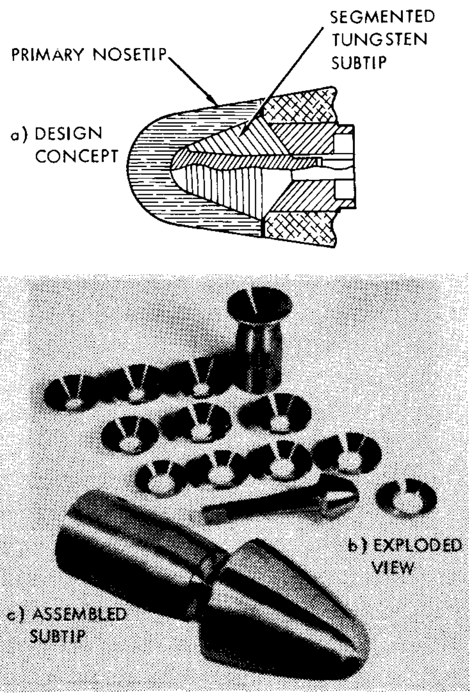

Fig. 1 Segmented tungsten subtip design concept.

lower stresses are a result of the smaller temperature gradients in the small components and the ability of the segments to deform independently under the influence of a temperature gradient. The probability of failure also is reduced by the smaller unit volumes of highly stressed material in individual components. Because a fracture in one segment cannot propagate across the segment boundaries, any fracture that does occur will not result in a catastrophic failure of the assembly; thus, a form of artificial fracture toughness is designed into the segmented unit. Finally, the segmented designs offer the additional important advantage of allowing the components to be fabricated from higher-strength material, such as cross-rolled plate or other highly worked fine-grain tungsten. Since the plate material is manufactured in very large sizes, this also eliminates the nose-tip size limitations associated with the use of cylindrical billets.

Several analytical and test programs were performed to develop and validate the segmented subtip concept.[1-3] These included 1) analyses to define the optimum segment configuration, 2) a material processing study to select the best method for forming the segments, 3) an ablation test program to determine if the segmented construction would affect the ablation performance of the assembly, and 4) a rocket motor test to evaluate the thermostructural performance in a simulated high-performance flight environment. The following sections describe each of these studies.

Segment Design Analyses

Analyses were performed to predict the thermal stress response of the subtip components in a flight environment. The calculations were performed with the SAAS III finite-element code[4] using material properties for extruded 2% thoriated tungsten.[5] These analyses accounted for temperature-dependent material properties with nonlinear stress-strain curves. Time-dependent external shapes, pressure loads, and internal temperatures were predicted from thermochemical ablation and heat-conduction codes and were included in the analyses.

The probability of crack initiation was estimated from the Priddy stress-based failure criterion for brittle anisotropic materials[6] using the predicted stress distributions. This criterion accounts for arbitrary stress states and multiaxial stress interactions using nine measurable strength parameters to describe the failure characteristics of a transversely isotropic material. The nine strengths required by the Priddy criterion were based on characterization results for extruded 2% thoriated tungsten.[5] The probability of crack initiation at any point in the subtip was calculated from the proximity of the stress vector to failure surfaces defined for various probabilities of failure by application of the Priddy criterion to uniaxial data reduced by Weibull statistics. Volume effect corrections also were applied to the failure probability of each volume of material represented by a finite element in the computer code.

The overall probability of crack initiation in any component of the subtip was calculated from the cumulative product of finite-element probabilities over all brittle elements (below 400°F) expressed by

$$P_F = 1 - \prod \left(1 - P_f\right)$$

where P_f is the probability of crack initiation in any single brittle element, and P_F is the overall probability of crack initiation for the entire component.

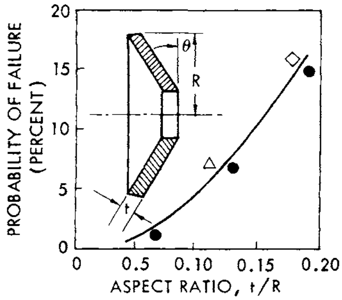

Fig. 2 Effect of aspect ratio on thermostructural response of subtip segments.

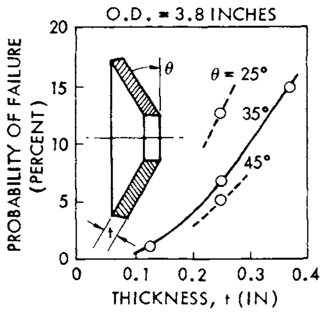

Fig. 3 Effect of segment cone angle on thermostructural response.

The thermostructural responses of the segment and stud components of the segmented subtip were calculated for the design shown in Fig. 1 and expressed in terms of component probability of failure. These results were compared to results for a monolithic design of the same external shape. Segment thicknesses and cone angles were varied to establish guidelines for optimizing the design. Thick segments with shallow cone angles are less costly to manufacture but generate higher thermal stresses because they are less compliant. That is, thermal gradients are accommodated more readily by deformation in thin segments with large cone angles. The computed failure probability is plotted in Fig. 2 as a function of segment aspect ratio (defined as the ratio of thickness t to the outside radius R) for forward, middle, and aft segments with 35-deg cone angles. All cases analyzed fall close to a common curve, illustrating the dominating effect of aspect ratio. Low failure probababilities can be achieved by designing all segments with aspect ratios of 0.10 or below. This permits thicker segments to be utilized near the aft end of the segment stack for economy.

Figure 3 shows the effect of cone angle on the thermostructural response of the middle segment with a thickness of 0.250 in. It can be seen that segment cone angles of about 35 deg are required to

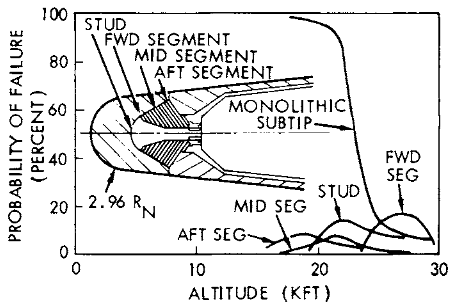

Fig. 4 Tungsten subtip thermostructural response summary.

achieve high survival probabilities by segment deformation under thermal gradients, but that larger cone angles have only marginal additional benefit. According to these figures, high segment survival probabilities are possible with reasonable design combinations of segment thickness and cone angle.

Calculated failure probability histories are shown in Fig. 4 for the segmented subtip components, including the stud, and for a monolithic subtip. Although the stud generates maximum stresses of the same order as the monolithic subtip, the critically stressed volume is much lower and, with the lower transverse stress, results in a much lower probability of failure. The subtip segments are 0.250 in. thick and have 35-deg half-cone angles. The improvement in survival probability offered by the segmented construction approach is apparent.

The failure probabilities shown in Figs. 2-4 were calculated using the failure characteristics of 3-in.-diam extruded 2% thoriated tungsten[5] to provide a direct evaluation of the effect of the segmented design relative to the monolithic design. However, significant material improvements are available for the segmented design which make this comparison very conservative. As described below, material process studies have shown that the segments can be either coined from cross-rolled tungsten plate or forged and coined from pressed and sintered billets. These processes yield segments with significantly higher failure strengths and lower ductile-brittle transition temperatures (DBTT) than the extruded product.[7] Consequently, much lower failure probabilities than those indicated in Figs. 2-4 can be achieved in the segmented design. In addition, the stud can be fabricated from a smaller-diameter extrusion or swaging and, consequently, can be upset-forged to improve the transverse properties by producing a more equiaxial grain structure. Since previously tested monolithic tungsten subtips[8] were critical with respect to transverse direction properties, it is reasonable to expect that upset-forging of the small-diameter billet will improve the thermostructural performance of the stud.

Artificial fracture toughness provided by segment boundaries, although not assessed quantitatively, is believed to be another major advantage of the segmented tungsten design concept. Cracks initiating in any component can propagate only to the nearest boundary. Because of the design geometry, several adjacent segments must fail to permit loss of material. Cracks in the stud resulting from transverse stresses can propagate only to the segment-stack boundary, and the segments should restrict the crack from opening.

Material Processing Study

Two methods of forming large-diameter, 35-deg half-cone angle segments were evaluated: 1) segments were coined in two steps from circular blanks cut from cross-rolled pure tungsten plate, and 2) segments were coined from upset-forgings of 92% dense pressed and sintered billets. Both methods successfully produced fully dense, flaw-free segments with minimal waste. Nine segments were formed by each method, with only one in-production failure.

Three segments were coined from each of three plate thicknesses: 0.250, 0.375, and 0.500 in. After clean-up machining, the resulting segments were 0.125, 0.250, and 0.375 in. thick, respectively, as illustrated in Fig. 5. Experience showed that this allowance for clean-up was nearly optimum. That is, only slightly thicker segments could have been obtained from the indicated plate thicknesses.

Cross-rolled commercially pure tungsten plate in 8-in. -diam blanks was purchased from Schwarzkopf Development Company

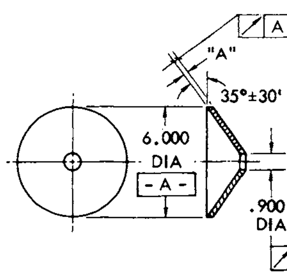

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Fig. 5 Tungsten segment for processing study.

(manufactured by Plansee Metallwerk of Austria). In a two-step operation, the plate first was coined into a hemispherical shape and then coined into the final segment shape. Coining temperatures of approximately 2100^{o}F were used. The segments then were single-point-machined to the final dimensions and inspected by dye penetrant procedures.

Ninety-two percent-dense pressed and sintered fine-grained billets were purchased from GTE-Sylvania. The billets were 4.0 in. diam and were cut into approximately 2.25-in. lengths for upset-forging. The short billets were upset-forged to 8.0-in.-diam x 0.5-in. thick plates, which produced favorable in-plane grain structure and consolidated the material to full density. Subsequent two-step coining operations at approximately 2100^{o}F produced the final shapes. All nine segments produced in this manner had the same thickness; large increases in forging power would have been required to produce large-diameter upset-forgings less than 0.5 in. thick after coining. The segments then were machined to the desired final thickness. The finish-machined specimens were identical in appearance to the specimens coined from cross-rolled plate.

Fifty specimens for four-point flexure testing were excised from the cross-rolled plate stock and from the segments fabricated by the

Table 1 Flexure test matrix

Material	Thickness, in.	Specimen orientation[a] 0 deg	45 deg	90 deg
Cross-rolled plate	0.250	1	1	...
	0.375	2	2	...
	0.500	1	1	...
Coined segments	0.125	3	2	3
	0.250	3	2	3
	0.375	3	2	3
Forged and coined segments	0.125	3	2	1
	0.250	3	2	1
	0.375	3	2	1

[a]Relative to last rolling direction for plate and segments coined from plate.

two processes. All segments were 2.750 in. long with 0.125-x 0.120-in. rectangular cross sections. Specimens from the coined segments were machined from 0-, 45-, and 90-deg locations relative to the last plate-rolling direction. (The 0-deg orientation was selected arbitrarily for the segments formed by forging and coining of sintered billets, since these components had no preferential grain direction.) The specimens were electropolished and inspected by dye penetrant prior to testing. The specimens prepared for flexure testing are tabulated in Table 1.

The tests were run either to failure or to maximum deformation in a standard four-point flexure fixture. The specimens were mounted with the short dimension vertical. The lower supports were 2.375 in. apart, and the load points were 1.000 in. apart on center. All specimens were strain-gaged on the lower surface in the maximum-moment region. Most of the tests were conducted at room temperature to compare processing methods and the distribution of properties relative to the plate-rolling directions. The stresses at failure were calculated from elastic beam theory using the actual specimen dimensions. Strains and deformations (at the midpoint) were recorded directly as functions of load during the test. Nine of the 50 specimens were tested at temperatures up to 300°F to obtain a limited indication of ductile-brittle transition temperature (DBTT).

The calculated stresses and measured strains at failure are plotted in Fig. 6. The data fall on a common stress-strain curve, which probably represents an ideal curve for unflawed fine-grained tungsten. Data from both processing methods represent clear improvements over the low-temperature design properties for large tungsten extrusions. By comparison, the segments produced by forging and coining of sintered billets are superior to the coined plate

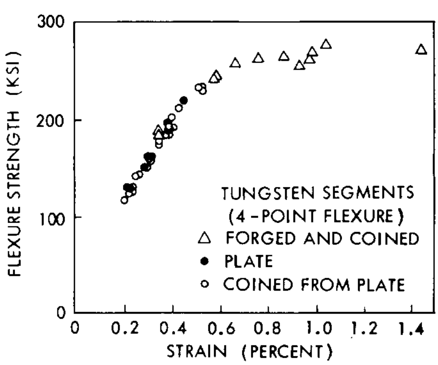


Fig. 6 Room-temperature flexure data from tungsten segments.

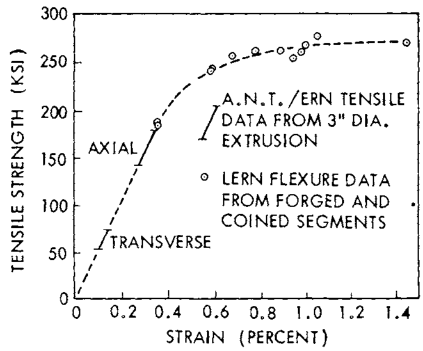


Fig. 7 Forged and segment strength compared to extruded material.

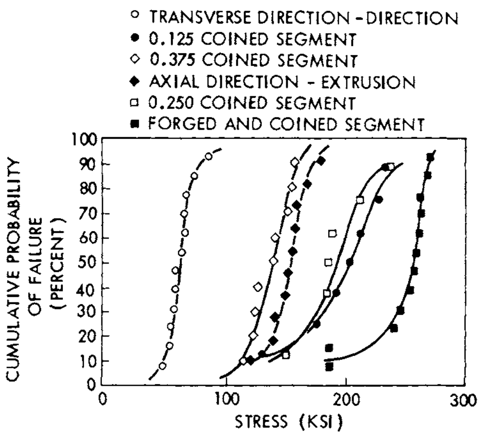


Fig. 8 Cumulative failure probability comparison.

material and exhibit significant nonlinear stress-strain behavior, representative of fine-grained highly worked tungsten.

A similar plot of calculated ultimate strengths and strains at failure is given in Fig. 7 for the forged and coined material only. Also shown in the figure are strength ranges in the axial and transverse directions for 3.0-in.-diam thoriated tungsten extrusions.[5] The potential for preventing thermostructural failures resulting from low transverse strengths in extruded products is obvious from the figure.

The room-temperature flexure strength data from each type of segment were ordered and plotted as cumulative failure probabilities vs stress, as shown in Fig. 8. Also shown for comparison are the cumulative data for extruded billets. No effect of specimen orientation relative to rolling direction can be observed in either the plate data or the data from the coined segments, indicating good in-plane isotropy in the rolled material.

Average results of the flexure tests are tabulated in Table 2. Comparison of the flexure strength data for the cross-rolled plate with the data for segments coined from the plates shows that the plate data fall inside the scatter band for each corresponding segment. Therefore, it is concluded that the coining operation has no detrimental effect on the strength of the plate stock. The 0.125-in.-thick segments (formed from 0.250-in. cross-rolled plate) and the 0.250-in. segments (formed from 0.375-in. plate) are clearly superior to the 0.375-in. segments (formed from 0.500-in. plate).

Although the elevated temperature data are very limited, there is evidence of ductile-brittle transition occurring around 200°F for the segment materials. The elevated temperature strain-at-failure data are summarized in Fig. 9 for all segments. All specimens (four) tested above 200°F deformed to the full extent of the test equipment, which corresponds to a tensile strain of at least 1.8%. Both coined and forged-and-coined materials were included in this population.

Limited microstructural examinations of the tungsten specimens by sectioning and polishing provided information regarding grain geometries. Grain structures in the coined segments were unchanged

Table 2 Results[a] from flexure tests at 70°F

Material	Plate or segment thickness, in.			
	0.125	0.250	0.375	0.500
Cross-rolled plate	...	209	158	146
Segments coined from cross-rolled plate	197	188	138	...
Segments forged and coined from pressed and sintered billets	242	251	248	...

[a]Average tensile strengths (ksi).

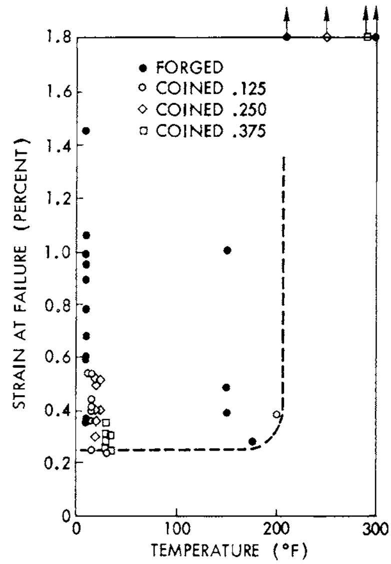


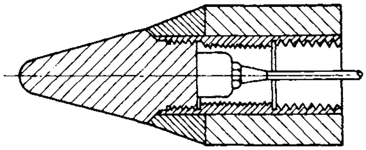
Fig. 9 Effect of temperature on strain-to-failure.

from the corresponding plate structures. The grain size ranged from 2 to 6 mils in the plate material. The grains were considerably finer in the forged material, with a size of about 1 mil. Grain structures had cold work factors (ratio of longitudinal to transverse grain dimensions) of about three to fifteen in the plate material, with the 0.250 plate having the largest amount of cold work. Rockwell hardness measurements are typical of fine-grained tungsten material (R_C = 40 to 50). No change in material hardness due to the plate coining operation is evident.

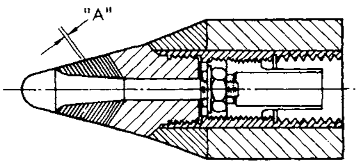
From these evaluations, it is concluded that acceptable 6.0-in.-diam tungsten segments can be formed by coining cross-rolled pure tungsten plates or by forging and coining sintered pure tungsten billets. Although the forged billet method produces the best material in terms of strength and ductility, both materials are far superior to extruded thoriated tungsten billets. Either method allows production of subtips of 6-in. (or larger) diameters, whereas 3-in. diameters are the practical limit from extrusions.

Ablation Test Program

One concern that has been expressed regarding the performance of segmented nose tips is the possibility of increased heating and surface recession at the segment boundaries. To investigate this possi-



a) MONOLITHIC TUNGSTEN TARE MODEL



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-2	0.045
-3	0.010

b) SEGMENTED TUNGSTEN MODELS

Fig. 10 50-MW test model configurations.

bility, a test program was run at the Air Force Flight Dynamics Laboratory (AFFDL) 50-MW arcjet facility.[1]

The ablation test models consisted of four different designs, illustrated in Fig. 10, which were selected to evaluate various aspects of segmented nose tip design and fabrication. All models had the same external sphere-cone geometry (0.25-in. nose radius with a 15-deg half-cone angle) to eliminate the effects of shape differences on ablation performance.

The solid (monolithic) configuration was included as a tare model to provide a reference basis for the ablation performance comparison. These models were machined from a 2-5/8-in.-diam billet of extruded commercially pure tungsten fabricated by the General Electric Company, (G.E.), Cleveland, Ohio.

All three types of segmented tungsten models were of the same basic design, as indicated in Fig. 10. Three different segment thicknesses were used to assess the effect of this parameter on thermal and structural performance. The segmented models all utilized a tapered center stud to provide positive retention of the segments during ablation. The studs all were machined from 3-in.-diam billets of extruded tungsten/2% thoria fabricated by G.E., Cleveland. The large aft segment on all models was machined from the same commercially pure tungsten used for the solid tare models.

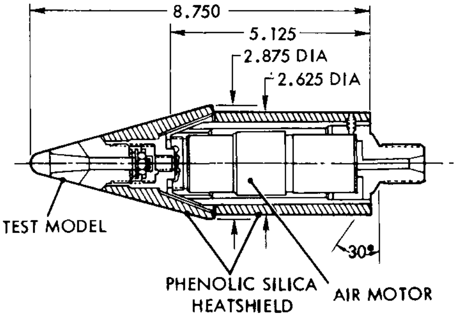

Fig. 11 Rotating sting assembly.

The 0.120-in.-thick segments were constructed by first cutting 0.25-in.-thick disks from the 2-5/8-in.-diam commercially pure tungsten billets. These disks then were upset-forged to 0.125-in. thickness, then coined to the final conical shape, and finish-machined to the 0.120-in. thickness. The 0.045-in.-thick segments were constructed from 0.060-in.-thick sheets of commercially pure tungsten supplied by Schwarzkopf Development Corporation. These segments were coined to the conical shape and then finish-machined to the final 0.045-in. thickness. The 0.010-in.-thick segments were coined from 0.010-in.-thick sheets of commercially pure tungsten supplied by AMAX, Inc. These segments were brazed together and then machined as a unit to the final conical shape. The small thickness made all handling and machining operations for these thin segments very difficult.

One of each type of ablation model was modified for installation on a rotating sting assembly.[9] In these tests, the models were rotated continuously throughout the test in an attempt to eliminate the effects of possible flowfield asymmetries on ablation and thermostructural performance. The rotating sting system is illustrated in Fig. 11.

All of the runs were made with the 1.38-in. exit diameter flared nozzle (10138F) with a reservoir pressure of 1500 psig. The nominal dwell time (model on flow axis) was 2 sec. Since the flow leaving the nozzle is expanding, the pressure continues to decline along the test-section centerline. At the test location of 0.1 in. from the nozzle exit plane, the model stagnation pressure is approximately 75 atm. In the mode of operation used in these tests, there is a strong radial variation of enthalpy, with the peak value occurring at the centerline. Since the flow at the edge of the model boundary layer comes from the hot central core, this produces an effective stagnation enthalpy of about 6000 Btu/lb.

Table 3 50-MW test recession data summary

Model no.	Model type	Rotating sting	Dwell time, sec	Total recession, in.	Steady-state recession rate, ips
1-1	Solid tungsten	Yes	1.88	0.765	0.483
2-2	Solid tungsten	No	1.82	0.607	0.427
4-2	0.010 segments	Yes	1.87	0.626	0.428
3-3	0.010 segments	No	1.90	0.808	0.433
4-5	0.010 segments	No	1.87	0.654	0.358
3-4	0.045 segments	Yes	1.98	0.750	0.432
2-3	0.045 segments	No	1.90	0.715	0.371
4-3	0.045 segments	No	1.86	0.629	0.343
3-2	0.120 segments	Yes	1.74	0.499	0.319
2-1	0.120 segments	No	1.85	0.553	0.350
4-1	0.120 segments	No	1.90	0.768	0.410
1-2	0.120 segments	No	1.81	0.781	0.465

A total of 12 models were tested, consisting of two solid tungsten (tare) models and 10 segmented models of the three configurations (segment thicknesses), as listed in Table 3. After completion of the tests, the models were inspected and photographed. Overall recession depths were obtained by comparisons of pretest and post-test measurements. Recession histories were obtained from the motion picture films; the films also were reviewed for loss of solid material and other significant events. The steady-state recession rates derived from the film data also are summarized in Table 3.

All of the models ablated satisfactorily in the test environment. However, the motion pictures showed that some distortion in the two nonspinning thin-sheet (0.010-in. segments) models occurred upon exit from the jet. During the dwell period at the jet centerline, the models ablated in a normal fashion. However, as they moved laterally through the jet boundary, the segments spread apart, and several segments were torn from each model. No similar spreading of segments occurred in the spinning version; however, the remnants of one 0.120-in. segment, which nearly had ablated away, were torn from the model at exit. No solid material loss was observed from any of the other models tested.

For a convenient comparison by model type, the steady-state axial recession rate data from Table 3 are plotted in bar-graph form in Fig. 12. The data from the rotating models are considered to be the more reliable indicators of the relative ablation performance expected in flight, since possible coupled effects of nose asymmetries

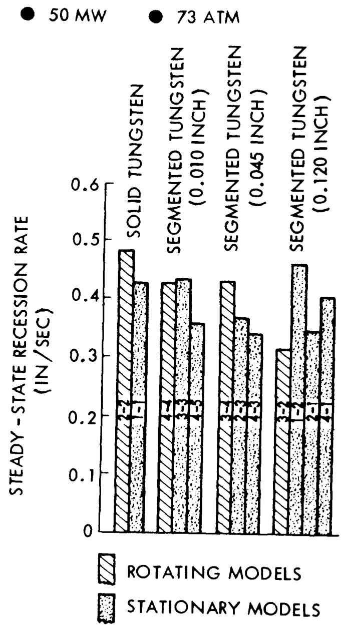

Fig. 12 Steady-state recession rates.

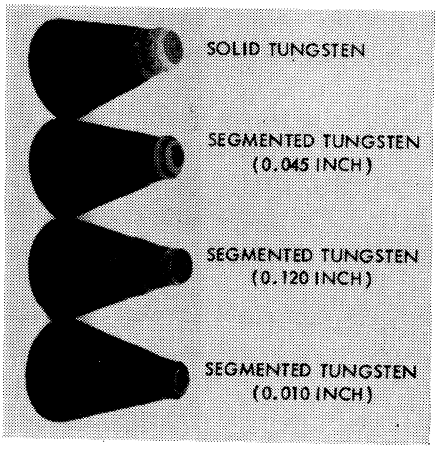

Fig. 13 Post-test photograph, solid and segmented tungsten 50-MW models.

on the test environment are precluded. The recession rates for all of the tests lie within a band of 0.4 ips $\pm$ 20%. The data suggest that the $\pm$ 20% scatter is random in nature and unrelated to the model type (i.e., normal scatter for this test facility). The scatter in the recession depths of the four 0.120-in. segment models, for example, is nearly as great as the scatter for the entire data set. The spread in the data for the spinning models is also of the same order as the scatter for the entire population. Thus, it is concluded that the axial recession performance of the segmented models was the same as the solid tungsten specimens within experimental error.

A post-test photograph comparing the three types of segmented models with the solid tungsten tare model is shown in Fig. 13. It is evident that, for all practical purposes, the ablated shapes of all four models are identical. There is no evidence of increased sidewall recession on any of the models. Thus, the results of the 50-MW test program indicate that the segmented construction technique should have no detrimental effect on the ablation performance of tungsten subtips.

Rocket Motor Test

The combined thermostructural and ablation performance of the tungsten subtip concept was evaluated in a rocket motor test. A full-scale nose-tip model (sketched in Fig. 1) was tested in the Air Force Rocket Propulsion Laboratory (AFRPL) Pad 1-52-C high-pressure rocket exhaust using the benzonitrile/liquid oxygen propellant system at an oxidizer/fuel mixture ratio of 2.8. The chamber pressure was 3000 psia, which, with the 5.78-in. exit diameter nozzle, produced a model stagnation pressure of 100 atm.

The nose tip had a FWCC shell primary nose tip, as illustrated in Fig. 1. The segmented tungsten subtip was constructed from forward-facing segments coined from cross-rolled pure tungsten plate. The subtip stud was machined from a swaged billet of thoriated tungsten, which was upset-forged to improve the transverse strength.

The thermal stress environment produced in the RPL facility at the described test condition is representative of high-performance

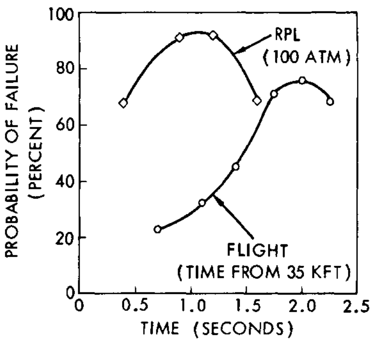

Fig. 14 Tungsten nosetip termostructural response.

Fig. 15 Post-test photograph, RPL test of monolithic tungsten subtip.

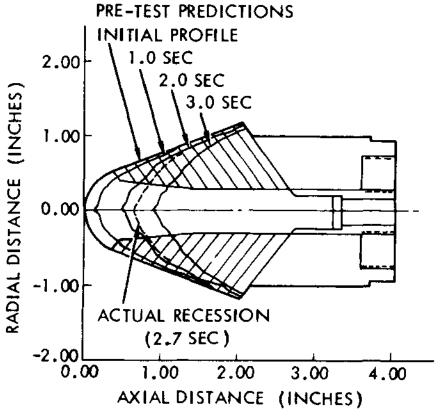

Fig. 16 Tungsten subtip ablation profiles, RPL test.

re-entry flight conditions, as illustrated in Fig. 14. The predicted probabilities of failure (fracture) as a function of time are shown for a monolithic tungsten subtip in flight and in the RPL test environment. (The analysis methodology was described previously.) As indicated, the RPL test condition is slightly more severe than the flight environment. This severity was verified in a series of tests[8] of monolithic tungsten subtips performed at the 100-atm test condition. A post-test photograph of a typical monolithic tungsten nose tip is shown in Fig. 15. The fractures were initiated during the period of high thermal stresses and became exposed as the external surface of the nose tip receded into the fracture zones. Exposure of the fracture surfaces in this manner eventually will result in loss of large pieces of material and subsequent catastrophic failure of the nose tip.

The segmented subtip was of nearly the same external configuration as the pretest shape of the monolithic nose tip and was exposed

Fig. 17 Post-test photograph, RPL test of segmented tungsten subtip.

to the same test environment. The model was tested for approximately 10.6 sec in the rocket exhaust flow at the 100-atm test condition. Removal of the FWCC primary nose tip shell occurred at about 7.9 sec due to burn-through in the stagnation region. The subtip recession during its 2.7 sec of exposure was in good agreement with the pretest predictions,[10] as illustrated in Fig. 16. The total axial recession depth was 0.63 in., and the first two segments were removed completely by ablation, in accordance with the pretest plans. The segmented tungsten subtip ablated as if it were a monolithic unit, with no indication of aggravated ablation at the segment boundaries. No post-test evidence of thermostructural failures was found in any components of the subtip. A photograph of the recovered model is shown in Fig. 17, clearly illustrating the improved performance of the segmented design compared to the monolithic design shown in Fig. 15.

Summary and Conclusions

Early erosion-resistant nose-tip (ERN) designs utilized monolithic subtips that were machined from extruded or swaged billets of tungsten. These subtips were found to be susceptible to thermal stress fractures, leading to possible catastrophic failure of the nose tips in high-performance re-entry flight. This behavior compromised the basic feasibility of the ERN concept and was due primarily to the low transverse strength of the tungsten in the billet forms. A secondary problem was the fact that the billets were available in a maximum diameter of 3 in., which severely limited the applicability of ERN's to some flight design applications.

A new design concept was developed which eliminated both of these problems by fabricating the subtips from a number of individual segments. This segmented design approach permits the use of much higher-strength tungsten products to fabricate the nose tip, and these products are available in essentially unlimited sizes. Thermal stresses developed in the segments are much lower than in a monolithic subtip because 1) the segments experience much lower temperature gradients as a result of their smaller size, and 2) stresses developed in the segments are relieved by compliance (i.e. deformation) of the segments. In addition, the segmented construction provides a form of artificial fracture toughness, since any cracks that might develop (due, for example, to an undetected flaw) can propagate only to a segment boundary and, therefore, would not induce an extensive failure of the nose tip assembly.

Two methods for fabricating the tungsten segments were investigated, and both were found to produce very high-strength material. The optimum method consists of forging disks from 92% dense pressed and sintered billets and then coining the disks into the final conical shape. The resultant material exhibited room-temperature tensile strengths approximately four to five times greater than the transverse strength of extruded tungsten. In addition, the critical ductile-brittle transition temperature (DBTT) of the forged and coined material is probably less than 250°F, compared to approximately 400°F for tungsten extrusions.

Two ground-test programs were conducted to validate the ablation and thermostructural performance of the segmented designs. An ablation test program was performed in the AFFDL 50-MW arcjet to determine if local increases in heating and recession would occur at the segment boundaries. Segmented models with three different segment thicknesses were tested and compared to a solid tungsten model of the same external configuration. The results indicated that the segmented construction had no detectable effect on the ablation performance in the 50-MW test environment.

A combined thermostructural and ablation test was performed in the AFRPL rocket motor facility. A full-size nose-tip configuration was tested which was nearly identical in size and shape to monolithic configurations tested previously at the same conditions. The test environment was representative of very high-performance re-entry flight conditions and had resulted in extensive thermostructural fractures in monolithic subtips. The segmented subtip experienced no detectable fractures, and, as in the case of the 50-MW tests, no increased heating or ablation was observed at the segment boundaries. Therefore, in view of these test results, it is concluded that the segmented tungsten subtip is a viable design concept for use in erosion-resistant nose tips on high-performance re-entry vehicles. It is recommended that final verification of the concept be obtained by means of a flight test on an ICBM test vehicle.

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2) Naval Surface Weapons Center (NSWC) Contract No. N60921-75-C-0236. Robert J. Edwards was the Contract Monitor for NSWC.

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All of the segment processing work was performed at Northwest Industries, Inc., Albany, Ore., under the direction of H. L. Kuebrich. The flexure test program was performed at Southern Research Institute, Birmingham, Ala., under the direction of H. S. Starrett.

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NOSE-TIP SHAPE OPTIMIZATION FOR MINIMUM TRANSPIRATION COOLANT REQUIREMENTS

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Abstract

Expressions for the total heat transfer to a nose tip for a realistic re-entry trajectory have been obtained by using local laminar and turbulent convective heat-transfer relationships and extending the trajectory integration procedures of earlier investigators. Then, employing approximate relationships for the boundary-layer edge properties along the nose-tip surface, we utilize a variational calculus approach to determine nose-tip shapes that minimize the total trajectory heat transfer (coolant requirements) for fixed fineness ratio and re-entry parameters. The relative heat-transfer rate to a particular one of the minimum heat-transfer shapes was found to be a factor of up to 3 times lower than that for other body shapes having the same fineness ratio.

I. Introduction

For some applications, a decrease in the amount of coolant required for a transpiration-cooled re-entry vehicle nose tip is desirable and potentially very beneficial. The work presented here seeks to optimize the shape of a re-entry vehicle nose tip in order to minimize the transpiration coolant requirements for a complete, realistic re-entry trajectory. The amount of coolant required is assumed to be the ideal amount, i.e., directly proportional to the total heat transfer. It is also assumed that the nose-tip shape remains constant throughout the trajectory.

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The majority of work devoted to the determination of optimum body shapes in a hypersonic flow environment has been concerned with the problem of finding minimum drag bodies.[1] Recently, however, work has appeared which considers the determination of minimum heat-transfer body shapes in a constant freestream environment. Laminar flow solutions were found by Belyanin[2] and Aihara.[3] Belyanin discussed the relative heat flux to his minimum heat-transfer solution bodies but did not give any details of the body shapes. Aihara's solution has been shown by Hull[4] not to be a minimum solution. Perminov and Solodkin[5] obtained nonslender body solutions for both minimum total drag and minimum heat transfer for both laminar and turbulent flow. They found the laminar and turbulent minimum heat-transfer shapes to be very similar to one another for a given fineness ratio. The work presented here extends this previous work to the case of a re-entry trajectory environment and seeks to minimize the total heat transfer to the nose tip for the entire trajectory.

Section II first presents a brief derivation of the total heat-transfer relationships used in this work and the approximate flowfield evaluation procedures utilized. Then, in Sec. III, the mathematical formulation of the minimum heat-transfer problem using a variational calculus approach is discussed. Calculated results then are given which discuss both minimum and maximum heat-transfer solutions and the relative heat-transfer rates to these shapes compared with several other body shapes.

II. Heat-Transfer Prediction Methods

Integrated Heat-Transfer Expressions

When expressions for the laminar and turbulent convective heat-transfer coefficient given by Vaglio-Laurin[6,7] are used and the procedure of Allen and Eggers[8] is followed, the rate of change of integrated heat transfer with altitude dQ/dy is given by

$$\frac{dQ}{dy} = \frac{C_n \rho_\infty V_\infty^2 \pi r_B^2}{Re_o^{n/n+1} \sin\theta_E} \int_0^{\bar{s}} \frac{\bar{\rho}_e \bar{u}_e \bar{\mu}_e \bar{r}^{n+1} d\bar{s}}{\left[\int_0^{\bar{s}} \bar{\rho} \bar{u}_e \bar{\mu}_e \bar{r}^{n+1} d\bar{s}\right]^{n/n+1}} \quad (1)$$

where n = 1 for laminar flow and n = 1/4 for turbulent flow. When the integration in nondimensional surface distance

$\bar{s} = s/r_B$ is performed, this expression becomes

$$\frac{dQ}{dy} = \frac{-(n+1)C_n \rho_\infty V_\infty^2 \pi r_B^2}{Re_o^{n/n+1} \sin\theta_E} I_Q^{1/n+1} \tag{2}$$

where the integral I_Q is given by

$$I_Q = \int_0^{\bar{s}} \bar{\rho}_e \bar{u}_e \bar{\mu}_e \bar{r}^{n+1} d\bar{s} \tag{3}$$

Allen and Eggers integrated an expression analogous to Eq. (2), representing ρ_∞ and V_∞ as functions of altitude y from their trajectory analysis and assuming Re_o to be constant. In the present analysis, the shock-layer Reynolds number Re_o is represented as a function of y. Then, following the general integration procedure of Brunner,[9] the total integrated heat transfer to a body throughout a trajectory is given for laminar flow by

$$Q^L = C^L f_E^L r_B^{3/2} \left[\beta I_Q^L\right]^{1/2} \operatorname{erf}\left(\sqrt{B^L}\right) \tag{4}$$

and for turbulent flow by

$$Q^T = C^T f_E^T r_B^{9/5} \left[\beta I_Q^T\right]^{4/5} \int_0^{[B^T]^{4/5}} e^{-t^{5/4}} dt \tag{5}$$

The complete derivation of Eqs. (4) and (5) is given in Ref. 10. The β and r_B in these equations are the ballistic coefficient and the base radius, respectively. The other parameters are given by

$$C^L = 7.39 \left\{ \frac{[(\gamma+1)/(\gamma-1)]}{(2\omega+3)\lambda} \right\}^{1/2} \tag{6a}$$

$$C^T = \frac{0.892}{\lambda^{1/5}} \left\{ \frac{[(\gamma+1)/(\gamma-1)]}{(2\omega+9)} \right\}^{4/5} \tag{6b}$$

$$f_E^L = \frac{\mu_E^{1/2}\, V_E^{3/2}}{(\sin\theta_E)^{1/2}} \tag{7a}$$

$$f_E^T = \frac{\mu_E^{1/5}\, V_E^{9/5}}{(\sin\theta_E)^{1/5}} \tag{7b}$$

$$I_Q^L = \int_0^{\bar{s}} \bar{\rho}_e \bar{u}_e \bar{\mu}_e \bar{r}^2 d\bar{s} \tag{8a}$$

$$I_Q^T = \int_0^{\bar{s}} \bar{\rho}_e \bar{u}_e \bar{\mu}_e \bar{r}^{5/4} d\bar{s} \tag{8b}$$

$$B = [\rho^o/(\lambda\beta\sin\theta_E)] \tag{9a}$$

$$B^L = [(2\omega+3)/4]B \tag{9b}$$

$$B^T = [(2\omega+9)/10]B \tag{9c}$$

The variables in these equations are the air specific heat ratio and viscosity law exponent γ and ω; the density scale height factor in the exponential atmosphere λ; the re-entry velocity and entry flight path angle V_E and θ_E; the air viscosity evaluated at the entry stagnation temperature μ_E; the ballistic factor B, and the heat-transfer integrals I_Q^L and I_Q^T, which depend upon an integration in body surface running length of the local boundary-layer edge density $\bar{\rho}_e$, velocity $\bar{u}_e$, viscosity $\bar{\mu}_e$, and local radial coordinate $\bar{r}$.

It now is evident that, if we fix f_E^L, r_B, β, and B^L in Eq. (4), then the total heat transfer Q^L will be minimized if we minimize the integral I_Q^L. Similarly, in the turbulent case Q^T is minimized by minimizing the integral I_Q^T.

Evaluation of I_Q^L and I_Q^T

The heat-transfer integrals I_Q^L and I_Q^T, given by Eqs. (8a) and (8b), respectively, depend intimately upon the nose-tip geometry, i.e., upon the shape contour of the nose tip.

As shown in Fig. 1, the local nose-tip coordinates r and z and the surface running length s are related to the local nose-tip surface angle α by the following simple trigonometric relationships:

$$dr/dz = \tan\alpha, \quad dr/ds = \sin\alpha \tag{10}$$

To a reasonably good approximation, the local nondimensional boundary-layer edge properties $\bar{\rho}_e$, $\bar{u}_e$, and $\bar{\mu}_e$ in Eqs. (8a) and (8b) may be represented as functions of α. In the present work, the simplest equations of this type have been used, i.e., the hypersonic Newtonian approximations for both pressure and velocity. Thus, the local pressure $\bar{p}_e = p_e/p_{t_2}$ is given by

$$\bar{p}_e = \sin^2\alpha \tag{11}$$

and the local velocity $\bar{u}_e = u_e/\sqrt{2H_o}$ is given by

$$\bar{u}_e = \cos\alpha \tag{12}$$

If the viscosity is assumed to be proportional to the temperature and independent of pressure, the density-viscosity product then is directly proportional to the pressure, i.e.,

$$\bar{\rho}_e\bar{\mu}_e = \rho_e\mu_e/\rho_o\mu_o = \sin^2\alpha \tag{13}$$

The utility rendered by the simplicity of these equations was felt to compensate for inherent inaccuracies associated with them for purposes of comparing relative heat-transfer rates to various nose-tip shapes as discussed herein. Work is currently in progress to reconsider the problem discussed in this paper and to replace the approximations given by Eqs. (11-13) with numerical inviscid flowfield calculation procedures. Numerical flowfield calculations for fixed body shapes somewhat similar to those discussed here, i.e., flat-face cones, are given in Ref. 10.

Local Properties on a Flat Face

In order to allow for solutions that may have a flat face, it is necessary to have an alternate expression for the boundary-layer edge velocity $\bar{u}_e$. This is because direct use of Eq. (12) would indicate that $\bar{u}_e$ and, therefore, I_Q^L and I_Q^T are all identically zero on a flat face. In Refs. 2 and 5, the following approximation for the $\rho_e u_e \mu_e$ product, obtained from a correlation of method of integral relations solutions, was used:

$$\rho_e u_e \mu_e / [\rho_e u_e \mu_e]^* = \bar{r}/\bar{r}^* \tag{14}$$

Equation (14) indicates that, for a flat-faced body, the normalized $\rho_e u_e \mu_e$ product varies linearly with distance to the sonic point. The sonic point values of the dependent variables may be obtained by isentropic expansion from the stagnation conditions. The terms ρ_e, u_e, and μ_e in Eq. (14) then may be renormalized by the stagnation values ρ_o, $\sqrt{2H_o}$, μ_o. Using ideal gas relationships, we obtain

$$\bar{\rho}_e \bar{u}_e \bar{\mu}_e = \frac{\rho_e u_e \mu_e}{\rho_o (2H_o)^{1/2} \mu_o} = F(\gamma) \frac{\bar{r}}{\bar{r}^*} \tag{15a}$$

$$F(\gamma) = \left[\frac{\gamma+1}{2}\right]^{\gamma/1-\gamma} \left[\frac{\gamma-1}{\gamma+1}\right]^{1/2} \tag{15b}$$

In Sec. III, Eqs. (10-15) are used to express the heat-transfer integral I_Q explicitly in terms of the body geometry.

III. Variational Problem

Mathematical Formulation

This section will be concerned with the determination of those shapes that in a limited sense produce extreme values of I_Q. Body shapes with a flat forward face of height r^*, possibly zero, will be considered (see Fig. 1). Normalizing all coordinates by r_B, define

$$x = z/r_B, \quad y = r/r_B \tag{16}$$

The Newtonian flow approximations, Eqs. (12) and (13), and the geometric relationships, Eqs. (10), may be substituted into the expressions for I_Q, Eqs. (8a) and (8b), to produce the following expression for I_Q along the body:

$$[I_Q]_{body} = \int_0^{2/\tau} \frac{y^{n+1} \dot{y}^2}{1 + \dot{y}^2} dx, \quad \dot{} = \frac{d}{dx} \tag{17}$$

Similarly, integrating Eq. (8), using approximation (15a), over a flat face yields

$$[I_Q]_{face} = [F(\gamma)/(n+3)] \, [y(0)]^{n+2} \tag{18}$$

Thus, the total value of I_Q may be expressed as

$$I_Q = \frac{F(\gamma)}{n+3}[y(0)]^{n+2} + \int_0^{2/\tau} \frac{y^{n+1}\dot{y}^2}{1+\dot{y}^2}\,dx \tag{19}$$

Since τ is defined to be the ratio of the diameter to the length of the body,

$$y(2/\tau) = 1 \tag{20}$$

We wish to determine the function y(x), chosen from a suitable class of functions on the interval $0 \leq x \leq 2/\tau$, which minimizes (or maximizes) the functional I_Q, Eq. (19), and which satisfies the condition (20). Any such minimum (or maximum) must satisfy the Euler equation associated with this problem:

$$\ddot{y}\, y[1-3\dot{y}^2] = -[(n+1)/2]\dot{y}^2[1-\dot{y}^4] \tag{21}$$

The boundary condition $y(2/\tau) = 1$ is given, but y(0) is not specified. Hence, this is a variable end-point problem of the calculus of variations, and the following transversality condition must be satisfied at x = 0:

$$y^{n+1}\left[\frac{\dot{y}}{(1+\dot{y}^2)^2} - \frac{n+2}{2(n+3)}F(\gamma)\right] = 0 \tag{22}$$

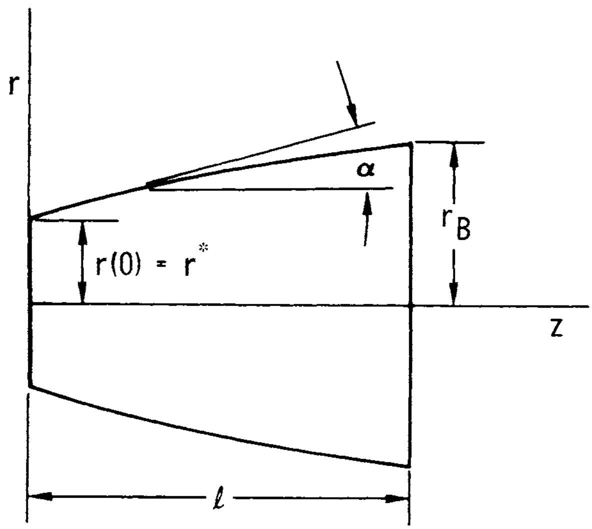

Fig. 1 Nose-tip geometry and coordinate system.

If a flat-faced, $y(0) \neq 0$, solution exists, then [from Eq. (22)] the initial slope must be a solution of the equation

$$\frac{\dot{y}(0)}{[1 + \dot{y}(0)^2]^2} = \frac{(n+2)}{2(n+3)} F(\gamma) \tag{23}$$

This equation is a quartic in $\dot{y}(0)$. An examination of this quartic for the range of parameters (n, γ) of interest reveals two real positive roots and a pair of complex conjugate roots. Thus, there are apparently two functions $y(x)$ which satisfy the Euler equation and the transversality condition. It should be observed that the values of the two permissible initial slopes are independent of τ.

One might expect that the two solutions for $y(x)$ discussed in the preceding paragraph represent a minimum and a maximum of I_Q. There are further checks that may be

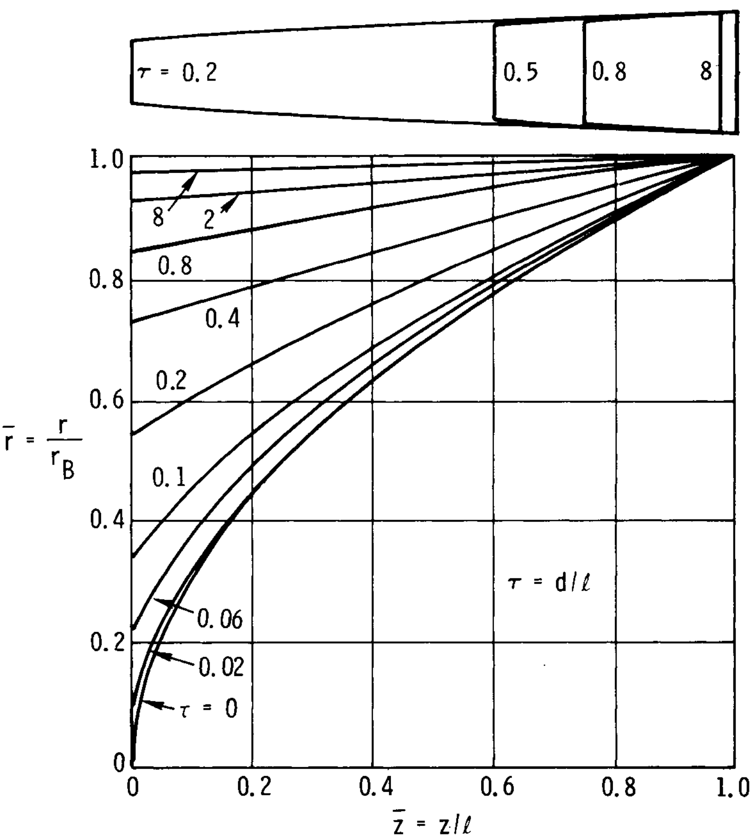

Fig. 2 Minimum laminar heat-transfer shapes.

made to verify or limit this conclusion. The necessary condition of Legendre implies that one of the two solutions for $y(x)$ is a minimum (maximum) if $y(x) \leq \sqrt{3}/3$ $[y(x) \geq \sqrt{3}/3]$ for all x in the interval $0 \leq x \leq 2/\tau$. The Jacobi condition is somewhat difficult to apply. However, it can be shown that the necessary condition of Weierstrass cannot be satisfied over the class of all possible shapes. Hence, neither of the preceding solutions corresponds to a minimum or a maximum of I_Q over the class of all piecewise continuously differentiable functions on the interval $0 \leq x \leq 2/\tau$ which satisfy the boundary condition, Eq. (20). Hull[4] has pointed out a similar occurrence in a related problem.

Interpretation of Solutions

The solutions discussed in the preceding subsection are insufficient as global extrema. The two solutions that satisfy the Euler equation [Eq. (21)], the transversality condition [Eq. (23)], and the boundary condition [Eq. (20)] will be referred to as a "minimum" solution or a "maximum" solution provided they satisfy the Legendre condition for a maximum or a minimum.

Examination of the Weierstrass condition reveals that these solutions represent extrema over a rather severely restricted class of functions. However, after obtaining the solutions, it was found that this class of functions could be enlarged. For instance, it was observed that the calculated maximum solutions had the property that $\dot{y}(x) \geq 1$ for all x. Examination of the Weierstrass excess function for this problem leads to the following conclusion. If a maximum solution is found whose derivative is greater than or equal to one for all x, then the I_Q corresponding to this solution will be larger than the I_Q corresponding to any piecewise continuously differentiable function $y(x)$ on the interval $0 \leq x \leq 2/\tau$ for which $y(2/\tau) = 1$ and for which $\dot{y}(x) \geq 0$ for all x in the interval.

When a minimum solution is found, its maximum derivative $\dot{y}_{max}$ over the interval $0 \leq x \leq 2/\tau$ may be determined. If the constant A is defined by

$$A = (1-\dot{y}^2_{max})/(2\dot{y}_{max}) \tag{24}$$

then examination of the Weierstrass E function for the problem leads to the following situation. If a minimum solution is found with maximum derivative $\dot{y}_{max}$ and constant A, then the I_Q corresponding to this solution will be smaller than the

I_Q corresponding to any piecewise continuously differentiable function $y(x)$ on the interval $0 \leq x \leq 2/\tau$ for which $y(2/\tau) = 1$ and for which $\dot{y}(x) \leq A$ for all x in the interval. It is in this limited sense that the solutions given in the following sections may be considered minima and maxima.

Computation of Solutions

The actual computation of a solution consists of solving a two-point boundary-value problem for a second-order nonlinear ordinary differential equation. The differential equation is the Euler equation (21). The boundary conditions are $y(2/\tau) = 1$ and at $x = 0$ the transversality condition (23). A program was written to solve a class of such problems, including the one discussed herein. In this program, the differential equations and boundary conditions were replaced by finite-difference approximations, and the resulting equations were solved by Newton's method (quasilinearization).

IV. Results and Discussion

Results obtained by numerical solution of the Euler equation with appropriate boundary conditions, as discussed previously, are presented here for the cases of both laminar and turbulent flow in the boundary layer, i.e., $n = 1$ and $n = 1/4$, respectively. In all cases, the specific heat ratio γ has been assumed to be the hypersonic value of 1.2. In the first subsection below, the body shapes and minimum heat-transfer integrals I_Q^L and I_Q^T are shown for a wide range of the fineness ratio τ. Then, in the next subsection, the minimum heat-transfer results are compared with heat-transfer rates to other selected body shapes and to some maximum heat-transfer solutions.

Minimum Heat-Transfer Solutions

Laminar flow minimum heat-transfer shapes for several values of the fineness ratio τ are shown in Fig. 2, where the shapes as they appear physically are shown in the upper part of the figure. In the lower part of the figure, the shapes are shown with the radial coordinate normalized by the base radius and the axial coordinate normalized by the body length. In this way, the maximum value of both $\bar{r} = r/r_B$ and $\bar{z} = z/\ell$ is unity, since two reference lengths have been used. In the remainder of this paper, all body shapes will be presented normalized as in the lower part of Fig. 2. It should be kept in mind that presenting results in this way masks the needlelike appearance of shapes with very small fineness ratios τ as well as the blunt disklike appearance of shapes with large fineness ratios.

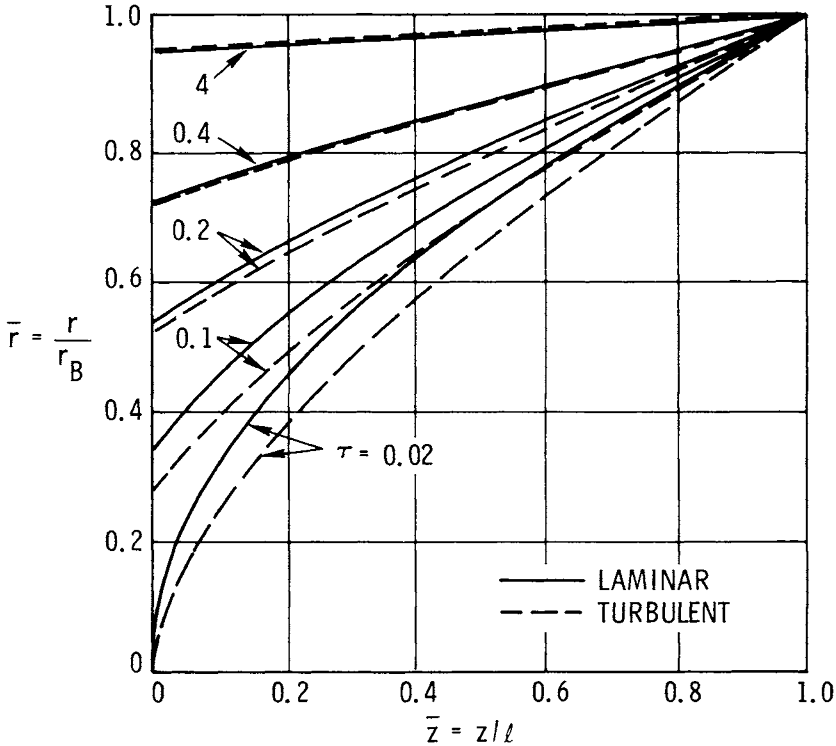

Fig. 3 Minimum laminar and turbulent heat-transfer shapes.

All of the shapes in Fig. 2, for nonzero values of τ, exhibit a flat face. The ratio of the radius of the flat face to that of the base increases with increasing τ. In the limit of τ approaching infinity, the shape becomes a flat-faced cylinder of zero length, i.e., a disk. In the opposite limit, as τ approaches zero, the flat-face height approaches zero, and the shape approaches the shape obtained invoking the slender-body approximation, i.e., a 1/2 power-law body.[4] However, since the slender-body power-law shape is approached only for fineness ratios less than 0.02, this shape appears to be of little practical interest.

In Fig. 3, turbulent flow minimum heat-transfer shapes are shown and compared with the laminar shapes for corresponding fineness ratios. As concluded by others,[5] the turbulent shapes are surprisingly similar to the laminar shapes for a given fineness ratio. It does not follow, however, that for the case of mixed laminar-turbulent flow over the nosetip, the shape would again be similar for given τ. In fact, from Eq. (19) for laminar-turbulent flow with transition at the corner, the flat-face height will be greater than that for the all laminar or all turbulent case. Similarly, for turbulent-laminar flow with relaminarization at the corner, the flat-face height will be less. No calculations of this type have been carried out.

In the limit of large τ, the turbulent minimum heat-transfer shape also approaches a flat-faced cylinder; in the limit of τ approaching zero, it can be shown easily that the turbulent minimum heat-transfer shape is an 8/13 power-law body.[11] Thus, in the slender-body limit the minimum laminar heat-transfer shape is a 1/2 power-law body, the minimum turbulent heat-transfer shape is an 8/13 power-law body, and[12] the minimum drag shape is a 3/4 power-law body. The nonslender-body minimum heat-transfer shapes found in the present work approach the slender-body minimum heat-transfer shapes only at very small fineness ratios ($\tau < 0.02$). However, the nonslender-body minimum drag shapes approach a 3/4 power-law body at much larger fineness ratios[12] ($\tau \approx 0.2$).

The question of how much reduction in the heat transfer is achieved by a minimum heat-transfer shape is considered in the results presented in Figs. 4-6 and in the following subsections. Recall that Eqs. (4) and (5) represent expressions for the integrated heat transfer to a nose tip throughout a trajectory. The parameter needed to obtain numerical

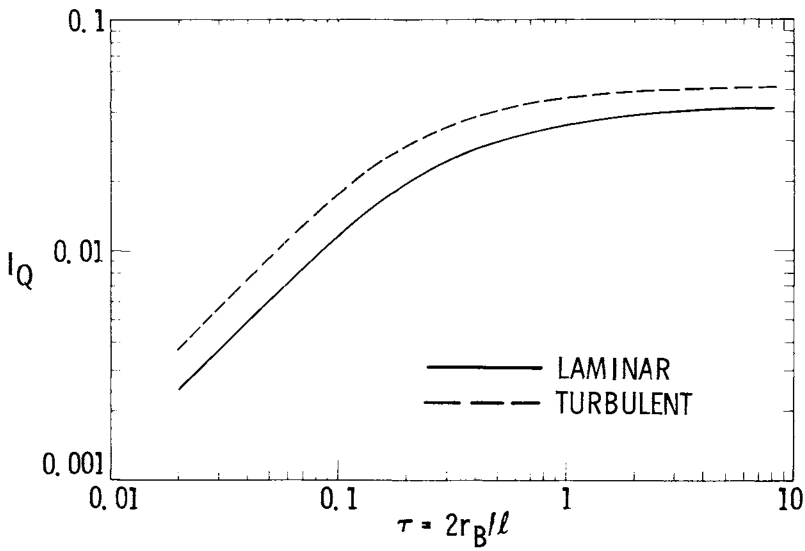

Fig. 4 Minimum I_Q as a function of fineness ratio.

predictions of heat transfer is the heat-transfer integral or body shape parameter I_Q.

In Fig. 4, I_Q for the minimum laminar and turbulent heat-transfer shapes is shown as a function of τ. The I_Q for both laminar and turbulent flow is quite insensitive to fineness ratio for $1 < \tau < 10$. This is because the ratio of flat-face height to base radius is greater than 0.9 for all $\tau > 1$. Thus, for $\tau > 1$, a large fraction of the total heat transfer occurs over the flat face. However, for $\tau < 1$, the I_Q^L and I_Q^T values decrease significantly as the body or nose tip becomes more slender, i.e., decreasing τ. This decrease is due in large part to a significant reduction in the flat-face height, which decreases the area associated with the relatively high flat-face heat transfer.

In the limit of τ approaching zero, I_Q^L and I_Q^T both go to zero. However, I_Q^L/τ and I_Q^T/τ are finite for $\tau = 0$, and the limiting values are given by slender-body theory.[11] They are

$$I_Q^L/\tau = 1/8 \tag{25a}$$

$$I_Q^T/\tau = 32/169 \tag{25b}$$

Comparing the numerical values of I_Q^L and I_Q^T from these equations with the nonslender-body calculated values in Fig. 4, we see that for $\tau \leq 0.02$ the slender- and nonslender-body calculated values are virtually identical. A quantitative comparison of minimum heat-transfer rates to those for other body shapes is given below.

Comparison with Other Body Shapes

The minimum heat-transfer body shapes and heat-transfer integrals just discussed,now will be compared with these quantities for a sphere, truncated spheres, sharp cones, minimum drag bodies, stable ablating shapes, and maximum heat-transfer bodies.

The minimum and maximum heat-transfer solutions were obtained directly by numerical solution of the Euler equation, as discussed previously. The minimum drag body solutions were obtained by application of the present numerical calculation procedure to the appropriate equations as given in Ref. 12. A stable ablating shape is one for which, even though it is ablating and its surface is receding, the

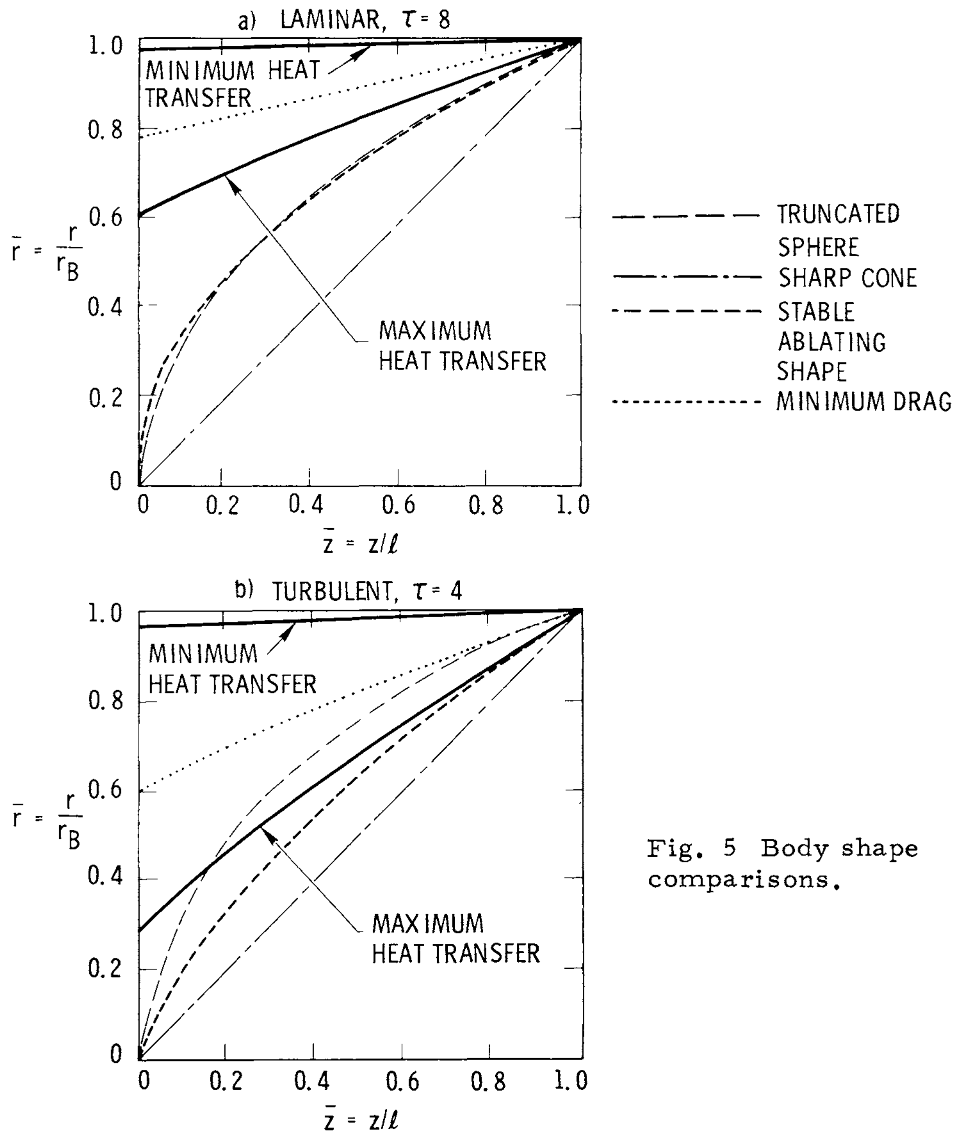

Fig. 5 Body shape comparisons.

body shape contour is independent of time.[13] The following analytic equations for laminar and turbulent stable ablating nose-tip shapes, using the approximations given by Eqs. (11-13) were derived in Ref. 11. For laminar flow,

$$\frac{2}{\tau}\bar{z} = \ell n\left[\frac{1 - \sqrt{1-\bar{r}^2}}{2}\right] - \sqrt{1-\bar{r}^2} + 1 \qquad (26a)$$

and for turbulent flow,

$$\frac{2}{\tau}\bar{z} = \frac{4}{3}\bar{r}^{3/4} + \left\{\frac{1}{2}\left[\bar{r}^{1/4}\sqrt{1-\bar{r}^{1/2}} - \sin^{-1}\left(\bar{r}^{1/4}\right)\right]\right\} - \bar{r}^{1/4}\left[1-\bar{r}^{1/2}\right]^{3/2} \tag{26b}$$

The shape contours for the six body types just listed are compared in Figs. 5a and 5b for fineness ratios of 8 and 4 for laminar and turbulent flow, respectively. The laminar stable ablating nose-tip shape in Fig. 5a most closely resembles the truncated sphere shape, whereas the turbulent stable shape in Fig. 5b is similar to the maximum heat-

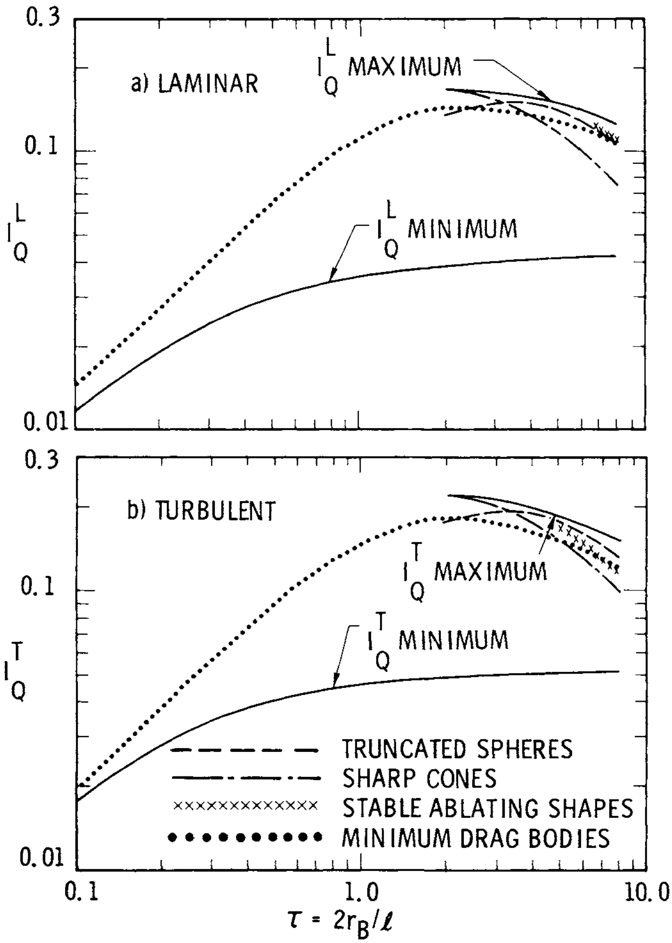

Fig. 6 Minimum and maximum I_Q values compared with other body shapes.

transfer shape. The flat-nose height is smallest for the maximum heat-transfer body, somewhat larger for the minimum drag body, and largest for the minimum heat-transfer body.

The heat-transfer integrals I_Q^L and I_Q^T are shown for these body shapes in Figs. 6a and 6b, respectively. The relative I_Q values for the different body shapes are generally similar for laminar and turbulent flows. The maximum solution I_Q values are up to four times greater than the minimum solution values. This means that the heat transfer for the laminar maximum solutions is up to two times greater than the laminar minimum solution values. The corresponding heat-transfer increase for the minimum to maximum turbulent solutions is up to three times greater.

The I_Q values for truncated spheres, sharp cones, stable ablating shapes, and minimum drag bodies all lie between the minimum and maximum solution values but tend to be much closer to the maximum value for a given τ. An exception is the I_Q for minimum drag bodies for small τ. The I_Q for these bodies peaks for $\tau = 2$ and decreases for all other fineness ratios. Maximum heat-transfer solutions could not be found for $\tau < 2$.

Solution Characteristics and Accuracy

Since many of the extremal solution body shapes discussed here have a flat nose followed by a body contour, it is of interest to compare these solutions in terms of the body angle immediately aft of the flat face and the ratio of flat-face height to base radius. For minimum heat transfer, minimum drag, and maximum heat-transfer bodies, the body angle aft of the flat face is about 4, 45, and 66 deg, respectively, independent of τ and essentially independent of the boundary-layer state, i.e., laminar or turbulent. This is

Table 1 Body angle aft of expansion corner

	Laminar	Turbulent
Minimum heat-transfer solution	$3^\circ 41'$	$3^\circ 24'$
Minimum drag solution	45°	45°
Maximum heat-transfer solution	$65^\circ 39'$	$66^\circ 23'$

shown in Table 1. Caution must be exercised concerning the maximum solutions. Although the local flow at the expansion aft of the flat face is indicated to be supersonic using the present approximations, more exact calculations could indicate subsonic flow at this location. The present formulation assumes the flow to be supersonic at the expansion corner. If this condition is not met, the proof of the existence of maximum solutions becomes much more difficult, and the present solutions are not valid.

The flat-face/base-radius ratio for minimum heat transfer, maximum heat transfer, and minimum drag shapes is shown in Fig. 7 as a function of fineness ratio. This ratio increases from essentially zero to 0.6-0.7 for τ increasing from 2 to 10 for the maximum heat-transfer solutions. For the minimum drag body, the flat-face height increases over the same range for fineness ratios from 0.2 to 10, and for the minimum heat-transfer body the τ range is about 0.02 to 10 for the same increase in flat-face height.

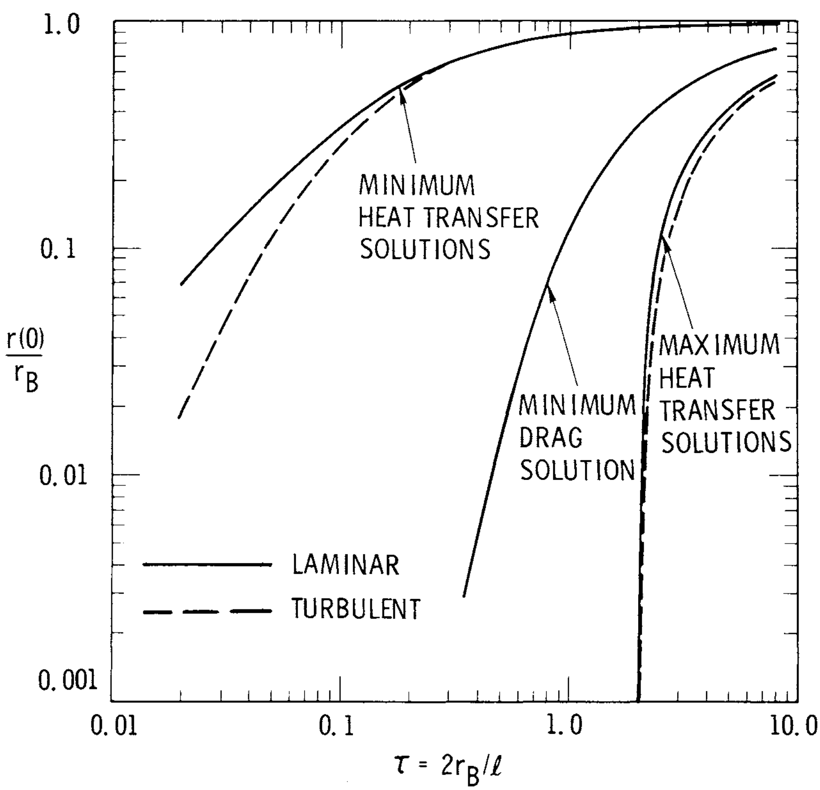

Fig. 7 Flat-face height vs fineness ratio.

A final word of caution is in order regarding the accuracy of the predicted reduction in heat transfer associated with the minimum heat-transfer shapes. Calculations using numerical inviscid flowfield computer codes for flat-face cones were carried out recently by Baker and Kramer.[10] These calculations indicate that, although the local flow along the surface is highly overexpanded immediately aft of the flat-face expansion corner, the flow rapidly recompresses to pressure levels considerably above those for the same station on a sphere-cone. Because of this behavior, the reduction in heat transfer attributable to minimum heat-transfer-type shapes, when calculated using numerical flowfield calculation methods, is expected to be a factor of up to 2 rather than the factor of 3 predicted by the present work.

V. Summary and Conclusions

Analytic expressions for total integrated heat transfer to a nose tip through a trajectory and variational calculus procedures have been used to determine nose-tip shape contours having minimum total heat transfer for specified fineness ratio. Solutions for both laminar and turbulent boundary-layer flow were obtained by numerical solution of the appropriate Euler equation and boundary conditions.

For fineness ratio τ greater than zero, all solutions found have a flat face. The ratio of flat-face height to the base radius increases as τ increases. As τ becomes large ($\tau > 10$), both the laminar and turbulent solutions approach a flat-faced cylinder of vanishing length, i.e., a disk. As τ approaches zero ($\tau < 0.02$), the solutions approach the slender-body ($\tau = 0$) solutions, which are a 1/2 power-law body for laminar flow and an 8/13 power-law body for turbulent flow. For a constant base radius, the relative heat-transfer rates to a family of minimum heat-transfer shapes decrease monotonically as τ decreases, i.e., as the bodies become more slender. For $\tau \geq 0.2$, the laminar and turbulent shapes are surprisingly similar to one another; i.e., the flat-face height to base radius ratio and the afterbody shape contour are essentially independent of the boundary-layer state.

Additional solutions to the Euler equation depicting maximum heat-transfer body shapes were found for $\tau \geq 2$. The heat transfer to the maximum solution shapes is up to two times greater than for the minimum solution shapes for laminar flow and up to three times greater for turbulent flow for $2 < \tau < 8$. Laminar and turbulent heat-transfer rates to sharp cones, spheres, truncated spheres, and stable ablating shapes generally were found to be much closer to the maximum solution values than to the minimum solution values. The laminar and turbulent heat transfer to a minimum drag body also was calculated.

The minimum and maximum solutions obtained could be shown mathematically to be minima and maxima only over restricted classes of body shapes. However, of all of the physically reasonable additional body shapes considered, none were found which result in heat-transfer rates lower than the minimum solution or greater than the maximum solution values.

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